

Fuzzy Domain Theory via Enriched Category Theory

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Part I

Fuzzy Domain Theory

Chapter 1

Fuzzy Predicates and Cut Subobjects

1.1 Introduction

This chapter develops the categorical framework for fuzzy predicates and their associated cut subobjects. The guiding idea is that a system of fuzzy truth values should be treated not merely as a set of degrees, but as an enriched base. A fuzzy predicate on a fuzzy preorder X is then a contravariant truth-valued enriched functor

$$X^{\text{op}} \longrightarrow \Lambda,$$

and the collection of all such predicates forms the enriched presheaf object

$$P_{\Lambda}(X) = [X^{\text{op}}, \Lambda].$$

The chapter has four main purposes. First, we fix the base structures used throughout the manuscript: fuzzy cosmoses in the enriched-categorical setting and fuzzy truth-value objects in the posetal, quantale-enriched setting. Second, we construct tensor products, internal homs, and the truth-value fuzzy preorder, so that fuzzy predicates can be treated as ordinary enriched presheaves. Third, we show that the predicate object $P_{\Lambda}(X)$ behaves as a fuzzy power object: it carries a generic fuzzy predicate and classifies indexed fuzzy predicates by the tensor–hom adjunction. Finally, we prove a cut-subobject representation theorem, identifying fuzzy predicates with coherent systems of level cuts.

In the crisp two-valued case, this construction reduces to the familiar correspondence between lower subsets of a preorder and their characteristic maps. For standard chain-valued bases such as the Gödel, Łukasiewicz, product, and Lawvere quantales, the same formalism yields the corresponding notions of fuzzy lower predicates, cost-valued potentials, and level-cut filtrations. The point of the enriched formulation is that these examples, including non-chain truth-value bases, are treated uniformly.

1.2 Fuzzy cosmoses, truth values, and fuzzy preorders

This section fixes the categorical and order-theoretic framework used throughout the manuscript. We begin with the enriched-categorical setting: a fuzzy cosmos is an affine symmetric monoidal closed

base, regarded as a categorified object of fuzzy truth values. The underlying enriched category theory follows the standard conventions of Kelly [10, Sections 1.1–1.6]; the posetal specialization is the theory of quantale-enriched categories [16, Definitions 2.1 and 2.2].

We then specialize to the posetal case, where the enriched structure becomes a quantale of fuzzy truth values. The resulting formalism covers ordinary preorders, standard chain-valued semantics for fuzzy logic, Lawvere metric spaces, and non-chain bases within a single framework; see also Lawvere’s original treatment of metric spaces as enriched categories [12] and Stubbe’s survey [16, Examples 2.3 and 2.4].

Set-theoretic conventions. We work relative to a fixed tower of Grothendieck universes

$$\mathbb{U}_0 \in \mathbb{U}_1 \in \mathbb{U}_2.$$

Unless otherwise specified, **small** means $\mathbb{U}_0$ -small, **large** means $\mathbb{U}_1$ -small, and all constructions are formed inside $\mathbb{U}_2$.

All ordinary sets underlying fuzzy preorders are assumed small. The truth-value objects used in the posetal part of the manuscript are also assumed small. Thus, when L is the underlying set of truth values, joins such as

$$\bigvee_{a \in L} u_a$$

are small joins.

A categorical base $\mathbb{V}$ may be large, but is always assumed locally small: its ordinary hom-sets are small. Completeness and cocompleteness of a categorical base always mean existence of small limits and small colimits.

If X is a possibly large $\mathbb{V}$ -category, we write

$$\mathcal{P}(X)$$

for the $\mathbb{V}$ -category of small $\mathbb{V}$ -presheaves on X , that is, those presheaves

$$X^{\text{op}} \longrightarrow \mathbb{V}$$

obtained as small weighted colimits of representables. Accordingly, the Yoneda embedding is always interpreted as a functor

$$Y : X \longrightarrow \mathcal{P}(X)$$

into small presheaves. This is the universe-bounded version of the presheaf and Yoneda formalism for enriched categories [10, Sections 1.6 and 1.9].

Accordingly, a coend indexed by a possibly large $\mathbb{V}$ -category is used only when the corresponding weight is small. In particular, for the self-enriched truth-value object $\underline{\mathbb{V}}$, an expression of the form

$$\int^{A \in \underline{\mathbb{V}}} F(A) \otimes A$$

means the weighted colimit of the identity functor

$$\underline{\mathbb{V}} \longrightarrow \mathbb{V}$$

by a small presheaf $F \in \mathcal{P}(\underline{\mathbb{V}})$.

We shall also use the tensor–hom adjunction in the following universe-bounded form. If X and S are small $\mathbb{V}$ -categories and Y is a locally small possibly large $\mathbb{V}$ -category for which the functor category $[X, Y]$ is defined by small ends, then there is a natural bijection

$$\mathbb{V}\text{-}\mathbf{CAT}(S \otimes X, Y) \cong \mathbb{V}\text{-}\mathbf{CAT}(S, [X, Y]).$$

Here $\mathbb{V}\text{-}\mathbf{CAT}$ denotes the ordinary category whose objects are the locally small $\mathbb{V}$ -categories in the chosen universe and whose morphisms are $\mathbb{V}$ -functors. The notation

$$\mathbb{V}\text{-}\mathbf{Cat}$$

will be reserved for the full subcategory on small $\mathbb{V}$ -categories.

1.2.1 Fuzzy cosmoses and enriched categories

We first record the categorical setting whose posetal specialization gives the fuzzy truth-value bases used below.

Definition 1.2.1 (Fuzzy cosmos). *A **fuzzy cosmos** is a locally small complete and cocomplete symmetric monoidal closed category*

$$(\mathbb{V}, \otimes, I, [-, -])$$

*such that the monoidal unit I is terminal. We call this condition **affinity**.*

The terminology is adapted to the present work. The standard background notions are those of symmetric monoidal categories and closed monoidal categories in Kelly [10, Sections 1.1, 1.4, and 1.5]. Since $\mathbb{V}$ is symmetric monoidal closed, for every object $A \in \mathbb{V}$ the functor

$$A \otimes - : \mathbb{V} \rightarrow \mathbb{V}$$

is left adjoint to $[A, -]$. Hence $A \otimes -$ preserves all small colimits. We shall refer to this consequence as **distributivity**. In the posetal case, it becomes the quantale distributivity law

$$a \otimes \left(\bigvee_i b_i \right) = \bigvee_i (a \otimes b_i),$$

for small joins. The affinity condition categorifies integrality: in a truth-value quantale, the monoidal unit is the top element.

Definition 1.2.2 (Self-enrichment). *Let $\mathbb{V}$ be a fuzzy cosmos. We write*

$$\underline{\mathbb{V}}$$

for the $\mathbb{V}$ -enriched category obtained by self-enriching $\mathbb{V}$. Its objects are the objects of $\mathbb{V}$, and its hom-objects are

$$\underline{\mathbb{V}}(A, B) := [A, B].$$

Composition is induced by the closed structure, using the evaluation maps

$$[B, C] \otimes [A, B] \otimes A \longrightarrow [B, C] \otimes B \longrightarrow C,$$

and identities are induced by the transposes of the unit isomorphisms

$$I \otimes A \cong A.$$

This is the standard self-enrichment of a symmetric monoidal closed category [10, Section 1.6, especially formula (1.28)]. Thus $\underline{\mathbb{V}}$ denotes the categorical analogue of the object of truth values. In the posetal case, its hom-object

$$\underline{\mathbb{V}}(a, b)$$

is the residual implication

$$a \multimap b.$$

Remark 1.2.1 (Co-Yoneda reconstruction). *For every $B \in \mathbb{V}$, the enriched co-Yoneda formula gives a canonical reconstruction*

$$B \cong \int^{A \in \underline{\mathbb{V}}} \underline{\mathbb{V}}(A, B) \otimes A = \int^{A \in \underline{\mathbb{V}}} [A, B] \otimes A.$$

This is the enriched density, or co-Yoneda, formula associated to the Yoneda embedding [10, Section 1.9, Section 3.1, especially formula (3.17), Section 4.2, formula (4.31), Section 5.1]. By the set-theoretic conventions above, this coend is interpreted as the weighted colimit of the identity functor

$$\underline{\mathbb{V}} \rightarrow \mathbb{V}$$

by the small representable presheaf

$$Y(B) = \underline{\mathbb{V}}(-, B).$$

We now recall the corresponding enriched notion of category.

Definition 1.2.3 (Fuzzy category). *Let $\mathbb{V}$ be a fuzzy cosmos. A **fuzzy category**, or **small $\mathbb{V}$ -category**, X consists of:*

1. *a small set of objects, also denoted X ;*
2. *for each $x, y \in X$, a hom-object*

$$X(x, y) \in \mathbb{V};$$

3. *identity morphisms*

$$I \longrightarrow X(x, x);$$

4. *composition morphisms*

$$X(y, z) \otimes X(x, y) \longrightarrow X(x, z);$$

satisfying the usual associativity and unit axioms.

This is the definition of a $\mathbb{V}$ -category [10, Section 1.2], specialized only by the choice of the affine base $\mathbb{V}$.

Definition 1.2.4 (Fuzzy functors and opposites). *Let X and Y be fuzzy categories.*

1. *A **fuzzy functor**, or **$\mathbb{V}$ -functor**, $F : X \rightarrow Y$ consists of a function on objects together with morphisms*

$$X(x, x') \longrightarrow Y(Fx, Fx')$$

compatible with identities and composition.

2. *The **opposite** fuzzy category X^{op} has the same objects as X and hom-objects*

$$X^{\text{op}}(x, y) := X(y, x),$$

with composition induced using the symmetry of $\mathbb{V}$.

The definition of $\mathbb{V}$ -functor is standard [10, Section 1.2]; the construction of the opposite $\mathbb{V}$ -category in the symmetric case is given in Kelly [10, formula (1.19)].

1.2.2 Fuzzy truth values as posetal bases

We now specialize the fuzzy-cosmic setting to the case where the base $\mathbb{V}$ is a posetal symmetric monoidal closed category. The underlying order-theoretic structure is a commutative integral quantale; compare Stubbe's definition of quantale [16, Definition 2.1]. We impose complete distributivity as a standing structural condition so that the induced predicate algebras are frames and the level-cut constructions below interact well with arbitrary joins and meets.

Definition 1.2.5 (Fuzzy truth-value object). *A **fuzzy truth-value object** is a small structure*

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top)$$

such that:

1. $(L, \leq, \wedge, \vee, \perp, \top)$ is a completely distributive complete lattice;
2. $(L, \otimes, \top)$ is a commutative monoid;
3. for all $a, b, c \in L$,

$$a \otimes b \leq c \iff b \leq a \multimap c. \quad (1.1)$$

Equivalently, Λ is a small completely distributive complete commutative integral residuated lattice, where integrality means that the monoidal unit coincides with the top element $\top$.

The monoidal product $\otimes$ encodes conjunction or composition of degrees, and the operation $a \multimap c$ is the associated residual implication. Equivalently, for each fixed $a \in L$, residuation says that there is an adjunction

$$L \begin{array}{c} \xrightarrow{a \otimes -} \\ \xleftarrow{a \multimap -} \end{array} L \quad a \otimes - \dashv a \multimap -.$$

This adjunction is the order-theoretic source of quantale distributivity.

Proposition 1.2.1 (Posetal fuzzy cosmoses and fuzzy truth values). *Every fuzzy truth-value object Λ determines a posetal fuzzy cosmos. Its underlying category is the thin category associated to $(L, \leq)$, its tensor product is $\otimes$, its unit is $\top$, and its internal hom is the residual $\multimap$.*

Conversely, every small posetal fuzzy cosmos whose object-poset is a completely distributive complete lattice determines a fuzzy truth-value object.

Proof. Let Λ be a fuzzy truth-value object. Regard L as a thin category, with a unique morphism $a \rightarrow b$ precisely when $a \leq b$. Completeness and cocompleteness of this thin category are exactly completeness of the lattice L . The monoidal structure is given by $\otimes$, with unit $\top$. Affinity follows from integrality, since $\top$ is terminal in the thin category. The residuation law (1.1) says exactly that the tensor is closed, with internal hom $a \multimap b$. Since the monoidal structure is commutative, the resulting posetal category is symmetric monoidal closed.

The self-enriched truth-value object $\underline{\Lambda}$ has object set L and hom-values

$$\underline{\Lambda}(a, b) = a \multimap b.$$

The canonical levels are therefore the truth values $a \in L$ themselves, represented by the Yoneda presheaves

$$Y(a) = \underline{\Lambda}(-, a).$$

No level system is chosen.

Conversely, let $\mathbb{V}$ be a small posetal fuzzy cosmos whose object-poset is a completely distributive complete lattice L . The affine symmetric monoidal closed structure gives a commutative monoid $(L, \otimes, \top)$, and closedness gives a residual operation $\multimap$ satisfying (1.1). Thus the underlying posetal symmetric monoidal closed category determines a fuzzy truth-value object. $\square$

Proposition 1.2.2 (Tensor preserves joins). *For every $a \in L$ and every small family $(b_i)_{i \in I}$ in L ,*

$$a \otimes \left(\bigvee_{i \in I} b_i \right) = \bigvee_{i \in I} (a \otimes b_i), \quad \left(\bigvee_{i \in I} b_i \right) \otimes a = \bigvee_{i \in I} (b_i \otimes a).$$

Consequently, every fuzzy truth-value object determines a commutative unital quantale in the sense of [16, Definition 2.1].

Proof. Fix $a \in L$. By (1.1), the map

$$a \otimes - : L \rightarrow L$$

is left adjoint to

$$a \multimap - : L \rightarrow L.$$

Since left adjoints between complete lattices preserve joins,

$$a \otimes \left(\bigvee_{i \in I} b_i \right) = \bigvee_{i \in I} (a \otimes b_i).$$

The second identity follows by commutativity of $\otimes$. $\square$

Definition 1.2.5 includes both the standard truth-value algebras of fuzzy logic and other familiar bases for degree-valued order. The use of $[0, 1]$ -valued and L -valued truth degrees goes back to Zadeh and Goguen [17, 3]; the connection with left-continuous t -norms, residual implications, and quantale bases is standard in mathematical fuzzy logic [16, Example 2.4] [1, Sections 1, 2, and 7].

Example 1.2.1 (The crisp quantale). *Let*

$$L = \{0 < 1\}, \quad a \otimes b := a \wedge b, \quad a \multimap b := (a \Rightarrow b),$$

where $\Rightarrow$ denotes ordinary two-valued implication. With

$$\perp = 0, \quad \top = 1,$$

the structure

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object. This is the two-valued case and recovers ordinary order theory, equivalently enrichment over the two-element Boolean quantale [16, Definition 2.2 and the paragraph following it].

Example 1.2.2 (Standard t -norm quantales). *Let $*$ be a left-continuous t -norm on $[0, 1]$, and set*

$$L = [0, 1], \quad a \otimes b := a * b, \quad a \multimap b := \sup\{c \in [0, 1] \mid a * c \leq b\}, \quad \perp := 0, \quad \top := 1.$$

The complete lattice $[0, 1]$ is completely distributive because it is a complete chain. For a left-continuous t -norm, each map

$$a * - : [0, 1] \rightarrow [0, 1]$$

preserves arbitrary suprema in the complete chain $[0, 1]$: if $s = \sup_i b_i$, then monotonicity gives

$$\sup_i (a * b_i) \leq a * s,$$

while left-continuity gives the reverse inequality by approximating s from below. Hence $a * -$ has the displayed right adjoint, namely its residuum [16, Example 2.4]. Therefore

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object. This yields the algebraic semantics of left-continuous t -norm logics [1, Sections 1, 2, and 7] [2].

Example 1.2.3 (The Gödel quantale). Let $L = [0, 1]$ with the order, and define

$$a \otimes b := a \wedge b, \quad a \multimap b := \begin{cases} 1, & a \leq b, \\ b, & a > b, \end{cases} \quad \perp := 0, \quad \top := 1.$$

Then

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object. This is the quantale associated to the minimum t -norm [16, Example 2.4]. Composition is governed by the weaker of the two input degrees.

Example 1.2.4 (The Łukasiewicz quantale). Let $L = [0, 1]$ with the order, and define

$$a \otimes b := \max\{0, a + b - 1\}, \quad a \multimap b := \min\{1, 1 - a + b\}, \quad \perp := 0, \quad \top := 1.$$

Then

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object. This is the quantale associated to the Łukasiewicz t -norm [16, Example 2.4]. Composition is bounded-additive rather than bottleneck-type.

Example 1.2.5 (The product quantale). Let $L = [0, 1]$ with the order, and define

$$a \otimes b := ab, \quad a \multimap b := \begin{cases} 1, & a \leq b, \\ \frac{b}{a}, & a > b, \end{cases} \quad \perp := 0, \quad \top := 1.$$

Then

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object. This is the quantale associated to the product t -norm [16, Example 2.4]. Composition is multiplicative and therefore attenuates non-maximal degrees along chains.

Example 1.2.6 (The nilpotent minimum quantale). Let $L = [0, 1]$ with the order, and define

$$a \otimes b := \begin{cases} a \wedge b, & a + b > 1, \\ 0, & a + b \leq 1, \end{cases} \quad a \multimap b := \begin{cases} 1, & a \leq b, \\ \max\{1 - a, b\}, & a > b, \end{cases}$$

with

$$\perp := 0, \quad \top := 1.$$

Then

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object. The operation $\otimes$ is the nilpotent minimum t-norm, a standard left-continuous example [16, Example 2.4, footnote 5]. Here composition is possible only above a threshold of joint support.

Example 1.2.7 (Finite many-valued chains). For each integer $n \geq 2$, let

$$L_n = \left\{ 0, \frac{1}{n-1}, \frac{2}{n-1}, \dots, \frac{n-2}{n-1}, 1 \right\} \subseteq [0, 1].$$

Restricting either the Gödel operations or the Łukasiewicz operations from $[0, 1]$ to L_n yields a finite fuzzy truth-value object

$$\Lambda_n = (L_n, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top).$$

Example 1.2.8 (The Lawvere quantale). Let $L = [0, \infty]$ with the reverse of the usual order:

$$a \leq_L b \quad :\Longleftrightarrow \quad a \geq b \quad \text{in the usual order.}$$

Define

$$a \otimes b := a + b, \quad a \multimap c := c \ominus a := \inf_{\text{usual}} \{r \in [0, \infty] \mid a + r \geq c\}, \quad \top := 0, \quad \perp := \infty.$$

Equivalently,

$$c \ominus a = \begin{cases} 0, & a \geq c, \\ c - a, & a < c < \infty, \\ \infty, & a < \infty, c = \infty. \end{cases}$$

Thus $a \multimap c$ is the truncated difference $c - a$, computed with respect to the usual order, with truncation at 0 and with the usual extended-value conventions. The reversed order on a completely distributive complete chain is again completely distributive. Hence

$$\Lambda = (L, \leq_L, \wedge, \vee, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object. Here $\wedge$ and $\vee$ are computed with respect to the reversed order, so $\wedge$ is the supremum and $\vee$ is the infimum on $[0, \infty]$. This is Lawvere's base for generalized metric spaces [12] [16, Example 2.3].

Under the change of variables

$$p = e^{-a}, \quad e^{-\infty} = 0,$$

this base is isomorphic, as a commutative integral residuated quantale, to the product quantale on $[0, 1]$ [16, Example 2.4]. The compatibility of tensor products is expressed by the commutative diagram

$$\begin{array}{ccc} [0, \infty]^{\text{op}} \times [0, \infty]^{\text{op}} & \xrightarrow{+} & [0, \infty]^{\text{op}} \\ e^{-(\cdot)} \times e^{-(\cdot)} \downarrow & & \downarrow e^{-(\cdot)} \\ \times [0, 1] & \xrightarrow{\cdot} & \cdot \end{array}$$

Thus addition of costs corresponds to multiplication of similarities.

1.2.3 Fuzzy preorders

We now specialize Definition 1.2.3 to the posetal fuzzy cosmos determined by a fuzzy truth-value object Λ . In the fuzzy-set literature, closely related L -valued and quantale-valued preorder structures occur throughout the work on fuzzy preorders and fuzzy topologies; see, for example, [11, 13].

Definition 1.2.6 (Fuzzy preorder). *A **fuzzy preorder over** Λ is a pair (X, α) consisting of a small set X and a map*

$$\alpha : X \times X \rightarrow L$$

such that for all $x, y, z \in X$,

$$\top \leq \alpha(x, x), \tag{1.2}$$

$$\alpha(x, y) \otimes \alpha(y, z) \leq \alpha(x, z). \tag{1.3}$$

We usually suppress α from the notation and write $\alpha(x, y)$ simply as $X(x, y)$.

Since $\top$ is the greatest element of L , condition (1.2) is equivalent to

$$X(x, x) = \top \quad \text{for all } x \in X.$$

Thus $X(x, y)$ records a degree-valued comparison whose interpretation depends on the chosen base Λ .

Remark 1.2.2 (Posetal enrichment). *Definition 1.2.6 is precisely Definition 1.2.3 in the posetal case $\mathbb{V} = \Lambda$. The identity morphism*

$$I \longrightarrow X(x, x)$$

becomes the inequality

$$\top \leq X(x, x),$$

and composition becomes the transitivity inequality

$$X(x, y) \otimes X(y, z) \leq X(x, z).$$

Definition 1.2.7 (Λ -functors, opposites, and fuzzy orders). *Let X and Y be fuzzy preorders over Λ .*

- (1) *A **Λ -functor** $f : X \rightarrow Y$ is a function $f : X \rightarrow Y$ such that*

$$X(x, x') \leq Y(fx, fx') \quad \text{for all } x, x' \in X.$$

- (2) *The **opposite** fuzzy preorder X^{op} has the same underlying set as X and hom-values*

$$X^{\text{op}}(x, y) := X(y, x).$$

- (3) *A fuzzy preorder X is called **separated**¹ if*

$$X(x, y) = \top = X(y, x) \implies x = y.$$

*A **fuzzy order over** Λ is a separated fuzzy preorder over Λ .*

¹This is the separatedness condition for generalized metric spaces [16, Example 2.3].

Proposition 1.2.3. *Fuzzy preorders over Λ and Λ -functors form a category, denoted*

$$\mathbf{Pre}_\Lambda.$$

Moreover, fuzzy orders over Λ and Λ -functors form a full subcategory

$$\mathbf{Ord}_\Lambda \subseteq \mathbf{Pre}_\Lambda.$$

Proof. The identity map on a fuzzy preorder is clearly a Λ -functor. If

$$f : X \rightarrow Y, \quad g : Y \rightarrow Z$$

are Λ -functors, then for all $x, x' \in X$,

$$X(x, x') \leq Y(fx, fx') \leq Z(g(fx), g(fx')).$$

Hence $g \circ f : X \rightarrow Z$ is again a Λ -functor. Associativity of composition and the identity laws are inherited from the category of sets.

The second statement is immediate: $\mathbf{Ord}_\Lambda$ is the full subcategory of $\mathbf{Pre}_\Lambda$ on the separated objects. $\square$

The inclusion of fuzzy orders into fuzzy preorders will be denoted by

$$\mathbf{Ord}_\Lambda \hookrightarrow \mathbf{Pre}_\Lambda.$$

Once the base Λ has been fixed, the abstract definition specializes in familiar ways.

Example 1.2.9 (Ordinary preorders). *For Λ equal to the crisp quantale of Example 1.2.1, a fuzzy preorder over Λ is precisely an ordinary preorder. Indeed, if we write*

$$x \preceq y \quad :\Longleftrightarrow \quad X(x, y) = 1,$$

then (1.2) and (1.3) become reflexivity and transitivity of $\preceq$. This is the two-element case of quantale-enriched categories [16, Definition 2.2 and the paragraph following it].

Example 1.2.10 (Lawvere metric spaces). *For the Lawvere quantale of Example 1.2.8, all inequalities defining fuzzy preorders are interpreted in the reversed order $\leq_L$. Translating them back to the usual order, a fuzzy preorder is exactly a Lawvere metric space, equivalently an extended quasipseudometric space [12] [16, Example 2.3]. Writing $d(x, y)$ for $X(x, y)$, the axioms become*

$$d(x, x) = 0, \quad d(x, z) \leq d(x, y) + d(y, z)$$

in the usual order on $[0, \infty]$. In this case a Λ -functor is exactly a nonexpansive map:

$$d_Y(fx, fx') \leq d_X(x, x').$$

Example 1.2.11 (Chain-valued fuzzy preorders). *For the Gödel, Łukasiewicz, and product quantales of Examples 1.2.3, 1.2.4, and 1.2.5, a fuzzy preorder is a map*

$$X : X \times X \rightarrow [0, 1]$$

satisfying $X(x, x) = 1$ together with, respectively,

$$X(x, z) \geq \min\{X(x, y), X(y, z)\},$$

$$X(x, z) \geq \max\{0, X(x, y) + X(y, z) - 1\},$$

or

$$X(x, z) \geq X(x, y) X(y, z),$$

for all $x, y, z \in X$. Thus the same definition interpolates between bottleneck, bounded-additive, and multiplicative transitivity laws. Quantale-enriched categories over left-continuous t -norms are often described as fuzzy preorders relative to the chosen t -norm [16, Example 2.4].

1.2.4 Non-chain fuzzy truth values

The preceding examples are chain-valued. Since one aim of this chapter is to work uniformly over arbitrary bases, it is useful to record that genuinely non-chain examples already arise at the level of truth values. The following examples are elementary finite or product constructions; they illustrate that the theory is not intrinsically tied to linearly ordered truth values, in the spirit of L -fuzzy set theory [3].

Example 1.2.12 (A five-element non-chain truth-value algebra). Let $P = \{p, q, r\}$ be the poset with

$$p < r, \quad q < r,$$

and with p and q incomparable. Let $L = \text{Down}(P)$ be the set of downsets of P :

$$L = \{\emptyset, \{p\}, \{q\}, \{p, q\}, \{p, q, r\}\},$$

ordered by inclusion. Define

$$U \otimes V := U \cap V, \quad U \multimap V := \bigcup \{W \in L \mid U \cap W \subseteq V\}, \quad \perp := \emptyset, \quad \top := P.$$

Since unions of downsets are again downsets, the right-hand side indeed lies in L . Moreover, the displayed formula is precisely the Heyting implication associated to the meet operation $U \cap V$. Since every finite distributive lattice is completely distributive, L satisfies the lattice condition in Definition 1.2.5. Therefore

$$\Lambda = (L, \subseteq, \cap, \cup, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object. This is not a chain, since

$$\{p\} \not\subseteq \{q\}, \quad \{q\} \not\subseteq \{p\}.$$

Example 1.2.13 (A seven-element non-chain truth-value algebra). Let $P = \{p, q, r, s\}$ be the poset with

$$p < r, \quad p < s, \quad q < r, \quad q < s,$$

and with p and q incomparable and r and s incomparable. Let $L = \text{Down}(P)$ be the set of downsets of P :

$$L = \{\emptyset, \{p\}, \{q\}, \{p, q\}, \{p, q, r\}, \{p, q, s\}, \{p, q, r, s\}\},$$

ordered by inclusion. Define

$$U \otimes V := U \cap V, \quad U \multimap V := \bigcup \{W \in L \mid U \cap W \subseteq V\}, \quad \perp := \emptyset, \quad \top := P.$$

Again, this is the Heyting implication associated to the meet operation $U \cap V$. Since every finite distributive lattice is completely distributive, L satisfies the lattice condition in Definition 1.2.5. Hence

$$\Lambda = (L, \subseteq, \cap, \cup, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object. This base is not a chain: for example,

$$\{p\} \not\subseteq \{q\}, \quad \{q\} \not\subseteq \{p\},$$

and also

$$\{p, q, r\} \not\subseteq \{p, q, s\}, \quad \{p, q, s\} \not\subseteq \{p, q, r\}.$$

Example 1.2.14 (A non-chain product of chain-based truth-value objects). *Let $L = [0, 1] \times [0, 1]$, ordered coordinatewise:*

$$(a_1, a_2) \leq (b_1, b_2) \quad :\Longleftrightarrow \quad a_1 \leq b_1 \text{ and } a_2 \leq b_2.$$

Define the operations coordinatewise using the Lukasiewicz formulas:

$$(a_1, a_2) \otimes (b_1, b_2) := (\max\{0, a_1 + b_1 - 1\}, \max\{0, a_2 + b_2 - 1\}),$$

$$(a_1, a_2) \multimap (b_1, b_2) := (\min\{1, 1 - a_1 + b_1\}, \min\{1, 1 - a_2 + b_2\}),$$

with

$$\perp := (0, 0), \quad \top := (1, 1).$$

Complete distributivity is inherited by finite products, since the complete distributivity law is checked coordinatewise. Therefore

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top)$$

is a fuzzy truth-value object, where $\wedge$ and $\vee$ are taken coordinatewise. This is not a chain, since

$$(1, 0) \not\leq (0, 1), \quad (0, 1) \not\leq (1, 0).$$

Thus even when one begins with standard chain-based truth-value objects, their coordinatewise combination typically produces a non-chain fuzzy truth-value object.

1.3 Tensor products, internal homs, and truth-value objects

This section records the monoidal closed structure used throughout the manuscript. We first recall the standard tensor product and internal hom for categories enriched in a fuzzy cosmos, following the construction of the symmetric monoidal closed structure on $\mathbb{V}$ -categories [10, Section 1.4 and Sections 2.2–2.3]. We then specialize these constructions to fuzzy preorders over a small fuzzy truth-value object Λ . In that specialization, the general enriched formulas become the familiar pointwise formulas for tensor products, internal homs, and the truth-value fuzzy preorder.

The final corollary isolates the object

$$[X^{\text{op}}, \Lambda],$$

which will be studied in the next section as the fuzzy preorder of Λ -valued contravariant predicates on X .

1.3.1 Tensor products and internal homs in a fuzzy cosmos

Let

$$(\mathbb{V}, \otimes, I, [-, -])$$

be a fuzzy cosmos. We write

$$\mathbb{V}\text{-}\mathbf{Cat}$$

for the ordinary category of small $\mathbb{V}$ -categories and $\mathbb{V}$ -functors. Since the object sets of the enriched categories under consideration are small and $\mathbb{V}$ is locally small, the ordinary hom-sets of $\mathbb{V}\text{-}\mathbf{Cat}$ are small. This is the underlying ordinary category associated to the enriched functor category formalism of Kelly [10, Section 2.2, especially formula (2.16)].

Definition 1.3.1 (Tensor product and unit). *Let X and Y be small $\mathbb{V}$ -categories. Their **tensor product***

$$X \otimes Y$$

is the $\mathbb{V}$ -category whose objects are pairs $(x, y) \in X \times Y$, with hom-objects

$$(X \otimes Y)((x, y), (x', y')) := X(x, x') \otimes Y(y, y').$$

The identity and composition morphisms are induced from those of X and Y , using the unit, associativity, and symmetry of the tensor product in $\mathbb{V}$.

*The **unit $\mathbb{V}$ -category***

$$\mathbf{1}$$

has one object $$ and hom-object*

$$\mathbf{1}(*, *) = I.$$

This is the tensor product of enriched categories over a symmetric monoidal base [10, Section 1.4, formulas (1.17)–(1.18)]. The use of the symmetry in the definition of composition is the middle-four interchange construction.

Definition 1.3.2 (Enriched internal hom). *Let X and Y be small $\mathbb{V}$ -categories. The **internal hom**, or **enriched functor category**,*

$$[X, Y]$$

is the $\mathbb{V}$ -category whose objects are $\mathbb{V}$ -functors

$$F : X \rightarrow Y,$$

and whose hom-objects are given by the end formula [10, Section 2.2, formula (2.10)]

$$[X, Y](F, G) := \int_{x \in X} Y(Fx, Gx).$$

The end exists because X is small and $\mathbb{V}$ is complete. Composition and identities are induced pointwise from those of Y .

The composition and identity maps are those of the enriched functor category [10, Section 2.2, formulas (2.14)–(2.15)].

There is a canonical evaluation $\mathbb{V}$ -functor

$$\text{ev} : [X, Y] \otimes X \longrightarrow Y, \quad \text{ev}(F, x) := F(x),$$

as in the enriched functor-category construction [10, Section 2.2, formula (2.17)]. The tensor–hom adjunction says that every $\mathbb{V}$ -functor

$$h : Z \otimes X \rightarrow Y$$

has a unique transpose

$$\tilde{h} : Z \rightarrow [X, Y], \quad \tilde{h}(z)(x) := h(z, x),$$

equivalently making the diagram

$$\begin{array}{ccc} Z \otimes X & \xrightarrow{\quad h \quad} & Y \\ & \searrow \tilde{h} \otimes 1_X & \nearrow \text{ev} \\ & [X, Y] \otimes X & \end{array}$$

commute.

Proposition 1.3.1 (Tensor–hom adjunction in a fuzzy cosmos). *For small $\mathbb{V}$ -categories X, Y, Z , there is a natural bijection*

$$\mathbb{V}\text{-}\mathbf{Cat}(Z \otimes X, Y) \cong \mathbb{V}\text{-}\mathbf{Cat}(Z, [X, Y]).$$

Hence the ordinary category $\mathbb{V}\text{-}\mathbf{Cat}$ is symmetric monoidal closed under the tensor product of Definition 1.3.1.

Proof. This is the tensor–hom adjunction for enriched categories [10, Section 2.3, especially formula (2.20)]. Explicitly, a $\mathbb{V}$ -functor

$$h : Z \otimes X \rightarrow Y$$

is transposed to the $\mathbb{V}$ -functor

$$\tilde{h} : Z \rightarrow [X, Y]$$

defined on objects by

$$\tilde{h}(z)(x) := h(z, x).$$

Conversely, a $\mathbb{V}$ -functor

$$k : Z \rightarrow [X, Y]$$

is transposed to the $\mathbb{V}$ -functor

$$\check{k} : Z \otimes X \rightarrow Y$$

defined by

$$\check{k}(z, x) := k(z)(x).$$

The enriched functor axioms for these two transposes are equivalent by the universal property of the end defining $[X, Y]$. The two assignments are inverse to one another, and naturality in X, Y, Z is immediate. $\square$

Remark 1.3.1 (Universe-bounded tensor–hom use). *Later we apply the same tensor–hom construction when the codomain is a locally small possibly large $\mathbb{V}$ -category, such as the self-enriched truth-value object $\underline{\mathbb{V}}$ or a small-presheaf category $\mathcal{P}(\underline{\mathbb{V}})$. This is justified by the universe-bounded convention stated in Section 1.2: all ends are indexed by small domains, and all hom-objects of the codomain are small enough for the relevant end formulas.*

1.3.2 The posetal specialization

We now specialize the preceding constructions to the posetal fuzzy cosmos determined by a small fuzzy truth-value object Λ .

Recall that if X is a fuzzy preorder, then its opposite X^{op} has the same underlying set as X and hom-values

$$X^{\text{op}}(x, y) := X(y, x).$$

Definition 1.3.3 (Tensor product and unit). *Let X and Y be fuzzy preorders over Λ . Their **tensor product***

$$X \otimes Y$$

is the fuzzy preorder with underlying set $X \times Y$ and hom-values

$$(X \otimes Y)((x, y), (x', y')) := X(x, x') \otimes Y(y, y').$$

If

$$f : X \rightarrow X', \quad g : Y \rightarrow Y'$$

are Λ -functors, define

$$f \otimes g : X \otimes Y \rightarrow X' \otimes Y'$$

by

$$(f \otimes g)(x, y) := (f(x), g(y)).$$

The **unit fuzzy preorder**

$$1$$

is the one-element fuzzy preorder with unique element $*$ and hom-value

$$1(*, *) = \top.$$

This is the specialization of the enriched tensor product of Definition 1.3.1 to the quantale-enriched case [10, Section 1.4, formulas (1.17)–(1.18)].

Proposition 1.3.2. *The tensor product of Definition 1.3.3 makes*

$$\mathbf{Pre}_\Lambda$$

into a symmetric monoidal category with unit object 1.

Proof. This is the posetal instance of the tensor product of enriched categories [10, Section 1.4]. Directly, reflexivity follows from

$$\top = \top \otimes \top \leq X(x, x) \otimes Y(y, y),$$

and transitivity follows from associativity, commutativity, monotonicity of $\otimes$, and transitivity in X and Y :

$$(X(x, x') \otimes Y(y, y')) \otimes (X(x', x'') \otimes Y(y', y'')) \leq X(x, x'') \otimes Y(y, y'').$$

The statement for tensor products of Λ -functors follows from monotonicity of $\otimes$. The associativity, unit, and symmetry isomorphisms are induced by the corresponding bijections of cartesian products of sets. $\square$

Lemma 1.3.1 (Pullbacks in $\mathbf{Pre}_\Lambda$). *The category $\mathbf{Pre}_\Lambda$ has conical pullbacks. The pullback of*

$$X \xrightarrow{f} Z \xleftarrow{g} Y$$

has underlying set the ordinary pullback

$$P = \{(x, y) \in X \times Y \mid f(x) = g(y)\},$$

and hom-values

$$P((x, y), (x', y')) := X(x, x') \wedge Y(y, y').$$

Proof. Reflexivity follows from reflexivity in X and Y . For transitivity, use monotonicity of $\otimes$, the inequalities

$$u \wedge v \leq u, \quad u \wedge v \leq v,$$

and transitivity in X and Y :

$$\begin{aligned} & (X(x, x') \wedge Y(y, y')) \otimes (X(x', x'') \wedge Y(y', y'')) \\ & \leq (X(x, x') \otimes X(x', x'')) \wedge (Y(y, y') \otimes Y(y', y'')) \\ & \leq X(x, x'') \wedge Y(y, y''). \end{aligned}$$

The two projections are Λ -functors. Given $T \rightarrow X$ and $T \rightarrow Y$ with equal composites into Z , the induced ordinary function $T \rightarrow P$ is a Λ -functor because its hom condition is exactly the conjunction of the two hom conditions for $T \rightarrow X$ and $T \rightarrow Y$. This gives the usual pullback universal property. $\square$

1.3.3 The truth-value fuzzy preorder

The self-enriched base $\underline{\mathbb{V}}$ specializes, in the posetal case, to the fuzzy preorder whose hom-values are residual implications. This is the thin, quantale-enriched instance of the self-enrichment of a symmetric monoidal closed category [10, Section 1.6, especially formula (1.28)].

Definition 1.3.4 (The truth-value fuzzy preorder). *By abuse of notation, we also write*

$$\Lambda$$

for the fuzzy preorder whose underlying set is L and whose hom-values are

$$\Lambda(a, b) := a \multimap b.$$

Equivalently, this is the self-enriched Λ -category $\underline{\Lambda}$.

Proposition 1.3.3. *The structure Λ of Definition 1.3.4 is a separated fuzzy preorder.*

Proof. This is the posetal form of the self-enrichment of $\mathbb{V}$ [10, Section 1.6]. Directly, residuation gives

$$\top \leq a \multimap a$$

because $a \otimes \top = a \leq a$. For transitivity, the inequalities

$$a \otimes (a \multimap b) \leq b, \quad b \otimes (b \multimap c) \leq c$$

imply

$$a \otimes ((a \multimap b) \otimes (b \multimap c)) \leq c,$$

hence

$$(a \multimap b) \otimes (b \multimap c) \leq a \multimap c.$$

If $a \multimap b = \top = b \multimap a$, then residuation gives $a \leq b$ and $b \leq a$, hence $a = b$. $\square$

1.3.4 Internal homs

The general enriched internal hom of Definition 1.3.2 specializes to the pointwise fuzzy preorder of Λ -functors.

Definition 1.3.5 (Internal hom). *Let X and Y be fuzzy preorders over Λ . Define*

$$[X, Y]$$

to be the fuzzy preorder whose underlying set is the set of Λ -functors $f : X \rightarrow Y$, and whose hom-values are

$$[X, Y](f, g) := \bigwedge_{x \in X} Y(f(x), g(x)).$$

The meet in this formula is the posetal form of the enriched end

$$\int_{x \in X} Y(fx, gx)$$

appearing in the enriched functor-category formula [10, Section 2.2, formula (2.10)]. In a thin enriched base, this end is computed as the greatest lower bound of the pointwise hom-values satisfying the usual wedge conditions. These wedge conditions follow from functoriality of f and g .

Proposition 1.3.4. *For fuzzy preorders X and Y , the structure $[X, Y]$ of Definition 1.3.5 is a fuzzy preorder. If Y is separated, then $[X, Y]$ is separated.*

Proof. This is the posetal instance of the enriched functor-category construction [10, Section 2.2, formulas (2.10), (2.14), and (2.15)]. Reflexivity and transitivity are checked pointwise:

$$\top \leq Y(fx, fx) \quad \text{and} \quad [X, Y](f, g) \otimes [X, Y](g, h) \leq Y(fx, hx)$$

for every $x \in X$. Taking the meet over x gives the fuzzy preorder axioms.

If Y is separated and

$$[X, Y](f, g) = \top = [X, Y](g, f),$$

then for every $x \in X$,

$$Y(fx, gx) = \top = Y(gx, fx).$$

Hence $f(x) = g(x)$ for all x , and therefore $f = g$. $\square$

The enriched evaluation functor specializes to an evaluation Λ -functor [10, Section 2.2, formula (2.17)]

$$\text{ev} : [X, Y] \otimes X \rightarrow Y, \quad \text{ev}(f, x) := f(x).$$

Proposition 1.3.5 (Tensor–hom adjunction). *For fuzzy preorders X, Y, Z , there is a natural bijection*

$$\text{Pre}_\Lambda(Z \otimes X, Y) \cong \text{Pre}_\Lambda(Z, [X, Y]).$$

Consequently, the ordinary category Pre_Λ is symmetric monoidal closed.

Proof. This is Proposition 1.3.1 specialized to the posetal fuzzy cosmos determined by Λ . Explicitly, a Λ -functor

$$h : Z \otimes X \rightarrow Y$$

is sent to

$$\tilde{h} : Z \rightarrow [X, Y], \quad \tilde{h}(z)(x) := h(z, x),$$

and a Λ -functor

$$k : Z \rightarrow [X, Y]$$

is sent to

$$\check{k} : Z \otimes X \rightarrow Y, \quad \check{k}(z, x) := k(z)(x).$$

The enriched proof shows that these are mutually inverse and natural; in the posetal case the required inequalities are precisely the pointwise functoriality inequalities induced by Definition 1.3.5. $\square$

Corollary 1.3.1. *The symmetric monoidal closed structure on Pre_Λ restricts to a symmetric monoidal closed structure on the full subcategory*

$$\text{Ord}_\Lambda \subseteq \text{Pre}_\Lambda.$$

Proof. The unit 1 is separated. If X and Y are separated, then $X \otimes Y$ is separated because integrality gives $a \otimes b \leq a$ and $a \otimes b \leq b$. Hence equality to $\top$ in both directions forces equality in both coordinates. Finally, if Y is separated, then $[X, Y]$ is separated by Proposition 1.3.4. Since Ord_Λ is full, the tensor–hom adjunction restricts. $\square$

Corollary 1.3.2 (Predicate transposition). *For all fuzzy preorders S and X , there is a natural bijection*

$$\text{Pre}_\Lambda(S, [X^{\text{op}}, \Lambda]) \cong \text{Pre}_\Lambda(S \otimes X^{\text{op}}, \Lambda).$$

Proof. This is the special case of the enriched tensor–hom adjunction [10, Section 2.3, formula (2.20)] with $Y = \Lambda$ and X replaced by X^{op} . $\square$

Under this bijection, a Λ -functor

$$\Phi : S \rightarrow [X^{\text{op}}, \Lambda]$$

corresponds to its uncurried form

$$\widehat{\Phi} : S \otimes X^{\text{op}} \rightarrow \Lambda, \quad \widehat{\Phi}(s, x) := \Phi(s)(x).$$

Equivalently, $\widehat{\Phi}$ is the composite

$$S \otimes X^{\text{op}} \xrightarrow{\Phi \otimes 1_{X^{\text{op}}}} [X^{\text{op}}, \Lambda] \otimes X^{\text{op}} \xrightarrow{\text{ev}} \Lambda.$$

Remark 1.3.2. *Dually, one may consider the covariant internal hom*

$$[X, \Lambda].$$

We shall not need that construction in what follows, so we now turn to the contravariant object

$$[X^{\text{op}}, \Lambda].$$

1.4 Fuzzy presheaves, fuzzy predicates, and predicate algebras

We now specialize the internal-hom construction to contravariant truth-valued functors.

1.4.1 Fuzzy presheaves in a fuzzy cosmos

Let

$$(\mathbb{V}, \otimes, I, [-, -])$$

be a fuzzy cosmos, and let X be a small $\mathbb{V}$ -category.

In this subsection we use the same functor-category notation when the domain is small and the codomain is possibly large. Thus, if X is small and Y is a locally small possibly large $\mathbb{V}$ -category, the $\mathbb{V}$ -category $[X, Y]$ has as objects the $\mathbb{V}$ -functors $X \rightarrow Y$, with hom-objects computed by the small end

$$[X, Y](F, G) = \int_{x \in X} Y(Fx, Gx).$$

This convention applies in particular to functor categories whose codomain is the self-enriched truth-value object $\underline{\mathbb{V}}$ or the small-presheaf category $\mathcal{P}(\underline{\mathbb{V}})$.

Definition 1.4.1 (Fuzzy presheaf). *A **fuzzy presheaf** on X is a fuzzy-functor*

$$\Phi : X^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

We write

$$\text{Psh}_{\mathbb{V}}(X) := [X^{\text{op}}, \underline{\mathbb{V}}]$$

for the possibly large $\mathbb{V}$ -category of fuzzy presheaves on X .

This is the usual enriched presheaf construction, expressed using the self-enrichment of the base $\mathbb{V}$ [10, Sections 1.6 and 1.9]. Since X is small, every $\mathbb{V}$ -presheaf

$$X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

is small in the sense of the set-theoretic conventions of Section 1.2. Indeed, by the enriched co-Yoneda formula every presheaf

$$\Phi : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

is canonically a small weighted colimit of representables:

$$\Phi \cong \int^{x \in X} \Phi x \otimes y_X(x).$$

Since X is small, this is a small colimit. Thus this notation is compatible with the earlier small-presheaf convention.

Unpacked, a fuzzy presheaf Φ assigns to each $x \in X$ an object

$$\Phi x \in \mathbb{V},$$

together with action morphisms

$$X(x', x) \otimes \Phi x \longrightarrow \Phi x'$$

satisfying the evident unit and associativity axioms. Thus fuzzy presheaves are contravariant $\mathbb{V}$ -valued functors.

Proposition 1.4.1 (Presheaf hom-objects). *For fuzzy presheaves $\Phi, \Psi \in \text{Psh}_{\mathbb{V}}(X)$, the hom-object in $\text{Psh}_{\mathbb{V}}(X)$ is*

$$\text{Psh}_{\mathbb{V}}(X)(\Phi, \Psi) = \int_{x \in X} [\Phi x, \Psi x].$$

Proof. This is the enriched functor-category hom formula [10, Section 2.2, formula (2.10)], applied to

$$\mathbf{Psh}_{\mathbb{V}}(X) = [X^{\text{op}}, \underline{\mathbb{V}}],$$

together with the identity

$$\underline{\mathbb{V}}(\Phi x, \Psi x) = [\Phi x, \Psi x].$$

The end is small because X is small. $\square$

Definition 1.4.2 (Representable fuzzy presheaves). *For each $x \in X$, the **representable fuzzy presheaf** at x is the $\mathbb{V}$ -functor*

$$y_X(x) : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

defined by

$$y_X(x)(z) := X(z, x).$$

These assemble into a $\mathbb{V}$ -functor

$$y_X : X \rightarrow \mathbf{Psh}_{\mathbb{V}}(X),$$

*called the **Yoneda embedding**.*

This is the enriched Yoneda embedding for the self-enriched base $\underline{\mathbb{V}}$ [10, Section 1.9]. The presheaf structure on $y_X(x)$ is induced by composition in X :

$$X(z', z) \otimes X(z, x) \longrightarrow X(z', x).$$

Proposition 1.4.2 (Enriched Yoneda formula). *For every small $\mathbb{V}$ -category X , every $x \in X$, and every fuzzy presheaf*

$$\Phi : X^{\text{op}} \rightarrow \underline{\mathbb{V}},$$

there is a canonical isomorphism in $\mathbb{V}$

$$\mathbf{Psh}_{\mathbb{V}}(X)(y_X(x), \Phi) \cong \Phi x.$$

Proof. By Proposition 1.4.1,

$$\mathbf{Psh}_{\mathbb{V}}(X)(y_X(x), \Phi) = \int_{z \in X} [X(z, x), \Phi z].$$

This end is the usual enriched object of natural transformations from the representable $X(-, x)$ to Φ . The enriched Yoneda lemma identifies it canonically with Φx [10, Section 1.9]. $\square$

Corollary 1.4.1 (Yoneda embedding). *The Yoneda embedding*

$$y_X : X \rightarrow \mathbf{Psh}_{\mathbb{V}}(X)$$

is fully faithful. Equivalently, for all $x, x' \in X$, there is a canonical isomorphism

$$X(x, x') \cong \mathbf{Psh}_{\mathbb{V}}(X)(y_X(x), y_X(x')).$$

Proof. Apply Proposition 1.4.2 to the representable fuzzy presheaf

$$\Phi = y_X(x').$$

Then

$$\mathbf{Psh}_{\mathbb{V}}(X)(y_X(x), y_X(x')) \cong y_X(x')(x) = X(x, x').$$

$\square$

The fuzzy predicate theory below is the posetal, truth-valued specialization of this enriched presheaf picture.

1.4.2 Fuzzy predicates

Let Λ be a fuzzy truth-value object, regarded as a posetal fuzzy cosmos, and let X be a fuzzy preorder over Λ . Then a fuzzy predicate

$$X^{\text{op}} \rightarrow \Lambda$$

is exactly a Λ -valued contravariant predicate on X . This is the posetal, quantale-enriched specialization of the enriched presheaf construction above [16, Definition 2.2]; closely related L -valued and GL -monoid-valued presheaf constructions occur in [6, 4, 7].

Definition 1.4.3 (Fuzzy predicate). *Let X be a fuzzy preorder over Λ . A **fuzzy predicate** on X is a Λ -functor*

$$\phi : X^{\text{op}} \rightarrow \Lambda.$$

Equivalently, by Lemma 1.4.1 below, it is a map

$$\phi : X \rightarrow L$$

satisfying

$$X(x', x) \otimes \phi(x) \leq \phi(x') \tag{1.4}$$

for all $x, x' \in X$.

We write

$$\text{Pred}_\Lambda(X) := [X^{\text{op}}, \Lambda]$$

for the fuzzy preorder of fuzzy predicates on X .

Lemma 1.4.1 (Pointwise characterization of fuzzy predicates). *Let X be a fuzzy preorder over Λ , and let $\phi : X \rightarrow L$ be a map. Then ϕ defines a Λ -functor*

$$X^{\text{op}} \rightarrow \Lambda$$

if and only if, for all $x, x' \in X$,

$$X(x', x) \otimes \phi(x) \leq \phi(x').$$

Proof. A map $\phi : X \rightarrow L$ is a Λ -functor $X^{\text{op}} \rightarrow \Lambda$ if and only if for all $x, x' \in X$,

$$X^{\text{op}}(x, x') \leq \Lambda(\phi(x), \phi(x')).$$

Since

$$X^{\text{op}}(x, x') = X(x', x) \quad \text{and} \quad \Lambda(\phi(x), \phi(x')) = \phi(x) \multimap \phi(x'),$$

this is equivalent, by residuation, to

$$X(x', x) \otimes \phi(x) \leq \phi(x').$$

□

Proposition 1.4.3. *For every fuzzy preorder X , the collection $\text{Pred}_\Lambda(X)$ of fuzzy predicates on X carries a natural separated fuzzy preorder, with hom-values*

$$\text{Pred}_\Lambda(X)(\phi, \psi) := \bigwedge_{x \in X} (\phi(x) \multimap \psi(x)).$$

Its induced crisp order is the pointwise order:

$$\phi \leq \psi \quad :\Longleftrightarrow \quad (\forall x \in X) \phi(x) \leq \psi(x).$$

Proof. This is the posetal instance of the enriched functor-category hom formula [10, Section 2.2, formula (2.10)], together with Proposition 1.4.1 and Proposition 1.3.3. The enriched end becomes the small meet

$$\bigwedge_{x \in X} \Lambda(\phi(x), \psi(x)) = \bigwedge_{x \in X} (\phi(x) \multimap \psi(x)).$$

The description of the induced crisp order follows because

$$\top \leq \phi(x) \multimap \psi(x) \iff \phi(x) \leq \psi(x).$$

Since the truth-value fuzzy preorder Λ is separated, Proposition 1.3.4 implies that $\text{Pred}_\Lambda(X) = [X^{\text{op}}, \Lambda]$ is separated. $\square$

Thus the crisp order underlying $\text{Pred}_\Lambda(X)$ is simply the order inherited pointwise from L , even though the fuzzy hom-value between predicates records the stronger degree

$$\bigwedge_{x \in X} (\phi(x) \multimap \psi(x)).$$

Proposition 1.4.4 (Representable fuzzy predicates). *For each $x \in X$, the map*

$$y_X(x) : X \rightarrow L, \quad y_X(x)(z) := X(z, x),$$

is a fuzzy predicate on X . Thus there is a canonical Λ -functor

$$y_X : X \rightarrow \text{Pred}_\Lambda(X), \quad x \mapsto y_X(x).$$

Proof. This is the posetal instance of the enriched representable presheaf and Yoneda embedding [10, Section 1.9]. Directly,

$$X(z', z) \otimes y_X(x)(z) = X(z', z) \otimes X(z, x) \leq X(z', x) = y_X(x)(z')$$

by transitivity in X . Hence $y_X(x)$ is a fuzzy predicate.

The fact that $y_X : X \rightarrow \text{Pred}_\Lambda(X)$ is a Λ -functor follows by applying transitivity:

$$X(x, x') \otimes X(z, x) \leq X(z, x')$$

for every z , then transposing by residuation and taking the meet over z . $\square$

Proposition 1.4.5 (Yoneda formula for fuzzy predicates). *For every fuzzy preorder X , every $x \in X$, and every $\phi \in \text{Pred}_\Lambda(X)$,*

$$\text{Pred}_\Lambda(X)(y_X(x), \phi) = \phi(x).$$

Proof. This is the posetal form of the enriched Yoneda lemma [10, Section 1.9]. Directly,

$$\text{Pred}_\Lambda(X)(y_X(x), \phi) = \bigwedge_{z \in X} (X(z, x) \multimap \phi(z)).$$

The $z = x$ term gives the inequality

$$\text{Pred}_\Lambda(X)(y_X(x), \phi) \leq \phi(x).$$

Conversely, since ϕ is a fuzzy predicate,

$$X(z, x) \otimes \phi(x) \leq \phi(z),$$

so residuation gives

$$\phi(x) \leq X(z, x) \multimap \phi(z)$$

for every z . Taking the meet over z gives the reverse inequality. $\square$

Corollary 1.4.2 (Yoneda embedding). *For every fuzzy preorder X , the map*

$$y_X : X \rightarrow \text{Pred}_\Lambda(X)$$

is fully faithful, that is,

$$X(x, x') = \text{Pred}_\Lambda(X)(y_X(x), y_X(x')) \quad \text{for all } x, x' \in X.$$

In particular, if X is separated, then y_X is injective.

Proof. Apply Proposition 1.4.5 to the representable fuzzy predicate

$$\phi = y_X(x').$$

If X is separated and $y_X(x) = y_X(x')$, then

$$\top \leq X(x, x') \quad \text{and} \quad \top \leq X(x', x),$$

so $x = x'$. □

1.4.3 Fuzzy predicate algebras

We now use the additional order-theoretic structure present in the posetal case. The fuzzy predicate object $\text{Pred}_\Lambda(X)$ carries more structure than its defining fuzzy preorder. Its induced crisp order is complete, finite meets distribute over small joins, and the truth-value object acts on it by pointwise scalar operations. Related algebraic treatments of fuzzy-valued sets, GL -monoids, and quantale-valued sets appear in [5, 6, 4, 7].

Proposition 1.4.6. *Let X be a fuzzy preorder over Λ , and let $(\phi_i)_{i \in I}$ be a small family of fuzzy predicates on X . Define maps*

$$\left(\bigvee_{i \in I} \phi_i \right)(x) := \bigvee_{i \in I} \phi_i(x), \quad \left(\bigwedge_{i \in I} \phi_i \right)(x) := \bigwedge_{i \in I} \phi_i(x)$$

for each $x \in X$. Then both

$$\bigvee_{i \in I} \phi_i \quad \text{and} \quad \bigwedge_{i \in I} \phi_i$$

are fuzzy predicates on X . Consequently, the induced crisp order of $\text{Pred}_\Lambda(X)$ is a complete lattice, with small joins and small meets computed pointwise.

Proof. For joins, Proposition 1.2.2 gives

$$X(x', x) \otimes \left(\bigvee_{i \in I} \phi_i(x) \right) = \bigvee_{i \in I} (X(x', x) \otimes \phi_i(x)) \leq \bigvee_{i \in I} \phi_i(x').$$

For meets, the empty meet is the constant predicate with value $\top$. If $I \neq \emptyset$, then for each $j \in I$,

$$X(x', x) \otimes \left(\bigwedge_{i \in I} \phi_i(x) \right) \leq X(x', x) \otimes \phi_j(x) \leq \phi_j(x').$$

Hence the left-hand side is bounded above by $\bigwedge_i \phi_i(x')$. Thus joins and meets are computed pointwise. □

Proposition 1.4.7. *For every fuzzy preorder X , the induced crisp order of $\text{Pred}_\Lambda(X)$ is a frame.*

Proof. By Proposition 1.4.6, small joins and meets in $\text{Pred}_\Lambda(X)$ are computed pointwise. Since L is completely distributive, it is in particular a frame, and finite meets distribute over small joins pointwise. $\square$

This frame structure is a structure on the induced crisp order; it is not the same datum as the Λ -valued hom of the fuzzy preorder $\text{Pred}_\Lambda(X)$.

Definition 1.4.4 (Pointwise scalar operations). *Let X be a fuzzy preorder over Λ , let $a \in L$, and let $\phi \in \text{Pred}_\Lambda(X)$. Define maps*

$$(a \odot \phi)(x) := a \otimes \phi(x), \quad (a \Rightarrow \phi)(x) := a \multimap \phi(x)$$

for each $x \in X$.

Proposition 1.4.8. *For every $a \in L$ and every fuzzy predicate ϕ on X , the maps*

$$a \odot \phi \quad \text{and} \quad a \Rightarrow \phi$$

of Definition 1.4.4 are again fuzzy predicates.

Proof. For $a \odot \phi$, associativity, commutativity, and the predicate law for ϕ give

$$X(x', x) \otimes (a \odot \phi)(x) = a \otimes (X(x', x) \otimes \phi(x)) \leq a \otimes \phi(x') = (a \odot \phi)(x').$$

For $a \Rightarrow \phi$, by residuation it suffices to prove

$$a \otimes (X(x', x) \otimes (a \Rightarrow \phi)(x)) \leq \phi(x').$$

The left-hand side is

$$X(x', x) \otimes (a \otimes (a \Rightarrow \phi)(x)) \leq X(x', x) \otimes \phi(x) \leq \phi(x'),$$

so $a \Rightarrow \phi$ is a fuzzy predicate. $\square$

Proposition 1.4.9. *Let X be a fuzzy preorder over Λ . For every $a \in L$ and all $\phi, \psi \in \text{Pred}_\Lambda(X)$,*

$$a \odot \phi \leq \psi \iff \phi \leq a \Rightarrow \psi.$$

Proof. By Proposition 1.4.3, the order on $\text{Pred}_\Lambda(X)$ is pointwise. Therefore the result is exactly the pointwise residuation law in L . $\square$

Thus, for each $a \in L$, there is an adjunction on the crisp order underlying $\text{Pred}_\Lambda(X)$:

$$\text{Pred}_\Lambda(X) \xrightleftharpoons[a \Rightarrow -]{a \odot -} \text{Pred}_\Lambda(X) \quad a \odot - \dashv a \Rightarrow -.$$

Corollary 1.4.3. *For every $a \in L$, the operation*

$$a \odot (-) : \text{Pred}_\Lambda(X) \rightarrow \text{Pred}_\Lambda(X)$$

preserves small joins, and the operation

$$a \Rightarrow (-) : \text{Pred}_\Lambda(X) \rightarrow \text{Pred}_\Lambda(X)$$

preserves small meets.

Proof. This is immediate from Proposition 1.4.9, since left adjoints preserve joins and right adjoints preserve meets. $\square$

Proposition 1.4.10. *Let X be a fuzzy preorder over Λ . For all $a, b \in L$ and all $\phi \in \text{Pred}_\Lambda(X)$,*

$$\top \odot \phi = \phi, \quad (a \otimes b) \odot \phi = a \odot (b \odot \phi).$$

Moreover, for every small family $(\phi_i)_{i \in I}$ in $\text{Pred}_\Lambda(X)$,

$$a \odot \left(\bigvee_{i \in I} \phi_i \right) = \bigvee_{i \in I} (a \odot \phi_i).$$

Proof. All three identities are pointwise consequences of the corresponding laws in L . For the last identity, use Proposition 1.2.2. $\square$

Proposition 1.4.11 (Constant fuzzy predicates). *Let X be a fuzzy preorder over Λ . For every $a \in L$, the constant map*

$$\underline{a} : X \rightarrow L, \quad \underline{a}(x) := a,$$

is a fuzzy predicate.

Proof. For all $x, x' \in X$, since $X(x', x) \leq \top$ and $\otimes$ is monotone,

$$X(x', x) \otimes a \leq \top \otimes a = a.$$

$\square$

Definition 1.4.5 (Fuzzy predicate algebra). *Let X be a fuzzy preorder over Λ . The **fuzzy predicate algebra** of X is the structure carried by $\text{Pred}_\Lambda(X)$ consisting of:*

1. *its frame structure from Proposition 1.4.7;*
2. *the family of pointwise scalar actions*

$$a \odot (-) : \text{Pred}_\Lambda(X) \rightarrow \text{Pred}_\Lambda(X) \quad (a \in L);$$

3. *the family of pointwise scalar residuals*

$$a \Rightarrow (-) : \text{Pred}_\Lambda(X) \rightarrow \text{Pred}_\Lambda(X) \quad (a \in L).$$

1.4.4 Principal examples

We now specialize the preceding constructions to several representative truth-value objects. In each case the abstract predicate object

$$\text{Pred}_\Lambda(X) = [X^{\text{op}}, \Lambda]$$

admits a concrete description, and the pointwise scalar operations induced by the base become explicit. The examples below include the ordinary crisp case, the standard chain-valued bases of mathematical fuzzy logic, and the Lawvere metric base [3, 12] [16, Example 2.4][1, Sections 1, 2, and 7].

Proposition 1.4.12 (Discrete predicate object). *Let X be a discrete fuzzy preorder over Λ . Then every map*

$$\phi : X \rightarrow L$$

is a fuzzy predicate. Hence

$$\text{Pred}_\Lambda(X) \cong L^X$$

as sets. Under this identification, the induced crisp order, small joins, small meets, and scalar operations

$$a \odot (-), \quad a \Rightarrow (-)$$

are all computed pointwise.

Proof. If $x' = x$, then $X(x, x) = \top$, so the predicate condition is immediate. If $x' \neq x$, then $X(x', x) = \perp$, and tensor preserves the empty join, so

$$\perp \otimes \phi(x) = \perp \leq \phi(x').$$

Thus every map $X \rightarrow L$ is a fuzzy predicate. The final statement follows from the general pointwise descriptions above. $\square$

In the quantale-valued setting, such full L -valued function objects are the basic powerset-like objects; compare the quantale-valued powerset perspective in [15].

The crisp case

The two-valued case is the ordinary posetal specialization of enriched presheaves and the Yoneda embedding [10, Section 1.9]. Let

$$\Lambda = \{0 < 1\}, \quad a \otimes b = a \wedge b.$$

If X is a fuzzy preorder over this base, then X is just an ordinary preorder. A fuzzy predicate on X is a map

$$\phi : X \rightarrow \{0, 1\}$$

such that

$$X(x', x) \wedge \phi(x) \leq \phi(x') \quad \text{for all } x, x' \in X.$$

Proposition 1.4.13. *For a preorder X , the assignment*

$$\phi \longmapsto A_\phi := \{x \in X \mid \phi(x) = 1\}$$

identifies $\text{Pred}_\Lambda(X)$ with the set of lower sets of X . Under this identification, the order on $\text{Pred}_\Lambda(X)$ is inclusion, small joins are unions, small meets are intersections, and the representable fuzzy predicate $y_X(x)$ corresponds to the principal lower set

$$\downarrow x = \{z \in X \mid z \preceq x\}.$$

Proof. The fuzzy predicate condition says precisely that

$$x' \preceq x \text{ and } x \in A_\phi \implies x' \in A_\phi.$$

Hence fuzzy predicates are exactly lower sets. The order, joins, meets, and representables are then identified pointwise. $\square$

Corollary 1.4.4. *If X is a discrete preorder, then*

$$\text{Pred}_\Lambda(X) \cong 2^X,$$

the ordinary powerset Boolean algebra. Under this identification, joins are unions, meets are intersections, and $y_X(x)$ corresponds to the singleton $\{x\}$.

Proof. In a discrete preorder, every subset is lower, and $\downarrow x = \{x\}$. □

The Gödel case

The Gödel base is the quantale associated to the minimum t -norm, one of the standard chain-valued bases in mathematical fuzzy logic [16, Example 2.4] [1, Sections 1, 2, and 7]. Let $\Lambda = [0, 1]$ with the Gödel operations

$$a \otimes b = a \wedge b, \quad a \multimap_G b = \begin{cases} 1, & a \leq b, \\ b, & a > b. \end{cases}$$

Then a fuzzy predicate on a fuzzy preorder X is a map $\phi : X \rightarrow [0, 1]$ such that

$$X(x', x) \wedge \phi(x) \leq \phi(x') \quad \text{for all } x, x' \in X.$$

Proposition 1.4.14. *For every fuzzy preorder X over the Gödel base, the induced crisp order on $\text{Pred}_\Lambda(X)$ is a frame, hence a complete Heyting algebra. Moreover, for every $a \in [0, 1]$, the constant map*

$$\underline{a}(x) := a$$

is a fuzzy predicate, and the scalar operations

$$(a \odot \phi)(x) := a \wedge \phi(x), \quad (a \Rightarrow \phi)(x) := a \multimap_G \phi(x)$$

are again fuzzy predicates. They satisfy

$$a \wedge \phi \leq \psi \iff \phi \leq a \Rightarrow \psi.$$

Proof. This is the Gödel instance of Propositions 1.4.7, 1.4.11, 1.4.8, and 1.4.9. □

Corollary 1.4.5. *If X is discrete, then*

$$\text{Pred}_\Lambda(X) \cong [0, 1]^X$$

with pointwise order and pointwise Gödel operations

$$(\phi \wedge \psi)(x) := \min\{\phi(x), \psi(x)\}, \quad (\phi \vee \psi)(x) := \max\{\phi(x), \psi(x)\},$$

$$(\phi \rightarrow_G \psi)(x) := \phi(x) \multimap_G \psi(x).$$

Under these operations $\text{Pred}_\Lambda(X)$ is a complete Gödel algebra, hence in particular a complete BL-algebra.

Proof. This follows from Proposition 1.4.12, since every pointwise operation on maps $X \rightarrow [0, 1]$ remains a fuzzy predicate when X is discrete. □

The natural structure here is therefore a frame equipped with an external Gödel action. In the discrete case, one recovers the familiar complete Gödel algebra of all $[0, 1]$ -valued functions on X . What fails in general is closure under the pointwise Gödel implication inherited from the base algebra.

Proposition 1.4.15. *There exists a fuzzy preorder X over the Gödel base and fuzzy predicates $\phi, \psi \in \text{Pred}_\Lambda(X)$ such that the pointwise Gödel implication*

$$(\phi \rightarrow_G \psi)(x) := \phi(x) \multimap_G \psi(x)$$

is not a fuzzy predicate. Consequently, $\text{Pred}_\Lambda(X)$ is not, in general, closed under pointwise Gödel implication.

Proof. Let $X = \{x', x\}$ with hom-values

$$X(x, x) = X(x', x') = 1, \quad X(x', x) = \frac{1}{2}, \quad X(x, x') = 0.$$

This is a fuzzy preorder. Define fuzzy predicates

$$\phi(x) = 0, \quad \phi(x') = \frac{1}{2},$$

and

$$\psi(x) = 0, \quad \psi(x') = 0.$$

Both satisfy the predicate law. But

$$(\phi \rightarrow_G \psi)(x) = 1, \quad (\phi \rightarrow_G \psi)(x') = 0.$$

The predicate law would require

$$\frac{1}{2} \wedge 1 \leq 0,$$

which is false. □

Remark 1.4.1. *This does not contradict Proposition 1.4.7. Since $\text{Pred}_\Lambda(X)$ is a frame, it has an internal Heyting implication. What fails here is not the existence of implication in the frame-theoretic sense, but closure under the **pointwise** Gödel residual inherited from the base algebra $[0, 1]$. In general, these two operations do not coincide.*

The Łukasiewicz case

The Łukasiewicz base is the quantale associated to the standard Łukasiewicz t -norm and residual implication, and underlies the algebraic semantics of Łukasiewicz and MV-valued fuzzy logic [16, Example 2.4] [2]. Let $\Lambda = [0, 1]$ with the Łukasiewicz operations

$$a \otimes b = \max\{0, a + b - 1\}, \quad a \multimap_{\text{Luk}} b = \min\{1, 1 - a + b\}.$$

Then a fuzzy predicate on a fuzzy preorder X is a map $\phi : X \rightarrow [0, 1]$ such that

$$\max\{0, X(x', x) + \phi(x) - 1\} \leq \phi(x') \quad \text{for all } x, x' \in X.$$

Proposition 1.4.16. *For every fuzzy preorder X over the Lukasiewicz base, the induced crisp order on $\text{Pred}_\Lambda(X)$ is a frame. Moreover, for every $a \in [0, 1]$, the constant map*

$$\underline{a}(x) := a$$

is a fuzzy predicate, and the scalar operations

$$(a \odot \phi)(x) := \max\{0, a + \phi(x) - 1\}, \quad (a \Rightarrow \phi)(x) := \min\{1, 1 - a + \phi(x)\}$$

are again fuzzy predicates. If one defines

$$(a \oplus \phi)(x) := \min\{1, a + \phi(x)\},$$

equivalently

$$a \oplus \phi = (1 - a) \Rightarrow \phi,$$

then

$$a \odot \phi \leq \psi \iff \phi \leq a \Rightarrow \psi.$$

Proof. This is the Lukasiewicz instance of Propositions 1.4.7, 1.4.11, 1.4.8, and 1.4.9. The formula for $a \oplus \phi$ follows from

$$(1 - a) \multimap_{\text{Luk}} u = \min\{1, a + u\}.$$

□

Corollary 1.4.6. *If X is discrete, then*

$$\text{Pred}_\Lambda(X) \cong [0, 1]^X$$

with the pointwise Lukasiewicz operations

$$(\phi \odot \psi)(x) := \max\{0, \phi(x) + \psi(x) - 1\}, \quad (\phi \oplus \psi)(x) := \min\{1, \phi(x) + \psi(x)\},$$

$$(\neg\phi)(x) := 1 - \phi(x), \quad (\phi \multimap_{\text{Luk}} \psi)(x) := \min\{1, 1 - \phi(x) + \psi(x)\}.$$

Under these operations $\text{Pred}_\Lambda(X)$ is a complete MV-algebra.

Proof. This follows from Proposition 1.4.12, since every pointwise operation on maps $X \rightarrow [0, 1]$ remains a fuzzy predicate when X is discrete. □

Remark 1.4.2. *Again, this does not contradict the fact that $\text{Pred}_\Lambda(X)$ is a frame. The frame $\text{Pred}_\Lambda(X)$ has its own internal Heyting negation, whereas the operation considered above is the **pointwise** Lukasiewicz negation inherited from the base algebra $[0, 1]$. In general these are different operations, and only the former is automatically available on every frame.*

Proposition 1.4.17. *There exists a fuzzy preorder X over the Lukasiewicz base and a fuzzy predicate $\phi \in \text{Pred}_\Lambda(X)$ such that the pointwise Lukasiewicz negation*

$$(\neg\phi)(x) := \phi(x) \multimap_{\text{Luk}} 0 = 1 - \phi(x)$$

is not a fuzzy predicate. Consequently, $\text{Pred}_\Lambda(X)$ is not, in general, closed under pointwise Lukasiewicz negation.

Proof. Let $X = \{x', x\}$ with hom-values

$$X(x, x) = X(x', x') = 1, \quad X(x', x) = \frac{1}{2}, \quad X(x, x') = 0.$$

Define

$$\phi(x) = 0, \quad \phi(x') = \frac{3}{4}.$$

Then ϕ is a fuzzy predicate, since

$$\max\{0, \frac{1}{2} + 0 - 1\} = 0 \leq \frac{3}{4}.$$

Its pointwise negation is

$$(\neg\phi)(x) = 1, \quad (\neg\phi)(x') = \frac{1}{4}.$$

The predicate law would require

$$\max\{0, \frac{1}{2} + 1 - 1\} \leq \frac{1}{4},$$

which is false. □

The Lawvere case

The Lawvere base is the quantale-enriched presentation of generalized metric spaces [12] [16, Example 2.3]. Let $\Lambda = [0, \infty]$ with the reverse of the usual order and operations

$$a \otimes b = a + b, \quad a \multimap c = c \ominus a.$$

If X is a fuzzy preorder over this base, we write $d(x, y) := X(x, y)$. Then a fuzzy predicate on X is a map $\phi : X \rightarrow [0, \infty]$ such that

$$\phi(x') \leq d(x', x) + \phi(x)$$

in the usual order. Thus fuzzy predicates are precisely Lawvere weights, i.e. contravariant modules into the Lawvere base in this enriched sense [12] [16, Example 2.3]. The terminology “predicate” reflects that they are contravariant maps into the chosen truth-value object.

Proposition 1.4.18. *For every Lawvere metric space X , equivalently every extended quasi-pseudometric space X , the induced crisp order on $\text{Pred}_\Lambda(X)$ is a frame. In the usual order on $[0, \infty]$, this amounts to reverse pointwise order:*

$$\phi \leq \psi \quad :\Longleftrightarrow \quad \phi(x) \geq \psi(x) \text{ for all } x \in X.$$

Accordingly, joins in $\text{Pred}_\Lambda(X)$ are pointwise infima in the usual order, and meets are pointwise suprema.

Moreover, for every $a \in [0, \infty]$, the constant map

$$\underline{a}(x) := a$$

is a fuzzy predicate, and the scalar operations

$$(a \odot \phi)(x) := a + \phi(x), \quad (a \Rightarrow \phi)(x) := \phi(x) \ominus a$$

are again fuzzy predicates. They satisfy

$$a \odot \phi \leq \psi \quad \Longleftrightarrow \quad \phi \leq a \Rightarrow \psi,$$

where $\leq$ denotes the crisp order on $\text{Pred}_\Lambda(X)$. Equivalently, in the usual pointwise order on $[0, \infty]$, this says

$$a + \phi(x) \geq \psi(x) \text{ for all } x \in X \quad \Longleftrightarrow \quad \phi(x) \geq \psi(x) \ominus a \text{ for all } x \in X.$$

Proof. This is the Lawvere instance of the general predicate-frame, constant-predicate, scalar-closure, and scalar-adjunction results, translated through the reversed order on $[0, \infty]$. $\square$

Corollary 1.4.7. *If X is discrete, then*

$$\text{Pred}_\Lambda(X) \cong [0, \infty]^X$$

with reverse pointwise order and pointwise operations

$$(\phi + \psi)(x) := \phi(x) + \psi(x), \quad (\psi \ominus \phi)(x) := \psi(x) \ominus \phi(x).$$

In particular, with reverse pointwise order and pointwise addition, $\text{Pred}_\Lambda(X)$ is the usual commutative unital Lawvere quantale of extended cost functions on X .

Proof. This follows from Proposition 1.4.12. $\square$

Proposition 1.4.19. *There exists a Lawvere metric space X and fuzzy predicates $\phi, \psi \in \text{Pred}_\Lambda(X)$ such that the pointwise sum*

$$(\phi + \psi)(x) := \phi(x) + \psi(x)$$

is not a fuzzy predicate.

Proof. Let $X = \{x', x\}$ with

$$d(x, x) = d(x', x') = 0, \quad d(x', x) = 1, \quad d(x, x') = \infty.$$

Define

$$\phi(x) = 0, \quad \phi(x') = 1,$$

and let $\psi = \phi$. Then ϕ and ψ are fuzzy predicates, since

$$\phi(x') = 1 \leq d(x', x) + \phi(x) = 1 + 0.$$

But

$$(\phi + \psi)(x) = 0, \quad (\phi + \psi)(x') = 2,$$

and the predicate law would require

$$2 \leq d(x', x) + 0 = 1,$$

which is false. $\square$

1.5 Fuzzy power objects and generic fuzzy predicates

We now regard the presheaf object as the **fuzzy power object** of X :

$$P_{\mathbb{V}}(X) := \text{Psh}_{\mathbb{V}}(X).$$

The term “power object” is used here in the enriched classifying sense: $P_{\mathbb{V}}(X)$ classifies indexed fuzzy presheaves by the tensor–hom adjunction. It is not asserted to be a subobject-classifier power object in an ambient topos.

1.5.1 Fuzzy power objects in a fuzzy cosmos

Let

$$(\mathbb{V}, \otimes, I, [-, -])$$

be a fuzzy cosmos, and let X be a small $\mathbb{V}$ -category.

Definition 1.5.1 (Fuzzy power object in a fuzzy cosmos). *The **fuzzy power object** of X is the possibly large $\mathbb{V}$ -category*

$$P_{\mathbb{V}}(X) := \text{Psh}_{\mathbb{V}}(X) = [X^{\text{op}}, \underline{\mathbb{V}}].$$

Thus an object of $P_{\mathbb{V}}(X)$ is a fuzzy presheaf

$$\Phi : X^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

Unpacked, an object $\Phi \in P_{\mathbb{V}}(X)$ assigns an object

$$\Phi x \in \mathbb{V}$$

to each $x \in X$, together with action morphisms

$$X(x', x) \otimes \Phi x \longrightarrow \Phi x'$$

satisfying the usual unit and associativity axioms. By Proposition 1.4.1, the hom-object between two fuzzy presheaves is

$$P_{\mathbb{V}}(X)(\Phi, \Psi) = \int_{x \in X} [\Phi x, \Psi x].$$

The fuzzy power-object construction is contravariantly functorial by reindexing.

Proposition 1.5.1 (Reindexing of fuzzy presheaves). *For every $\mathbb{V}$ -functor*

$$F : X \rightarrow Y,$$

precomposition with F^{op} defines a $\mathbb{V}$ -functor

$$P_{\mathbb{V}}(F) : P_{\mathbb{V}}(Y) \rightarrow P_{\mathbb{V}}(X), \quad P_{\mathbb{V}}(F)(\Psi) := \Psi \circ F^{\text{op}}.$$

Hence $P_{\mathbb{V}}$ is contravariantly functorial on the ordinary category of small $\mathbb{V}$ -categories.

Proof. This is the standard functoriality of enriched functor categories under precomposition [10, Section 2.2]. On hom-objects, precomposition is induced by the universal property of the end:

$$\int_{y \in Y} [\Phi y, \Psi y] \longrightarrow \int_{x \in X} [\Phi(Fx), \Psi(Fx)].$$

The identity and composition laws follow from functoriality of precomposition. $\square$

The truth-value object itself is recovered as the fuzzy power object of the enriched unit.

Proposition 1.5.2 (The fuzzy power object of the unit). *There is a canonical isomorphism of $\mathbb{V}$ -categories*

$$P_{\mathbb{V}}(\mathbf{1}) \cong \underline{\mathbb{V}}.$$

Proof. Since $\mathbf{1}^{\text{op}} \cong \mathbf{1}$, a $\mathbb{V}$ -functor

$$\mathbf{1}^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

is uniquely determined by the image of the unique object of $\mathbf{1}$. For objects $A, B \in \mathbb{V}$, the hom-object is

$$[\mathbf{1}^{\text{op}}, \underline{\mathbb{V}}](A, B) = \int_{* \in \mathbf{1}} \underline{\mathbb{V}}(A, B) = [A, B] = \underline{\mathbb{V}}(A, B).$$

$\square$

1.5.2 The generic fuzzy presheaf

A $\mathbb{V}$ -functor

$$\Phi : S \otimes X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

will be called an **S -indexed fuzzy presheaf on X** . The fuzzy power object $P_{\mathbb{V}}(X)$ classifies such indexed fuzzy presheaves through a universal evaluation fuzzy presheaf.

Definition 1.5.2 (Generic fuzzy presheaf). *Let X be a small $\mathbb{V}$ -category. The **generic fuzzy presheaf** on X is the evaluation $\mathbb{V}$ -functor*

$$G_X : P_{\mathbb{V}}(X) \otimes X^{\text{op}} \longrightarrow \underline{\mathbb{V}}.$$

Equivalently, G_X is the transpose of the identity

$$1_{P_{\mathbb{V}}(X)} : P_{\mathbb{V}}(X) \rightarrow P_{\mathbb{V}}(X)$$

under the universe-bounded tensor-hom adjunction.

Thus G_X is displayed as

$$P_{\mathbb{V}}(X) \otimes X^{\text{op}} \xrightarrow{G_X = \text{ev}} \underline{\mathbb{V}}.$$

Proposition 1.5.3 (Evaluation formula). *For every small $\mathbb{V}$ -category X , every fuzzy presheaf*

$$\Phi \in P_{\mathbb{V}}(X),$$

and every $x \in X$, the generic fuzzy presheaf is given on objects by

$$G_X(\Phi, x) = \Phi x.$$

Proof. This is the evaluation functor associated to the internal hom

$$P_{\mathbb{V}}(X) = [X^{\text{op}}, \underline{\mathbb{V}}].$$

Under the universe-bounded tensor-hom adjunction, the transpose of the identity on $P_{\mathbb{V}}(X)$ evaluates a presheaf at an object. $\square$

Proposition 1.5.4 (Naturality of the generic fuzzy presheaf). *For every $\mathbb{V}$ -functor $F : X \rightarrow Y$, the diagram*

$$\begin{array}{ccc} P_{\mathbb{V}}(Y) \otimes X^{\text{op}} & \xrightarrow{P_{\mathbb{V}}(F) \otimes 1_{X^{\text{op}}}} & P_{\mathbb{V}}(X) \otimes X^{\text{op}} \\ \downarrow 1_{P_{\mathbb{V}}(Y)} \otimes F^{\text{op}} & & \downarrow G_X \\ P_{\mathbb{V}}(Y) \otimes Y^{\text{op}} & \xrightarrow{G_Y} & \underline{\mathbb{V}} \end{array}$$

commutes.

Proof. Both composites send an object (Ψ, x) to

$$\Psi(Fx).$$

Equivalently, under the tensor-hom adjunction, both transpose to the reindexing functor $P_{\mathbb{V}}(F)$. $\square$

Proposition 1.5.5 (Principal fuzzy presheaves). *For every small $\mathbb{V}$ -category X , the Yoneda embedding*

$$y_X : X \rightarrow P_{\mathbb{V}}(X)$$

satisfies

$$G_X(y_X(x), x') = X(x', x)$$

on objects. Thus $y_X(x)$ is the principal fuzzy presheaf generated by x .

Proof. This follows from

$$y_X(x)(x') = X(x', x)$$

and the evaluation formula for G_X . □

Theorem 1.5.1 (Universal property of the fuzzy power object). *For all small $\mathbb{V}$ -categories S and X , composition with the generic fuzzy presheaf induces a natural bijection*

$$\mathbb{V}\text{-CAT}(S, P_{\mathbb{V}}(X)) \cong \mathbb{V}\text{-CAT}(S \otimes X^{\text{op}}, \underline{\mathbb{V}}),$$

given by

$$F \mapsto G_X \circ (F \otimes 1_{X^{\text{op}}}).$$

Equivalently, $\mathbb{V}$ -functors

$$S \rightarrow P_{\mathbb{V}}(X)$$

are in natural bijection with S -indexed fuzzy presheaves on X .

Proof. Since

$$P_{\mathbb{V}}(X) = [X^{\text{op}}, \underline{\mathbb{V}}],$$

this is exactly the universe-bounded tensor-hom adjunction for enriched functor categories. Explicitly, a functor

$$F : S \rightarrow P_{\mathbb{V}}(X)$$

corresponds to

$$\hat{F} : S \otimes X^{\text{op}} \rightarrow \underline{\mathbb{V}}, \quad \hat{F}(s, x) = F(s)(x),$$

and the inverse sends $H : S \otimes X^{\text{op}} \rightarrow \underline{\mathbb{V}}$ to

$$\tilde{H} : S \rightarrow P_{\mathbb{V}}(X), \quad \tilde{H}(s)(x) = H(s, x).$$

□

Remark 1.5.1. *Taking $S = \mathbf{1}$ in Theorem 1.5.1 shows that global elements*

$$\mathbf{1} \rightarrow P_{\mathbb{V}}(X)$$

are precisely fuzzy presheaves on X .

1.5.3 The fuzzy power object

We now specialize the preceding construction to the posetal fuzzy cosmos determined by a fuzzy truth-value object Λ . Let X be a fuzzy preorder over Λ .

Definition 1.5.3 (Fuzzy power object). *The **fuzzy power object** of X is*

$$P_\Lambda(X) := \text{Pred}_\Lambda(X) = [X^{\text{op}}, \Lambda].$$

Thus an element of $P_\Lambda(X)$ is precisely a fuzzy predicate on X , equivalently a map $\phi : X \rightarrow L$ satisfying

$$X(x', x) \otimes \phi(x) \leq \phi(x') \quad \text{for all } x, x' \in X.$$

Again, the term “power object” is used in the enriched classifying sense: $P_\Lambda(X)$ classifies indexed fuzzy predicates by the tensor–hom adjunction. It is not meant to identify $P_\Lambda(X)$ with a topos-theoretic power object determined by an ambient subobject classifier.

By Proposition 1.4.3, Proposition 1.4.6, and Proposition 1.4.7, the object $P_\Lambda(X)$ is a separated fuzzy preorder whose induced crisp order is the pointwise order, in fact a frame, and whose pointwise scalar operations are those of Definition 1.4.4. We shall use this structure freely.

Proposition 1.5.6. *For every Λ -functor $f : X \rightarrow Y$, the assignment*

$$P_\Lambda(f) : P_\Lambda(Y) \rightarrow P_\Lambda(X), \quad P_\Lambda(f)(\psi) := \psi \circ f,$$

defines a Λ -functor. Hence

$$P_\Lambda : \text{Pre}_\Lambda^{\text{op}} \rightarrow \text{Ord}_\Lambda$$

is a contravariant functor.

Proof. This is the posetal instance of Proposition 1.5.1. Directly, if $\psi : Y^{\text{op}} \rightarrow \Lambda$ is a fuzzy predicate, then $\psi \circ f^{\text{op}}$ is a fuzzy predicate on X . For $\phi, \psi \in P_\Lambda(Y)$, one has

$$P_\Lambda(Y)(\phi, \psi) = \bigwedge_{y \in Y} (\phi(y) \multimap \psi(y)) \leq \bigwedge_{x \in X} (\phi(fx) \multimap \psi(fx)),$$

so $P_\Lambda(f)$ is a Λ -functor. Functoriality is ordinary functoriality of precomposition. $\square$

Proposition 1.5.7 (Reindexing preserves fuzzy predicate algebra structure). *Let $f : X \rightarrow Y$ be a Λ -functor. Then the reindexing map*

$$P_\Lambda(f) : P_\Lambda(Y) \rightarrow P_\Lambda(X)$$

preserves small joins, small meets, and commutes with the pointwise scalar operations. Explicitly, for every small family $(\psi_i)_{i \in I}$ in $P_\Lambda(Y)$, every $a \in L$, and every $\psi \in P_\Lambda(Y)$,

$$\begin{aligned} P_\Lambda(f)\left(\bigvee_{i \in I} \psi_i\right) &= \bigvee_{i \in I} P_\Lambda(f)(\psi_i), & P_\Lambda(f)\left(\bigwedge_{i \in I} \psi_i\right) &= \bigwedge_{i \in I} P_\Lambda(f)(\psi_i), \\ P_\Lambda(f)(a \odot \psi) &= a \odot P_\Lambda(f)(\psi), & P_\Lambda(f)(a \Rightarrow \psi) &= a \Rightarrow P_\Lambda(f)(\psi). \end{aligned}$$

Proof. All identities are pointwise after evaluating at $x \in X$. $\square$

Proposition 1.5.8 (The fuzzy power object of the unit). *There is a canonical isomorphism of fuzzy orders*

$$P_\Lambda(1) \cong \Lambda.$$

Under this identification, a fuzzy predicate

$$\phi : 1^{\text{op}} \rightarrow \Lambda$$

corresponds to its unique value $\phi() \in L$.*

Proof. This is the posetal instance of Proposition 1.5.2. A predicate on 1 is uniquely determined by its value $a \in L$, and the hom-value between a and b is $a \multimap b$. $\square$

Corollary 1.5.1 (Constant fuzzy predicates by reindexing). *Let*

$$!_X : X \rightarrow 1$$

be the unique Λ -functor. Under the identification $P_\Lambda(1) \cong \Lambda$, the reindexing map

$$P_\Lambda(!_X) : \Lambda \rightarrow P_\Lambda(X)$$

sends $a \in L$ to the constant fuzzy predicate $\underline{a}$ on X .

Proof. It sends the unique-value predicate $a : 1^{\text{op}} \rightarrow \Lambda$ to $a \circ !_X$, the constant predicate with value a . $\square$

1.5.4 The generic fuzzy predicate

A Λ -functor

$$\Phi : S \otimes X^{\text{op}} \rightarrow \Lambda$$

will be called an **S -indexed fuzzy predicate on X** . Equivalently, by transposition, it is a Λ -functor

$$S \rightarrow P_\Lambda(X),$$

that is, an S -indexed compatible family of fuzzy predicates on X . The fuzzy power object $P_\Lambda(X)$ classifies such indexed fuzzy predicates through the posetal specialization of the generic fuzzy presheaf.

Definition 1.5.4 (Generic fuzzy predicate). *Let X be a fuzzy preorder over Λ . The **generic fuzzy predicate** on X is the evaluation Λ -functor*

$$G_X : P_\Lambda(X) \otimes X^{\text{op}} \rightarrow \Lambda.$$

Equivalently, G_X is the transpose of the identity

$$1_{P_\Lambda(X)} : P_\Lambda(X) \rightarrow P_\Lambda(X)$$

under the tensor-hom adjunction.

Proposition 1.5.9. *For every fuzzy preorder X , every $\phi \in P_\Lambda(X)$, and every $x \in X$,*

$$G_X(\phi, x) = \phi(x).$$

Thus G_X is the ordinary evaluation map.

Proof. This is the posetal instance of Proposition 1.5.3. $\square$

Remark 1.5.2 (Direct functoriality of evaluation). *Directly, the evaluation map $G_X(\phi, x) = \phi(x)$ is a Λ -functor because, for fuzzy predicates $\phi, \psi \in P_\Lambda(X)$ and elements $x, x' \in X$,*

$$P_\Lambda(X)(\phi, \psi) \otimes X^{\text{op}}(x, x') \otimes \phi(x) = P_\Lambda(X)(\phi, \psi) \otimes X(x', x) \otimes \phi(x).$$

Since ϕ is a fuzzy predicate,

$$X(x', x) \otimes \phi(x) \leq \phi(x'),$$

and since

$$P_\Lambda(X)(\phi, \psi) \leq \phi(x') \multimap \psi(x'),$$

we obtain

$$P_\Lambda(X)(\phi, \psi) \otimes X(x', x) \otimes \phi(x) \leq \psi(x').$$

By residuation,

$$P_\Lambda(X)(\phi, \psi) \otimes X^{\text{op}}(x, x') \leq \phi(x) \multimap \psi(x'),$$

which is exactly the functoriality condition for

$$G_X : P_\Lambda(X) \otimes X^{\text{op}} \rightarrow \Lambda.$$

Proposition 1.5.10 (Naturality of the generic fuzzy predicate). *For every Λ -functor $f : X \rightarrow Y$, the diagram*

$$\begin{array}{ccc} P_\Lambda(Y) \otimes X^{\text{op}} & \xrightarrow{P_\Lambda(f) \otimes 1_{X^{\text{op}}}} & P_\Lambda(X) \otimes X^{\text{op}} \\ \downarrow 1_{P_\Lambda(Y)} \otimes f^{\text{op}} & & \downarrow G_X \\ P_\Lambda(Y) \otimes Y^{\text{op}} & \xrightarrow{G_Y} & \Lambda \end{array}$$

commutes. Equivalently, for every $\psi \in P_\Lambda(Y)$ and every $x \in X$,

$$G_X(P_\Lambda(f)(\psi), x) = G_Y(\psi, f(x)).$$

Proof. This is the posetal instance of Proposition 1.5.4; objectwise both sides are $\psi(fx)$. $\square$

Proposition 1.5.11 (Principal fuzzy predicates). *For every fuzzy preorder X , the Yoneda map*

$$y_X : X \rightarrow P_\Lambda(X)$$

satisfies

$$G_X(y_X(x), x') = X(x', x) \quad \text{for all } x, x' \in X.$$

Thus $y_X(x)$ is the principal fuzzy predicate generated by x . If X is separated, then y_X is injective.

Proof. This is the posetal instance of Proposition 1.5.5. Injectivity follows from Corollary 1.4.2. $\square$

Theorem 1.5.2 (Universal property of the fuzzy power object). *For all fuzzy preorders S and X , composition with the generic fuzzy predicate induces a natural bijection*

$$\text{Pre}_\Lambda(S, P_\Lambda(X)) \cong \text{Pre}_\Lambda(S \otimes X^{\text{op}}, \Lambda),$$

given by

$$f \mapsto G_X \circ (f \otimes 1_{X^{\text{op}}}).$$

Equivalently, Λ -functors $S \rightarrow P_\Lambda(X)$ are in natural bijection with S -indexed fuzzy predicates on X , with inverse assignment

$$\Phi \mapsto \tilde{\Phi}, \quad \tilde{\Phi}(s)(x) := \Phi(s, x).$$

Proof. This is the posetal instance of Theorem 1.5.1, equivalently Corollary 1.3.2. The displayed formulas are exactly the objectwise formulas for currying and evaluation. $\square$

Remark 1.5.3. Taking $S = 1$ in Theorem 1.5.2 shows that global elements

$$1 \rightarrow P_\Lambda(X)$$

are the same thing as fuzzy predicates on X .

1.5.5 Running examples

We now record the form taken by the fuzzy power object and the generic fuzzy predicate for the main bases considered so far.

Example 1.5.1 (The crisp power object). Let $\Lambda = \{0 < 1\}$, so that fuzzy preorders are ordinary preorders. By Proposition 1.4.13,

$$P_\Lambda(X) = \text{Pred}_\Lambda(X)$$

is the preorder of lower sets of X , ordered by inclusion. For a lower set $A \subseteq X$, the generic fuzzy predicate is ordinary membership:

$$G_X(A, x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Thus monotone maps

$$S \rightarrow P_\Lambda(X)$$

are precisely S -indexed compatible families of lower sets of X . If X is discrete, then $P_\Lambda(X) \cong 2^X$, and G_X is the ordinary element-of relation.

Example 1.5.2 (The Gödel power object). Let $\Lambda = [0, 1]$ with the Gödel operations. Then

$$P_\Lambda(X) = \text{Pred}_\Lambda(X)$$

consists of all maps $\phi : X \rightarrow [0, 1]$ satisfying

$$X(x', x) \wedge \phi(x) \leq \phi(x') \quad \text{for all } x, x' \in X.$$

These are the lower fuzzy predicates on X in the Gödel sense. The generic fuzzy predicate is evaluation:

$$G_X(\phi, x) = \phi(x).$$

If X is discrete, then $P_\Lambda(X) \cong [0, 1]^X$.

Example 1.5.3 (The Łukasiewicz power object). Let $\Lambda = [0, 1]$ with the Łukasiewicz operations. Then

$$P_\Lambda(X) = \text{Pred}_\Lambda(X)$$

consists of all maps $\phi : X \rightarrow [0, 1]$ satisfying

$$\max\{0, X(x', x) + \phi(x) - 1\} \leq \phi(x') \quad \text{for all } x, x' \in X.$$

These are the lower fuzzy predicates compatible with Łukasiewicz transitivity. As before,

$$G_X(\phi, x) = \phi(x).$$

If X is discrete, then $P_\Lambda(X) \cong [0, 1]^X$, now equipped with the pointwise Łukasiewicz operations.

Example 1.5.4 (The Lawvere power object). *Let $\Lambda = [0, \infty]$ with the reverse of the usual order, so that fuzzy preorders are Lawvere metric spaces. Writing $d(x, y) = X(x, y)$, the fuzzy power object*

$$P_\Lambda(X) = \text{Pred}_\Lambda(X)$$

consists of all maps $\phi : X \rightarrow [0, \infty]$ satisfying

$$\phi(x') \leq d(x', x) + \phi(x) \quad \text{in the usual order, for all } x, x' \in X.$$

These are precisely the Lawvere weights on X , read here as Lawvere-valued fuzzy predicates. The generic fuzzy predicate is again evaluation:

$$G_X(\phi, x) = \phi(x).$$

If X is discrete, then $P_\Lambda(X) \cong [0, \infty]^X$, the quantale of extended cost functions on X with reverse pointwise order.

Example 1.5.5 (A non-chain coordinatewise power object). *Let*

$$\Lambda = [0, 1] \times [0, 1]$$

with the coordinatewise Lukasiewicz operations, ordered coordinatewise. If X is a fuzzy preorder over Λ , write

$$X(x, x') = (X_1(x, x'), X_2(x, x')).$$

Then a fuzzy predicate

$$\phi : X \rightarrow [0, 1] \times [0, 1]$$

may be written as

$$\phi(x) = (\phi_1(x), \phi_2(x)),$$

and the fuzzy predicate condition is equivalent to the pair of coordinatewise conditions

$$\max\{0, X_i(x', x) + \phi_i(x) - 1\} \leq \phi_i(x') \quad (i = 1, 2).$$

Thus $P_\Lambda(X)$ consists exactly of pairs of Lukasiewicz fuzzy predicates, ordered and acted on coordinatewise, and the generic fuzzy predicate is

$$G_X((\phi_1, \phi_2), x) = (\phi_1(x), \phi_2(x)).$$

This illustrates that the fuzzy power-object construction is not restricted to chain-valued bases.

1.6 Cut subobjects and the representation theorem

In the preceding sections we defined the fuzzy power object of a small $\mathbb{V}$ -category X by

$$P_{\mathbb{V}}(X) = [X^{\text{op}}, \underline{\mathbb{V}}],$$

together with its generic fuzzy presheaf

$$G_X : P_{\mathbb{V}}(X) \otimes X^{\text{op}} \longrightarrow \underline{\mathbb{V}}.$$

We now show that this power object represents enriched cut subobjects.

1.6.1 The fuzzy cut classifier

Let

$$(\mathbb{V}, \otimes, I, [-, -])$$

be a fuzzy cosmos, and let $\underline{\mathbb{V}}$ be its self-enriched truth-value object. Thus the objects of $\underline{\mathbb{V}}$ are the objects of $\mathbb{V}$, and the hom-object from A to B is

$$\underline{\mathbb{V}}(A, B) = [A, B].$$

Definition 1.6.1 (Cut classifier). *The **cut classifier** is the self-enriched hom-functor*

$$\text{Cut}_{\mathbb{V}} : \underline{\mathbb{V}}^{\text{op}} \otimes \underline{\mathbb{V}} \longrightarrow \underline{\mathbb{V}}$$

defined on objects by

$$\text{Cut}_{\mathbb{V}}(A, B) := [A, B].$$

Equivalently, $\text{Cut}_{\mathbb{V}}$ is the Yoneda probe of the truth value B by the level A .

Proposition 1.6.1. *The assignment*

$$(A, B) \longmapsto [A, B]$$

defines a $\mathbb{V}$ -functor

$$\text{Cut}_{\mathbb{V}} : \underline{\mathbb{V}}^{\text{op}} \otimes \underline{\mathbb{V}} \longrightarrow \underline{\mathbb{V}}.$$

Proof. For objects (A, B) and (A', B') , functoriality requires a morphism

$$[A', A] \otimes [B, B'] \longrightarrow [[A, B], [A', B']].$$

By closedness, this is equivalently a morphism

$$[A', A] \otimes [B, B'] \otimes [A, B] \otimes A' \longrightarrow B'.$$

Using symmetry and associativity, followed by the evaluation maps, we define it by the composite

$$\begin{aligned} [A', A] \otimes [B, B'] \otimes [A, B] \otimes A' &\longrightarrow [B, B'] \otimes [A, B] \otimes [A', A] \otimes A' \\ &\longrightarrow [B, B'] \otimes [A, B] \otimes A \\ &\longrightarrow [B, B'] \otimes B \\ &\longrightarrow B'. \end{aligned}$$

The identity and composition axioms are the unit and associativity axioms for the self-enrichment of $\mathbb{V}$. $\square$

1.6.2 Fuzzy cut profiles and cut nerves

Let X be a small $\mathbb{V}$ -category. Let

$$\mathcal{P}(\underline{\mathbb{V}})$$

denote the $\mathbb{V}$ -category of small presheaves on the possibly large $\mathbb{V}$ -category $\underline{\mathbb{V}}$. The Yoneda embedding is

$$Y : \underline{\mathbb{V}} \longrightarrow \mathcal{P}(\underline{\mathbb{V}}), \quad B \longmapsto \underline{\mathbb{V}}(-, B) = [-, B].$$

$$G_X^{\text{cut}}(A, \Phi, x) = [A, \Phi x].$$

Proposition 1.6.2 (Universality of the generic cut). *Let*

$$F : S \rightarrow P_{\mathbb{V}}(X)$$

classify an S -indexed fuzzy presheaf

$$\widehat{F} : S \otimes X^{\text{op}} \longrightarrow \underline{\mathbb{V}}.$$

Then the associated S -indexed cut profile is obtained from the generic cut by substitution:

$$\begin{array}{ccc} \underline{\mathbb{V}}^{\text{op}} \otimes S \otimes X^{\text{op}} & \xrightarrow{1 \otimes F \otimes 1} & \underline{\mathbb{V}}^{\text{op}} \otimes P_{\mathbb{V}}(X) \otimes X^{\text{op}} \xrightarrow{G_X^{\text{cut}}} \underline{\mathbb{V}}. \\ & \searrow & \nearrow \\ & N_{S,X}(\widehat{F}) & \end{array}$$

Objectwise,

$$N_{S,X}(\widehat{F})(A, s, x) = [A, \widehat{F}(s, x)] = [A, F(s)(x)].$$

Proof. Since F classifies $\widehat{F}$, one has

$$\widehat{F}(s, x) = F(s)(x).$$

Substitution into the cut classifier gives

$$\text{Cut}_{\mathbb{V}}(A, \widehat{F}(s, x)) = [A, \widehat{F}(s, x)] = [A, F(s)(x)].$$

□

1.6.3 Realization and closed cut profiles

The Yoneda embedding

$$Y : \underline{\mathbb{V}} \rightarrow \mathcal{P}(\underline{\mathbb{V}})$$

has a left adjoint

$$\sup_{\mathbb{V}} : \mathcal{P}(\underline{\mathbb{V}}) \longrightarrow \underline{\mathbb{V}},$$

given by the canonical weighted colimit

$$\sup_{\mathbb{V}}(F) = \int^{A \in \underline{\mathbb{V}}} F(A) \otimes A.$$

This notation denotes the weighted colimit of the identity functor on $\underline{\mathbb{V}}$ by the small presheaf F . Thus

$$\sup_{\mathbb{V}} \dashv Y.$$

The counit of this adjunction is the enriched co-Yoneda isomorphism

$$\int^{A \in \underline{\mathbb{V}}} [A, B] \otimes A \cong B$$

[10, Section 1.9].

Definition 1.6.5 (Realization of a cut profile). *Let*

$$C : X^{\text{op}} \rightarrow \mathcal{P}(\underline{\mathbb{V}})$$

*be a cut profile. Its **realization** is the fuzzy presheaf*

$$R_X(C) : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

defined pointwise by

$$R_X(C)(x) := \sup_{\mathbb{V}}(Cx).$$

Equivalently,

$$R_X(C)(x) = \int^{A \in \underline{\mathbb{V}}} Cx(A) \otimes A.$$

Since $R_X(C) = \sup_{\mathbb{V}} \circ C$, functoriality of $R_X(C)$ follows from functoriality of C and of $\sup_{\mathbb{V}}$. Therefore realization defines a $\mathbb{V}$ -functor

$$R_X : \text{FCut}_{\mathbb{V}}(X) \longrightarrow P_{\mathbb{V}}(X).$$

Proposition 1.6.3 (Realization–nerve adjunction). *For every small $\mathbb{V}$ -category X , there is an adjunction of $\mathbb{V}$ -categories*

$$R_X \dashv N_X : \text{FCut}_{\mathbb{V}}(X) \rightleftarrows P_{\mathbb{V}}(X).$$

Equivalently, for every cut profile C and every fuzzy presheaf Φ , there is a natural isomorphism

$$P_{\mathbb{V}}(X)(R_X(C), \Phi) \cong \text{FCut}_{\mathbb{V}}(X)(C, N_X(\Phi)).$$

Proof. This is the pointwise adjunction induced by

$$\sup_{\mathbb{V}} \dashv Y.$$

Using the enriched functor-category hom formula over the small category X , we compute

$$\begin{aligned} P_{\mathbb{V}}(X)(R_X(C), \Phi) &= \int_{x \in X} \underline{\mathbb{V}}(\sup_{\mathbb{V}}(Cx), \Phi x) \\ &\cong \int_{x \in X} \mathcal{P}(\underline{\mathbb{V}})(Cx, Y(\Phi x)) \\ &= \text{FCut}_{\mathbb{V}}(X)(C, N_X(\Phi)). \end{aligned}$$

□

Proposition 1.6.4 (Co-Yoneda counit). *For every fuzzy presheaf*

$$\Phi : X^{\text{op}} \rightarrow \underline{\mathbb{V}},$$

the counit of the adjunction

$$\varepsilon_{\Phi} : R_X N_X(\Phi) \longrightarrow \Phi$$

is an isomorphism.

Proof. Evaluating at $x \in X$, we have

$$R_X N_X(\Phi)(x) = \sup_{\mathbb{V}}(Y(\Phi x)) = \int^{A \in \mathbb{V}} [A, \Phi x] \otimes A.$$

By the enriched co-Yoneda lemma, this coend is canonically isomorphic to Φx . These isomorphisms are natural in x . $\square$

Corollary 1.6.1. *The intrinsic cut nerve functor*

$$N_X : P_{\mathbb{V}}(X) \longrightarrow \mathbf{FCut}_{\mathbb{V}}(X)$$

is fully faithful.

Proof. In an adjunction $R_X \dashv N_X$, the right adjoint N_X is fully faithful precisely when the counit $R_X N_X \rightarrow 1$ is an isomorphism. This is Proposition 1.6.4. $\square$

Definition 1.6.6 (Closed cut profile). *A cut profile*

$$C : X^{\text{op}} \rightarrow \mathcal{P}(\underline{\mathbb{V}})$$

*is called **closed** if the unit of the adjunction*

$$\eta_C : C \longrightarrow N_X R_X(C)$$

is an isomorphism.

Objectwise, the unit is the canonical map

$$\eta_{C,x} : Cx \longrightarrow Y(\sup_{\mathbb{V}}(Cx)).$$

Equivalently, in uncurried notation it is the map

$$\eta_{C,A,x} : C(A, x) \longrightarrow [A, R_X(C)(x)] = \left[A, \int^{B \in \mathbb{V}} C(B, x) \otimes B \right],$$

obtained by transposing the canonical cowedge map

$$C(A, x) \otimes A \longrightarrow \int^{B \in \mathbb{V}} C(B, x) \otimes B.$$

Thus C is closed precisely when, for every $x \in X$, the small level profile

$$C(-, x)$$

is represented by its realization

$$R_X(C)(x).$$

Definition 1.6.7 (Cut subobjects). *The $\mathbb{V}$ -category of **enriched cut subobjects** of X is the full $\mathbb{V}$ -subcategory*

$$\mathbf{Sub}_{\mathbb{V}}^{\text{cut}}(X) \subseteq \mathbf{FCut}_{\mathbb{V}}(X)$$

spanned by the closed cut profiles.

Theorem 1.6.1 (Enriched cut representation theorem). *For every small $\mathbb{V}$ -category X , the cut nerve and realization restrict to an equivalence of $\mathbb{V}$ -categories*

$$P_{\mathbb{V}}(X) \simeq \text{Sub}_{\mathbb{V}}^{\text{cut}}(X).$$

More explicitly, the functors

$$N_X : P_{\mathbb{V}}(X) \longrightarrow \text{Sub}_{\mathbb{V}}^{\text{cut}}(X)$$

and

$$R_X : \text{Sub}_{\mathbb{V}}^{\text{cut}}(X) \longrightarrow P_{\mathbb{V}}(X)$$

are inverse up to the unit and counit isomorphisms of the adjunction

$$R_X \dashv N_X.$$

Proof. By Proposition 1.6.4, for every fuzzy presheaf Φ , the counit

$$R_X N_X(\Phi) \longrightarrow \Phi$$

is an isomorphism. Hence N_X is fully faithful.

The essential image of a fully faithful right adjoint consists precisely of those objects for which the unit is an isomorphism. In the present case, these are exactly the closed cut profiles. Restricting N_X and R_X gives the claimed equivalence. $\square$

1.6.4 Top-support cuts in the posetal case

We now specialize the above construction to the posetal fuzzy cosmos determined by a fuzzy truth-value object

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp, \top).$$

Write $\underline{\Lambda}$ for the associated Λ -preorder of truth values, with object set L and hom-values

$$\underline{\Lambda}(a, b) = a \multimap b.$$

The cut classifier becomes the residual:

$$\text{Cut}_{\Lambda}(a, b) = a \multimap b.$$

Its top support recovers the underlying order relation on truth values.

Proposition 1.6.5. *For all $a, b \in L$,*

$$\top \leq a \multimap b \iff a \leq b.$$

Proof. By residuation,

$$\top \leq a \multimap b \iff a \otimes \top \leq b.$$

Since Λ is integral, $a \otimes \top = a$. $\square$

Definition 1.6.8 (Principal upper cut classifiers). *For $a \in L$, let*

$$\uparrow a := \{b \in L \mid a \leq b\},$$

regarded as the induced full subpreorder of $\underline{\Lambda}$, and let

$$j_a : \uparrow a \hookrightarrow \underline{\Lambda}$$

be the inclusion.

The inclusion j_a is the top-support shadow of the enriched probe

$$b \mapsto a \multimap b.$$

Indeed,

$$b \in \uparrow a \iff a \leq b \iff \top \leq a \multimap b.$$

1.6.5 Level cuts of fuzzy predicates

Let X be a fuzzy preorder over Λ .

Definition 1.6.9 (Level cuts). *Let*

$$\phi : X^{\text{op}} \rightarrow \underline{\Lambda}$$

be a fuzzy predicate on X . For $a \in L$, the a -cut of ϕ is the induced fuzzy subpreorder

$$X_{\phi,a} \hookrightarrow X$$

on the underlying subset

$$|X_{\phi,a}| := \{x \in X \mid a \leq \phi(x)\} = \phi^{-1}(\uparrow a).$$

We write

$$m_{\phi,a} : X_{\phi,a} \hookrightarrow X$$

for the inclusion.

When working with induced fuzzy subpreorders of X , we identify them with their underlying subsets whenever no confusion can arise.

Proposition 1.6.6 (Level cuts as classifier pullbacks). *Let*

$$\phi : X^{\text{op}} \rightarrow \underline{\Lambda}$$

be a fuzzy predicate. For each $a \in L$, the a -cut

$$m_{\phi,a} : X_{\phi,a} \hookrightarrow X$$

is classified by

$$j_a : \uparrow a \hookrightarrow \underline{\Lambda}$$

through the pullback square

$$\begin{array}{ccc} X_{\phi,a}^{\text{op}} & \xrightarrow{\phi_a} & \uparrow a \\ m_{\phi,a}^{\text{op}} \downarrow & & \downarrow j_a \\ X^{\text{op}} & \xrightarrow{\phi} & \underline{\Lambda}. \end{array}$$

Proof. We use the conical pullbacks in Pre_{Λ} described in Lemma 1.3.1. The pullback of j_a along ϕ has underlying set

$$\{x \in X \mid \phi(x) \in \uparrow a\} = \{x \in X \mid a \leq \phi(x)\}.$$

For $x, y \in X_{\phi,a}$, the hom-value in the pullback is

$$X^{\text{op}}(x, y) \wedge \underline{\Lambda}(\phi x, \phi y) = X(y, x) \wedge (\phi(x) \multimap \phi(y)).$$

Since $\phi : X^{\text{op}} \rightarrow \underline{\Lambda}$ is a Λ -functor,

$$X^{\text{op}}(x, y) \leq \phi(x) \multimap \phi(y).$$

Therefore the meet is just $X^{\text{op}}(x, y)$, so the pullback carries exactly the induced full fuzzy subpreorder on the displayed subset. $\square$

Proposition 1.6.7 (Basic properties of level cuts). *Let*

$$\phi : X^{\text{op}} \rightarrow \underline{\Lambda}$$

be a fuzzy predicate. Then:

$$(F1) \quad X_{\phi, \perp} = X;$$

$$(F2) \quad \text{if } a \leq b, \text{ then}$$

$$X_{\phi, b} \subseteq X_{\phi, a};$$

$$(F3) \quad \text{for every small family } (a_i)_{i \in I} \text{ in } L,$$

$$X_{\phi, \bigvee_{i \in I} a_i} = \bigcap_{i \in I} X_{\phi, a_i}$$

as induced fuzzy subpreorders of X ;

$$(F4) \quad \text{if } x \in X_{\phi, a}, \text{ then for every } x' \in X,$$

$$x' \in X_{\phi, X(x', x) \otimes a}.$$

Proof. The first three statements follow immediately from the definition of the cuts and the lattice order on L . For the fourth, if $x \in X_{\phi, a}$, then $a \leq \phi(x)$. Since ϕ is a fuzzy predicate,

$$X(x', x) \otimes \phi(x) \leq \phi(x').$$

Hence

$$X(x', x) \otimes a \leq \phi(x'),$$

which is exactly the asserted membership. $\square$

Proposition 1.6.8 (Recovery from level cuts). *Let*

$$\phi : X^{\text{op}} \rightarrow \underline{\Lambda}$$

be a fuzzy predicate. Then for every $x \in X$,

$$\phi(x) = \bigvee \{a \in L \mid x \in X_{\phi, a}\}.$$

Proof. By definition,

$$x \in X_{\phi, a} \iff a \leq \phi(x).$$

Therefore the displayed join is the join of all elements below $\phi(x)$, hence is $\phi(x)$. $\square$

1.6.6 Fuzzy cut subobjects and classifying predicates

The preceding properties motivate the intrinsic posetal definition. In the posetal case we now pass from enriched closed cut profiles to their top-support presentation. Thus the notation

$$\text{Sub}_\Lambda^{\text{cut}}(X)$$

below denotes the levelwise cut-family presentation of the closed enriched cut profiles, not the raw Λ -enriched presheaf profiles themselves. These are cut-theoretic subobjects classified by fuzzy predicates; they should not be confused with arbitrary categorical subobjects of X in Pre_Λ .

Definition 1.6.10 (Fuzzy cut subobject). *A **fuzzy cut subobject** of X , or simply a **fuzzy cut subobject** in this section, is a family*

$$A = (m_{A,a} : A_a \hookrightarrow X)_{a \in L}$$

of induced fuzzy subpreorders of X such that:

$$(F1) \quad A_\perp = X;$$

$$(F2) \quad \text{if } a \leq b, \text{ then}$$

$$A_b \subseteq A_a;$$

$$(F3) \quad \text{for every small family } (a_i)_{i \in I} \text{ in } L,$$

$$A_{\bigvee_{i \in I} a_i} = \bigcap_{i \in I} A_{a_i}$$

as induced fuzzy subpreorders of X ;

$$(F4) \quad \text{if } x \in A_a, \text{ then for every } x' \in X,$$

$$x' \in A_{X(x',x) \otimes a}.$$

We write

$$\text{Sub}_\Lambda^{\text{cut}}(X)$$

for the collection of all fuzzy cut subobjects of X .

Definition 1.6.11 (Classifying fuzzy predicate). *Let*

$$A = (m_{A,a} : A_a \hookrightarrow X)_{a \in L} \in \text{Sub}_\Lambda^{\text{cut}}(X).$$

Define

$$\chi_A : X \rightarrow L, \quad \chi_A(x) := \bigvee \{a \in L \mid x \in A_a\}.$$

*We call χ_A the **classifying fuzzy predicate** of A .*

This formula is the top-support shadow of the enriched co-Yoneda reconstruction. The enriched nerve of a fuzzy predicate ϕ is

$$N_X(\phi)(a, x) = a \multimap \phi(x).$$

Passing to top support retains precisely the levels satisfying

$$\top \leq a \multimap \phi(x),$$

equivalently

$$a \leq \phi(x).$$

The reconstructed truth value is the join of the retained levels.

Proposition 1.6.9. *For every fuzzy cut subobject $A \in \text{Sub}_\Lambda^{\text{cut}}(X)$, the map*

$$\chi_A : X^{\text{op}} \rightarrow \underline{\Lambda}$$

is a fuzzy predicate.

Proof. Let $x, x' \in X$, and set

$$S_x := \{a \in L \mid x \in A_a\}.$$

Then

$$\chi_A(x) = \bigvee S_x.$$

If $a \in S_x$, then $x \in A_a$, so by Definition 1.6.10(F4)

$$x' \in A_{X(x',x) \otimes a}.$$

Hence

$$X(x',x) \otimes a \leq \chi_A(x').$$

Taking the join over all $a \in S_x$, and using preservation of joins by $X(x',x) \otimes -$, gives

$$X(x',x) \otimes \chi_A(x) = \bigvee_{a \in S_x} (X(x',x) \otimes a) \leq \chi_A(x').$$

□

Proposition 1.6.10. *Let*

$$A = (m_{A,a} : A_a \hookrightarrow X)_{a \in L} \in \text{Sub}_\Lambda^{\text{cut}}(X).$$

Then for every $a \in L$,

$$A_a = \{x \in X \mid a \leq \chi_A(x)\}$$

as induced fuzzy subpreorders of X . Equivalently,

$$A_a = X_{\chi_A, a}.$$

In particular, every fuzzy cut subobject is uniquely determined by its classifying fuzzy predicate.

Proof. Fix $a \in L$ and $x \in X$.

If $x \in A_a$, then a belongs to the set whose join defines $\chi_A(x)$, so $a \leq \chi_A(x)$.

Conversely, suppose $a \leq \chi_A(x)$. Let

$$S_x := \{b \in L \mid x \in A_b\}.$$

Since $A_\perp = X$, the set S_x is nonempty. By definition,

$$\chi_A(x) = \bigvee S_x.$$

By Definition 1.6.10(F3),

$$A_{\chi_A(x)} = A_{\bigvee S_x} = \bigcap_{b \in S_x} A_b.$$

Since $x \in A_b$ for every $b \in S_x$, we have $x \in A_{\chi_A(x)}$. Now $a \leq \chi_A(x)$, so Definition 1.6.10(F2) gives

$$A_{\chi_A(x)} \subseteq A_a.$$

Hence $x \in A_a$. Since all subpreorders are induced from X , equality of underlying subsets gives equality as induced fuzzy subpreorders. □

1.6.7 The fuzzy preorder of fuzzy cut subobjects

We next put a fuzzy preorder structure on $\text{Sub}_\Lambda^{\text{cut}}(X)$. Instead of merely transporting the hom-values from $P_\Lambda(X)$ by the classifying predicate, we define the hom value directly in terms of cut containment.

Definition 1.6.12 (Fuzzy hom between cut subobjects). *For $A, B \in \text{Sub}_\Lambda^{\text{cut}}(X)$, define*

$$\text{Sub}_\Lambda^{\text{cut}}(X)(A, B) := \bigvee \{r \in L \mid \forall a \in L, A_a \subseteq B_{r \otimes a}\}.$$

Thus the degree to which A is contained in B is the greatest truth value r such that every a -cut of A is contained in the $(r \otimes a)$ -cut of B .

Lemma 1.6.1 (Cut-containment hom formula). *For all $A, B \in \text{Sub}_\Lambda^{\text{cut}}(X)$,*

$$\text{Sub}_\Lambda^{\text{cut}}(X)(A, B) = \bigwedge_{x \in X} (\chi_A(x) \multimap \chi_B(x)).$$

Proof. Let

$$q := \bigwedge_{x \in X} (\chi_A(x) \multimap \chi_B(x)).$$

We show that, for $r \in L$,

$$r \leq q$$

if and only if

$$\forall a \in L, A_a \subseteq B_{r \otimes a}.$$

By residuation, $r \leq q$ is equivalent to

$$r \otimes \chi_A(x) \leq \chi_B(x) \quad \text{for every } x \in X.$$

Since

$$\chi_A(x) = \bigvee \{a \in L \mid x \in A_a\}$$

and $r \otimes -$ preserves joins, this is equivalent to requiring that whenever $x \in A_a$, one has

$$r \otimes a \leq \chi_B(x).$$

By Proposition 1.6.10 applied to B , this is equivalent to

$$x \in B_{r \otimes a}.$$

Thus $r \leq q$ is equivalent to the displayed containment condition. Therefore the set whose join defines $\text{Sub}_\Lambda^{\text{cut}}(X)(A, B)$ is the principal downset of q , and its join is q . $\square$

Proposition 1.6.11. *The hom-values of Definition 1.6.12 make $\text{Sub}_\Lambda^{\text{cut}}(X)$ into a fuzzy preorder. Moreover, the classifying-predicate assignment*

$$\text{Cl}_X : \text{Sub}_\Lambda^{\text{cut}}(X) \longrightarrow P_\Lambda(X), \quad A \longmapsto \chi_A$$

is fully faithful.

Proof. By Lemma 1.6.1,

$$\text{Sub}_\Lambda^{\text{cut}}(X)(A, B) = \bigwedge_{x \in X} (\chi_A(x) \multimap \chi_B(x)) = P_\Lambda(X)(\chi_A, \chi_B).$$

Since $P_\Lambda(X)$ is a fuzzy preorder, the displayed equality implies the fuzzy preorder axioms for $\text{Sub}_\Lambda^{\text{cut}}(X)$. It also says exactly that Cl_X is fully faithful. $\square$

1.6.8 The fuzzy cut-subobject representation theorem

For $\phi \in P_\Lambda(X)$, regarded as a fuzzy predicate on X , write

$$X_\phi := (m_{\phi,a} : X_{\phi,a} \hookrightarrow X)_{a \in L}$$

for its cut family.

There are two canonical assignments:

$$\text{Cut}_X : P_\Lambda(X) \longrightarrow \text{Sub}_\Lambda^{\text{cut}}(X), \quad \phi \longmapsto X_\phi,$$

and

$$\text{Cl}_X : \text{Sub}_\Lambda^{\text{cut}}(X) \longrightarrow P_\Lambda(X), \quad A \longmapsto \chi_A.$$

Theorem 1.6.2 (Fuzzy cut-subobject representation theorem). *For every fuzzy preorder X , the assignments*

$$\text{Cut}_X : P_\Lambda(X) \longrightarrow \text{Sub}_\Lambda^{\text{cut}}(X), \quad \text{Cl}_X : \text{Sub}_\Lambda^{\text{cut}}(X) \longrightarrow P_\Lambda(X)$$

are mutually inverse isomorphisms of fuzzy preorders:

$$\text{Cl}_X \circ \text{Cut}_X = 1_{P_\Lambda(X)}, \quad \text{Cut}_X \circ \text{Cl}_X = 1_{\text{Sub}_\Lambda^{\text{cut}}(X)}.$$

Hence

$$P_\Lambda(X) \cong \text{Sub}_\Lambda^{\text{cut}}(X)$$

as fuzzy preorders.

Proof. Let $\phi \in P_\Lambda(X)$. By Proposition 1.6.7, the cut family X_ϕ is a fuzzy cut subobject of X , so Cut_X is well-defined on objects.

Conversely, if $A \in \text{Sub}_\Lambda^{\text{cut}}(X)$, then Proposition 1.6.9 shows that

$$\chi_A \in P_\Lambda(X),$$

so Cl_X is well-defined on objects.

For $\phi \in P_\Lambda(X)$, Proposition 1.6.8 gives

$$\chi_{X_\phi}(x) = \phi(x)$$

for every $x \in X$. Hence

$$\text{Cl}_X(\text{Cut}_X(\phi)) = \phi.$$

For $A \in \text{Sub}_\Lambda^{\text{cut}}(X)$, Proposition 1.6.10 gives

$$(X_{\chi_A})_a = A_a$$

for every $a \in L$. Hence

$$\text{Cut}_X(\text{Cl}_X(A)) = A.$$

Finally, Proposition 1.6.11 gives

$$\text{Sub}_\Lambda^{\text{cut}}(X)(A, B) = P_\Lambda(X)(\chi_A, \chi_B)$$

for all $A, B \in \text{Sub}_\Lambda^{\text{cut}}(X)$. Thus Cl_X is fully faithful. Since Cut_X and Cl_X are inverse on objects, they are mutually inverse isomorphisms of fuzzy preorders. $\square$

1.6.9 Examples of the cut-subobject representation theorem

We finish by spelling out the representation theorem in the principal examples. In each case, the word “subobject” refers to the coherent cut-subobjects of Definition 1.6.10, not to arbitrary monomorphism subobjects of X in Pre_Λ . Thus the theorem says that fuzzy predicates on X are equivalently coherent systems of levelwise induced subpreorders of X .

Example 1.6.1 (The crisp case). *Let*

$$\Lambda = 2 = \{0 < 1\}$$

with tensor product $a \otimes b = a \wedge b$. A fuzzy preorder over Λ is an ordinary preorder. A fuzzy predicate

$$\phi : X^{\text{op}} \rightarrow 2$$

is exactly a lower subset of X . Indeed, if

$$U := \{x \in X \mid \phi(x) = 1\},$$

then the predicate law says that whenever $x \in U$ and $x' \leq x$, one has $x' \in U$.

The cut family of ϕ has only two levels:

$$X_{\phi,0} = X, \quad X_{\phi,1} = U.$$

Conversely, a fuzzy cut subobject

$$A = (A_0, A_1)$$

is determined by A_1 , since axiom (F1) gives

$$A_0 = X.$$

Axiom (F4) says exactly that A_1 is lower: if $x \in A_1$ and $x' \leq x$, then $x' \in A_1$. Hence the classifying predicate is simply the characteristic map

$$\chi_A(x) = \begin{cases} 1, & x \in A_1, \\ 0, & x \notin A_1. \end{cases}$$

The fuzzy hom between cut subobjects also reduces to ordinary inclusion. Indeed,

$$\text{Sub}_2^{\text{cut}}(X)(A, B) = 1 \iff A_1 \subseteq B_1.$$

Thus Theorem 1.6.2 specializes to the usual correspondence between lower subsets of a preorder and their characteristic maps. If X is discrete, this is the ordinary correspondence between subsets and characteristic functions.

Example 1.6.2 (The Gödel case). *Let*

$$\Lambda = ([0, 1], \leq, \wedge, \vee, \otimes, \multimap, 0, 1)$$

be the Gödel truth-value object, with

$$a \otimes b = a \wedge b, \quad a \multimap b = \begin{cases} 1, & a \leq b, \\ b, & a > b. \end{cases}$$

A fuzzy predicate

$$\phi : X^{\text{op}} \rightarrow \underline{\Lambda}$$

is a map $\phi : X \rightarrow [0, 1]$ satisfying

$$X(x', x) \wedge \phi(x) \leq \phi(x')$$

for all $x, x' \in X$.

Its a -cut is

$$X_{\phi, a} = \{x \in X \mid a \leq \phi(x)\}.$$

Thus a fuzzy cut subobject is a decreasing family

$$(A_a)_{a \in [0, 1]}$$

of induced fuzzy subpreorders satisfying

$$A_{\bigvee_i a_i} = \bigcap_i A_{a_i}$$

and the Gödel stability condition

$$x \in A_a \implies x' \in A_{X(x', x) \wedge a}.$$

The classifying predicate is recovered by

$$\chi_A(x) = \sup\{a \in [0, 1] \mid x \in A_a\}.$$

For $A, B \in \text{Sub}_{\Lambda}^{\text{cut}}(X)$, the hom-value is

$$\text{Sub}_{\Lambda}^{\text{cut}}(X)(A, B) = \sup\{r \in [0, 1] \mid \forall a \in [0, 1], A_a \subseteq B_{r \wedge a}\}.$$

Equivalently, by Lemma 1.6.1,

$$\text{Sub}_{\Lambda}^{\text{cut}}(X)(A, B) = \inf_{x \in X} (\chi_A(x) \multimap \chi_B(x)).$$

Thus A is contained in B to degree r precisely when every a -level of A is contained in the $(r \wedge a)$ -level of B . In particular, containment to degree 1 is ordinary pointwise containment of classifying predicates:

$$\text{Sub}_{\Lambda}^{\text{cut}}(X)(A, B) = 1 \iff \chi_A \leq \chi_B \iff A_a \subseteq B_a \text{ for all } a.$$

Example 1.6.3 (The Łukasiewicz case). Let

$$\Lambda = ([0, 1], \leq, \wedge, \vee, \otimes, \multimap, 0, 1)$$

be the Łukasiewicz truth-value object, with

$$a \otimes b = \max\{0, a + b - 1\}, \quad a \multimap b = \min\{1, 1 - a + b\}.$$

A fuzzy predicate

$$\phi : X^{\text{op}} \rightarrow \underline{\Lambda}$$

is a map $\phi : X \rightarrow [0, 1]$ satisfying

$$\max\{0, X(x', x) + \phi(x) - 1\} \leq \phi(x')$$

for all $x, x' \in X$.

The a -cut is again

$$X_{\phi,a} = \{x \in X \mid a \leq \phi(x)\}.$$

A fuzzy cut subobject is therefore a decreasing, join-continuous family

$$(A_a)_{a \in [0,1]}$$

of induced fuzzy subpreorders satisfying the Lukasiewicz stability condition

$$x \in A_a \implies x' \in A_{\max\{0, X(x',x)+a-1\}}.$$

The classifying predicate is recovered by

$$\chi_A(x) = \sup\{a \in [0, 1] \mid x \in A_a\}.$$

For $A, B \in \text{Sub}_\Lambda^{\text{cut}}(X)$, the cut-containment hom is

$$\text{Sub}_\Lambda^{\text{cut}}(X)(A, B) = \sup \{r \in [0, 1] \mid \forall a \in [0, 1], A_a \subseteq B_{\max\{0, r+a-1\}}\}.$$

Equivalently,

$$\text{Sub}_\Lambda^{\text{cut}}(X)(A, B) = \inf_{x \in X} \min\{1, 1 - \chi_A(x) + \chi_B(x)\}.$$

Thus containment to degree r permits a uniform loss of $1 - r$ in the level parameter: an a -level point of A is required to lie in B only at level

$$\max\{0, a - (1 - r)\}.$$

In particular, the hom-value can be written as

$$\text{Sub}_\Lambda^{\text{cut}}(X)(A, B) = 1 - \sup_{x \in X} \max\{0, \chi_A(x) - \chi_B(x)\}.$$

Hence the Lukasiewicz hom records the worst pointwise deficit of χ_B below χ_A .

Example 1.6.4 (The Lawvere case). Let

$$\Lambda = ([0, \infty], \leq_L, \wedge, \vee, +, \multimap, \infty, 0)$$

be the Lawvere truth-value object, where

$$a \leq_L b \iff a \geq b \text{ in the usual order.}$$

Thus the top element is 0, the bottom element is ∞ , and the tensor product is addition.

A fuzzy preorder over this base is a Lawvere metric space, or extended quasi-pseudometric space. Writing

$$d(x, y) := X(x, y),$$

the preorder axioms become

$$d(x, x) = 0, \quad d(x, z) \leq d(x, y) + d(y, z)$$

in the usual order on $[0, \infty]$.

A fuzzy predicate

$$\phi : X^{\text{op}} \rightarrow \underline{\Lambda}$$

is a map $\phi : X \rightarrow [0, \infty]$ satisfying

$$d(x', x) + \phi(x) \geq \phi(x')$$

in the usual order. Equivalently,

$$\phi(x') \leq \phi(x) + d(x', x).$$

Thus ϕ is a one-sided nonexpansive potential.

Because the order on L is reversed, the a -cut is the usual sublevel set

$$X_{\phi, a} = \{x \in X \mid a \leq_L \phi(x)\} = \{x \in X \mid \phi(x) \leq a\}.$$

A fuzzy cut subobject is therefore a family of induced fuzzy subpreorders

$$(A_a)_{a \in [0, \infty]}$$

satisfying

$$A_\infty = X,$$

monotonicity in the usual threshold parameter,

$$b \leq a \implies A_b \subseteq A_a,$$

and the closure condition

$$A_{\inf_i a_i} = \bigcap_i A_{a_i}.$$

The stability axiom becomes the metric expansion condition

$$x \in A_a \implies x' \in A_{d(x', x) + a}.$$

Thus a point within cost a forces every point x' to lie within cost $a + d(x', x)$.

The classifying predicate of a cut family is

$$\chi_A(x) = \bigvee_L \{a \in [0, \infty] \mid x \in A_a\}.$$

Since $\bigvee_L$ is infimum in the usual order, this says

$$\chi_A(x) = \inf\{a \in [0, \infty] \mid x \in A_a\}.$$

So $\chi_A(x)$ is the least usual threshold at which x enters the filtration.

Finally, for $A, B \in \text{Sub}_\Lambda^{\text{cut}}(X)$, the hom-value is

$$\text{Sub}_\Lambda^{\text{cut}}(X)(A, B) = \bigvee_L \{r \in [0, \infty] \mid \forall a \in [0, \infty], A_a \subseteq B_{r+a}\}.$$

Since joins in Λ are infima in the usual order, this is the least usual cost r such that every a -sublevel of A lies inside the $(a + r)$ -sublevel of B . Equivalently, written in the usual order on $[0, \infty]$,

$$\text{Sub}_\Lambda^{\text{cut}}(X)(A, B) = \sup_{x \in X} \max\{0, \chi_B(x) - \chi_A(x)\}.$$

Thus the Lawvere hom is the directed pointwise excess of χ_B over χ_A : it is the least additive shift r for which

$$\chi_B(x) \leq \chi_A(x) + r$$

for all $x \in X$.

Chapter 2

Fuzzy Frames, Fuzzy Locales, and the Fuzzy Sierpiński Classifier

In the previous chapter we constructed fuzzy predicates as enriched presheaves. For a small $\mathbb{V}$ -category X , the fuzzy presheaf category was

$$P_{\mathbb{V}}(X) = [X^{\text{op}}, \mathbb{V}],$$

and, after passing to the posetal specialization determined by the fuzzy truth-value object

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp_L, \top_L),$$

the corresponding fuzzy predicate object was

$$P_{\Lambda}(X) = [X^{\text{op}}, \Lambda].$$

This chapter gives these presheaf objects their localic interpretation.

There are two structures on $P_{\Lambda}(X)$ which should be kept distinct. First, $P_{\Lambda}(X)$ is a separated Λ -preorder, with hom-values

$$P_{\Lambda}(X)(\phi, \psi) = \bigwedge_{x \in X} (\phi(x) \multimap \psi(x)).$$

Second, its induced crisp order is a frame, with joins and meets computed pointwise. In this chapter, when we call $P_{\Lambda}(X)$ a frame, we mean this induced crisp frame of predicates, equipped with the pointwise Λ -scalar action

$$(r \otimes \phi)(x) = r \otimes \phi(x).$$

The enriched hom-values remain present, but the frame order is the induced crisp order.

The guiding idea is that fuzzy frames are obtained by localizing fuzzy presheaves. At the categorical level this is expressed by enriched left-exact reflective localizations of

$$P_{\mathbb{V}}(X).$$

At the posetal level this becomes a scalar-compatible nucleus on

$$P_{\Lambda}(X).$$

We prove below that these two descriptions agree in the posetal specialization: left-exact Λ -enriched reflective localizations of $P_\Lambda(X)$ correspond, up to equivalence over $P_\Lambda(X)$, to fuzzy nuclei on $P_\Lambda(X)$. Fuzzy locales are then defined by formal duality.

The fuzzy Sierpiński classifier is treated by its classifying universal property. This is an important point. In the crisp case the Sierpiński frame can be realized as the presheaf frame on the two-element preorder

$$U \leq 1.$$

For general fuzzy truth-value bases this raw presheaf construction does not, in general, have the required free-open property. In particular, for non-idempotent tensors such as the Łukasiewicz and product tensors, the presheaf frame on the two-point subterminal shape need not classify arbitrary fuzzy opens by finite-meet-preserving scalar frame homomorphisms.

Accordingly, the fuzzy Sierpiński frame is defined in this chapter as the free fuzzy frame on one fuzzy open:

$$\Omega_\Lambda = F_{\text{op}}^\Lambda(1),$$

with generic fuzzy open

$$\sigma_\Lambda \in \Omega_\Lambda.$$

Its dual fuzzy locale

$$\mathbb{S}_\Lambda$$

classifies fuzzy opens by the universal property

$$\mathbf{FLoc}_\Lambda(\mathcal{X}, \mathbb{S}_\Lambda) \cong |\mathcal{O}_\Lambda(\mathcal{X})|.$$

Thus the Sierpiński object is not identified with a raw presheaf object unless an additional presentation theorem has been proved for the chosen base.

At the enriched-cosmos level, the analogous construction is bicategorical. The free fuzzy frame on an S -indexed family of cosmos-level fuzzy opens represents a hom-category of models:

$$\mathbf{FFrm}_\mathbb{V}(F_{\text{op}}(S), \mathcal{A}) \simeq \text{Sub}_\mathcal{A}(1_\mathcal{A})^S.$$

Equivalently, if S_{disc} is the discrete category on S , the right-hand side is

$$\mathbf{Cat}(S_{\text{disc}}, \text{Sub}_\mathcal{A}(1_\mathcal{A})).$$

After passing to the posetal reflection, these hom-categories become ordinary hom-sets and the corresponding classification becomes a strict set-level universal property.

All large categories and collections of subobjects are taken relative to the fixed universe convention of the previous chapter. If $\mathcal{A}$ is an ordinary or enriched category with terminal object, we write

$$\text{Sub}_\mathcal{A}(1_\mathcal{A})$$

for the ordinary category of subobjects of $1_\mathcal{A}$. Its objects are monomorphisms

$$U \hookrightarrow 1_\mathcal{A},$$

and its morphisms are commuting triangles over $1_\mathcal{A}$. When only isomorphism classes are needed, we write

$$\text{Iso Sub}_\mathcal{A}(1_\mathcal{A}) := \text{Ob Sub}_\mathcal{A}(1_\mathcal{A}) / \cong.$$

In the posetal reflection, $\text{Sub}_\mathcal{A}(\top_\mathcal{A})$ is equivalent to the thin category underlying the frame $\mathcal{A}$, and its set of objects is simply $|A|$.

Finite-limit convention for this chapter. We fix once and for all a class $\mathcal{F}$ of finite weighted limits in $\mathbb{V}$ -categories. The class $\mathcal{F}$ contains the terminal-object and binary-product weights and is the finite-limit class used in all terms such as “left exact”, “finite-limit preserving”, and $\text{Lex}_{\mathbb{V}}$. Thus

$$\text{Lex}_{\mathbb{V}}(\mathcal{C}, \mathcal{A})$$

denotes the ordinary category of $\mathbb{V}$ -functors

$$\mathcal{C} \rightarrow \mathcal{A}$$

preserving the chosen $\mathcal{F}$ -limits, with enriched natural transformations.

In the posetal results below, the only finite limits used explicitly are the terminal element and binary meets of the underlying crisp frame. Thus left exactness in the posetal comparison theorem means preservation of $\top$ and binary meets.

The overall progression is:

$$\text{fuzzy preorders} \xrightarrow{P_{\Lambda}} \text{fuzzy presheaves} \xrightarrow{\text{localization}} \text{fuzzy frames} \xrightarrow{\text{op}} \text{fuzzy locales} \xrightarrow{\mathbf{FLoc}_{\Lambda}(-, \mathbb{S}_{\Lambda})} \text{fuzzy opens}.$$

The notation $P_{\mathbb{V}}(X)$ and $P_{\Lambda}(X)$ is retained from the previous chapter. The new notation introduced in this chapter is:

$$\mathbf{FFrm}_{\Lambda}$$

for the category of fuzzy frames over Λ ,

$$\mathbf{FLoc}_{\Lambda}$$

for the category of fuzzy locales over Λ ,

$$\Omega_{\Lambda}$$

for the fuzzy Sierpiński frame,

$$\sigma_{\Lambda} \in \Omega_{\Lambda}$$

for its generic fuzzy open, and

$$\mathbb{S}_{\Lambda}$$

for the corresponding fuzzy Sierpiński locale.

2.1 Fuzzy presheaves recalled

We begin by recalling the presheaf constructions from the previous chapter. The definition–theorem–proof development of these objects has already been given there.

For a small $\mathbb{V}$ -category X , the fuzzy presheaf category is

$$P_{\mathbb{V}}(X) = [X^{\text{op}}, \mathbb{V}].$$

This is generally a locally small, possibly large $\mathbb{V}$ -category. Its objects are fuzzy presheaves on X . The Yoneda embedding is the $\mathbb{V}$ -functor

$$y_X : X \rightarrow P_{\mathbb{V}}(X), \quad x \mapsto X(-, x).$$

By the enriched Yoneda lemma,

$$P_{\mathbb{V}}(X)(y_X x, \Phi) \cong \Phi x$$

for every fuzzy presheaf $\Phi \in P_V(X)$. Equivalently, the variance-correct Yoneda comparison says that the composite

$$\begin{array}{ccc} X^{\text{op}} & \xrightarrow{y_X^{\text{op}}} & P_V(X)^{\text{op}} \\ & \searrow \Phi & \downarrow P_V(X)(-, \Phi) \\ & & \underline{V} \end{array}$$

is isomorphic to Φ . These are the standard enriched presheaf and Yoneda constructions [10].

The density formula says that every fuzzy presheaf is a weighted colimit of representables:

$$\Phi \cong \Phi \star y_X \cong \int^{x \in X} \Phi x \otimes y_X x.$$

Thus the values of Φ give the coefficients with which the principal presheaves $y_X x$ are joined to reconstruct Φ . This is the usual enriched density formula for the Yoneda embedding [10].

After passing to the posetal specialization determined by

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp_L, \top_L),$$

a fuzzy preorder X has fuzzy predicate object

$$P_\Lambda(X) = [X^{\text{op}}, \Lambda].$$

An element of $P_\Lambda(X)$ is a map

$$\phi : X \rightarrow L$$

satisfying

$$X(x', x) \otimes \phi(x) \leq \phi(x')$$

for all $x, x' \in X$. Joins, meets, and scalar multiplication in the induced crisp frame are computed pointwise:

$$\begin{aligned} \left(\bigvee_i \phi_i \right) (x) &= \bigvee_i \phi_i(x), \\ \left(\bigwedge_i \phi_i \right) (x) &= \bigwedge_i \phi_i(x), \end{aligned}$$

and

$$(r \otimes \phi)(x) = r \otimes \phi(x).$$

This is the form in which the presheaf construction will be localized below.

2.2 Localizing fuzzy presheaves

We now perform the localic operation. Instead of postulating an arbitrary algebraic notion of fuzzy frame, we obtain fuzzy frames by localizing fuzzy presheaves.

At the categorical level, this means a left-exact enriched localization of

$$P_V(X).$$

At the posetal level, this becomes a scalar-compatible nucleus on

$$P_\Lambda(X).$$

The fixed points of such a nucleus form the corresponding fuzzy frame.

2.2.1 Categorical form

Let X be a small $\mathbb{V}$ -category. A localization of the fuzzy presheaf category $P_{\mathbb{V}}(X)$ is given by a $\mathbb{V}$ -adjunction

$$a \dashv i : P_{\mathbb{V}}(X) \rightleftarrows \mathcal{A}$$

with i fully faithful. The associated idempotent monad is

$$J := ia.$$

We require this localization to be left exact with respect to the chosen $\mathcal{F}$ -limits.

Definition 2.2.1 (Categorical fuzzy-frame presentation). *A **categorical fuzzy-frame presentation** is a $\mathbb{V}$ -adjunction*

$$a \dashv i : P_{\mathbb{V}}(X) \rightleftarrows \mathcal{A}$$

between locally small, possibly large $\mathbb{V}$ -categories such that:

1. *i is fully faithful as a $\mathbb{V}$ -functor;*
2. *$\mathcal{A}$ has the chosen finite weighted limits;*
3. *a preserves the chosen finite weighted limits.*

The fuzzy frame presented by this localization has category of opens $\mathcal{A}$.

The localization is displayed by

$$P_{\mathbb{V}}(X) \begin{array}{c} \xrightarrow{a} \\ \leftarrow i \end{array} \mathcal{A}$$

and the associated idempotent monad is

$$P_{\mathbb{V}}(X) \xrightarrow{a} \mathcal{A} \xrightarrow{i} P_{\mathbb{V}}(X).$$

Weighted limits and colimits are understood in the standard enriched sense [10].

Definition 2.2.2 (Local object). *Let*

$$J = ia : P_{\mathbb{V}}(X) \rightarrow P_{\mathbb{V}}(X)$$

*be the idempotent monad associated to a categorical fuzzy-frame presentation. A fuzzy presheaf $\Phi \in P_{\mathbb{V}}(X)$ is called **J -local** if the unit*

$$\Phi \rightarrow J\Phi$$

is an isomorphism.

Proposition 2.2.1 (Local objects). *Let*

$$a \dashv i : P_{\mathbb{V}}(X) \rightleftarrows \mathcal{A}$$

be a categorical fuzzy-frame presentation, and let

$$J = ia.$$

Then the full subcategory of J -local objects in $P_{\mathbb{V}}(X)$ is equivalent to $\mathcal{A}$.

Proof. Since i is fully faithful, the counit

$$aiA \rightarrow A$$

is an isomorphism for every $A \in \mathcal{A}$. Hence

$$J(iA) = iaia \cong iA,$$

so every object in the image of i is J -local.

Conversely, if Φ is J -local, then

$$\Phi \cong J\Phi = ia\Phi,$$

so Φ is isomorphic to an object in the image of i . Therefore the local objects are precisely the objects of $\mathcal{A}$, up to equivalence. $\square$

The local-object subcategory sits inside the presheaf category as

$$\begin{array}{ccc} \mathcal{A} & \xhookrightarrow{i} & P_{\mathbb{V}}(X) \\ & \searrow \simeq & \uparrow \\ & & \text{Fix}(J) \end{array}$$

where $\text{Fix}(J)$ denotes the full subcategory of J -local objects.

Proposition 2.2.2 (Closed colimits). *Let*

$$a \dashv i : P_{\mathbb{V}}(X) \rightleftarrows \mathcal{A}$$

be a categorical fuzzy-frame presentation. Let K be small, let

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be a weight, and let

$$D : K \rightarrow \mathcal{A}$$

be a diagram. Whenever the weighted colimit $W \star iD$ exists in $P_{\mathbb{V}}(X)$, the weighted colimit of D exists in $\mathcal{A}$ and is computed by

$$W \star_{\mathcal{A}} D \cong a(W \star iD).$$

Proof. For $A \in \mathcal{A}$, the enriched adjunction and the universal property of the weighted colimit give

$$\begin{aligned} \mathcal{A}(a(W \star iD), A) &\cong P_{\mathbb{V}}(X)(W \star iD, iA) \\ &\cong [K^{\text{op}}, \underline{\mathbb{V}}](W, P_{\mathbb{V}}(X)(iD-, iA)) \\ &\cong [K^{\text{op}}, \underline{\mathbb{V}}](W, \mathcal{A}(D-, A)). \end{aligned}$$

This is the enriched universal property of the weighted colimit of D in $\mathcal{A}$. $\square$

Proposition 2.2.3 (Finite limits created by the inclusion). *Let*

$$a \dashv i : P_{\mathbb{V}}(X) \rightleftarrows \mathcal{A}$$

be a categorical fuzzy-frame presentation. Let K and W belong to the chosen finite-limit class $\mathcal{F}$, and let

$$D : K \rightarrow \mathcal{A}$$

be a diagram. Then the corresponding finite weighted limits in $\mathcal{A}$ are created by the inclusion

$$i : \mathcal{A} \rightarrow P_{\mathbb{V}}(X).$$

Proof. Let

$$M = \{W, iD\}$$

be the corresponding finite weighted limit in $P_V(X)$. Since a preserves the chosen finite weighted limits,

$$aM \cong a\{W, iD\} \cong \{W, aiD\} \cong \{W, D\}$$

in $\mathcal{A}$.

Since i is a right adjoint, it preserves limits. Hence iaM is also a limit of iD in $P_V(X)$. But M is already the limit of iD . The unit

$$M \rightarrow iaM$$

is the unique comparison map between two limiting cones over iD . Since both objects satisfy the same finite-limit universal property in $P_V(X)$, this comparison is an isomorphism. Thus M is local, and the chosen finite limits in $\mathcal{A}$ are created by i . $\square$

Thus, categorically, localization has the expected frame-theoretic behavior: colimits are formed in the ambient presheaf category and then closed, while finite limits are inherited by the local objects.

2.2.2 Posetal reflection and fuzzy nuclei

We now pass to the posetal specialization. The categorical localization above becomes a closure operation on

$$P_\Lambda(X),$$

compatible with finite meets and with the scalar action inherited from Λ . The first three axioms below are the usual nucleus axioms from frame theory [9]. The final axiom is the additional compatibility with fuzzy truth-value rescaling.

Definition 2.2.3 (Fuzzy nucleus). *Let X be a fuzzy preorder. A **fuzzy nucleus** on $P_\Lambda(X)$ is a map*

$$j : P_\Lambda(X) \rightarrow P_\Lambda(X)$$

such that, for all $\phi, \psi \in P_\Lambda(X)$ and all $r \in L$,

$$\phi \leq j(\phi),$$

$$j(j(\phi)) = j(\phi),$$

$$j(\phi \wedge \psi) = j(\phi) \wedge j(\psi),$$

and

$$j(r \otimes j(\phi)) = j(r \otimes \phi).$$

The first three equations are equivalent to the usual nucleus axioms on the frame $P_\Lambda(X)$; monotonicity follows from finite-meet preservation and is recorded below. The final equation says that localization commutes with fuzzy truth-value rescaling after closure. This scalar-compatible form is close in spirit to nucleus constructions in quantale theory [14].

Lemma 2.2.1 (Fuzzy nuclei are monotone). *Every fuzzy nucleus is monotone.*

Proof. If $\phi \leq \psi$, then

$$\phi = \phi \wedge \psi.$$

Therefore

$$j(\phi) = j(\phi \wedge \psi) = j(\phi) \wedge j(\psi) \leq j(\psi).$$

□

Lemma 2.2.2 (Fuzzy nuclei preserve top). *Every fuzzy nucleus satisfies*

$$j(\top_{P_\Lambda(X)}) = \top_{P_\Lambda(X)}.$$

Proof. Inflationarity gives

$$\top_{P_\Lambda(X)} \leq j(\top_{P_\Lambda(X)}).$$

Since $\top_{P_\Lambda(X)}$ is the greatest element of $P_\Lambda(X)$, this forces

$$j(\top_{P_\Lambda(X)}) = \top_{P_\Lambda(X)}.$$

□

Lemma 2.2.3 (Scalars as enriched tensors). *Let $P = P_\Lambda(X)$. For every $r \in L$, the pointwise scalar operation*

$$r \otimes (-) : P \rightarrow P$$

is the tensor of the Λ -category P by the scalar r . Explicitly, for all $\phi, \psi \in P$,

$$P(r \otimes \phi, \psi) = r \multimap P(\phi, \psi).$$

Proof. By the scalar-closure result for fuzzy predicates from the previous chapter, $r \otimes \phi$ is again an element of P . By definition,

$$P(r \otimes \phi, \psi) = \bigwedge_{x \in X} ((r \otimes \phi(x)) \multimap \psi(x)).$$

For $q \in L$, residuation gives

$$\begin{aligned} q \leq P(r \otimes \phi, \psi) &\iff q \otimes r \otimes \phi(x) \leq \psi(x) \quad \text{for all } x \\ &\iff q \otimes r \leq \phi(x) \multimap \psi(x) \quad \text{for all } x \\ &\iff q \otimes r \leq \bigwedge_{x \in X} (\phi(x) \multimap \psi(x)) \\ &\iff q \otimes r \leq P(\phi, \psi) \\ &\iff q \leq r \multimap P(\phi, \psi). \end{aligned}$$

Hence

$$P(r \otimes \phi, \psi) = r \multimap P(\phi, \psi).$$

□

Lemma 2.2.4 (Tensors in a reflective localization). *Let*

$$a \dashv i : P_\Lambda(X) \rightleftarrows \mathcal{A}$$

be a Λ -enriched reflective localization with i fully faithful. Then $\mathcal{A}$ is tensored over Λ . For $r \in L$ and $A \in \mathcal{A}$, the tensor is computed by

$$r \otimes_{\mathcal{A}} A \cong a(r \otimes iA).$$

Moreover,

$$a(r \otimes \phi) \cong r \otimes_{\mathcal{A}} a\phi$$

for all $\phi \in P_\Lambda(X)$.

Proof. For $B \in \mathcal{A}$, using the enriched adjunction and Lemma 2.2.3, we compute

$$\begin{aligned} \mathcal{A}(a(r \otimes iA), B) &\cong P_\Lambda(X)(r \otimes iA, iB) \\ &\cong r \multimap P_\Lambda(X)(iA, iB) \\ &\cong r \multimap \mathcal{A}(A, B). \end{aligned}$$

Thus $a(r \otimes iA)$ satisfies the universal property of $r \otimes_{\mathcal{A}} A$.

Taking $A = a\phi$ and using the counit $aia\phi \cong a\phi$, we have

$$r \otimes_{\mathcal{A}} a\phi \cong a(r \otimes ia\phi).$$

Also, since a is an enriched left adjoint, it preserves tensors. Hence

$$a(r \otimes \phi) \cong r \otimes_{\mathcal{A}} a\phi.$$

□

Theorem 2.2.1 (Posetal comparison for localizations and fuzzy nuclei). *Let X be a fuzzy preorder, and write*

$$P = P_\Lambda(X).$$

There is a correspondence, up to equivalence of reflective localizations over P , between:

- (1) *left-exact Λ -enriched reflective localizations*

$$a \dashv i : P \rightleftarrows \mathcal{A}$$

with i fully faithful, where left exactness in the posetal reflection means preservation of the terminal element and binary meets in the underlying frame;

- (2) *fuzzy nuclei*

$$j : P \rightarrow P.$$

Under this correspondence, a localization determines the fuzzy nucleus

$$j = ia,$$

and a fuzzy nucleus j determines the reflective localization

$$q_j \dashv i_j : P \rightleftarrows P_j,$$

where

$$P_j = \{\phi \in P \mid j(\phi) = \phi\}, \quad q_j(\phi) = j(\phi),$$

and $i_j : P_j \hookrightarrow P$ is the inclusion. Moreover,

$$\mathcal{A} \simeq P_j$$

over P .

In the posetal specialization, the underlying crisp order of P is a genuine poset. Since P is separated as a Λ -preorder, any ordinary isomorphism between objects of P is an equality. Thus, after replacing a localization by its equivalent full subpreorder of local objects, the idempotent monad ia may be regarded strictly as a closure operator on the underlying predicate frame.

Proof. First let

$$a \dashv i : P \rightleftarrows \mathcal{A}$$

be a left-exact Λ -enriched reflective localization with i fully faithful, and define

$$j = ia.$$

The unit of the adjunction gives

$$\phi \leq ia\phi = j(\phi),$$

so j is inflationary. Since i is fully faithful, the counit

$$aiA \rightarrow A$$

is an isomorphism for every $A \in \mathcal{A}$. Hence

$$j(j(\phi)) = iaia\phi \cong ia\phi = j(\phi).$$

Since P is separated, this isomorphism may be read as the strict equality

$$j(j(\phi)) = j(\phi).$$

Thus j is idempotent.

Because a preserves the finite meets of the underlying frame and i , as a right adjoint, preserves them, we have

$$\begin{aligned} j(\phi \wedge \psi) &= ia(\phi \wedge \psi) \\ &\cong i(a\phi \wedge a\psi) \\ &\cong ia\phi \wedge ia\psi \\ &= j(\phi) \wedge j(\psi). \end{aligned}$$

Again using separatedness of P , this comparison is a strict equality:

$$j(\phi \wedge \psi) = j(\phi) \wedge j(\psi).$$

Similarly,

$$j(\top_P) = \top_P.$$

It remains to prove scalar compatibility. By Lemma 2.2.4,

$$a(r \otimes \phi) \cong r \otimes_{\mathcal{A}} a\phi \cong a(r \otimes ia\phi).$$

Therefore

$$\begin{aligned} j(r \otimes j(\phi)) &= ia(r \otimes ia\phi) \\ &\cong ia(r \otimes \phi) \\ &= j(r \otimes \phi). \end{aligned}$$

Using separatedness once more, this gives the strict equality

$$j(r \otimes j(\phi)) = j(r \otimes \phi).$$

Thus j is a fuzzy nucleus.

Conversely, let $j : P \rightarrow P$ be a fuzzy nucleus, and let

$$P_j = \{\phi \in P \mid j(\phi) = \phi\}$$

be the full Λ -subpreorder of fixed points. Define

$$q_j : P \rightarrow P_j, \quad q_j(\phi) = j(\phi),$$

and let

$$i_j : P_j \hookrightarrow P$$

be the inclusion.

First q_j is a Λ -functor. Let $r \leq P(\phi, \psi)$. Equivalently,

$$r \otimes \phi \leq \psi.$$

By monotonicity of j ,

$$j(r \otimes \phi) \leq j(\psi).$$

By inflationarity applied to $r \otimes j(\phi)$, followed by scalar compatibility,

$$r \otimes j(\phi) \leq j(r \otimes j(\phi)) = j(r \otimes \phi).$$

Hence

$$r \otimes j(\phi) \leq j(\psi),$$

and so

$$r \leq P(j\phi, j\psi).$$

Therefore

$$P(\phi, \psi) \leq P_j(q_j\phi, q_j\psi),$$

so q_j is a Λ -functor.

We now prove that

$$q_j \dashv i_j.$$

Let $\psi \in P_j$. Since P_j is full,

$$P_j(q_j\phi, \psi) = P(j\phi, \psi).$$

We prove

$$P(j\phi, \psi) = P(\phi, \psi).$$

For $r \in L$, if

$$r \leq P(j\phi, \psi),$$

then

$$r \otimes j\phi \leq \psi.$$

Since $\phi \leq j\phi$, this implies

$$r \otimes \phi \leq \psi,$$

so

$$r \leq P(\phi, \psi).$$

Conversely, if

$$r \otimes \phi \leq \psi,$$

then

$$j(r \otimes \phi) \leq j(\psi) = \psi.$$

By inflationarity and scalar compatibility,

$$r \otimes j\phi \leq j(r \otimes j\phi) = j(r \otimes \phi) \leq \psi.$$

Thus

$$r \leq P(j\phi, \psi).$$

Therefore

$$P(j\phi, \psi) = P(\phi, \psi),$$

and hence $q_j \dashv i_j$ as a Λ -adjunction.

The inclusion i_j is fully faithful by definition. The reflector q_j preserves terminal element and binary meets because

$$j(\top_P) = \top_P$$

and

$$j(\phi \wedge \psi) = j(\phi) \wedge j(\psi).$$

Thus

$$q_j \dashv i_j : P \rightleftarrows P_j$$

is a left-exact Λ -enriched reflective localization.

Finally, starting from a localization and applying the construction above gives the fixed-point subcategory of the idempotent monad $j = ia$, which is equivalent to $\mathcal{A}$ by Proposition 2.2.1. Starting from a fuzzy nucleus j , the induced monad is exactly $i_j q_j = j$. Hence the two constructions are inverse up to equivalence over P . $\square$

2.2.3 Fixed points of fuzzy nuclei

Lemma 2.2.5 (Intersections of fuzzy nuclei). *Let $(j_\alpha)_{\alpha \in I}$ be a family of fuzzy nuclei on $P_\Lambda(X)$. Then the pointwise meet*

$$j(\phi) := \bigwedge_{\alpha \in I} j_\alpha(\phi)$$

is again a fuzzy nucleus. If $I = \emptyset$, this is the constant-top nucleus.

Proof. Inflationarity follows because $\phi \leq j_\alpha(\phi)$ for every α . Since every j_α is monotone, j is monotone.

For idempotence, inflationarity gives

$$j(\phi) \leq j(j(\phi)).$$

Conversely, since

$$j(\phi) \leq j_\alpha(\phi)$$

for every α , monotonicity of j_α gives

$$j_\alpha(j(\phi)) \leq j_\alpha(j_\alpha(\phi)) = j_\alpha(\phi).$$

Taking the meet over α yields

$$j(j(\phi)) \leq j(\phi).$$

Thus

$$j(j(\phi)) = j(\phi).$$

For finite meets,

$$\begin{aligned} j(\phi \wedge \psi) &= \bigwedge_{\alpha} j_{\alpha}(\phi \wedge \psi) \\ &= \bigwedge_{\alpha} (j_{\alpha}(\phi) \wedge j_{\alpha}(\psi)) \\ &= \left(\bigwedge_{\alpha} j_{\alpha}(\phi) \right) \wedge \left(\bigwedge_{\alpha} j_{\alpha}(\psi) \right) \\ &= j(\phi) \wedge j(\psi). \end{aligned}$$

Finally, for scalar compatibility, inflationarity gives

$$r \otimes \phi \leq r \otimes j(\phi),$$

and hence, by monotonicity of j ,

$$j(r \otimes \phi) \leq j(r \otimes j(\phi)).$$

For the reverse inequality, since $j(\phi) \leq j_\alpha(\phi)$, we have

$$r \otimes j(\phi) \leq r \otimes j_\alpha(\phi).$$

Thus

$$j_\alpha(r \otimes j(\phi)) \leq j_\alpha(r \otimes j_\alpha(\phi)) = j_\alpha(r \otimes \phi).$$

Taking the meet over α gives

$$j(r \otimes j(\phi)) \leq j(r \otimes \phi).$$

Therefore

$$j(r \otimes j(\phi)) = j(r \otimes \phi).$$

□

Definition 2.2.4 (Fixed points of a fuzzy nucleus). *Let j be a fuzzy nucleus on $P_\Lambda(X)$. Define*

$$P_\Lambda(X)_j := \{\phi \in P_\Lambda(X) \mid j(\phi) = \phi\}.$$

For fixed points $\phi_i, \phi, \psi \in P_\Lambda(X)_j$, define

$$\begin{aligned} \bigvee_j \phi_i &= j\left(\bigvee_i \phi_i\right), \\ \phi \wedge_j \psi &= \phi \wedge \psi, \end{aligned}$$

and

$$r \otimes_j \phi = j(r \otimes \phi).$$

Proposition 2.2.4 (Fixed points of a fuzzy nucleus). *Let j be a fuzzy nucleus on $P_\Lambda(X)$. Then*

$$P_\Lambda(X)_j$$

is a frame. Its joins are closed joins,

$$\bigvee_j \phi_i = j\left(\bigvee_i \phi_i\right),$$

its finite meets are inherited from $P_\Lambda(X)$, and it carries the closed scalar action

$$r \otimes_j \phi = j(r \otimes \phi).$$

The closed scalar action preserves arbitrary joins in each variable.

Proof. First, $\top_{P_\Lambda(X)}$ is fixed by Lemma 2.2.2. If ϕ, ψ are fixed, then

$$j(\phi \wedge \psi) = j(\phi) \wedge j(\psi) = \phi \wedge \psi,$$

so finite meets are inherited from $P_\Lambda(X)$.

For a family (ϕ_i) of fixed points, define

$$c = j\left(\bigvee_i \phi_i\right).$$

Then c is fixed by idempotence. Since

$$\phi_i \leq \bigvee_i \phi_i \leq j\left(\bigvee_i \phi_i\right),$$

the element c is an upper bound of the ϕ_i . If ψ is any fixed upper bound of the ϕ_i , then

$$\bigvee_i \phi_i \leq \psi.$$

By monotonicity,

$$j\left(\bigvee_i \phi_i\right) \leq j(\psi) = \psi.$$

Thus c is the least fixed upper bound. Hence joins in $P_\Lambda(X)_j$ are closed joins. In particular, the bottom element is

$$\perp_j = j(\perp_{P_\Lambda(X)}).$$

Frame distributivity follows from frame distributivity in $P_\Lambda(X)$ and finite-meet preservation by j . If ϕ and all ψ_i are fixed, then

$$\begin{aligned} \phi \wedge_j \bigvee_j \psi_i &= \phi \wedge j\left(\bigvee_i \psi_i\right) \\ &= j(\phi) \wedge j\left(\bigvee_i \psi_i\right) \\ &= j\left(\phi \wedge \bigvee_i \psi_i\right) \\ &= j\left(\bigvee_i (\phi \wedge \psi_i)\right) \\ &= \bigvee_j (\phi \wedge_j \psi_i). \end{aligned}$$

The closed scalar action is fixed by definition:

$$r \otimes_j \phi = j(r \otimes \phi).$$

The scalar unit law follows from the unit law in $P_\Lambda(X)$ and the fact that ϕ is fixed:

$$\top_L \otimes_j \phi = j(\top_L \otimes \phi) = j(\phi) = \phi.$$

For associativity, use the scalar-compatibility equation:

$$\begin{aligned} r \otimes_j (s \otimes_j \phi) &= j(r \otimes j(s \otimes \phi)) \\ &= j(r \otimes (s \otimes \phi)) \\ &= j((r \otimes s) \otimes \phi) \\ &= (r \otimes s) \otimes_j \phi. \end{aligned}$$

It remains to verify that the closed scalar action preserves arbitrary joins. In the frame variable,

$$\begin{aligned} r \otimes_j \bigvee_j \phi_i &= j\left(r \otimes j\left(\bigvee_i \phi_i\right)\right) \\ &= j\left(r \otimes \bigvee_i \phi_i\right) \\ &= j\left(\bigvee_i (r \otimes \phi_i)\right). \end{aligned}$$

For any family (η_i) in $P_\Lambda(X)$, the closure identity

$$j\left(\bigvee_i \eta_i\right) = j\left(\bigvee_i j(\eta_i)\right)$$

follows from inflationarity, monotonicity, and idempotence of j . Applying this to $\eta_i = r \otimes \phi_i$, we get

$$\begin{aligned} j \left(\bigvee_i (r \otimes \phi_i) \right) &= j \left(\bigvee_i j(r \otimes \phi_i) \right) \\ &= \bigvee_j (r \otimes_j \phi_i). \end{aligned}$$

Therefore

$$r \otimes_j \bigvee_j \phi_i = \bigvee_j (r \otimes_j \phi_i).$$

The scalar variable is similar. For a family (r_i) in L ,

$$\begin{aligned} \left(\bigvee_i r_i \right) \otimes_j \phi &= j \left(\left(\bigvee_i r_i \right) \otimes \phi \right) \\ &= j \left(\bigvee_i (r_i \otimes \phi) \right) \\ &= j \left(\bigvee_i j(r_i \otimes \phi) \right) \\ &= \bigvee_j (r_i \otimes_j \phi). \end{aligned}$$

Thus the closed scalar action preserves arbitrary joins in each variable. □

Remark 2.2.1 (Bottom scalars). *Taking the empty join in the scalar variable gives*

$$\perp_L \otimes_j \phi = \perp_j.$$

Taking the empty join in the frame variable gives

$$r \otimes_j \perp_j = \perp_j.$$

Thus each scalar operation preserves bottom, and the bottom scalar acts as the constant-bottom operator. Notice that

$$\perp_j = j(\perp_{P_\Lambda(X)}),$$

which need not equal the bottom element of $P_\Lambda(X)$.

Lemma 2.2.6 (Enriched hom recovered from the scalar action). *Let*

$$A = P_\Lambda(X)_j$$

be the fixed-point frame of a fuzzy nucleus. For fixed points $\phi, \psi \in A$, one has

$$A(\phi, \psi) = \bigvee \{ r \in L \mid r \otimes_j \phi \leq \psi \}.$$

Moreover this hom-value agrees with the hom inherited from $P_\Lambda(X)$:

$$A(\phi, \psi) = P_\Lambda(X)(\phi, \psi).$$

Proof. For $r \in L$, using the tensor-hom formula in $P_\Lambda(X)$, we have

$$r \leq P_\Lambda(X)(\phi, \psi) \iff r \otimes \phi \leq \psi.$$

Since ψ is fixed, this is equivalent to

$$j(r \otimes \phi) \leq \psi.$$

But

$$j(r \otimes \phi) = r \otimes_j \phi.$$

Hence

$$r \leq P_\Lambda(X)(\phi, \psi) \iff r \otimes_j \phi \leq \psi.$$

Taking the join over all such r gives the claim. $\square$

Definition 2.2.5 (Λ -fuzzy-frame presentation). A **Λ -fuzzy-frame presentation** is a pair

$$(X, j),$$

where X is a fuzzy preorder and

$$j : P_\Lambda(X) \rightarrow P_\Lambda(X)$$

is a fuzzy nucleus.

The frame presented by (X, j) is the fixed-point frame

$$P_\Lambda(X)_j.$$

Definition 2.2.6 (Fuzzy frame over Λ). A **fuzzy frame over Λ** is a frame presented by a Λ -fuzzy-frame presentation. Equivalently, a fuzzy frame is a nuclear quotient of a fuzzy presheaf frame:

$$P_\Lambda(X) \twoheadrightarrow P_\Lambda(X)_j.$$

Thus the phrase *nuclear quotient* is frame-theoretic. Under locale duality, the same structure is read as a fuzzy sublocale.

Definition 2.2.7 (Λ -scalar frame). A **Λ -scalar frame** is a frame A equipped with an action

$$L \times A \rightarrow A, \quad (r, a) \mapsto r \otimes_A a,$$

such that

$$\top_L \otimes_A a = a, \quad (r \otimes s) \otimes_A a = r \otimes_A (s \otimes_A a),$$

and such that the operation preserves arbitrary joins in each variable:

$$r \otimes_A \bigvee_i a_i = \bigvee_i (r \otimes_A a_i),$$

$$\left(\bigvee_i r_i \right) \otimes_A a = \bigvee_i (r_i \otimes_A a).$$

Every fixed-point frame $P_\Lambda(X)_j$ is a Λ -scalar frame by Proposition 2.2.4.

Definition 2.2.8 (Fuzzy-frame homomorphism). *Let*

$$P_\Lambda(X)_j \quad \text{and} \quad P_\Lambda(Y)_k$$

*be fuzzy frames presented by Λ -fuzzy-frame presentations. A **fuzzy-frame homomorphism***

$$h : P_\Lambda(X)_j \rightarrow P_\Lambda(Y)_k$$

is a frame homomorphism satisfying

$$h(r \otimes_j \phi) = r \otimes_k h(\phi)$$

for all $r \in L$ and all $\phi \in P_\Lambda(X)_j$.

Corollary 2.2.1 (Fuzzy-frame homomorphisms are enriched functors). *Let*

$$h : A \rightarrow B$$

be a fuzzy-frame homomorphism between fixed-point fuzzy frames. Then h is a Λ -functor for the enriched homs recovered from the scalar actions.

Proof. Suppose

$$r \leq A(a, b).$$

By Lemma 2.2.6, this means

$$r \otimes_A a \leq b.$$

Applying h and using scalar compatibility gives

$$r \otimes_B h(a) = h(r \otimes_A a) \leq h(b).$$

Again by Lemma 2.2.6, this implies

$$r \leq B(h(a), h(b)).$$

Therefore

$$A(a, b) \leq B(h(a), h(b)),$$

so h is a Λ -functor. □

Definition 2.2.9 (The category of fuzzy frames). *The category of fuzzy frames over Λ and fuzzy-frame homomorphisms is denoted*

$$\mathbf{FFrm}_\Lambda.$$

Remark 2.2.2 (Presentations). *Strictly speaking, one may keep the presentation (X, j) as part of the data, or identify presentations that present isomorphic fixed-point frames. In this chapter we write fuzzy frames as the presented frames $P_\Lambda(X)_j$, while remembering that they arise from some localization of a fuzzy presheaf frame.*

The definition is deliberately presentation-based. Fuzzy frames are not introduced as arbitrary frames equipped with a scalar action; they are the fixed-point frames of scalar-compatible nuclei on fuzzy presheaf frames.

Definition 2.2.10 (Nuclear quotient map). *Let j be a fuzzy nucleus on $P_\Lambda(X)$. The associated **nuclear quotient map** is the closure map*

$$q_j : P_\Lambda(X) \rightarrow P_\Lambda(X)_j, \quad q_j(\phi) = j(\phi).$$

Proposition 2.2.5 (The closure map is a quotient). *Let j be a fuzzy nucleus on $P_\Lambda(X)$. Then*

$$q_j : P_\Lambda(X) \rightarrow P_\Lambda(X)_j$$

is a surjective fuzzy-frame homomorphism.

Proof. The map q_j is surjective because every fixed point $\phi \in P_\Lambda(X)_j$ satisfies

$$j(\phi) = \phi.$$

It preserves arbitrary joins because joins in $P_\Lambda(X)_j$ are closed joins:

$$j\left(\bigvee_i \phi_i\right) = \bigvee_j j(\phi_i).$$

For the empty family this says

$$q_j(\perp_{P_\Lambda(X)}) = j(\perp_{P_\Lambda(X)}) = \perp_j,$$

so q_j also preserves bottom. It preserves finite meets because

$$j(\phi \wedge \psi) = j(\phi) \wedge j(\psi),$$

and it preserves top by Lemma 2.2.2.

It preserves scalars because

$$\begin{aligned} q_j(r \otimes \phi) &= j(r \otimes \phi) \\ &= j(r \otimes j(\phi)) \\ &= r \otimes_j j(\phi) \\ &= r \otimes_j q_j(\phi). \end{aligned}$$

Therefore q_j is a surjective fuzzy-frame homomorphism. $\square$

Conversely, every scalar-compatible quotient of a fuzzy presheaf frame determines a fuzzy nucleus. This is the scalar-compatible version of the usual correspondence between nuclei and quotient frames [9].

Proposition 2.2.6 (Quotients determine fuzzy nuclei). *Let B be a Λ -scalar frame, and let*

$$q : P_\Lambda(X) \twoheadrightarrow B$$

be a surjective frame homomorphism satisfying

$$q(r \otimes \phi) = r \otimes_B q(\phi)$$

for all $r \in L$ and $\phi \in P_\Lambda(X)$. Since q preserves arbitrary joins, it has a right adjoint

$$q_* : B \rightarrow P_\Lambda(X)$$

defined by

$$q_*(b) = \bigvee \{\phi \in P_\Lambda(X) \mid q(\phi) \leq b\}.$$

Then

$$j_q := q_* q$$

is a fuzzy nucleus on $P_\Lambda(X)$, and the fixed-point fuzzy frame

$$P_\Lambda(X)_{j_q}$$

is isomorphic to B as a fuzzy frame.

Proof. Since $q \dashv q_*$, the composite

$$j_q = q_*q$$

is inflationary. Since q is surjective, the counit

$$qq_* \leq \text{id}_B$$

is an equality. Indeed, if $b = q(\phi)$, then $\phi \leq q_*(b)$, so

$$b = q(\phi) \leq qq_*(b),$$

while $qq_*(b) \leq b$ follows from the counit inequality. Hence

$$qq_* = \text{id}_B.$$

Therefore

$$j_q j_q = q_* q q_* q = q_* q = j_q.$$

Because q preserves finite meets and q_* preserves all meets,

$$\begin{aligned} j_q(\phi \wedge \psi) &= q_*(q(\phi \wedge \psi)) \\ &= q_*(q(\phi) \wedge q(\psi)) \\ &= q_*q(\phi) \wedge q_*q(\psi) \\ &= j_q(\phi) \wedge j_q(\psi). \end{aligned}$$

For scalar compatibility, use that q preserves scalars and that $qq_* = \text{id}_B$:

$$\begin{aligned} j_q(r \otimes j_q(\phi)) &= q_*q(r \otimes q_*q(\phi)) \\ &= q_*(r \otimes_B qq_*q(\phi)) \\ &= q_*(r \otimes_B q(\phi)) \\ &= q_*q(r \otimes \phi) \\ &= j_q(r \otimes \phi). \end{aligned}$$

Therefore j_q is a fuzzy nucleus.

The restriction

$$q : P_\Lambda(X)_{j_q} \rightarrow B$$

is an isomorphism of frames with inverse q_* . First, for every $b \in B$,

$$j_q(q_*b) = q_*qq_*b = q_*b,$$

so q_*b is fixed. Moreover,

$$qq_*(b) = b,$$

and for $\phi \in P_\Lambda(X)_{j_q}$,

$$q_*q(\phi) = j_q(\phi) = \phi.$$

It remains to note that this isomorphism respects the scalar actions. The restriction of q preserves scalars by assumption. Conversely, for $b \in B$, write $b = q(\phi)$. Then

$$\begin{aligned} q(r \otimes_{j_q} q_*(b)) &= q(j_q(r \otimes q_*(b))) \\ &= q(r \otimes q_*(b)) \\ &= r \otimes_B qq_*(b) \\ &= r \otimes_B b. \end{aligned}$$

Since $q : P_\Lambda(X)_{j_q} \rightarrow B$ is an isomorphism, this implies

$$r \otimes_{j_q} q_*(b) = q_*(r \otimes_B b).$$

Thus

$$P_\Lambda(X)_{j_q} \cong B$$

as fuzzy frames. □

2.2.4 Examples

Example 2.2.1 (The identity localization). *For every fuzzy preorder X , the identity map*

$$\text{id}_{P_\Lambda(X)} : P_\Lambda(X) \rightarrow P_\Lambda(X)$$

is a fuzzy nucleus. Hence

$$P_\Lambda(X)_{\text{id}} = P_\Lambda(X).$$

Thus every fuzzy presheaf frame is itself a fuzzy frame.

In this case no predicates are identified or closed further. Joins, finite meets, and scalar multiplication are the pointwise operations inherited directly from Λ . Locally, this is the ambient presheaf fuzzy locale

$$\mathcal{P}_\Lambda X.$$

All other examples of fuzzy frames in this chapter may be read as sublocales of such presheaf fuzzy locales, equivalently as nuclear quotients of their frames of opens.

Example 2.2.2 (The trivial localization). *For every fuzzy preorder X , the constant-top map*

$$j_\top(\phi) = \top_{P_\Lambda(X)}$$

is a fuzzy nucleus. Inflationarity and idempotence are immediate, and

$$j_\top(\phi \wedge \psi) = \top_{P_\Lambda(X)} = \top_{P_\Lambda(X)} \wedge \top_{P_\Lambda(X)}.$$

For scalar compatibility,

$$j_\top(r \otimes j_\top(\phi)) = \top_{P_\Lambda(X)} = j_\top(r \otimes \phi).$$

The fixed-point frame has exactly one element:

$$P_\Lambda(X)_{j_\top} \cong 1.$$

This is the trivial fuzzy frame. On the locale side it corresponds to the initial, or empty, fuzzy sublocale of the presheaf fuzzy locale. The identity nucleus gives the whole sublocale, while the constant-top nucleus gives the least sublocale.

Example 2.2.3 (The crisp case). *Let*

$$\Lambda = 2.$$

Then fuzzy preorders are ordinary preorders and, for an ordinary preorder X ,

$$P_2(X) = [X^{\text{op}}, 2] \cong \text{Down}(X),$$

the frame of downsets of X . A fuzzy nucleus on $P_2(X)$ is exactly an ordinary nucleus on the frame $\text{Down}(X)$, because the scalar action of 2 is forced:

$$\top \otimes U = U, \quad \perp \otimes U = \perp.$$

Thus the scalar-compatibility equation adds no extra condition beyond the ordinary nucleus axioms. Concretely, a nucleus

$$j : \text{Down}(X) \rightarrow \text{Down}(X)$$

assigns to each downset $U \subseteq X$ a closed downset $j(U)$, satisfying

$$U \subseteq j(U), \quad j(j(U)) = j(U), \quad j(U \cap V) = j(U) \cap j(V).$$

The fixed points form the ordinary quotient frame

$$\text{Down}(X)_j.$$

Thus the crisp specialization recovers the standard description of frames as nuclear quotients of downset frames.

Theorem 2.2.2 (Ordinary frames as localized presheaves). *Let A be a small ordinary frame. Regard the underlying poset of A as a preorder. Then the map*

$$q : \text{Down}(A) \rightarrow A, \quad q(U) = \bigvee U,$$

is a surjective frame homomorphism. Consequently A is isomorphic to the fixed-point frame of the nucleus

$$j_A = q_* q$$

on $\text{Down}(A)$. Explicitly,

$$j_A(U) = \downarrow \bigvee U.$$

Proof. The map q preserves arbitrary joins because

$$\begin{aligned} q\left(\bigcup_i U_i\right) &= \bigvee \bigcup_i U_i \\ &= \bigvee_i \bigvee U_i \\ &= \bigvee_i q(U_i). \end{aligned}$$

It preserves top because

$$q(A) = \bigvee A = \top_A.$$

It remains to check finite meets. Since

$$U \cap V \subseteq U \quad \text{and} \quad U \cap V \subseteq V,$$

we have

$$q(U \cap V) \leq q(U) \wedge q(V).$$

Conversely, by frame distributivity,

$$\begin{aligned} q(U) \wedge q(V) &= \left(\bigvee U \right) \wedge \left(\bigvee V \right) \\ &= \bigvee_{u \in U, v \in V} (u \wedge v). \end{aligned}$$

Since U and V are downsets, for every $u \in U$ and $v \in V$, the element

$$u \wedge v$$

belongs to $U \cap V$. Hence

$$q(U) \wedge q(V) \leq q(U \cap V).$$

Therefore

$$q(U \cap V) = q(U) \wedge q(V).$$

Finally, q is surjective because

$$q(\downarrow a) = a$$

for every $a \in A$. Therefore q is a surjective frame homomorphism. Its right adjoint is

$$q_*(a) = \downarrow a.$$

Hence

$$j_A(U) = q_*q(U) = \downarrow \bigvee U.$$

Proposition 2.2.6 gives

$$\text{Down}(A)_{j_A} \cong A.$$

□

Theorem 2.2.3 (The crisp case recovers ordinary frames). *When*

$$\Lambda = 2,$$

the small fuzzy frames over 2 are exactly the small ordinary frames, up to isomorphism.

Proof. Every fuzzy frame over 2 is, by definition, a nuclear quotient of a frame

$$\text{Down}(X).$$

By Example 2.2.3, this is an ordinary nuclear quotient of an ordinary frame, and hence is an ordinary frame.

Conversely, every small ordinary frame is a nuclear quotient of an ordinary presheaf frame by Theorem 2.2.2. Therefore every small ordinary frame is a fuzzy frame over 2. □

This recovers the usual pointfree frame-theoretic localization picture [9].

Example 2.2.4 (Gödel-valued localizations). *Let*

$$\Lambda = [0, 1]$$

with Gödel tensor

$$r \otimes s = r \wedge s = \min(r, s).$$

A fuzzy predicate on a fuzzy preorder X is a map

$$\phi : X \rightarrow [0, 1]$$

satisfying

$$X(x', x) \wedge \phi(x) \leq \phi(x')$$

for all $x, x' \in X$. The scalar action is truncation by the level r :

$$(r \otimes \phi)(x) = \min(r, \phi(x)).$$

A Gödel-valued fuzzy nucleus closes predicates compatibly with truncation:

$$j(r \wedge j(\phi)) = j(r \wedge \phi).$$

Thus if a predicate has already been closed, then truncating it at level r and closing again gives the same result as truncating the original predicate and closing. The fixed-point frame consists of the predicates satisfying the chosen closure condition. Joins are closed pointwise suprema, finite meets are pointwise minima, and the scalar action is closed truncation:

$$r \otimes_j \phi = j(r \wedge \phi).$$

This gives a localic form of Gödel-valued fuzzy topology, obtained here by enriched presheaf localization rather than by postulating a topology of fuzzy opens directly [8].

Example 2.2.5 (Łukasiewicz-valued localizations). *Let*

$$\Lambda = [0, 1]$$

with Łukasiewicz tensor

$$r \otimes s = \max(0, r + s - 1).$$

A fuzzy predicate on X is a map

$$\phi : X \rightarrow [0, 1]$$

satisfying

$$X(x', x) \otimes \phi(x) \leq \phi(x').$$

The scalar action is no longer truncation. It shifts truth degrees downward:

$$(r \otimes \phi)(x) = \max(0, r + \phi(x) - 1).$$

For $r = 1$, this is the identity; for $r < 1$, all values are lowered, and values below $1 - r$ are sent to 0.

A Łukasiewicz-valued fuzzy nucleus is therefore a closure operator satisfying

$$j(r \otimes j(\phi)) = j(r \otimes \phi).$$

The fixed-point fuzzy frame has closed joins, inherited finite meets, and closed Łukasiewicz scalar action. This example emphasizes that the scalar structure is not merely order-theoretic: the same underlying lattice $[0, 1]$ produces different fuzzy-localic behavior depending on the chosen tensor.

Example 2.2.6 (Product-valued localizations). *Let*

$$\Lambda = [0, 1]$$

with product tensor

$$r \otimes s = rs.$$

Then a fuzzy predicate on X is a map

$$\phi : X \rightarrow [0, 1]$$

satisfying

$$X(x', x)\phi(x) \leq \phi(x')$$

for all $x, x' \in X$. Scalar multiplication is ordinary multiplication:

$$(r \otimes \phi)(x) = r\phi(x).$$

A product-valued fuzzy nucleus closes these predicates compatibly with product rescaling:

$$j(r \, j(\phi)) = j(r \, \phi).$$

In the fixed-point frame, multiplying a closed predicate by r may fail to remain closed, so the closed scalar action is

$$r \otimes_j \phi = j(r\phi).$$

This can be interpreted as first weakening the degree of membership by the factor r , and then imposing the local closure condition. Product-valued localizations therefore behave differently from Gödel-valued localizations: Gödel scalars cut truth values from above, whereas product scalars contract them proportionally.

Example 2.2.7 (Lawvere-valued localizations). *Let*

$$\Lambda = ([0, \infty], \leq_L, \wedge_L, \vee_L, +, \ominus, \infty, 0), \quad a \leq_L b \iff a \geq b \text{ in the usual order.}$$

Thus the tensor is addition, the top element is 0, and the bottom element is ∞ . Joins in Λ are infima in the usual order, and meets in Λ are suprema in the usual order.

A fuzzy preorder is a Lawvere metric space. A Lawvere-valued fuzzy predicate is a map

$$\phi : X \rightarrow [0, \infty]$$

satisfying, in the usual order,

$$\phi(x') \leq d(x', x) + \phi(x).$$

Thus $\phi(x)$ is better read as a cost, distance, or potential rather than as a truth degree. The scalar action adds a constant:

$$(r \otimes \phi)(x) = r + \phi(x).$$

Since the order is reversed, inflationarity of a nucleus

$$\phi \leq_L j(\phi)$$

means

$$j(\phi)(x) \leq \phi(x)$$

in the usual order. Thus a Lawvere-valued nucleus lowers costs, subject to the chosen closure condition.

The scalar-compatibility equation becomes

$$j(r + j(\phi)) = j(r + \phi),$$

where the displayed equation is written in the usual arithmetic notation on $[0, \infty]$. The fixed-point frame consists of the closed cost predicates. Closed joins are closures of pointwise infima, while finite meets are pointwise suprema. This is the metric analogue of passing from arbitrary presheaves to local objects [12].

2.3 Fuzzy locales

We now pass from fuzzy frames to fuzzy locales by formal duality. As in ordinary pointfree topology, frame homomorphisms are read contravariantly as inverse-image maps of locales [9].

2.3.1 Fuzzy locales as duals of fuzzy frames

Definition 2.3.1 (Fuzzy locale). *The category of fuzzy locales over Λ is*

$$\mathbf{FLoc}_\Lambda := \mathbf{FFrm}_\Lambda^{\text{op}}.$$

If $\mathcal{X}$ is a fuzzy locale, its fuzzy frame of opens is denoted

$$\mathcal{O}_\Lambda(\mathcal{X}).$$

A map of fuzzy locales

$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

is represented by a fuzzy-frame homomorphism in the opposite direction:

$$f^* : \mathcal{O}_\Lambda(\mathcal{Y}) \rightarrow \mathcal{O}_\Lambda(\mathcal{X}).$$

Thus the fundamental duality is

$$\mathcal{X} \xrightarrow{f} \mathcal{Y}$$

$$\mathcal{O}_\Lambda(\mathcal{X}) \xleftarrow{f^*} \mathcal{O}_\Lambda(\mathcal{Y})$$

and, categorically,

$$\mathbf{FLoc}_\Lambda \xrightarrow{\mathcal{O}_\Lambda} \mathbf{FFrm}_\Lambda^{\text{op}}.$$

2.3.2 Locales presented by fuzzy presheaves

For a fuzzy preorder X , the fuzzy presheaf frame $P_\Lambda(X)$ determines a fuzzy locale.

Definition 2.3.2 (Presheaf fuzzy locale). *For a fuzzy preorder X , the **presheaf fuzzy locale** associated to X is the fuzzy locale*

$$\mathcal{P}_\Lambda X$$

whose fuzzy frame of opens is

$$\mathcal{O}_\Lambda(\mathcal{P}_\Lambda X) = P_\Lambda(X).$$

Equivalently, $\mathcal{P}_\Lambda X$ is the fuzzy locale corresponding to the identity nucleus on $P_\Lambda(X)$.

Proposition 2.3.1 (Functoriality of presheaf fuzzy locales). *A Λ -functor*

$$f : X \rightarrow Y$$

induces a map of fuzzy locales

$$\mathcal{P}_\Lambda X \rightarrow \mathcal{P}_\Lambda Y.$$

Proof. Precomposition gives

$$f^* : P_\Lambda(Y) \rightarrow P_\Lambda(X), \quad f^*(\psi) = \psi f.$$

Explicitly, if $\psi : Y \rightarrow L$ is a fuzzy predicate on Y , then

$$\begin{aligned} X(x', x) \otimes (\psi f)(x) &= X(x', x) \otimes \psi(fx) \\ &\leq Y(fx', fx) \otimes \psi(fx) \\ &\leq \psi(fx') \\ &= (\psi f)(x'). \end{aligned}$$

Thus $f^*(\psi) \in P_\Lambda(X)$. Since precomposition preserves pointwise joins, finite meets, and scalars, f^* is a fuzzy-frame homomorphism. Therefore it defines a fuzzy-locale map

$$\mathcal{P}_\Lambda X \rightarrow \mathcal{P}_\Lambda Y.$$

□

The construction is covariant on fuzzy preorders:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \mathcal{P}_\Lambda \downarrow & & \downarrow \mathcal{P}_\Lambda \\ \mathcal{P}_\Lambda X & \longrightarrow & \mathcal{P}_\Lambda Y \end{array}$$

while the inverse-image map on opens is contravariant:

$$\begin{array}{ccc} \mathcal{O}_\Lambda(\mathcal{P}_\Lambda X) & \xleftarrow{f^*} & \mathcal{O}_\Lambda(\mathcal{P}_\Lambda Y) \\ \parallel & & \parallel \\ P_\Lambda(X) & \xleftarrow{f^*} & P_\Lambda(Y) \end{array}$$

2.3.3 Localized presheaves as fuzzy sublocales

A fuzzy nucleus has two equivalent readings. Frame-theoretically, it is a quotient of the fuzzy presheaf frame:

$$P_\Lambda(X) \twoheadrightarrow P_\Lambda(X)_j.$$

Locally, it is a sublocale of the presheaf fuzzy locale. This is the ordinary duality between nuclei, quotient frames, and sublocales, with the scalar compatibility carried along [9].

Definition 2.3.3 (Fuzzy sublocale presented by a nucleus). *Let j be a fuzzy nucleus on $P_\Lambda(X)$. The **fuzzy sublocale presented by j** is the fuzzy locale*

$$\mathcal{X}_j$$

whose frame of opens is

$$\mathcal{O}_\Lambda(\mathcal{X}_j) = P_\Lambda(X)_j.$$

The quotient and sublocale readings are displayed by

$$P_\Lambda(X) \xrightarrow{q_j} P_\Lambda(X)_j$$

$$\mathcal{P}_\Lambda X \longleftrightarrow \mathcal{X}_j$$

where the top row is in fuzzy frames and the bottom row is in fuzzy locales. The hook on the locale side is justified by the fact that q_j is a quotient frame homomorphism.

2.3.4 Examples

Example 2.3.1 (Ordinary presheaf locales). *Let*

$$\Lambda = 2.$$

For an ordinary preorder X ,

$$\mathcal{O}_2(\mathcal{P}_2 X) = P_2(X) = \text{Down}(X).$$

Thus $\mathcal{P}_2 X$ is the ordinary lower presheaf locale associated to X . Nuclei on $\text{Down}(X)$ correspond to ordinary sublocales of this locale [9].

If X is discrete, then $\text{Down}(X) \cong \mathcal{P}(X)$, so the corresponding locale is the ordinary powerset locale on the set X . If X has nontrivial order, then the opens are downward-closed subsets, and the locale remembers the Alexandrov topology determined by the preorder.

Example 2.3.2 (Gödel-valued presheaf locales). *For Gödel truth values,*

$$\mathcal{O}_\Lambda(\mathcal{P}_\Lambda X) = P_\Lambda(X)$$

is the frame of Gödel-valued fuzzy predicates on X . A fuzzy open is therefore a monotone fuzzy lower set, where monotonicity is measured by the Gödel tensor:

$$X(x', x) \wedge \phi(x) \leq \phi(x').$$

A fuzzy nucleus on this frame presents a Gödel-valued fuzzy sublocale. Its opens are exactly the closed Gödel predicates. Locally, the quotient

$$P_\Lambda(X) \twoheadrightarrow P_\Lambda(X)_j$$

is read as the inclusion of the fuzzy sublocale

$$\mathcal{X}_j \hookrightarrow \mathcal{P}_\Lambda X.$$

Example 2.3.3 (Łukasiewicz-valued presheaf locales). For Łukasiewicz truth values,

$$\mathcal{O}_\Lambda(\mathcal{P}_\Lambda X) = P_\Lambda(X)$$

is the frame of Łukasiewicz-valued fuzzy predicates on X . The enriched lower-set condition is

$$\max(0, X(x', x) + \phi(x) - 1) \leq \phi(x').$$

A fuzzy nucleus presents a localized fuzzy locale whose opens are the corresponding closed predicates. The scalar action lowers degrees by Łukasiewicz multiplication and then recloses them:

$$r \otimes_j \phi = j(\max(0, r + \phi - 1)),$$

where the formula is interpreted pointwise.

Example 2.3.4 (Product-valued presheaf locales). For product truth values,

$$\mathcal{O}_\Lambda(\mathcal{P}_\Lambda X) = P_\Lambda(X)$$

is the frame of product-valued fuzzy predicates satisfying

$$X(x', x)\phi(x) \leq \phi(x').$$

A fuzzy nucleus gives a product-valued fuzzy sublocale whose opens are closed under the chosen closure operation. The scalar action multiplies fuzzy predicates by constants and then closes:

$$r \otimes_j \phi = j(r\phi).$$

Thus scalar rescaling behaves like proportional attenuation of fuzzy membership.

Example 2.3.5 (Lawvere-valued presheaf locales). For a Lawvere metric space X , the presheaf fuzzy locale $\mathcal{P}_\Lambda X$ has frame of opens

$$\mathcal{O}_\Lambda(\mathcal{P}_\Lambda X) = P_\Lambda(X),$$

whose elements are maps

$$\phi : X \rightarrow [0, \infty]$$

satisfying, in the usual order,

$$\phi(x') \leq d(x', x) + \phi(x).$$

These are nonexpansive cost predicates. A nucleus imposes a local closure condition on such cost predicates. Since the order is reversed, closed joins correspond to closures of pointwise infima of costs, while finite meets correspond to pointwise suprema. This gives a metric-localic interpretation of Lawvere presheaves.

2.4 The fuzzy Sierpiński classifier

In ordinary locale theory, the Sierpiński locale classifies opens. On the frame side, this says that the Sierpiński frame is freely generated by one open. In the present enriched setting the same principle holds, but the free object is not, in general, the raw presheaf category on the two-object subterminal shape. For non-idempotent fuzzy tensors, such as the Łukasiewicz and product tensors, that raw two-point construction does not classify arbitrary fuzzy opens by finite-limit-preserving fuzzy-frame morphisms.

The correct construction uses a small enriched syntactic sketch of finite fuzzy-open formulae and then takes the associated left-exact enriched-sketch localization of its fuzzy presheaf category. The resulting object is characterized by its classifying universal property.

2.4.1 Categorical fuzzy frames and fuzzy opens

We recall the categorical form of fuzzy frames used in this chapter. A categorical fuzzy frame is a left-exact reflective localization

$$a \dashv i : P_{\mathbb{V}}(X) \rightleftarrows \mathcal{A}$$

of a fuzzy presheaf category, where X is a small $\mathbb{V}$ -category, i is fully faithful, and a preserves the chosen finite $\mathcal{F}$ -weighted limits.

A morphism of categorical fuzzy frames

$$H : \mathcal{A} \rightarrow \mathcal{B}$$

is a $\mathbb{V}$ -functor preserving small weighted colimits and the chosen finite $\mathcal{F}$ -weighted limits. Enriched natural transformations are the 2-cells. We write

$$\mathbf{FFrm}_{\mathbb{V}}$$

for the resulting 2-category: objects are categorical fuzzy frames, 1-cells are small-colimit-preserving and finite-limit-preserving $\mathbb{V}$ -functors, and 2-cells are enriched natural transformations. Thus

$$\mathbf{FFrm}_{\mathbb{V}}(\mathcal{A}, \mathcal{B})$$

denotes the ordinary hom-category of such morphisms and enriched natural transformations.

A cosmos-level fuzzy open of a categorical fuzzy frame $\mathcal{A}$ is a subobject

$$S \hookrightarrow 1_{\mathcal{A}}$$

of the terminal object in the underlying ordinary category of $\mathcal{A}$. We write

$$\mathrm{Sub}_{\mathcal{A}}(1_{\mathcal{A}})$$

for the ordinary category of such subobjects. Thus, if $\mathcal{A}$ is enriched, subobjects are computed in the underlying ordinary category $\mathcal{A}_0$.

Equivalently, a representative object S is subterminal precisely when the diagonal

$$S \longrightarrow S \times S$$

is an isomorphism. Since morphisms of categorical fuzzy frames preserve terminal objects and binary products, they send subterminal objects to subterminal objects.

We shall use the following tensor calculation for categorical fuzzy frames.

Lemma 2.4.1 (Tensors in categorical fuzzy frames). *Let*

$$a \dashv i : P_{\mathbb{V}}(X) \rightleftarrows \mathcal{A}$$

be a categorical fuzzy-frame presentation. Then $\mathcal{A}$ is tensored over $\mathbb{V}$. For $V \in \mathbb{V}$ and $A \in \mathcal{A}$, the tensor is computed by

$$V \otimes_{\mathcal{A}} A \cong a(V \otimes iA).$$

Proof. For $B \in \mathcal{A}$, using the enriched adjunction and the tensor–hom adjunction in $P_{\mathbb{V}}(X)$, we compute

$$\begin{aligned} \mathcal{A}(a(V \otimes iA), B) &\cong P_{\mathbb{V}}(X)(V \otimes iA, iB) \\ &\cong [V, P_{\mathbb{V}}(X)(iA, iB)] \\ &\cong [V, \mathcal{A}(A, B)]. \end{aligned}$$

Thus $a(V \otimes iA)$ represents the tensor of A by V in $\mathcal{A}$. □

2.4.2 The enriched syntactic sketch of one fuzzy open

We now construct the enriched syntactic sketch whose models are subterminal objects together with the formal finite-limit and scalar formulae generated from them.

The construction is made relative to the universe convention fixed earlier. In particular, the collection of scalar objects used in the sketch, the chosen class $\mathcal{F}$ of finite weights, and the generated syntax below are small in the ambient universe in which enriched sketches are formed. When the fuzzy cosmos $\mathbb{V}$ is not small in the lowest universe, this construction is understood in the next universe of the fixed tower, equivalently after choosing a small skeleton of the scalar objects used by the sketch.

Let

$$\mathbb{T}_{\Omega}$$

denote the small enriched finite-limit-and-scalar sketch freely generated by one subterminal object. More explicitly, $\mathbb{T}_{\Omega}$ is generated as follows.

First, there are two basic formal objects

$$1, \quad U.$$

The object 1 is declared terminal, and U is declared subterminal. Thus the sketch contains the terminal-object data for 1 , the canonical map

$$U \longrightarrow 1,$$

and the finite-limit data expressing that the diagonal comparison

$$U \longrightarrow U \times U$$

is an isomorphism.

Second, the sketch is closed under the chosen finite $\mathcal{F}$ -weighted limit operations. Thus, whenever a formal finite-limit expression

$$\{\varphi, D\}$$

is built from earlier formal objects, it is again a formal object of $\mathbb{T}_{\Omega}$, together with its designated limiting cone. In particular, binary products of formal objects are again formal objects.

Third, for every scalar object $V \in \mathbb{V}$ and every formal object $A \in \mathbb{T}_\Omega$, there is a formal scalar object

$$V \diamond A.$$

The symbol $V \diamond A$ is syntactic at this stage. It is not required to represent the tensor of A by V in the raw presheaf category, nor is it required to satisfy a tensor universal property inside $\mathbb{T}_\Omega$.

For each formal scalar object $V \diamond A$, the sketch contains a generating enriched morphism

$$\kappa_{V,A} : V \longrightarrow \mathbb{T}_\Omega(A, V \diamond A).$$

Equivalently, $\kappa_{V,A}$ is a formal tensor coevaluation map. Under Yoneda, it determines a canonical comparison map in the presheaf category

$$\alpha_{V,A} : V \otimes yA \longrightarrow y(V \diamond A),$$

where

$$y : \mathbb{T}_\Omega \rightarrow P_{\mathbb{V}}(\mathbb{T}_\Omega)$$

is the enriched Yoneda embedding.

The hom-objects and equations of $\mathbb{T}_\Omega$ are generated freely by the terminal-object data, the subterminality data for U , the specified finite-limit cones, the scalar-forming operations $V \diamond A$, and the generating maps $\kappa_{V,A}$, subject only to the coherence equations for the specified finite-limit cones and for functoriality of the displayed generating data. No tensor universal property for $V \diamond A$ is imposed inside $\mathbb{T}_\Omega$. The tensor universal property is imposed only on models, by requiring the transposes

$$V \otimes_A F(A) \longrightarrow F(V \diamond A)$$

to be isomorphisms.

Equivalently, $\mathbb{T}_\Omega$ is the free small enriched finite-limit-and-scalar sketch generated by one subterminal object U , where the scalar operations are formal operations whose intended interpretation is by tensors in the target category of models.

Remark 2.4.1 (Formal scalar objects). *One should not impose*

$$y(V \diamond A) \cong V \otimes yA$$

inside the raw presheaf category. In general, a scalar multiple of a representable presheaf need not be representable.

In the posetal specialization, if such an identity held representably, then for a scalar $v < \top_L$ it would force

$$\mathbb{T}_\Omega(Z, V \diamond A) = v \otimes \mathbb{T}_\Omega(Z, A).$$

Taking $Z = V \diamond A$ would give

$$\top_L \leq \mathbb{T}_\Omega(V \diamond A, V \diamond A) = v \otimes \mathbb{T}_\Omega(V \diamond A, A) \leq v,$$

which is impossible. Thus $V \diamond A$ is a formal scalar formula, and the comparison

$$V \otimes yA \rightarrow y(V \diamond A)$$

is imposed only after passing to the sketch localization, and only in the correct classifying variance.

2.4.3 Models of the Sierpiński sketch

Let $\mathcal{A}$ be a categorical fuzzy frame. By Lemma 2.4.1, $\mathcal{A}$ is tensored over $\mathbb{V}$. If

$$a \dashv i : P_{\mathbb{V}}(X) \rightleftarrows \mathcal{A}$$

is a presentation of $\mathcal{A}$, then the tensor of $A \in \mathcal{A}$ by $V \in \mathbb{V}$ is computed by

$$V \otimes_{\mathcal{A}} A \cong a(V \otimes iA).$$

A **model of the Sierpiński sketch** in $\mathcal{A}$ is a $\mathbb{V}$ -functor

$$F : \mathbb{T}_{\Omega} \rightarrow \mathcal{A}$$

which satisfies the following conditions:

1. F sends the specified finite $\mathcal{F}$ -limit cones to finite limits in $\mathcal{A}$;
2. $F(U)$ is subterminal;
3. for every scalar object $V \in \mathbb{V}$ and every formal object $A \in \mathbb{T}_{\Omega}$, the image of the generating morphism

$$\kappa_{V,A} : V \rightarrow \mathbb{T}_{\Omega}(A, V \diamond A)$$

has transpose

$$\bar{\kappa}_{V,A}^F : V \otimes_{\mathcal{A}} F(A) \rightarrow F(V \diamond A),$$

and this transpose is an isomorphism.

Thus scalar formulae are interpreted as actual tensors in $\mathcal{A}$, and the formal coevaluation maps are interpreted as the corresponding tensor coevaluations.

Remark 2.4.2 (Scalar formulae need not be subterminal). *Although the generator U is required to be subterminal, the formal scalar objects $V \diamond A$ are not declared subterminal. In a general categorical fuzzy frame, tensors of subterminal objects need not be subterminal. Thus the cosmos-level classifier remembers one distinguished fuzzy open*

$$U \hookrightarrow 1,$$

while the scalar and finite-limit formulae are formal objects generated from it. In the posetal reflection every object of a frame lies below the top element, so these formula objects become ordinary fuzzy opens.

We write

$$\text{Mod}_{\Omega}(\mathcal{A})$$

for the ordinary category of such models and enriched natural transformations.

Proposition 2.4.1 (Models are subterminal objects). *For every categorical fuzzy frame $\mathcal{A}$, evaluation at U induces an equivalence of ordinary categories*

$$\text{Mod}_{\Omega}(\mathcal{A}) \simeq \text{Sub}_{\mathcal{A}}(1_{\mathcal{A}}).$$

Proof. Let

$$F : \mathbb{T}_\Omega \rightarrow \mathcal{A}$$

be a model. Since the sketch declares U subterminal, $F(U)$ is subterminal in $\mathcal{A}$. The formal map $U \rightarrow 1$ is sent to a map

$$F(U) \rightarrow F(1) \cong 1_{\mathcal{A}}.$$

Thus evaluation at U gives a functor

$$\text{Mod}_\Omega(\mathcal{A}) \longrightarrow \text{Sub}_{\mathcal{A}}(1_{\mathcal{A}}).$$

Conversely, let

$$S \hookrightarrow 1_{\mathcal{A}}$$

be a subterminal object of $\mathcal{A}$. By the initiality of the free enriched finite-limit-and-scalar sketch generated by one subterminal object, giving a model

$$F : \mathbb{T}_\Omega \rightarrow \mathcal{A}$$

with $F(U) = S$ is equivalent to interpreting the generator U as the chosen subterminal object S . The values on finite-limit formulae, scalar formulae, and the generating maps $\kappa_{V,A}$ are then forced up to the unique coherent isomorphisms determined by the relevant universal properties.

Explicitly, the corresponding model F_S is defined recursively by

$$F_S(U) = S, \quad F_S(1) = 1_{\mathcal{A}},$$

by interpreting the specified finite-limit formulae as the corresponding finite limits in $\mathcal{A}$, and by interpreting scalar formulae as tensors:

$$F_S(V \diamond A) = V \otimes_{\mathcal{A}} F_S(A).$$

The generating morphism

$$\kappa_{V,A} : V \rightarrow \mathbb{T}_\Omega(A, V \diamond A)$$

is interpreted as the tensor coevaluation

$$V \rightarrow \mathcal{A}(F_S(A), V \otimes_{\mathcal{A}} F_S(A)).$$

Similarly, a morphism of models

$$F_S \rightarrow F_T$$

is determined by its component at the generator U . The components at finite-limit formulae are forced by the universal property of the corresponding limits, and the components at scalar formulae are forced by tensoring with the scalar object. The component at U is exactly a morphism

$$S \rightarrow T$$

over $1_{\mathcal{A}}$. Therefore evaluation at U is fully faithful and essentially surjective, hence an equivalence:

$$\text{Mod}_\Omega(\mathcal{A}) \simeq \text{Sub}_{\mathcal{A}}(1_{\mathcal{A}}).$$

□

2.4.4 The Sierpiński localization

Form the fuzzy presheaf category

$$P_{\mathbb{V}}(\mathbb{T}_{\Omega}) = [\mathbb{T}_{\Omega}^{\text{op}}, \underline{\mathbb{V}}].$$

Let

$$y : \mathbb{T}_{\Omega} \rightarrow P_{\mathbb{V}}(\mathbb{T}_{\Omega})$$

be the Yoneda embedding.

We use the standard enriched finite-limit sketch localization theorem in the following form.

Proposition 2.4.2 (Sierpiński sketch localization). *There is a left-exact reflective localization*

$$a_{\Omega} \dashv i_{\Omega} : P_{\mathbb{V}}(\mathbb{T}_{\Omega}) \rightleftarrows P_{\mathbb{V}}(\mathbb{T}_{\Omega})_{J_{\Omega}}$$

with the following universal property. For every categorical fuzzy frame $\mathcal{A}$, restriction along the composite

$$\mathbb{T}_{\Omega} \xrightarrow{y} P_{\mathbb{V}}(\mathbb{T}_{\Omega}) \xrightarrow{a_{\Omega}} P_{\mathbb{V}}(\mathbb{T}_{\Omega})_{J_{\Omega}}$$

induces a natural equivalence of ordinary categories

$$\mathbf{FFrm}_{\mathbb{V}}(P_{\mathbb{V}}(\mathbb{T}_{\Omega})_{J_{\Omega}}, \mathcal{A}) \simeq \text{Mod}_{\Omega}(\mathcal{A}).$$

Proof. This is the enriched finite-limit sketch localization theorem applied to the small sketch $\mathbb{T}_{\Omega}$. The localization is generated by the sketch data: the specified finite-limit cones, the subterminality equation for U , and the scalar comparison maps

$$\alpha_{V,A} : V \otimes yA \longrightarrow y(V \diamond A).$$

The universal property says exactly that small-colimit-preserving left-exact $\mathbb{V}$ -functors out of the localized presheaf category are the same as models of the sketch in the target categorical fuzzy frame. $\square$

The localization is generated by the enriched sketch data in the following sense: after reflection, the designated finite-limit cones become limiting cones, the subterminality equation for U holds, and the scalar comparison maps

$$\alpha_{V,A} : V \otimes yA \longrightarrow y(V \diamond A)$$

become isomorphisms in the correct classifying variance. Thus the localization is characterized by its universal property for small-colimit-preserving left-exact functors out of the localized presheaf category.

It is important not to describe the J_{Ω} -local presheaves themselves as presheaves satisfying the tensor equations. Orthogonality of a presheaf F to

$$V \otimes yA \rightarrow y(V \diamond A)$$

would instead identify

$$F(V \diamond A) \cong [V, F(A)].$$

The tensor equations are imposed in the correct variance by the universal property for cocontinuous left-exact functors out of the localization.

Definition 2.4.1 (Fuzzy Sierpiński frame). *The **fuzzy Sierpiński frame** over the fuzzy cosmos $\mathbb{V}$ is*

$$\Omega_{\mathbb{V}} := P_{\mathbb{V}}(\mathbb{T}_{\Omega})_{J_{\Omega}}.$$

*Its **generic fuzzy open** is*

$$\sigma_{\mathbb{V}} := a_{\Omega}(yU).$$

By construction,

$$\Omega_{\mathbb{V}}$$

is a left-exact reflective localization of a fuzzy presheaf category. Hence it is a categorical fuzzy frame in the sense of this chapter.

Proposition 2.4.3 (The generic fuzzy open is subterminal). *The object*

$$\sigma_{\mathbb{V}} \in \Omega_{\mathbb{V}}$$

is subterminal.

Proof. The localization $a_{\Omega} \dashv i_{\Omega}$ is generated so that the image of the designated subterminality data for U becomes valid in $\Omega_{\mathbb{V}}$. Since the localization also forces the specified product cone for $U \times U$ to become a product cone, the object

$$a_{\Omega}y(U \times U)$$

is canonically isomorphic to

$$\sigma_{\mathbb{V}} \times \sigma_{\mathbb{V}}.$$

Moreover, the image of the diagonal comparison

$$U \rightarrow U \times U$$

becomes an isomorphism after applying $a_{\Omega}y$. Hence

$$\sigma_{\mathbb{V}} \cong \sigma_{\mathbb{V}} \times \sigma_{\mathbb{V}}.$$

Since $\Omega_{\mathbb{V}}$ has a terminal object, this says precisely that $\sigma_{\mathbb{V}}$ is subterminal. □

2.4.5 The classifying theorem

Theorem 2.4.1 (Fuzzy Sierpiński classification, frame form). *For every categorical fuzzy frame $\mathcal{A}$, evaluation at the generic fuzzy open induces a natural equivalence of ordinary categories*

$$\mathbf{FFrm}_{\mathbb{V}}(\Omega_{\mathbb{V}}, \mathcal{A}) \simeq \mathbf{Sub}_{\mathcal{A}}(1_{\mathcal{A}}).$$

Under this equivalence, a fuzzy-frame morphism

$$H : \Omega_{\mathbb{V}} \rightarrow \mathcal{A}$$

corresponds to the subterminal object

$$H(\sigma_{\mathbb{V}}) \hookrightarrow 1_{\mathcal{A}}.$$

Proof. By Proposition 2.4.2, restriction along

$$a_{\Omega}y : \mathbb{T}_{\Omega} \rightarrow \Omega_{\mathbb{V}}$$

induces a natural equivalence

$$\mathbf{FFrm}_{\mathbb{V}}(\Omega_{\mathbb{V}}, \mathcal{A}) \simeq \mathrm{Mod}_{\Omega}(\mathcal{A}).$$

Under this equivalence, a morphism

$$H : \Omega_{\mathbb{V}} \rightarrow \mathcal{A}$$

corresponds to the model

$$Ha_{\Omega}y : \mathbb{T}_{\Omega} \rightarrow \mathcal{A}.$$

Its value at the generator U is

$$H(a_{\Omega}yU) = H(\sigma_{\mathbb{V}}).$$

By Proposition 2.4.1, evaluation at U gives an equivalence

$$\mathrm{Mod}_{\Omega}(\mathcal{A}) \simeq \mathrm{Sub}_{\mathcal{A}}(1_{\mathcal{A}}).$$

Combining these equivalences gives

$$\mathbf{FFrm}_{\mathbb{V}}(\Omega_{\mathbb{V}}, \mathcal{A}) \simeq \mathrm{Sub}_{\mathcal{A}}(1_{\mathcal{A}}).$$

Under the composite equivalence, H is sent to

$$H(\sigma_{\mathbb{V}}) \hookrightarrow 1_{\mathcal{A}}.$$

Since H preserves terminal objects and binary products, $H(\sigma_{\mathbb{V}})$ is subterminal. $\square$

Definition 2.4.2 (Characteristic morphism of a fuzzy open). *Let $\mathcal{A}$ be a categorical fuzzy frame, and let*

$$S \hookrightarrow 1_{\mathcal{A}}$$

*be a fuzzy open. The **characteristic morphism** of S is the fuzzy-frame morphism*

$$\chi_S^* : \Omega_{\mathbb{V}} \rightarrow \mathcal{A}$$

corresponding to S under the equivalence of Theorem 2.4.1. Equivalently,

$$\chi_S^*(\sigma_{\mathbb{V}}) \cong S$$

as subobjects of $1_{\mathcal{A}}$.

2.4.6 The fuzzy Sierpiński locale

Define

$$\mathbf{FLoc}_{\mathbb{V}} := \mathbf{FFrm}_{\mathbb{V}}^{\mathrm{op}}.$$

Definition 2.4.3 (Fuzzy Sierpiński locale). *The **fuzzy Sierpiński locale** over $\mathbb{V}$ is the fuzzy locale*

$$\mathbb{S}_{\mathbb{V}}$$

whose fuzzy frame of opens is

$$\mathcal{O}_{\mathbb{V}}(\mathbb{S}_{\mathbb{V}}) = \Omega_{\mathbb{V}}.$$

Theorem 2.4.2 (Fuzzy Sierpiński classification, localic form). *For every categorical fuzzy locale $\mathcal{X}$, there is a natural equivalence of ordinary categories*

$$\mathbf{FLoc}_{\mathbb{V}}(\mathcal{X}, \mathbb{S}_{\mathbb{V}}) \simeq \text{Sub}_{\mathcal{O}_{\mathbb{V}}(\mathcal{X})}(1_{\mathcal{O}_{\mathbb{V}}(\mathcal{X})}).$$

Thus maps into the fuzzy Sierpiński locale classify cosmos-level fuzzy opens.

Proof. By definition of fuzzy locales,

$$\begin{aligned} \mathbf{FLoc}_{\mathbb{V}}(\mathcal{X}, \mathbb{S}_{\mathbb{V}}) &= \mathbf{FFrm}_{\mathbb{V}}(\mathcal{O}_{\mathbb{V}}(\mathbb{S}_{\mathbb{V}}), \mathcal{O}_{\mathbb{V}}(\mathcal{X})) \\ &= \mathbf{FFrm}_{\mathbb{V}}(\Omega_{\mathbb{V}}, \mathcal{O}_{\mathbb{V}}(\mathcal{X})). \end{aligned}$$

Applying Theorem 2.4.1 to

$$\mathcal{A} = \mathcal{O}_{\mathbb{V}}(\mathcal{X})$$

gives

$$\mathbf{FFrm}_{\mathbb{V}}(\Omega_{\mathbb{V}}, \mathcal{O}_{\mathbb{V}}(\mathcal{X})) \simeq \text{Sub}_{\mathcal{O}_{\mathbb{V}}(\mathcal{X})}(1_{\mathcal{O}_{\mathbb{V}}(\mathcal{X})}).$$

Therefore

$$\mathbf{FLoc}_{\mathbb{V}}(\mathcal{X}, \mathbb{S}_{\mathbb{V}}) \simeq \text{Sub}_{\mathcal{O}_{\mathbb{V}}(\mathcal{X})}(1_{\mathcal{O}_{\mathbb{V}}(\mathcal{X})}).$$

□

The characteristic map of a fuzzy open

$$S \hookrightarrow 1_{\mathcal{O}_{\mathbb{V}}(\mathcal{X})}$$

is the fuzzy-locale map

$$\chi_S : \mathcal{X} \rightarrow \mathbb{S}_{\mathbb{V}}$$

whose inverse-image morphism

$$\chi_S^* : \Omega_{\mathbb{V}} \rightarrow \mathcal{O}_{\mathbb{V}}(\mathcal{X})$$

satisfies

$$\chi_S^*(\sigma_{\mathbb{V}}) \cong S.$$

2.4.7 Relation with the raw two-point presheaf

The construction above explains why the fuzzy Sierpiński frame should not be identified, in general, with the raw presheaf frame on the two-point subterminal shape.

Let

$$C = (U \leq 1)$$

be the ordinary two-element preorder, regarded as a Λ -preorder by taking

$$C(U, 1) = \top_L, \quad C(1, U) = \perp_L,$$

and identity homs equal to $\top_L$. In the crisp case, the presheaf frame

$$P_2(C)$$

is the ordinary Sierpiński frame. For a general fuzzy truth-value base

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp_L, \top_L),$$

however, the raw presheaf frame

$$P_\Lambda(C)$$

does not have the required free finite-meet behavior among scalar-generated fuzzy open formulae.

In the posetal reflection, an element of

$$P_\Lambda(C)$$

is a pair

$$(u, t) \in L \times L$$

with

$$t \leq u,$$

where u is the value at U and t is the value at 1. The representable corresponding to U is

$$\sigma = (\top_L, \perp_L),$$

and the top predicate is

$$\top = (\top_L, \top_L).$$

Every element decomposes as

$$(u, t) = (u \otimes \sigma) \vee (t \otimes \top).$$

Indeed,

$$u \otimes \sigma = (u, \perp_L), \quad t \otimes \top = (t, t),$$

and $t \leq u$.

Therefore any scalar-compatible frame homomorphism

$$H : P_\Lambda(C) \rightarrow A$$

into a Λ -scalar frame A , with

$$H(\sigma) = a,$$

is forced to satisfy

$$H(u, t) = (u \otimes_A a) \vee (t \otimes_A \top_A).$$

In the special case $A = \Lambda$, this becomes

$$H_a(u, t) = (u \otimes a) \vee t.$$

For non-idempotent tensors, this forced map need not preserve binary meets.

For example, let

$$\Lambda = [0, 1]$$

with the Łukasiewicz tensor

$$r \otimes s = \max(0, r + s - 1).$$

Take

$$a = \frac{1}{2}, \quad p = (0.8, 0.1), \quad q = (0.4, 0.3).$$

Then

$$p \wedge q = (0.4, 0.1).$$

The forced map

$$H_a(u, t) = (u \otimes a) \vee t$$

satisfies

$$H_a(p) = 0.3, \quad H_a(q) = 0.3.$$

Thus

$$H_a(p) \wedge H_a(q) = 0.3.$$

But

$$H_a(p \wedge q) = H_a(0.4, 0.1) = 0.1.$$

Hence

$$H_a(p \wedge q) \neq H_a(p) \wedge H_a(q).$$

So H_a is not a frame homomorphism.

Similarly, for the product tensor

$$r \otimes s = rs,$$

take the same values

$$a = \frac{1}{2}, \quad p = (0.8, 0.1), \quad q = (0.4, 0.3).$$

Then

$$H_a(u, t) = \max(au, t).$$

Thus

$$H_a(p) = \max(0.4, 0.1) = 0.4, \quad H_a(q) = \max(0.2, 0.3) = 0.3,$$

so

$$H_a(p) \wedge H_a(q) = 0.3.$$

But

$$H_a(p \wedge q) = H_a(0.4, 0.1) = \max(0.2, 0.1) = 0.2.$$

Hence

$$H_a(p \wedge q) \neq H_a(p) \wedge H_a(q).$$

Therefore the raw two-point presheaf frame does not classify arbitrary fuzzy opens for these non-idempotent bases.

The enriched syntactic site $\mathbb{T}_\Omega$ corrects this defect by adding formal scalar formulae and their finite-limit combinations before freely adjoining presheaves and then imposing the sketch localization. Thus the fuzzy Sierpiński frame is

$$\Omega_{\mathbb{V}} = P_{\mathbb{V}}(\mathbb{T}_\Omega)_{J_\Omega},$$

not the raw presheaf object on the two-point shape.

Corollary 2.4.1 (Crisp specialization). *When*

$$\Lambda = 2,$$

the fuzzy Sierpiński frame Ω_2 is isomorphic to the ordinary three-element Sierpiński frame. Equivalently, it is isomorphic to the downset frame of the two-element preorder

$$U \leq 1.$$

Proof. For $\Lambda = 2$, the scalar action is forced:

$$1 \otimes a = a, \quad 0 \otimes a = 0.$$

Thus scalar compatibility adds no structure beyond the ordinary finite-meet and join structure. The free fuzzy frame on one fuzzy open is therefore the ordinary free frame on one open, namely the Sierpiński frame. This is the downset frame of the preorder $U \leq 1$, whose generator is the open $\{U\}$, and which has three elements:

$$\emptyset, \quad \{U\}, \quad \{U, 1\}.$$

□

2.4.8 Posetal reflection

When the fuzzy cosmos is the thin cosmos associated to

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp_L, \top_L),$$

we write

$$\Omega_\Lambda, \quad \sigma_\Lambda, \quad \mathbb{S}_\Lambda$$

for the corresponding posetal reflections of

$$\Omega_V, \quad \sigma_V, \quad \mathbb{S}_V.$$

The enriched syntactic sketch has a transparent posetal shadow. In the posetal comparison where the only finite limits used explicitly are terminal objects and binary meets, its formal objects are finite fuzzy-open formulae generated by

$$\top, \quad \sigma, \quad r \otimes \varphi \quad (r \in L), \quad \varphi \wedge \psi.$$

Here σ denotes the formal generator corresponding to U . Arbitrary joins are not part of the finite syntactic sketch; they are freely adjoined by passing to the presheaf frame and then imposing the sketch localization.

The fuzzy hom-value

$$\mathbb{T}_\Omega(\varphi, \psi)$$

is the degree to which the sequent

$$\varphi \Rightarrow \psi$$

is derivable from the finite-limit, subterminality, and scalar rules of the sketch.

The posetal fuzzy Sierpiński frame is

$$\Omega_\Lambda = P_\Lambda(\mathbb{T}_\Omega)_{J_\Omega},$$

with generic fuzzy open

$$\sigma_\Lambda = a_\Omega(y\sigma).$$

For every fuzzy frame A over Λ , the classification theorem specializes to the set-level universal property

$$\mathbf{FFrm}_\Lambda(\Omega_\Lambda, A) \cong |A|.$$

Here the right-hand side is the underlying set of the frame A , because in the posetal reflection every element $a \in A$ determines a unique subobject

$$a \hookrightarrow \top_A,$$

and every subobject of $\top_A$ arises this way. The element corresponding to a fuzzy-frame homomorphism

$$H : \Omega_A \rightarrow A$$

is

$$H(\sigma_A) \in |A|.$$

Dualizing, for every fuzzy locale $\mathcal{X}$,

$$\mathbf{FLoc}_A(\mathcal{X}, \mathbb{S}_A) \cong |\mathcal{O}_A(\mathcal{X})|.$$

Thus maps into the fuzzy Sierpiński locale classify fuzzy opens exactly as in the ordinary localic case, but the construction of the classifier uses the enriched syntactic site rather than the naive two-point presheaf object.

Chapter 3

The Fuzzy Ideal Monad, Fuzzy Domains, and Fuzzy Scott Opens

In the previous chapters we developed fuzzy predicates as enriched presheaves and then localized their predicate frames to obtain fuzzy frames and fuzzy locales. We now apply the same predicate construction to X^{op} . Thus the upper fuzzy predicate object of a small $\mathbb{V}$ -category X is

$$P_{\mathbb{V}}(X^{\text{op}}) = [X, \underline{\mathbb{V}}],$$

and, in the posetal specialization determined by

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp_L, \top_L),$$

it is

$$P_{\Lambda}(X^{\text{op}}) = [X, \Lambda].$$

Its elements are upper fuzzy predicates on X .

The central construction of this chapter is the character adjunction

$$\mathcal{P}_{\mathbb{V}}^{\uparrow} \dashv \text{Char}_{\mathbb{V}}.$$

Here $\mathcal{P}_{\mathbb{V}}^{\uparrow}X$ is the upper fuzzy Alexandrov locale whose frame of opens is $[X, \underline{\mathbb{V}}]$, while $\text{Char}_{\mathbb{V}}$ sends a fuzzy locale to its $\mathbb{V}$ -category of fuzzy-frame morphisms into the truth-value fuzzy frame $\underline{\mathbb{V}}$. Thus characters do not target the fuzzy Sierpiński frame. The fuzzy Sierpiński frame classifies fuzzy opens; the truth-value frame receives fuzzy characters.

The induced monad on fuzzy preorders is the fuzzy ideal monad:

$$T_{\mathbb{V}}X \simeq \text{Idl}_{\mathbb{V}}(X),$$

where $\text{Idl}_{\mathbb{V}}(X)$ is the $\mathbb{V}$ -category of finite-flat weights

$$X^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

In the posetal specialization,

$$T_{\Lambda}X \cong \text{Idl}_{\Lambda}(X),$$

whose elements are fuzzy ideals of X .

A weak fuzzy domain is a fuzzy preorder equipped with a chosen operation

$$\alpha : \text{Idl}_\Lambda(X) \rightarrow X$$

assigning candidate joins to fuzzy ideals. A fuzzy domain is a strict algebra for the fuzzy ideal monad. In the crisp case $\Lambda = 2$, this recovers the ordinary ideal-completion monad, and strict algebras on separated preorders are precisely deposes.

The same monad also contains the algebraic and continuous layers of domain theory. Free fuzzy domains

$$\text{Idl}_\Lambda(B)$$

are the presented fuzzy algebraic domains. Fuzzy continuous domains are fuzzy domains for which the ideal-join map

$$\alpha : \text{Idl}_\Lambda(X) \rightarrow X$$

admits an enriched left adjoint

$$\beta : X \rightarrow \text{Idl}_\Lambda(X).$$

In the crisp case, $\beta(x)$ is the ideal of elements way-below x .

Finally, fuzzy Scott opens are upper fuzzy predicates preserving fuzzy ideal joins. For a weak fuzzy domain

$$\alpha : \text{Idl}_\Lambda(X) \rightarrow X,$$

a fuzzy Scott open is an upper fuzzy predicate

$$U : X \rightarrow L$$

satisfying

$$U(\alpha(I)) = \bigvee_{x \in X} I(x) \otimes U(x)$$

for every fuzzy ideal I .

Remark 3.0.1 (Universe and doctrine conventions). *We use the same universe convention as in the preceding chapters. If $\mathcal{A}$ is not small, then*

$$[\mathcal{A}, \mathbb{V}] \quad \text{and} \quad \text{Char}_\mathbb{V}(\mathcal{A})$$

are understood in the appropriate larger universe. In particular, for small X , the category

$$T_\mathbb{V}X = \text{Char}_\mathbb{V}(P_\mathbb{V}(X^{\text{op}}))$$

is taken in the universe where the corresponding category of finite-flat weights exists.

At the categorical-cosmos level, $T_\mathbb{V}$ is therefore a universe-relative monad on the chosen $\mathbb{V}$ -CAT of locally small $\mathbb{V}$ -categories for which the finite-flat-weight category exists in the ambient universe. In the small posetal specialization it restricts to an honest monad

$$T_\Lambda : \text{Pre}_\Lambda \rightarrow \text{Pre}_\Lambda.$$

Throughout the chapter, “finite-flat” means flat with respect to the chosen finite-limit doctrine used in the definition of categorical fuzzy frames. In the posetal scalar-frame specialization, this doctrine consists of the terminal element and binary meets. Thus finite-flatness specializes to preservation of top and binary meets.

Remark 3.0.2 (Strictification of the ideal monad). *The enriched character construction gives*

$$T_{\mathbb{V}}X = \text{Char}_{\mathbb{V}}(P_{\mathbb{V}}(X^{\text{op}})) \simeq \text{Idl}_{\mathbb{V}}(X).$$

After the finite-flat-weight representation is established, we freely replace this monad by the transported equivalent monad whose value on X is $\text{Idl}_{\mathbb{V}}(X)$, with unit and multiplication given by the Yoneda and flattening formulas below. In the posetal specialization this identification is an isomorphism of fuzzy preorders.

3.1 Upper fuzzy predicates and upper fuzzy Alexandrov locales

3.1.1 The enriched construction

Definition 3.1.1 (Upper fuzzy predicates). *Let X be a small $\mathbb{V}$ -category. The **upper fuzzy predicate category** of X is*

$$P_{\mathbb{V}}(X^{\text{op}}) = [X, \underline{\mathbb{V}}].$$

Its objects are $\mathbb{V}$ -functors

$$U : X \rightarrow \underline{\mathbb{V}}.$$

Definition 3.1.2 (Upper fuzzy Alexandrov locale). *The **upper fuzzy Alexandrov locale** associated to X is the fuzzy locale*

$$\mathcal{P}_{\mathbb{V}}^{\uparrow}X$$

whose fuzzy frame of opens is

$$\text{Opens}_{\mathbb{V}}(\mathcal{P}_{\mathbb{V}}^{\uparrow}X) = P_{\mathbb{V}}(X^{\text{op}}) = [X, \underline{\mathbb{V}}].$$

Since $P_{\mathbb{V}}(X^{\text{op}})$ is itself a fuzzy presheaf category, it is a categorical fuzzy frame via the identity left-exact localization

$$1 \dashv 1 : P_{\mathbb{V}}(X^{\text{op}}) \rightleftarrows P_{\mathbb{V}}(X^{\text{op}}).$$

Thus it defines a fuzzy locale by duality.

Proposition 3.1.1 (Functoriality of upper fuzzy predicates). *Let*

$$f : X \rightarrow Y$$

be a $\mathbb{V}$ -functor. Precomposition defines a fuzzy-frame morphism

$$f^* : P_{\mathbb{V}}(Y^{\text{op}}) = [Y, \underline{\mathbb{V}}] \longrightarrow [X, \underline{\mathbb{V}}] = P_{\mathbb{V}}(X^{\text{op}}), \quad f^*(V) = V f.$$

Consequently there is a fuzzy-locale map

$$\mathcal{P}_{\mathbb{V}}^{\uparrow}X \longrightarrow \mathcal{P}_{\mathbb{V}}^{\uparrow}Y,$$

and hence a covariant functor

$$\mathcal{P}_{\mathbb{V}}^{\uparrow} : \mathbb{V}\text{-Cat} \longrightarrow \mathbf{FLoc}_{\mathbb{V}}.$$

Proof. All fuzzy-frame operations in the predicate frame are computed pointwise. Precomposition preserves them. Passing to frame opposites gives the displayed locale map. $\square$

3.1.2 The posetal specialization

Definition 3.1.3 (Upper fuzzy predicate over Λ). *Let X be a fuzzy preorder over Λ . An **upper fuzzy predicate** on X is a map*

$$U : X \rightarrow L$$

satisfying

$$X(x, x') \otimes U(x) \leq U(x')$$

for all $x, x' \in X$.

Proposition 3.1.2 (The upper Alexandrov fuzzy frame). *The upper fuzzy predicates on X form the fuzzy frame*

$$P_\Lambda(X^{\text{op}}) = [X, \Lambda].$$

Its hom-values are

$$P_\Lambda(X^{\text{op}})(U, V) = \bigwedge_{x \in X} (U(x) \multimap V(x)),$$

and its induced crisp order is pointwise. Joins, meets, and scalar multiplication are also computed pointwise:

$$\left(\bigvee_i U_i \right)(x) = \bigvee_i U_i(x), \quad \left(\bigwedge_i U_i \right)(x) = \bigwedge_i U_i(x), \quad (r \otimes U)(x) = r \otimes U(x).$$

Proof. This is the predicate-frame construction from the previous chapters applied to X^{op} . $\square$

Example 3.1.1 (The crisp case). *For $\Lambda = 2$, a fuzzy preorder is an ordinary preorder and*

$$P_2(X^{\text{op}}) \cong \text{Up}(X).$$

Indeed, the upper-predicate law becomes

$$x \leq x' \quad \text{and} \quad U(x) \implies U(x').$$

Later, Theorem 3.10.1 identifies this frame with the Scott frame of the free dcpo $\text{Idl}(X)$.

Example 3.1.2 (Standard truth-value objects). *For the Gödel tensor $r \otimes s = r \wedge s$, an upper fuzzy predicate satisfies*

$$X(x, x') \wedge U(x) \leq U(x').$$

For the Łukasiewicz tensor $r \otimes s = \max(0, r + s - 1)$, it satisfies

$$\max(0, X(x, x') + U(x) - 1) \leq U(x').$$

For the product tensor $r \otimes s = rs$, it satisfies

$$X(x, x') U(x) \leq U(x').$$

For the Lawvere quantale $[0, \infty]$, with the Lawvere order $a \leq_\Lambda b$ iff $a \geq b$ in the usual order, tensor $+$, top 0 , and bottom ∞ , the upper-predicate law becomes

$$U(x') \leq d(x, x') + U(x)$$

in the usual numerical order.

3.2 Fuzzy characters

3.2.1 The enriched construction

The truth-value object $\mathbb{V}$ is itself a categorical fuzzy frame. Indeed, if $\mathbf{1}$ denotes the terminal small $\mathbb{V}$ -category, then

$$P_{\mathbb{V}}(\mathbf{1}) = [\mathbf{1}^{\text{op}}, \mathbb{V}] \simeq \mathbb{V}.$$

Thus $\mathbb{V}$ is the presheaf fuzzy frame on $\mathbf{1}$.

Definition 3.2.1 (Fuzzy character). *Let $\mathcal{A}$ be a categorical fuzzy frame. Its **fuzzy character category** is the full $\mathbb{V}$ -subcategory*

$$\text{Char}_{\mathbb{V}}(\mathcal{A}) \subseteq [\mathcal{A}, \mathbb{V}]$$

spanned by those $\mathbb{V}$ -functors

$$\chi : \mathcal{A} \rightarrow \mathbb{V}$$

that are fuzzy-frame morphisms, that is, that preserve colimits, the chosen finite limits, and scalars.

Definition 3.2.2 (Characters of a fuzzy locale). *If $\mathcal{X}$ is a fuzzy locale, define*

$$\text{Char}_{\mathbb{V}}(\mathcal{X}) := \text{Char}_{\mathbb{V}}(\text{Opens}_{\mathbb{V}}(\mathcal{X})).$$

Remark 3.2.1 (Presentation-wise characters). *If a fuzzy frame is presented as a reflective localization*

$$a \dashv i : P_{\mathbb{V}}(X) \rightleftarrows \mathcal{A},$$

then characters of $\mathcal{A}$ may be described as characters of $P_{\mathbb{V}}(X)$ satisfying the descent condition imposed by the localization. For upper predicate frames, this descent-free case becomes the finite-flat-weight description of Section 3.4.

3.2.2 The posetal specialization

Definition 3.2.3 (Fuzzy character over Λ). *Let $A \in \mathbf{FFrm}_{\Lambda}$. Its **fuzzy character preorder***

$$\text{Char}_{\Lambda}(A)$$

is the full fuzzy subpreorder of $[A, \Lambda]$ spanned by the scalar-frame homomorphisms

$$\chi : A \rightarrow \Lambda.$$

Equivalently, its underlying set is

$$|\text{Char}_{\Lambda}(A)| = \mathbf{FFrm}_{\Lambda}(A, \Lambda).$$

Thus an element of $\text{Char}_{\Lambda}(A)$ is a map preserving arbitrary joins, finite meets, and scalar multiplication:

$$\chi(r \otimes_A a) = r \otimes \chi(a).$$

Proposition 3.2.1 (The fuzzy preorder of characters). *The fuzzy preorder on $\text{Char}_{\Lambda}(A)$ is inherited from the pointwise functor preorder:*

$$\text{Char}_{\Lambda}(A)(\chi, \psi) = \bigwedge_{a \in A} (\chi(a) \multimap \psi(a)).$$

Its induced crisp order is pointwise.

Proof. The character preorder is a full fuzzy subpreorder of $[A, \Lambda]$. $\square$

Definition 3.2.4 (Characters of a fuzzy locale over Λ). *For a fuzzy locale $\mathcal{X}$, define*

$$\text{Char}_\Lambda(\mathcal{X}) := \text{Char}_\Lambda(\mathcal{O}_\Lambda(\mathcal{X})).$$

Proposition 3.2.2 (Characters of the truth-value frame). *There is exactly one scalar-frame character*

$$\Lambda \rightarrow \Lambda,$$

namely the identity. Hence

$$\text{Char}_\Lambda(\Lambda)$$

is the terminal fuzzy preorder.

Proof. Let $\chi : \Lambda \rightarrow \Lambda$ be a scalar-frame homomorphism. Since χ preserves top,

$$\chi(\top_L) = \top_L.$$

For every $r \in L$,

$$\chi(r) = \chi(r \otimes \top_L) = r \otimes \chi(\top_L) = r \otimes \top_L = r.$$

Thus $\chi = 1_\Lambda$. $\square$

Example 3.2.1 (The crisp case). *For $\Lambda = 2$, a character of a frame A is a frame homomorphism*

$$A \rightarrow 2,$$

equivalently a completely prime filter of A . Thus, in the crisp localic case, characters are points.

3.3 The upper fuzzy character adjunction

3.3.1 The enriched adjunction

Theorem 3.3.1 (Upper fuzzy character adjunction). *Let X be a small $\mathbb{V}$ -category and let $\mathcal{X}$ be a fuzzy locale. There is a natural equivalence*

$$\mathbf{FLoc}_\mathbb{V}(\mathcal{P}_\mathbb{V}^\uparrow X, \mathcal{X}) \simeq \mathbb{V}\text{-}\mathbf{Cat}(X, \text{Char}_\mathbb{V}(\mathcal{X})).$$

Equivalently, writing

$$A = \text{Opens}_\mathbb{V}(\mathcal{X}),$$

there is a natural equivalence

$$\mathbf{FFrm}_\mathbb{V}(A, P_\mathbb{V}(X^{\text{op}})) \simeq \mathbb{V}\text{-}\mathbf{Cat}(X, \text{Char}_\mathbb{V}(A)).$$

Proof. A fuzzy-locale map

$$\mathcal{P}_\mathbb{V}^\uparrow X \rightarrow \mathcal{X}$$

is frame-dually a fuzzy-frame morphism

$$h : A \rightarrow [X, \underline{\mathbb{V}}].$$

By the enriched tensor–hom adjunction and the symmetry of $\otimes$, this is equivalently a $\mathbb{V}$ -functor

$$\widehat{h} : X \rightarrow [A, \underline{\mathbb{V}}], \quad \widehat{h}(x)(a) = h(a)(x).$$

Explicitly, this is the composite transposition

$$\mathbb{V}\text{-}\mathbf{CAT}(A, [X, \underline{\mathbb{V}}]) \cong \mathbb{V}\text{-}\mathbf{CAT}(A \otimes X, \underline{\mathbb{V}}) \cong \mathbb{V}\text{-}\mathbf{CAT}(X, [A, \underline{\mathbb{V}}]).$$

Since evaluation at x preserves the pointwise fuzzy-frame structure on $[X, \underline{\mathbb{V}}]$, each

$$\widehat{h}(x) : A \rightarrow \underline{\mathbb{V}}$$

is a fuzzy character of A . Hence $\widehat{h}$ factors through

$$\text{Char}_{\mathbb{V}}(A) \subseteq [A, \underline{\mathbb{V}}].$$

Conversely, a $\mathbb{V}$ -functor

$$k : X \rightarrow \text{Char}_{\mathbb{V}}(A)$$

defines

$$\widetilde{k} : A \rightarrow [X, \underline{\mathbb{V}}], \quad \widetilde{k}(a)(x) = k(x)(a).$$

For each a , this is an upper predicate on X ; for each x , the map $a \mapsto k(x)(a)$ preserves the fuzzy-frame structure. Therefore $\widetilde{k}$ is a fuzzy-frame morphism. The two constructions are inverse by the enriched tensor–hom adjunction, and naturality is inherited from the naturality of that adjunction. $\square$

3.3.2 The posetal specialization

Theorem 3.3.2 (Posetal upper fuzzy character adjunction). *Let X be a fuzzy preorder and let $A \in \mathbf{FFrm}_{\Lambda}$. There is a natural isomorphism*

$$\mathbf{FFrm}_{\Lambda}(A, P_{\Lambda}(X^{\text{op}})) \cong \text{Pre}_{\Lambda}(X, \text{Char}_{\Lambda}(A)).$$

Proof. For a fuzzy-frame homomorphism

$$h : A \rightarrow P_{\Lambda}(X^{\text{op}}),$$

define

$$\widehat{h}(x)(a) = h(a)(x).$$

Each $\widehat{h}(x)$ is a character because evaluation at x preserves the pointwise scalar-frame structure. Moreover, for all $x, x' \in X$ and $a \in A$, the upper-predicate law for $h(a)$ gives

$$X(x, x') \otimes h(a)(x) \leq h(a)(x'),$$

or equivalently

$$X(x, x') \leq h(a)(x) \multimap h(a)(x').$$

Taking the meet over a shows that

$$X(x, x') \leq \text{Char}_{\Lambda}(A)(\widehat{h}(x), \widehat{h}(x')).$$

Thus $\widehat{h}$ is a Λ -functor.

Conversely, a Λ -functor

$$k : X \rightarrow \text{Char}_\Lambda(A)$$

defines

$$\tilde{k}(a)(x) = k(x)(a).$$

Functoriality of k gives the upper-predicate law in x , while the character condition on each $k(x)$ gives preservation of joins, finite meets, and scalars in a . Hence $\tilde{k}$ is a fuzzy-frame homomorphism. The two assignments are inverse. $\square$

Example 3.3.1 (The crisp case). *For $\Lambda = 2$, the adjunction becomes*

$$\mathbf{Frm}(A, \text{Up}(X)) \cong \mathbf{Pre}(X, \text{Char}_2(A)).$$

Thus frame maps into an upper Alexandrov frame are the same as monotone point-valued families indexed by X .

Example 3.3.2 (Discrete fuzzy preorders). *If X is discrete, then*

$$P_\Lambda(X^{\text{op}}) \cong L^X.$$

A fuzzy-frame homomorphism

$$A \rightarrow L^X$$

is precisely an X -indexed family of characters $A \rightarrow \Lambda$.

3.4 Fuzzy ideals as finite-flat weights

3.4.1 The enriched construction

Definition 3.4.1 (Finite-flat weight). *Let X be a small $\mathbb{V}$ -category. A **finite-flat weight** on X is a weight*

$$I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

such that the weighted-colimit functional

$$\chi_I : [X, \underline{\mathbb{V}}] \rightarrow \underline{\mathbb{V}}, \quad \chi_I(U) = \int^{x \in X} I(x) \otimes U(x),$$

preserves the chosen finite limits of the fuzzy-frame doctrine. Equivalently, I -weighted colimits in $\underline{\mathbb{V}}$ commute with those finite limits.

Definition 3.4.2 (Fuzzy ideal). *A **fuzzy ideal** of a small $\mathbb{V}$ -category X is a finite-flat weight*

$$I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

We write

$$\text{Idl}_{\mathbb{V}}(X)$$

for the full $\mathbb{V}$ -subcategory of

$$[X^{\text{op}}, \underline{\mathbb{V}}]$$

spanned by the finite-flat weights.

Proposition 3.4.1 (Flat-weight representation of cocontinuous functionals). *Let X be small. Cocontinuous $\mathbb{V}$ -functors*

$$[X, \mathbb{V}] \rightarrow \underline{\mathbb{V}}$$

are, up to canonical isomorphism, precisely the weighted-colimit functionals

$$U \mapsto \int^{x \in X} I(x) \otimes U(x)$$

for weights

$$I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

Such a functional preserves the chosen finite limits precisely when I is finite-flat.

Proof. Let

$$\chi : [X, \mathbb{V}] \rightarrow \underline{\mathbb{V}}$$

be cocontinuous. Define

$$I_\chi(x) = \chi(X(x, -)).$$

The assignment

$$x \mapsto X(x, -)$$

is the Yoneda embedding of X^{op} into $[X, \mathbb{V}]$. Since χ is a $\mathbb{V}$ -functor, I_χ is a weight

$$I_\chi : X^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

For every

$$U : X \rightarrow \mathbb{V},$$

the enriched co-Yoneda formula in the copresheaf category $[X, \mathbb{V}]$ gives

$$U \cong \int^{x \in X} U(x) \otimes X(x, -).$$

Applying the cocontinuous functor χ , we obtain

$$\begin{aligned} \chi(U) &\cong \chi \left(\int^{x \in X} U(x) \otimes X(x, -) \right) \\ &\cong \int^{x \in X} U(x) \otimes \chi(X(x, -)) \\ &= \int^{x \in X} U(x) \otimes I_\chi(x) \\ &\cong \int^{x \in X} I_\chi(x) \otimes U(x). \end{aligned}$$

Thus χ is represented by I_χ .

Conversely, every weight $I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$ determines the weighted-colimit functional

$$\chi_I(U) = \int^{x \in X} I(x) \otimes U(x).$$

This functional is cocontinuous because it is built from tensors and small colimits in $\underline{\mathbb{V}}$, and tensors preserve colimits in each variable. The two constructions are inverse by the same co-Yoneda calculation, since

$$\chi_I(X(x, -)) \cong \int^{y \in X} I(y) \otimes X(x, y) \cong I(x).$$

Finally, preservation of the chosen finite limits by χ_I is exactly the definition of finite-flatness of the weight I . $\square$

Theorem 3.4.1 (Characters of upper predicate frames). *For every small $\mathbb{V}$ -category X , there is a canonical equivalence*

$$\text{Char}_{\mathbb{V}}(P_{\mathbb{V}}(X^{\text{op}})) \simeq \text{Idl}_{\mathbb{V}}(X).$$

Under this equivalence, a fuzzy ideal

$$I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

corresponds to the character

$$\chi_I(U) = \int^{x \in X} I(x) \otimes U(x).$$

Proof. Since

$$P_{\mathbb{V}}(X^{\text{op}}) = [X, \underline{\mathbb{V}}],$$

a character is a cocontinuous, finite-limit-preserving, scalar-preserving $\mathbb{V}$ -functor

$$[X, \underline{\mathbb{V}}] \rightarrow \underline{\mathbb{V}}.$$

By Proposition 3.4.1, the cocontinuous finite-limit-preserving functors are exactly the finite-flat weighted-colimit functionals

$$U \mapsto \int^{x \in X} I(x) \otimes U(x).$$

Scalar preservation is automatic for these functionals because tensoring distributes over the coends defining the weighted colimits:

$$\begin{aligned} \chi_I(r \otimes U) &= \int^{x \in X} I(x) \otimes r \otimes U(x) \\ &\cong r \otimes \int^{x \in X} I(x) \otimes U(x) \\ &= r \otimes \chi_I(U). \end{aligned}$$

Thus characters of the upper predicate frame are precisely finite-flat weights on X . $\square$

3.4.2 The posetal specialization

Definition 3.4.3 (Fuzzy ideal over Λ). *Let X be a fuzzy preorder over Λ . A **fuzzy ideal** of X is a lower fuzzy predicate*

$$I : X^{\text{op}} \rightarrow \Lambda$$

such that

$$\chi_I(U) = \bigvee_{x \in X} I(x) \otimes U(x)$$

preserves top and binary meets as a map

$$P_\Lambda(X^{\text{op}}) \rightarrow \Lambda.$$

Equivalently, I satisfies the lower-predicate law

$$X(x', x) \otimes I(x) \leq I(x')$$

and the finite-flatness equations

$$\bigvee_{x \in X} I(x) = \top_L$$

and

$$\bigvee_{x \in X} I(x) \otimes (U(x) \wedge V(x)) = \left(\bigvee_{x \in X} I(x) \otimes U(x) \right) \wedge \left(\bigvee_{x \in X} I(x) \otimes V(x) \right)$$

for all upper fuzzy predicates U, V .

Definition 3.4.4 (The fuzzy preorder of fuzzy ideals). *For a fuzzy preorder X , define*

$$\text{Idl}_\Lambda(X)$$

to be the full fuzzy subpreorder of $P_\Lambda(X)$ spanned by the fuzzy ideals. Thus, for fuzzy ideals I, J ,

$$\text{Idl}_\Lambda(X)(I, J) = \bigwedge_{x \in X} (I(x) \multimap J(x)).$$

Theorem 3.4.2 (Characters of upper fuzzy Alexandrov frames). *For every fuzzy preorder X , there is an isomorphism of fuzzy preorders*

$$\text{Char}_\Lambda(P_\Lambda(X^{\text{op}})) \cong \text{Idl}_\Lambda(X).$$

Under this isomorphism, a fuzzy ideal I corresponds to the character

$$\chi_I(U) = \bigvee_{x \in X} I(x) \otimes U(x).$$

Proof. The coend formula in Theorem 3.4.1 specializes to the displayed join formula, and finite-flatness specializes to preservation of top and binary meets.

It remains to identify hom-values. If

$$r \leq \bigwedge_x (I(x) \multimap J(x)),$$

then $r \otimes I(x) \leq J(x)$ for every x , and therefore

$$r \otimes \chi_I(U) = r \otimes \bigvee_x I(x) \otimes U(x) \leq \bigvee_x J(x) \otimes U(x) = \chi_J(U)$$

for every upper predicate U . Hence

$$r \leq \bigwedge_U (\chi_I(U) \multimap \chi_J(U)).$$

Conversely, evaluate at the upper representable $X(x, -)$. The enriched Yoneda calculation gives

$$\chi_I(X(x, -)) = \bigvee_y I(y) \otimes X(x, y) = I(x),$$

and similarly for J . Hence the character hom is bounded above by

$$\bigwedge_x (I(x) \multimap J(x)).$$

Thus the hom-values agree. □

Example 3.4.1 (The crisp case). *For $\Lambda = 2$, a fuzzy ideal is an ordinary ideal*

$$I \subseteq X.$$

Thus I is a lower set, is inhabited, and is directed. Therefore

$$\text{Char}_2(\text{Up}(X)) \cong \text{Idl}(X).$$

Example 3.4.2 (Standard truth-value objects). *For Gödel truth values,*

$$\chi_I(U) = \bigvee_x \min(I(x), U(x)).$$

For Łukasiewicz truth values,

$$\chi_I(U) = \bigvee_x \max(0, I(x) + U(x) - 1).$$

For product truth values,

$$\chi_I(U) = \bigvee_x I(x)U(x).$$

For the Lawvere quantale, joins in Λ are usual infima and tensor is addition, so

$$\chi_I(U) = \inf_x (I(x) + U(x)).$$

3.5 The fuzzy ideal monad

3.5.1 The enriched monad

Definition 3.5.1 (Fuzzy ideal monad). *The **fuzzy ideal monad** is the monad*

$$T_{\mathbb{V}} = \text{Char}_{\mathbb{V}} \mathcal{P}_{\mathbb{V}}^{\uparrow}$$

induced by the adjunction

$$\mathcal{P}_{\mathbb{V}}^{\uparrow} \dashv \text{Char}_{\mathbb{V}}.$$

Thus

$$T_{\mathbb{V}}X = \text{Char}_{\mathbb{V}}(P_{\mathbb{V}}(X^{\text{op}})) \simeq \text{Idl}_{\mathbb{V}}(X).$$

After the strictification convention of Remark 3.0.2, we write simply

$$T_{\mathbb{V}}X = \text{Idl}_{\mathbb{V}}(X).$$

Proposition 3.5.1 (Unit and multiplication). *Under the finite-flat-weight description, the unit*

$$\eta_X : X \rightarrow \text{Idl}_{\mathbb{V}}(X)$$

is the Yoneda embedding into finite-flat weights:

$$\eta_X(x) = X(-, x).$$

The multiplication

$$\mu_X : \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X)) \rightarrow \text{Idl}_{\mathbb{V}}(X)$$

is flattening:

$$\mu_X(\mathcal{I})(x) = \int^{I \in \text{Idl}_{\mathbb{V}}(X)} \mathcal{I}(I) \otimes I(x).$$

Proof. The unit of the character adjunction sends x to evaluation at x :

$$U \mapsto U(x).$$

The corresponding weight evaluates covariant representables by

$$X(y, -)(x) = X(y, x),$$

so $\eta_X(x) = X(-, x)$. The multiplication is the monad multiplication induced by the adjunction, transported through the finite-flat-weight representation; it is the weighted colimit of an ideal of ideals. $\square$

3.5.2 The posetal specialization

Definition 3.5.2 (The posetal fuzzy ideal monad). *In the posetal specialization,*

$$T_{\Lambda}(X) = \text{Idl}_{\Lambda}(X).$$

The unit is

$$\eta_X : X \rightarrow \text{Idl}_{\Lambda}(X), \quad \eta_X(x)(y) = X(y, x),$$

and the multiplication is

$$\mu_X : \text{Idl}_{\Lambda}(\text{Idl}_{\Lambda}(X)) \rightarrow \text{Idl}_{\Lambda}(X),$$

where

$$\mu_X(\mathcal{I})(x) = \bigvee_{I \in \text{Idl}_{\Lambda}(X)} \mathcal{I}(I) \otimes I(x).$$

Proposition 3.5.2 (Functorial action of the fuzzy ideal monad). *Let*

$$f : X \rightarrow Y$$

be a Λ -functor. Then

$$T_{\Lambda}(f) : \text{Idl}_{\Lambda}(X) \rightarrow \text{Idl}_{\Lambda}(Y)$$

is given by

$$T_{\Lambda}(f)(I)(y) = \bigvee_{x \in X} I(x) \otimes Y(y, f(x)).$$

This is the fuzzy lower closure of the direct image of I along f .

Proof. This is the posetal specialization of enriched left Kan extension of a weight along f . Since it is the functorial action of the monad induced by the character adjunction, it preserves fuzzy ideals and is functorial in f . $\square$

Example 3.5.1 (The crisp ideal monad). For $\Lambda = 2$,

$$T_2(X) = \text{Idl}(X).$$

The unit sends

$$x \mapsto \downarrow x,$$

multiplication sends an ideal of ideals to its union,

$$\mu_X(\mathcal{I}) = \bigcup_{I \in \mathcal{I}} I,$$

and for a monotone map $f : X \rightarrow Y$,

$$T_2(f)(I) = \downarrow f[I].$$

Thus T_2 is the ordinary ideal-completion monad.

Example 3.5.2 (Standard direct-image formulas). For Gödel truth values,

$$T_\Lambda(f)(I)(y) = \bigvee_x \min(I(x), Y(y, f(x))).$$

For Łukasiewicz truth values,

$$T_\Lambda(f)(I)(y) = \bigvee_x \max(0, I(x) + Y(y, f(x)) - 1).$$

For product truth values,

$$T_\Lambda(f)(I)(y) = \bigvee_x I(x)Y(y, f(x)).$$

For the Lawvere quantale,

$$T_\Lambda(f)(I)(y) = \inf_x (I(x) + d_Y(y, f(x))).$$

3.6 Weak fuzzy domains and fuzzy domains

3.6.1 The enriched notion

Definition 3.6.1 (Weak fuzzy domain). A **weak fuzzy domain** is a small $\mathbb{V}$ -category X equipped with a chosen $\mathbb{V}$ -functor

$$\alpha : T_{\mathbb{V}}X \rightarrow X.$$

Using

$$T_{\mathbb{V}}X = \text{Idl}_{\mathbb{V}}(X),$$

we regard α as assigning to each fuzzy ideal

$$I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

a chosen candidate fuzzy ideal-join

$$\alpha(I) \in X.$$

Definition 3.6.2 (Fuzzy domain). A **fuzzy domain** is a weak fuzzy domain whose structure map

$$\alpha : T_{\mathbb{V}}X \rightarrow X$$

is a strict algebra for the fuzzy ideal monad:

$$\alpha\eta_X = 1_X, \quad \alpha\mu_X = \alpha T_{\mathbb{V}}(\alpha).$$

Remark 3.6.1. Weak fuzzy domains carry chosen candidate ideal-joins. Strict fuzzy domains are those for which these choices are coherent with principal ideals and iterated fuzzy ideals.

3.6.2 The posetal specialization

Definition 3.6.3 (Weak fuzzy domain over Λ). A **weak fuzzy domain over Λ** is a fuzzy preorder X equipped with a Λ -functor

$$\alpha : \text{Idl}_{\Lambda}(X) \rightarrow X.$$

Definition 3.6.4 (Fuzzy domain over Λ). A **fuzzy domain over Λ** is a weak fuzzy domain satisfying

$$\alpha(\eta_X x) = x$$

and

$$\alpha(\mu_X \mathcal{I}) = \alpha(T_{\Lambda}\alpha(\mathcal{I})).$$

Equivalently,

$$\alpha(X(-, x)) = x$$

and

$$\alpha \left(x \mapsto \bigvee_{I \in \text{Idl}_{\Lambda}(X)} \mathcal{I}(I) \otimes I(x) \right) = \alpha \left(x \mapsto \bigvee_{I \in \text{Idl}_{\Lambda}(X)} \mathcal{I}(I) \otimes X(x, \alpha(I)) \right).$$

Theorem 3.6.1 (Crisp algebras are dcpos). For $\Lambda = 2$, strict algebras for the ideal monad on separated preorders are precisely dcpos, where dcpos are understood to have suprema of nonempty directed subsets.

Proof. Let

$$\alpha : \text{Idl}(X) \rightarrow X$$

be a strict algebra. The unit law says

$$\alpha(\downarrow x) = x.$$

If $I \in \text{Idl}(X)$ and $x \in I$, then

$$\downarrow x \subseteq I,$$

so by monotonicity

$$x = \alpha(\downarrow x) \leq \alpha(I).$$

Thus $\alpha(I)$ is an upper bound of I . If y is any upper bound of I , then

$$I \subseteq \downarrow y,$$

and again by monotonicity,

$$\alpha(I) \leq \alpha(\downarrow y) = y.$$

Hence $\alpha(I) = \bigvee I$. Every nonempty directed subset generates an ideal with the same supremum, so X is a dcpo when separated.

Conversely, if X is a dcpo, define

$$\alpha(I) = \bigvee I.$$

The unit law is immediate, and the multiplication law says that the supremum of a directed family of directed suprema is the supremum of its union. Thus X is a strict algebra. $\square$

Example 3.6.1 (Gödel, product, and Lawvere weak fuzzy domains). *A Gödel-valued weak fuzzy domain assigns candidate joins to finite-flat lower weights governed by the minimum tensor. A product-valued weak fuzzy domain assigns candidate joins to finite-flat lower weights with multiplicative membership. For the Lawvere quantale, a weak fuzzy domain assigns colimits to finite-flat Lawvere weights, giving a metric-enriched domain-like completion structure.*

3.7 The category of fuzzy domains

We now record the categorical structure carried by fuzzy domains. The point is not to add extra structure to a fuzzy domain, but to observe that the finite-flat doctrine already used to define fuzzy ideals also determines the colimits, morphisms, tensor products, and internal homs of fuzzy domains.

Throughout this section, finite-flatness is understood relative to the finite-limit doctrine fixed at the beginning of the chapter. Thus, in each predicate category $[X, \mathbb{V}]$, the specified finite limits are computed pointwise. In the posetal scalar-frame specialization, this means preservation of the terminal predicate and binary meets.

Definition 3.7.1 (The category of fuzzy domains). *A morphism of fuzzy domains*

$$f : (X, \alpha_X) \longrightarrow (Y, \alpha_Y)$$

is a $\mathbb{V}$ -functor

$$f : X \rightarrow Y$$

such that

$$f\alpha_X = \alpha_Y T_{\mathbb{V}}(f).$$

The ordinary category of fuzzy domains and fuzzy-domain morphisms is denoted

$$\mathbf{FDom}_{\mathbb{V}}.$$

Thus $\mathbf{FDom}_{\mathbb{V}}$ is the Eilenberg–Moore category of strict algebras for the fuzzy ideal monad. The rest of this section identifies these strict algebras as finite-flat-cocomplete $\mathbb{V}$ -categories and then applies Kelly’s tensor-product theorem for saturated commutative classes of weights.

3.7.1 Finite-flat colimits in fuzzy domains

Lemma 3.7.1 (Representables are finite-flat). *Let X be a small $\mathbb{V}$ -category. For every $x \in X$, the representable weight*

$$X(-, x) : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

is finite-flat.

Proof. The colimit functional associated to $X(-, x)$ is

$$\chi_{X(-, x)}(U) = \int^{y \in X} X(y, x) \otimes U(y).$$

By enriched co-Yoneda,

$$\chi_{X(-, x)}(U) \cong U(x).$$

Thus $\chi_{X(-, x)}$ is evaluation at x . Since the specified finite limits in $[X, \underline{\mathbb{V}}]$ are computed pointwise, evaluation at x preserves them. Hence $X(-, x)$ is finite-flat. $\square$

Lemma 3.7.2 (Finite-flat direct images). *Let*

$$D : K \rightarrow X$$

be a $\mathbb{V}$ -functor, and let

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be finite-flat. Define the lower direct image

$$D_! W : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

by

$$(D_! W)(x) = \int^{k \in K} W(k) \otimes X(x, Dk).$$

Then $D_! W$ is finite-flat.

Proof. Let

$$U : X \rightarrow \underline{\mathbb{V}}$$

be an upper predicate on X . Then

$$\begin{aligned} \chi_{D_! W}(U) &= \int^{x \in X} (D_! W)(x) \otimes U(x) \\ &= \int^{x \in X} \left(\int^{k \in K} W(k) \otimes X(x, Dk) \right) \otimes U(x) \\ &\cong \int^{k \in K} W(k) \otimes \left(\int^{x \in X} X(x, Dk) \otimes U(x) \right) \\ &\cong \int^{k \in K} W(k) \otimes U(Dk) \\ &= \chi_W(UD). \end{aligned}$$

The first isomorphism is Fubini for coends, and the second is enriched co-Yoneda.

Precomposition

$$D^* : [X, \underline{\mathbb{V}}] \rightarrow [K, \underline{\mathbb{V}}], \quad U \mapsto UD,$$

preserves the specified finite limits because they are computed pointwise. Since W is finite-flat, χ_W preserves those finite limits. Hence

$$\chi_{D_! W} \cong \chi_W D^*$$

preserves them, so $D_! W$ is finite-flat. $\square$

Proposition 3.7.1 (Finite-flat saturation). *Finite-flat weights are closed under finite-flat weighted colimits in presheaf categories.*

Explicitly, let

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be finite-flat, and let

$$H : K \rightarrow \text{Idl}_{\mathbb{V}}(X)$$

be a $\mathbb{V}$ -functor. Define

$$J : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

pointwise by

$$J(x) = \int^{k \in K} W(k) \otimes Hk(x).$$

Then J is finite-flat.

Proof. For every upper predicate

$$U : X \rightarrow \underline{\mathbb{V}},$$

we have

$$\begin{aligned} \chi_J(U) &= \int^{x \in X} J(x) \otimes U(x) \\ &= \int^{x \in X} \left(\int^{k \in K} W(k) \otimes Hk(x) \right) \otimes U(x) \\ &\cong \int^{k \in K} W(k) \otimes \left(\int^{x \in X} Hk(x) \otimes U(x) \right) \\ &= \int^{k \in K} W(k) \otimes \chi_{Hk}(U). \end{aligned}$$

Thus

$$\chi_J(U) \cong \chi_W(\overline{H}(U)),$$

where

$$\overline{H} : [X, \underline{\mathbb{V}}] \rightarrow [K, \underline{\mathbb{V}}]$$

is defined by

$$\overline{H}(U)(k) = \chi_{Hk}(U).$$

We first check that $\overline{H}(U)$ is an upper predicate on K . Since H is a $\mathbb{V}$ -functor, for every $k, k' \in K$ there is a morphism

$$K(k, k') \longrightarrow \text{Idl}_{\mathbb{V}}(X)(Hk, Hk').$$

Functoriality of the coend construction gives, naturally in U ,

$$\text{Idl}_{\mathbb{V}}(X)(Hk, Hk') \longrightarrow \underline{\mathbb{V}}(\chi_{Hk}(U), \chi_{Hk'}(U)).$$

Composing these maps gives

$$K(k, k') \longrightarrow \underline{\mathbb{V}}(\overline{H}(U)(k), \overline{H}(U)(k')),$$

so $\overline{H}(U)$ is indeed a $\mathbb{V}$ -functor

$$K \rightarrow \underline{\mathbb{V}}.$$

Now let (U_i) be a finite diagram of upper predicates whose specified limit exists in the finite-limit doctrine. Since each Hk is finite-flat,

$$\chi_{Hk}(\lim_i U_i) \cong \lim_i \chi_{Hk}(U_i).$$

Since specified finite limits in $[K, \mathbb{V}]$ are computed pointwise, this gives

$$\overline{H}(\lim_i U_i) \cong \lim_i \overline{H}(U_i).$$

Thus $\overline{H}$ preserves the specified finite limits. Since W is finite-flat, χ_W preserves them. Hence

$$\chi_J \cong \chi_W \overline{H}$$

preserves the specified finite limits, so J is finite-flat. $\square$

Corollary 3.7.1 (Flattening is finite-flat). *If*

$$\mathcal{I} : \text{Idl}_{\mathbb{V}}(X)^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

is finite-flat, then the flattened weight

$$\mu_X(\mathcal{I})(x) = \int^{I \in \text{Idl}_{\mathbb{V}}(X)} \mathcal{I}(I) \otimes I(x)$$

is finite-flat.

Proof. Apply Proposition 3.7.1 to the identity diagram

$$1_{\text{Idl}_{\mathbb{V}}(X)} : \text{Idl}_{\mathbb{V}}(X) \rightarrow \text{Idl}_{\mathbb{V}}(X).$$

$\square$

Proposition 3.7.2 (The KZ comparison for the ideal monad). *For every small $\mathbb{V}$ -category X , there is an enriched natural transformation*

$$T_{\mathbb{V}}\eta_X \Longrightarrow \eta_{T_{\mathbb{V}}X}.$$

Equivalently, for every finite-flat weight

$$I : X^{\text{op}} \rightarrow \underline{\mathbb{V}},$$

there is a canonical morphism of weights

$$(T_{\mathbb{V}}\eta_X)(I) \longrightarrow \text{Idl}_{\mathbb{V}}(X)(-, I).$$

Together with the monad unit and multiplication, these comparisons make the ideal monad a KZ, or lax-idempotent, monad.

Proof. Let

$$I, J \in \text{Idl}_{\mathbb{V}}(X).$$

The value of $(T_{\mathbb{V}}\eta_X)(I)$ at J is

$$(T_{\mathbb{V}}\eta_X)(I)(J) = \int^{x \in X} I(x) \otimes \text{Idl}_{\mathbb{V}}(X)(J, X(-, x)).$$

By enriched Yoneda,

$$I(x) \cong [X^{\text{op}}, \mathbb{V}](X(-, x), I).$$

Composition in the presheaf category gives morphisms

$$\text{Idl}_{\mathbb{V}}(X)(J, X(-, x)) \otimes [X^{\text{op}}, \mathbb{V}](X(-, x), I) \longrightarrow \text{Idl}_{\mathbb{V}}(X)(J, I).$$

Tensoring with the Yoneda identification and integrating over x gives

$$(T_{\mathbb{V}}\eta_X)(I)(J) \longrightarrow \text{Idl}_{\mathbb{V}}(X)(J, I).$$

This is natural in J , and therefore defines a morphism of weights

$$(T_{\mathbb{V}}\eta_X)(I) \longrightarrow \text{Idl}_{\mathbb{V}}(X)(-, I).$$

It is also enriched-natural in I , since it is induced by composition in $[X^{\text{op}}, \mathbb{V}]$.

We now record the KZ consequences explicitly. The monad laws give

$$\mu_X T_{\mathbb{V}}\eta_X = 1_{T_{\mathbb{V}}X}, \quad \mu_X \eta_{T_{\mathbb{V}}X} = 1_{T_{\mathbb{V}}X}.$$

The comparison just constructed gives

$$T_{\mathbb{V}}\eta_X \Longrightarrow \eta_{T_{\mathbb{V}}X}.$$

Composing with μ_X and using the first monad law gives the identity on $T_{\mathbb{V}}X$. Equivalently, the comparison exhibits

$$\mu_X \dashv \eta_{T_{\mathbb{V}}X}$$

in the enriched order of endomorphisms. The remaining triangle inequality is the statement that every finite-flat ideal of finite-flat ideals is contained in the principal ideal generated by its flattening:

$$1_{T_{\mathbb{V}}T_{\mathbb{V}}X} \Longrightarrow \eta_{T_{\mathbb{V}}X}\mu_X.$$

On a finite-flat weight

$$\mathcal{I} : \text{Idl}_{\mathbb{V}}(X)^{\text{op}} \rightarrow \mathbb{V},$$

this is the morphism

$$\mathcal{I}(J) \longrightarrow \text{Idl}_{\mathbb{V}}(X)(J, \mu_X \mathcal{I})$$

obtained by composing

$$\mathcal{I}(J) \otimes J(x) \longrightarrow \int^I \mathcal{I}(I) \otimes I(x) = \mu_X(\mathcal{I})(x)$$

and transposing in the presheaf hom. Naturality and the triangle identities are the associativity and unitality of composition in $[X^{\text{op}}, \mathbb{V}]$, together with Fubini for the coends defining flattening. Hence the ideal monad is KZ. $\square$

Theorem 3.7.1 (The algebra map is left adjoint to Yoneda). *Let*

$$\alpha : \text{Idl}_{\mathbb{V}}(X) \rightarrow X$$

be a strict algebra structure for the fuzzy ideal monad. Then

$$\alpha \dashv \eta_X$$

with identity counit

$$\alpha\eta_X = 1_X.$$

Consequently, enriched-naturally in $I \in \text{Idl}_{\mathbb{V}}(X)$ and $x \in X$, there is an isomorphism

$$X(\alpha I, x) \cong \text{Idl}_{\mathbb{V}}(X)(I, X(-, x)).$$

Proof. The algebra unit law gives

$$\alpha\eta_X = 1_X,$$

so the counit of the desired adjunction is the identity.

It remains to construct the unit

$$1_{\text{Idl}_{\mathbb{V}}(X)} \longrightarrow \eta_X \alpha.$$

By Proposition 3.7.2, we have

$$T_{\mathbb{V}}\eta_X \Longrightarrow \eta_{T_{\mathbb{V}}X}.$$

Applying $T_{\mathbb{V}}\alpha$ gives

$$T_{\mathbb{V}}\alpha T_{\mathbb{V}}\eta_X \longrightarrow T_{\mathbb{V}}\alpha \eta_{T_{\mathbb{V}}X}.$$

The left-hand side is

$$T_{\mathbb{V}}(\alpha\eta_X) = T_{\mathbb{V}}(1_X) = 1_{\text{Idl}_{\mathbb{V}}(X)}.$$

By naturality of η , the right-hand side is

$$\eta_X \alpha.$$

Thus we obtain

$$1_{\text{Idl}_{\mathbb{V}}(X)} \longrightarrow \eta_X \alpha.$$

The KZ coherence supplies the triangle identities. Therefore

$$\alpha \dashv \eta_X.$$

The displayed hom-isomorphism is the enriched hom-form of this adjunction. $\square$

Theorem 3.7.2 (Fuzzy domains and finite-flat colimits). *Let X be a small $\mathbb{V}$ -category. To give X the structure of a fuzzy domain is equivalently to give a strict coherent choice of all finite-flat weighted colimits in X .*

Under this equivalence, the algebra map

$$\alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X$$

sends a finite-flat weight $I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$ to the I -weighted colimit of the identity diagram:

$$\alpha_X(I) = I * 1_X.$$

More generally, if

$$D : K \rightarrow X$$

is a diagram and

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

is finite-flat, then

$$W * D = \alpha_X(D!W),$$

where

$$(D!W)(x) = \int^{k \in K} W(k) \otimes X(x, Dk).$$

A $\mathbb{V}$ -functor between fuzzy domains is a fuzzy-domain morphism if and only if it preserves finite-flat weighted colimits.

Proof. Suppose first that

$$\alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X$$

is a strict algebra. By Theorem 3.7.1,

$$\alpha_X \dashv \eta_X.$$

Hence, for every finite-flat weight I and every $x \in X$,

$$\begin{aligned} X(\alpha_X I, x) &\cong \text{Idl}_{\mathbb{V}}(X)(I, \eta_X x) \\ &= [X^{\text{op}}, \underline{\mathbb{V}}](I, X(-, x)). \end{aligned}$$

This is the universal property of the I -weighted colimit of the identity diagram. Thus

$$\alpha_X(I) = I * 1_X.$$

Now let $D : K \rightarrow X$, and let $W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}$ be finite-flat. By Lemma 3.7.2, $D_! W$ is finite-flat. For every $x \in X$,

$$\begin{aligned} X(\alpha_X(D_! W), x) &\cong [X^{\text{op}}, \underline{\mathbb{V}}](D_! W, X(-, x)) \\ &\cong [K^{\text{op}}, \underline{\mathbb{V}}](W, X(D-, x)). \end{aligned}$$

The second isomorphism is the weighted left-Kan-extension adjunction defining $D_! W$. This is the universal property of the weighted colimit $W * D$. Therefore

$$W * D = \alpha_X(D_! W)$$

under the strictification convention.

Conversely, suppose X is equipped with a strict coherent choice of all finite-flat weighted colimits. Define

$$\alpha_X(I) = I * 1_X.$$

The representing isomorphisms

$$X(\alpha_X I, x) \cong [X^{\text{op}}, \underline{\mathbb{V}}](I, X(-, x))$$

are enriched-natural in I and x . Hence $I \mapsto \alpha_X I$ extends uniquely to a $\mathbb{V}$ -functor

$$\alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X$$

left adjoint to the Yoneda embedding.

For a representable weight,

$$X(-, x) * 1_X = x,$$

so

$$\alpha_X \eta_X = 1_X.$$

For a finite-flat weight

$$\mathcal{I} : \text{Idl}_{\mathbb{V}}(X)^{\text{op}} \rightarrow \underline{\mathbb{V}},$$

the left-hand side of the multiplication law is

$$\alpha_X \mu_X(\mathcal{I}) = \mu_X(\mathcal{I}) * 1_X.$$

The right-hand side is

$$\alpha_X T_{\mathbb{V}}(\alpha_X)(\mathcal{I}) = (T_{\mathbb{V}} \alpha_X)(\mathcal{I}) * 1_X.$$

The weight $T_{\mathbb{V}}(\alpha_X)(\mathcal{I})$ is the lower direct image of $\mathcal{I}$ along

$$\alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X.$$

Hence its colimit against 1_X is the same weighted colimit as $\mathcal{I}$ against the diagram α_X :

$$(T_{\mathbb{V}}\alpha_X)(\mathcal{I}) * 1_X \cong \mathcal{I} * \alpha_X.$$

On the other hand,

$$\mu_X(\mathcal{I})(x) = \int^I \mathcal{I}(I) \otimes I(x)$$

is the flattened weight computing the same iterated finite-flat colimit. Strict coherence of the chosen finite-flat colimits gives

$$\mu_X(\mathcal{I}) * 1_X = \mathcal{I} * (I \mapsto I * 1_X) = \mathcal{I} * \alpha_X.$$

Therefore

$$\alpha_X \mu_X = \alpha_X T_{\mathbb{V}}(\alpha_X),$$

so α_X is a strict algebra.

Finally, let

$$f : (X, \alpha_X) \rightarrow (Y, \alpha_Y)$$

be a $\mathbb{V}$ -functor. If f is a fuzzy-domain morphism, then for every finite-flat $I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$,

$$f(I * 1_X) = f\alpha_X(I) = \alpha_Y T_{\mathbb{V}}(f)(I) = T_{\mathbb{V}}(f)(I) * 1_Y.$$

Thus f preserves finite-flat colimits of identity diagrams, and hence, by the direct-image formula, all finite-flat weighted colimits.

Conversely, if f preserves finite-flat weighted colimits, then for every finite-flat I ,

$$f\alpha_X(I) = f(I * 1_X) = T_{\mathbb{V}}(f)(I) * 1_Y = \alpha_Y T_{\mathbb{V}}(f)(I).$$

Therefore

$$f\alpha_X = \alpha_Y T_{\mathbb{V}}(f),$$

so f is a fuzzy-domain morphism. □

3.7.2 Free fuzzy domains and limits

Theorem 3.7.3 (Free fuzzy domains). *The forgetful functor*

$$U : \mathbf{FDom}_{\mathbb{V}} \rightarrow \mathbb{V}\text{-}\mathbf{Cat}$$

has a left adjoint

$$F : \mathbb{V}\text{-}\mathbf{Cat} \rightarrow \mathbf{FDom}_{\mathbb{V}}, \quad F(X) = \text{Idl}_{\mathbb{V}}(X),$$

where $F(X)$ has algebra structure

$$\mu_X : \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X)) \rightarrow \text{Idl}_{\mathbb{V}}(X).$$

Thus, for every small $\mathbb{V}$ -category X and every fuzzy domain A , there is a natural bijection

$$\mathbf{FDom}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X), A) \cong \mathbb{V}\text{-}\mathbf{Cat}(X, UA).$$

Proof. This is the free-algebra adjunction of the monad $T_{\mathbb{V}}$. Explicitly, if

$$f : X \rightarrow UA$$

is a $\mathbb{V}$ -functor and

$$\alpha_A : \text{Idl}_{\mathbb{V}}(A) \rightarrow A$$

is the algebra map of A , define

$$\bar{f} : \text{Idl}_{\mathbb{V}}(X) \rightarrow A$$

by

$$\bar{f}(I) = \alpha_A(T_{\mathbb{V}}f(I)).$$

Equivalently,

$$\bar{f}(I) = I * f.$$

By Theorem 3.7.2, $\bar{f}$ preserves finite-flat weighted colimits, so it is a fuzzy-domain morphism. Moreover,

$$\bar{f}(\eta_X x) = \alpha_A(\eta_A(fx)) = fx.$$

Thus

$$\bar{f}\eta_X = f.$$

Conversely, if

$$g : \text{Idl}_{\mathbb{V}}(X) \rightarrow A$$

is a fuzzy-domain morphism, then g preserves finite-flat weighted colimits. For every finite-flat weight $I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$, the object $I \in \text{Idl}_{\mathbb{V}}(X)$ is the I -weighted colimit of the Yoneda diagram:

$$I \cong I * \eta_X.$$

Therefore

$$g(I) \cong g(I * \eta_X) \cong I * (g\eta_X).$$

Hence

$$g = \overline{g\eta_X}$$

under the strictification convention. The two constructions are inverse and natural. $\square$

Corollary 3.7.2 (Monadicity). *The forgetful functor*

$$U : \mathbf{FDom}_{\mathbb{V}} \rightarrow \mathbb{V}\text{-}\mathbf{Cat}$$

is monadic, and the induced monad is the fuzzy ideal monad

$$T_{\mathbb{V}}.$$

Thus

$$\mathbf{FDom}_{\mathbb{V}} = (\mathbb{V}\text{-}\mathbf{Cat})^{T_{\mathbb{V}}}.$$

Proof. By definition, $\mathbf{FDom}_{\mathbb{V}}$ is the Eilenberg–Moore category of strict algebras for $T_{\mathbb{V}}$. The adjunction of Theorem 3.7.3 has underlying monad $T_{\mathbb{V}}$, with Yoneda unit and flattening multiplication. $\square$

Theorem 3.7.4 (Small conical limits). *The category $\mathbf{FDom}_{\mathbb{V}}$ has all small conical limits whose underlying limits exist in*

$$\mathbb{V}\text{-}\mathbf{Cat}.$$

These limits are created by the forgetful functor

$$U : \mathbf{FDom}_{\mathbb{V}} \rightarrow \mathbb{V}\text{-}\mathbf{Cat}.$$

Proof. Eilenberg–Moore categories have all limits that exist in the base category, and the forgetful functor creates them.

Explicitly, let

$$D : J \rightarrow \mathbf{FDom}_{\mathbb{V}}$$

be a small diagram, and suppose that

$$L = \lim_{j \in J} U D_j$$

exists in $\mathbb{V}\text{-}\mathbf{Cat}$, with projections

$$p_j : L \rightarrow U D_j.$$

There is a unique algebra structure

$$\alpha_L : T_{\mathbb{V}} L \rightarrow L$$

such that

$$p_j \alpha_L = \alpha_j T_{\mathbb{V}}(p_j)$$

for every j . With this algebra structure, L satisfies the universal property of the limit in $\mathbf{FDom}_{\mathbb{V}}$. $\square$

3.7.3 Tensor products and internal homs

Definition 3.7.2 (External product of weights). *Let*

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}, \quad Q : J^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be weights. Their external product is the weight

$$W \boxtimes Q : (K \otimes J)^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

defined by

$$(W \boxtimes Q)(k, j) = W(k) \otimes Q(j).$$

Proposition 3.7.3 (External products of finite-flat weights). *If W and Q are finite-flat, then*

$$W \boxtimes Q$$

is finite-flat.

Proof. Let

$$E : K \otimes J \rightarrow \underline{\mathbb{V}}$$

be an upper predicate. Then

$$\begin{aligned} \chi_{W \boxtimes Q}(E) &= \int^{(k, j) \in K \otimes J} W(k) \otimes Q(j) \otimes E(k, j) \\ &\cong \int^{k \in K} W(k) \otimes \left(\int^{j \in J} Q(j) \otimes E(k, j) \right). \end{aligned}$$

For each k , the inner expression is

$$\chi_Q(E(k, -)).$$

Since $E : K \otimes J \rightarrow \underline{\mathbb{V}}$ is a $\mathbb{V}$ -functor, the assignment

$$k \mapsto E(k, -)$$

defines a $\mathbb{V}$ -functor

$$K \rightarrow [J, \underline{\mathbb{V}}].$$

Composing with

$$\chi_Q : [J, \underline{\mathbb{V}}] \rightarrow \underline{\mathbb{V}}$$

gives an upper predicate

$$k \mapsto \chi_Q(E(k, -))$$

on K . Since Q is finite-flat, this construction preserves the specified finite limits in the J -variable. Since W is finite-flat, applying χ_W preserves the specified finite limits in the remaining K -variable. Hence $\chi_{W \boxtimes Q}$ preserves the specified finite limits, so $W \boxtimes Q$ is finite-flat. $\square$

Proposition 3.7.4 (Finite-flat Fubini). *Let*

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}, \quad Q : J^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be finite-flat. For every diagram

$$E : K \otimes J \rightarrow C$$

in a fuzzy domain C , there are canonical isomorphisms

$$W * (k \mapsto Q * (j \mapsto E(k, j))) \cong (W \boxtimes Q) * E \cong Q * (j \mapsto W * (k \mapsto E(k, j))).$$

Proof. These are the Fubini isomorphisms for enriched weighted colimits. By Proposition 3.7.3, the external product $W \boxtimes Q$ is finite-flat. Since C is a fuzzy domain, all three finite-flat weighted colimits appearing above exist. $\square$

Definition 3.7.3 (Internal hom fuzzy domain). *For fuzzy domains B and C , define*

$$[B, C]_{\text{ff}}$$

to be the full $\mathbb{V}$ -subcategory of the enriched functor category $[B, C]$ spanned by the finite-flat-colimit-preserving $\mathbb{V}$ -functors

$$B \rightarrow C.$$

Proposition 3.7.5 (Internal homs are fuzzy domains). *For fuzzy domains B and C , the $\mathbb{V}$ -category*

$$[B, C]_{\text{ff}}$$

is a fuzzy domain. Its finite-flat weighted colimits are computed pointwise in C .

Proof. Let

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be finite-flat, and let

$$H : K \rightarrow [B, C]_{\text{ff}}$$

be a diagram. Since C is a fuzzy domain, the pointwise formula

$$(W * H)(b) = W * (k \mapsto Hk(b))$$

defines the W -weighted colimit of H in the ambient functor category $[B, C]$.

It remains to show that $W * H$ preserves finite-flat weighted colimits in B . Let

$$Q : J^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be finite-flat, and let

$$D : J \rightarrow B$$

be a diagram. Then

$$\begin{aligned} (W * H)(Q * D) &\cong W * (k \mapsto Hk(Q * D)) \\ &\cong W * (k \mapsto Q * (j \mapsto Hk(Dj))) \\ &\cong Q * (j \mapsto W * (k \mapsto Hk(Dj))) \\ &\cong Q * ((W * H)D). \end{aligned}$$

The second isomorphism uses that each $Hk : B \rightarrow C$ preserves finite-flat weighted colimits. The third isomorphism is Proposition 3.7.4. Hence $W * H$ preserves finite-flat weighted colimits in B .

Therefore $[B, C]_{\text{ff}}$ is closed under finite-flat weighted colimits inside $[B, C]$. Since $[B, C]_{\text{ff}}$ is full in $[B, C]$, the ambient weighted-colimit universal property restricts to $[B, C]_{\text{ff}}$. By Theorem 3.7.2, $[B, C]_{\text{ff}}$ is a fuzzy domain. $\square$

Definition 3.7.4 (Finite-flat bimorphism). *Let A, B, C be fuzzy domains. A finite-flat bimorphism*

$$M : A \otimes B \rightarrow C$$

is a $\mathbb{V}$ -functor such that:

1. *for each $a \in A$, the partial functor*

$$M(a, -) : B \rightarrow C$$

preserves finite-flat weighted colimits;

2. *for each $b \in B$, the partial functor*

$$M(-, b) : A \rightarrow C$$

preserves finite-flat weighted colimits.

We write

$$\text{Bimor}_{\text{ff}}(A, B; C)$$

for the set of finite-flat bimorphisms from A, B to C .

Proposition 3.7.6 (Currying finite-flat bimorphisms). *For fuzzy domains A, B, C , there is a natural bijection*

$$\text{Bimor}_{\text{ff}}(A, B; C) \cong \mathbf{FDom}_{\mathbb{V}}(A, [B, C]_{\text{ff}}).$$

Proof. Let

$$M : A \otimes B \rightarrow C$$

be a finite-flat bimorphism. By enriched currying, M corresponds to a $\mathbb{V}$ -functor

$$\widehat{M} : A \rightarrow [B, C], \quad \widehat{M}(a)(b) = M(a, b).$$

Since M preserves finite-flat weighted colimits in the B -variable, each

$$\widehat{M}(a) : B \rightarrow C$$

lies in $[B, C]_{\text{ff}}$. Thus $\widehat{M}$ factors through

$$[B, C]_{\text{ff}} \subseteq [B, C].$$

Since M preserves finite-flat weighted colimits in the A -variable and finite-flat colimits in $[B, C]_{\text{ff}}$ are computed pointwise,

$$\widehat{M} : A \rightarrow [B, C]_{\text{ff}}$$

preserves finite-flat weighted colimits. Hence $\widehat{M}$ is a fuzzy-domain morphism.

Conversely, let

$$H : A \rightarrow [B, C]_{\text{ff}}$$

be a fuzzy-domain morphism. Uncurrying gives

$$\widetilde{H} : A \otimes B \rightarrow C, \quad \widetilde{H}(a, b) = H(a)(b).$$

For each a , the partial functor

$$\widetilde{H}(a, -) = H(a)$$

preserves finite-flat weighted colimits because $H(a)$ is an object of $[B, C]_{\text{ff}}$. For each b , the partial functor

$$\widetilde{H}(-, b) : A \rightarrow C$$

preserves finite-flat weighted colimits because H does and colimits in $[B, C]_{\text{ff}}$ are computed pointwise. Hence $\widetilde{H}$ is a finite-flat bimorphism.

The two constructions are inverse by the enriched tensor-hom adjunction. $\square$

Theorem 3.7.5 (Kelly tensor product for fuzzy domains). *For fuzzy domains A and B , there is a fuzzy domain*

$$A \boxtimes_{\text{ff}} B$$

and a finite-flat bimorphism

$$\kappa_{A,B} : A \otimes B \rightarrow A \boxtimes_{\text{ff}} B$$

such that, for every fuzzy domain C , composition with $\kappa_{A,B}$ induces a natural bijection

$$\mathbf{FDom}_{\mathbb{V}}(A \boxtimes_{\text{ff}} B, C) \cong \mathbf{Bimor}_{\text{ff}}(A, B; C).$$

Proof. By Lemma 3.7.1, Lemma 3.7.2, and Proposition 3.7.1, the assignment

$$X \mapsto \text{Idl}_{\mathbb{V}}(X)$$

contains the representables, is stable under lower direct image along $\mathbb{V}$ -functors, and is closed under finite-flat weighted colimits in presheaf categories. Hence the finite-flat weights form a saturated doctrine.

Moreover, Proposition 3.7.3 and Proposition 3.7.4 show that the finite-flat doctrine is commutative: finite-flat weighted colimits commute with finite-flat weighted colimits.

Kelly's tensor-product theorem for categories equipped with a saturated commutative class of specified weighted colimits therefore applies to the finite-flat doctrine. The representing object for separately finite-flat-colimit-preserving $\mathbb{V}$ -functors

$$A \otimes B \rightarrow C$$

is denoted

$$A \boxtimes_{\mathbb{F}} B.$$

Its universal map

$$\kappa_{A,B} : A \otimes B \rightarrow A \boxtimes_{\mathbb{F}} B$$

is a finite-flat bimorphism, and the displayed bijection is exactly the representing universal property. $\square$

Corollary 3.7.3 (Tensor product of free fuzzy domains). *For small $\mathbb{V}$ -categories X and Y , there is a canonical equivalence*

$$\mathrm{Idl}_{\mathbb{V}}(X) \boxtimes_{\mathbb{F}} \mathrm{Idl}_{\mathbb{V}}(Y) \simeq \mathrm{Idl}_{\mathbb{V}}(X \otimes Y).$$

Proof. For every fuzzy domain C , there are natural bijections

$$\begin{aligned} \mathbf{FDom}_{\mathbb{V}}(\mathrm{Idl}_{\mathbb{V}}(X) \boxtimes_{\mathbb{F}} \mathrm{Idl}_{\mathbb{V}}(Y), C) &\cong \mathrm{Bimor}_{\mathbb{F}}(\mathrm{Idl}_{\mathbb{V}}(X), \mathrm{Idl}_{\mathbb{V}}(Y); C) \\ &\cong \mathbb{V}\text{-}\mathbf{Cat}(X \otimes Y, UC) \\ &\cong \mathbf{FDom}_{\mathbb{V}}(\mathrm{Idl}_{\mathbb{V}}(X \otimes Y), C). \end{aligned}$$

The middle bijection sends a bimorphism to its restriction along

$$\eta_X \otimes \eta_Y : X \otimes Y \rightarrow \mathrm{Idl}_{\mathbb{V}}(X) \otimes \mathrm{Idl}_{\mathbb{V}}(Y).$$

Conversely, a $\mathbb{V}$ -functor

$$m : X \otimes Y \rightarrow UC$$

extends uniquely to the finite-flat bimorphism

$$\tilde{m} : \mathrm{Idl}_{\mathbb{V}}(X) \otimes \mathrm{Idl}_{\mathbb{V}}(Y) \rightarrow C$$

defined by

$$\tilde{m}(I, J) = I * (x \mapsto J * (y \mapsto m(x, y))).$$

By Proposition 3.7.3 and Proposition 3.7.4, this is equivalently

$$\tilde{m}(I, J) \cong (I \boxtimes J) * m.$$

Thus $\tilde{m}$ preserves finite-flat weighted colimits separately in I and J , and it is the unique such extension of m . By representability, the displayed equivalence follows. $\square$

Corollary 3.7.4 (Tensor unit). *Let*

$$\mathbf{1}$$

be the terminal small $\mathbb{V}$ -category. The tensor unit for fuzzy domains is

$$\mathrm{Idl}_{\mathbb{V}}(\mathbf{1}).$$

Proof. For every fuzzy domain C ,

$$\begin{aligned} \mathbf{FDom}_{\mathbb{V}}(\mathrm{Idl}_{\mathbb{V}}(\mathbf{1}) \boxtimes_{\mathbb{F}} A, C) &\cong \mathrm{Bimor}_{\mathbb{F}}(\mathrm{Idl}_{\mathbb{V}}(\mathbf{1}), A; C) \\ &\cong \mathbf{FDom}_{\mathbb{V}}(A, C). \end{aligned}$$

The second bijection uses that $\mathrm{Idl}_{\mathbb{V}}(\mathbf{1})$ is freely generated as a fuzzy domain by the unique object of $\mathbf{1}$. Hence

$$\mathrm{Idl}_{\mathbb{V}}(\mathbf{1}) \boxtimes_{\mathbb{F}} A \simeq A.$$

The right unit law is analogous. □

Theorem 3.7.6 (Symmetric monoidal closure of fuzzy domains). *The category*

$$\mathbf{FDom}_{\mathbb{V}}$$

is symmetric monoidal closed. Its tensor product is

$$A \boxtimes_{\mathbb{F}} B,$$

its tensor unit is

$$\mathrm{Idl}_{\mathbb{V}}(\mathbf{1}),$$

and its internal hom is

$$[B, C]_{\mathbb{F}}.$$

The closed structure is the natural bijection

$$\mathbf{FDom}_{\mathbb{V}}(A \boxtimes_{\mathbb{F}} B, C) \cong \mathbf{FDom}_{\mathbb{V}}(A, [B, C]_{\mathbb{F}}).$$

Proof. By Theorem 3.7.5,

$$\mathbf{FDom}_{\mathbb{V}}(A \boxtimes_{\mathbb{F}} B, C) \cong \mathrm{Bimor}_{\mathbb{F}}(A, B; C).$$

By Proposition 3.7.6,

$$\mathrm{Bimor}_{\mathbb{F}}(A, B; C) \cong \mathbf{FDom}_{\mathbb{V}}(A, [B, C]_{\mathbb{F}}).$$

Composing these bijections gives the closed structure.

The symmetry of $\boxtimes_{\mathbb{F}}$ follows from the symmetry of

$$A \otimes B \cong B \otimes A$$

and from the symmetric nature of the condition of preserving finite-flat weighted colimits separately in the two variables.

Associativity follows by representability. For every fuzzy domain D , both

$$(A \boxtimes_{\mathbb{F}} B) \boxtimes_{\mathbb{F}} C$$

and

$$A \boxtimes_{\text{ff}} (B \boxtimes_{\text{ff}} C)$$

represent trilinear $\mathbb{V}$ -functors

$$A \otimes B \otimes C \rightarrow D$$

that preserve finite-flat weighted colimits separately in each variable. Hence they are canonically equivalent.

The tensor unit is $\text{Idl}_{\mathbb{V}}(\mathbf{1})$ by Corollary 3.7.4. Therefore $\mathbf{FDom}_{\mathbb{V}}$ is symmetric monoidal closed. $\square$

Corollary 3.7.5 (Evaluation). *For fuzzy domains B and C , there is an evaluation morphism*

$$[B, C]_{\text{ff}} \boxtimes_{\text{ff}} B \longrightarrow C.$$

On generators it is given by

$$(f, b) \longmapsto f(b).$$

Proof. The ordinary enriched evaluation

$$[B, C] \otimes B \rightarrow C$$

restricts to a finite-flat bimorphism

$$[B, C]_{\text{ff}} \otimes B \rightarrow C.$$

It is finite-flat-colimit-preserving in the B -variable because every object of $[B, C]_{\text{ff}}$ preserves finite-flat weighted colimits. It is finite-flat-colimit-preserving in the $[B, C]_{\text{ff}}$ -variable because finite-flat colimits there are computed pointwise in C . By the universal property of $\boxtimes_{\text{ff}}$, this bimorphism induces the displayed fuzzy-domain morphism. $\square$

3.7.4 Posetal and crisp specializations

In the posetal specialization, a fuzzy domain over Λ is a fuzzy preorder X equipped with a strict algebra

$$\alpha_X : \text{Idl}_{\Lambda}(X) \rightarrow X.$$

The map α_X sends a fuzzy ideal I to its fuzzy ideal-join.

If

$$D : K \rightarrow X$$

is a Λ -functor and

$$W : K^{\text{op}} \rightarrow \Lambda$$

is a fuzzy ideal of K , that is, a finite-flat lower fuzzy predicate on K , then

$$W * D = \alpha_X(D_{!}W),$$

where

$$(D_{!}W)(x) = \bigvee_{k \in K} W(k) \otimes X(x, Dk).$$

A morphism of fuzzy domains

$$f : X \rightarrow Y$$

is a Λ -functor satisfying

$$f\alpha_X = \alpha_Y T_\Lambda(f).$$

Equivalently, for every fuzzy ideal I of X ,

$$f(\alpha_X(I)) = \alpha_Y(T_\Lambda f(I)).$$

Here the direct image $T_\Lambda f(I)$ is given by

$$(T_\Lambda f)(I)(y) = \bigvee_{x \in X} I(x) \otimes Y(y, fx).$$

For fuzzy domains B and C , the internal hom

$$[B, C]_{\text{ff}}$$

has as objects the fuzzy-domain morphisms

$$B \rightarrow C.$$

Its hom-values are inherited from the enriched functor preorder:

$$[B, C]_{\text{ff}}(f, g) = \bigwedge_{b \in B} C(fb, gb).$$

Finite-flat joins in $[B, C]_{\text{ff}}$ are computed pointwise in C .

For $\Lambda = 2$, finite-flat weights are ordinary ideals. On separated preorders, fuzzy domains are precisely dcpos, and fuzzy-domain morphisms are precisely Scott-continuous maps. Hence

$$\mathbf{FDom}_2^{\text{sep}} \simeq \mathbf{DCPO},$$

where $\mathbf{DCPO}$ denotes the category of dcpos and Scott-continuous maps, small in the chosen universe.

In this crisp separated case, finite-flat bimorphisms are monotone maps

$$A \times B \rightarrow C$$

between dcpos that are Scott-continuous in each variable separately. Such maps are Scott-continuous for the product dcpo. Indeed, let

$$D \subseteq A \times B$$

be nonempty and directed. Then $\pi_A D$ and $\pi_B D$ are nonempty directed subsets, and

$$\bigvee D = \left(\bigvee \pi_A D, \bigvee \pi_B D \right).$$

If

$$f : A \times B \rightarrow C$$

is separately Scott-continuous, then

$$\begin{aligned} f\left(\bigvee D\right) &= f\left(\bigvee \pi_A D, \bigvee \pi_B D\right) \\ &= \bigvee_{a \in \pi_A D} f\left(a, \bigvee \pi_B D\right) \\ &= \bigvee_{a \in \pi_A D} \bigvee_{b \in \pi_B D} f(a, b). \end{aligned}$$

Since D is directed, every pair

$$(a, b) \in \pi_A D \times \pi_B D$$

is bounded above by some element of D . By monotonicity,

$$\bigvee_{a \in \pi_A D, b \in \pi_B D} f(a, b) = \bigvee_{(a, b) \in D} f(a, b).$$

Hence

$$f\left(\bigvee D\right) = \bigvee_{(a, b) \in D} f(a, b).$$

Thus f is Scott-continuous on the product dcpo.

Consequently, in the separated crisp specialization, the Kelly tensor product $\boxtimes_{\mathbb{F}}$ agrees with the cartesian product of dcpos, and the internal hom agrees with the dcpo of Scott-continuous maps ordered pointwise. Hence, under the equivalence

$$\mathbf{FDom}_2^{\text{sep}} \simeq \mathbf{DCPO},$$

the symmetric monoidal closed structure constructed above specializes to the standard cartesian closed structure on the category of dcpos and Scott-continuous maps.

3.8 The Kleisli category and fuzzy approximable maps

3.8.1 The enriched Kleisli category

Definition 3.8.1 (Fuzzy Kleisli morphism). A *fuzzy Kleisli morphism*

$$X \rightsquigarrow Y$$

is a $\mathbb{V}$ -functor

$$X \rightarrow T_{\mathbb{V}} Y = \text{Idl}_{\mathbb{V}}(Y).$$

Equivalently, it is a profunctor-like object

$$R : X \otimes Y^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

whose fibre

$$R(x, -) : Y^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

is a fuzzy ideal of Y , functorially in x .

Proposition 3.8.1 (Kleisli composition). If

$$R : X \rightsquigarrow Y \quad \text{and} \quad S : Y \rightsquigarrow Z$$

are fuzzy Kleisli morphisms, their composite is

$$(S \circ R)(x, z) = \int^{y \in Y} R(x, y) \otimes S(y, z).$$

The identity Kleisli morphism on X is

$$1_X(x, y) = X(y, x).$$

Proof. This is Kleisli composition for the monad $T_{\mathbb{V}}$, written under the finite-flat-weight representation. The identity is the unit $\eta_X(x) = X(-, x)$. $\square$

Proposition 3.8.2 (Extension to free fuzzy domains). *A fuzzy Kleisli morphism*

$$R : X \rightsquigarrow Y$$

extends to a morphism between free fuzzy domains

$$\overline{R} : \text{Idl}_{\mathbb{V}}(X) \rightarrow \text{Idl}_{\mathbb{V}}(Y)$$

given by

$$\overline{R}(I)(y) = \int^{x \in X} I(x) \otimes R(x, y).$$

Proof. This is the Kleisli extension

$$T_{\mathbb{V}}X \xrightarrow{T_{\mathbb{V}}R} T_{\mathbb{V}}T_{\mathbb{V}}Y \xrightarrow{\mu_Y} T_{\mathbb{V}}Y.$$

The displayed formula is the resulting convolution. $\square$

3.8.2 The posetal specialization

Definition 3.8.2 (Fuzzy approximable map). *For fuzzy preorders X, Y , a **fuzzy approximable map***

$$R : X \rightsquigarrow Y$$

is a map

$$R : X \times Y \rightarrow L$$

such that each fibre

$$R(x, -) : Y^{\text{op}} \rightarrow \Lambda$$

is a fuzzy ideal of Y , and the assignment

$$x \mapsto R(x, -)$$

is a Λ -functor

$$X \rightarrow \text{Idl}_{\Lambda}(Y).$$

Proposition 3.8.3 (Posetal Kleisli formulas). *For fuzzy approximable maps*

$$R : X \rightsquigarrow Y, \quad S : Y \rightsquigarrow Z,$$

composition is

$$(S \circ R)(x, z) = \bigvee_{y \in Y} R(x, y) \otimes S(y, z).$$

The identity is

$$1_X(x, y) = X(y, x).$$

The extension of R to free fuzzy domains is

$$\overline{R}(I)(y) = \bigvee_{x \in X} I(x) \otimes R(x, y).$$

Proof. These are the posetal specializations of Propositions 3.8.1 and 3.8.2. $\square$

Example 3.8.1 (The crisp Kleisli category). *For $\Lambda = 2$, a Kleisli morphism*

$$X \rightsquigarrow Y$$

is a monotone map

$$X \rightarrow \text{Idl}(Y).$$

Equivalently, it is an ideal-valued relation

$$R \subseteq X \times Y$$

such that each fibre

$$R[x] = \{y \in Y \mid xRy\}$$

is an ideal of Y , and

$$x \leq x' \implies R[x] \subseteq R[x'].$$

Composition is

$$(S \circ R)[x] = \bigcup_{y \in R[x]} S[y],$$

and the Kleisli extension is

$$\bar{r}(I) = \bigcup_{x \in I} r(x).$$

These are the classical approximable maps between bases.

Example 3.8.2 (Lawvere Kleisli morphisms). *For the Lawvere quantale, Kleisli morphisms are flat-weight-valued metric relations. Composition is infimal convolution:*

$$(S \circ R)(x, z) = \inf_{y \in Y} (R(x, y) + S(y, z)).$$

3.9 Fuzzy algebraic and fuzzy continuous domains

3.9.1 Fuzzy algebraic domains

Definition 3.9.1 (Presented fuzzy algebraic domain). *A **presented fuzzy algebraic domain** is a free fuzzy domain*

$$\text{Idl}_{\mathbb{V}}(B)$$

for a small $\mathbb{V}$ -category B . The objects of B are regarded as basic elements, and the principal ideals

$$\eta_B(b) = B(-, b)$$

are their basic compact approximants.

Proposition 3.9.1 (Algebra structure on a free fuzzy domain). *The algebra structure on*

$$\text{Idl}_{\mathbb{V}}(B)$$

is the monad multiplication

$$\mu_B : \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(B)) \rightarrow \text{Idl}_{\mathbb{V}}(B).$$

In the posetal case,

$$\mu_B(\mathcal{I})(b) = \bigvee_{I \in \text{Idl}_{\Lambda}(B)} \mathcal{I}(I) \otimes I(b).$$

Proof. This is the free-algebra construction for a monad, with multiplication given by flattening. $\square$

Example 3.9.1 (Crisp algebraic domains). *For $\Lambda = 2$,*

$$\text{Idl}(B)$$

is an algebraic dcpo with compact elements the principal ideals

$$\downarrow b.$$

Every ideal is the directed supremum of its principal subideals:

$$I = \bigvee_{b \in I} \downarrow b.$$

Thus free algebras for the crisp ideal monad are algebraic dcpos equipped with a chosen basis of compact generators.

3.9.2 Fuzzy continuous domains

Definition 3.9.2 (Fuzzy continuous domain). *Let X be a fuzzy domain with algebra structure*

$$\alpha : \text{Idl}_{\mathbb{V}}(X) \rightarrow X.$$

*We say that X is a **fuzzy continuous domain** if α admits an enriched left adjoint*

$$\beta : X \rightarrow \text{Idl}_{\mathbb{V}}(X), \quad \beta \dashv \alpha.$$

Thus

$$\text{Idl}_{\mathbb{V}}(X)(\beta x, I) \cong X(x, \alpha I).$$

The object $\beta(x)$ is the fuzzy ideal of approximants to x .

Definition 3.9.3 (Fuzzy way-below degree). *In the posetal specialization, the associated **fuzzy way-below degree** is*

$$\text{wb}_X(y, x) = \beta(x)(y).$$

Proposition 3.9.2 (Adjunction and approximation). *Let X be a separated fuzzy domain over Λ , with algebra map*

$$\alpha : \text{Idl}_{\Lambda}(X) \rightarrow X.$$

If

$$\beta : X \rightarrow \text{Idl}_{\Lambda}(X)$$

satisfies

$$\beta \dashv \alpha,$$

then

$$\alpha\beta = 1_X$$

in the induced crisp order. Hence every element is recovered as the fuzzy join of its approximants.

Proof. The unit of the adjunction gives

$$1_X \leq \alpha\beta.$$

The counit gives

$$\beta\alpha \leq 1_{\text{Idl}_\Lambda(X)}.$$

Apply the counit inequality to the principal ideal $\eta_X(x)$. Since

$$\alpha\eta_X(x) = x,$$

we obtain

$$\beta(x) \leq \eta_X(x).$$

Applying α gives

$$\alpha\beta(x) \leq \alpha\eta_X(x) = x.$$

Together with $x \leq \alpha\beta(x)$, this gives equality in the induced crisp order. Since X is separated, this is equality of objects. $\square$

Example 3.9.2 (Crisp continuous domains). *Let D be a dcpo with ideal-supremum map*

$$\bigvee : \text{Idl}(D) \rightarrow D.$$

Then D is continuous precisely when this map has a left adjoint

$$\beta : D \rightarrow \text{Idl}(D).$$

The left adjoint is

$$\beta(x) = \{y \in D \mid y \ll x\},$$

and the adjunction says

$$\beta(x) \subseteq I \iff x \leq \bigvee I.$$

Example 3.9.3 (Split Kleisli idempotents). *In the crisp case, the map*

$$\beta : D \rightarrow \text{Idl}(D), \quad \beta(x) = \{y \mid y \ll x\},$$

is a Kleisli endomorphism

$$D \rightsquigarrow D.$$

Interpolation of the way-below relation gives

$$\beta \circ_{\text{Kl}} \beta = \beta.$$

Thus continuous dcpos appear as splittings of Kleisli idempotents, equivalently as retracts of algebraic dcpos.

Example 3.9.4 (Lawvere-valued approximation). *For the Lawvere quantale, a fuzzy continuity structure assigns to each point x a finite-flat weight*

$$\beta(x) : X^{\text{op}} \rightarrow [0, \infty].$$

The value $\beta(x)(y)$ is the cost by which y approximates x . The adjunction

$$\beta \dashv \alpha$$

says that approximation into x is controlled exactly by comparison with finite-flat-weight colimits.

3.10 Fuzzy Scott opens and fuzzy Scott locales

3.10.1 The enriched construction

Definition 3.10.1 (Fuzzy Scott open). *Let X be a weak fuzzy domain with structure map*

$$\alpha : T_{\mathbb{V}}X \rightarrow X.$$

*A **fuzzy Scott open** on X is an upper fuzzy predicate*

$$U : X \rightarrow \underline{\mathbb{V}}$$

such that, for every fuzzy ideal I ,

$$U(\alpha(I)) \cong \int^{x \in X} I(x) \otimes U(x).$$

Definition 3.10.2 (Enriched fuzzy Scott frame). *When the required equalizer exists in categorical fuzzy frames, define*

$$\text{Scott}_{\mathbb{V}}(X)$$

to be the equalizer

$$\text{Scott}_{\mathbb{V}}(X) \hookrightarrow P_{\mathbb{V}}(X^{\text{op}}) \begin{array}{c} \xrightarrow{U \mapsto U\alpha} \\ \xrightarrow{U \mapsto (I \mapsto \int^x I(x) \otimes U(x))} \end{array} P_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X)^{\text{op}}).$$

The second map is a fuzzy-frame morphism because, for each finite-flat weight I , the functional

$$U \longmapsto I * U$$

is a character, and functoriality in I gives an upper predicate on $\text{Idl}_{\mathbb{V}}(X)$.

Definition 3.10.3 (Fuzzy Scott locale). *The **fuzzy Scott locale** of X is the fuzzy locale*

$$\text{ScottLoc}_{\mathbb{V}}(X)$$

whose frame of opens is

$$\text{Opens}_{\mathbb{V}}(\text{ScottLoc}_{\mathbb{V}}(X)) = \text{Scott}_{\mathbb{V}}(X).$$

The inclusion

$$\text{Scott}_{\mathbb{V}}(X) \hookrightarrow P_{\mathbb{V}}(X^{\text{op}})$$

defines a fuzzy-locale map

$$\mathcal{P}_{\mathbb{V}}^{\uparrow} X \longrightarrow \text{ScottLoc}_{\mathbb{V}}(X).$$

3.10.2 The posetal specialization

Definition 3.10.4 (Fuzzy Scott open over Λ). *Let X be a weak fuzzy domain over Λ , with structure map*

$$\alpha : \text{Idl}_{\Lambda}(X) \rightarrow X.$$

*A **fuzzy Scott open** is an upper fuzzy predicate*

$$U : X \rightarrow L$$

satisfying

$$U(\alpha(I)) = \bigvee_{x \in X} I(x) \otimes U(x)$$

for every fuzzy ideal I .

Definition 3.10.5 (The fuzzy Scott frame). *The **fuzzy Scott frame** of X is the full fuzzy subpre-order*

$$\text{Scott}_\Lambda(X) \hookrightarrow P_\Lambda(X^{\text{op}})$$

spanned by the fuzzy Scott opens. Equivalently, it is the equalizer

$$\text{Scott}_\Lambda(X) \hookrightarrow P_\Lambda(X^{\text{op}}) \xrightarrow[\Theta]{U \mapsto U\alpha} P_\Lambda(\text{Idl}_\Lambda(X)^{\text{op}}),$$

where

$$\Theta(U)(I) = \bigvee_{x \in X} I(x) \otimes U(x).$$

Proposition 3.10.1 (Fuzzy Scott opens form a fuzzy frame). *Let X be a weak fuzzy domain over Λ . Then*

$$\text{Scott}_\Lambda(X)$$

is a fuzzy frame, and the inclusion

$$\text{Scott}_\Lambda(X) \hookrightarrow P_\Lambda(X^{\text{op}})$$

is a fuzzy-frame homomorphism.

Proof. First, $\Theta(U)$ is an upper fuzzy predicate on $\text{Idl}_\Lambda(X)$. If

$$r \leq \text{Idl}_\Lambda(X)(I, J),$$

then $r \otimes I(x) \leq J(x)$ for every x , and hence

$$r \otimes \Theta(U)(I) = r \otimes \bigvee_x I(x) \otimes U(x) \leq \bigvee_x J(x) \otimes U(x) = \Theta(U)(J).$$

Thus

$$\Theta(U) \in P_\Lambda(\text{Idl}_\Lambda(X)^{\text{op}}).$$

For each fuzzy ideal I , the map

$$U \mapsto \Theta(U)(I)$$

is the character χ_I , so it preserves arbitrary joins, finite meets, and scalars. Since fuzzy-frame structure in

$$P_\Lambda(\text{Idl}_\Lambda(X)^{\text{op}})$$

is computed pointwise in I , the map

$$\Theta : P_\Lambda(X^{\text{op}}) \rightarrow P_\Lambda(\text{Idl}_\Lambda(X)^{\text{op}})$$

preserves arbitrary joins, finite meets, and scalars. The map $U \mapsto U\alpha$ is precomposition by α , so it preserves the same structure. Therefore their equalizer is closed under arbitrary joins, finite meets, and scalars, and the inclusion is a fuzzy-frame homomorphism. $\square$

Definition 3.10.6 (Fuzzy Scott locale over Λ). *For a weak fuzzy domain X , the **fuzzy Scott locale** of X is the fuzzy locale*

$$\text{ScottLoc}_\Lambda(X)$$

whose frame of opens is

$$\mathcal{O}_\Lambda(\text{ScottLoc}_\Lambda(X)) = \text{Scott}_\Lambda(X).$$

The inclusion

$$\text{Scott}_\Lambda(X) \hookrightarrow P_\Lambda(X^{\text{op}})$$

gives a fuzzy-locale map

$$\mathcal{P}_\Lambda^\uparrow X \longrightarrow \text{ScottLoc}_\Lambda(X).$$

Theorem 3.10.1 (Scott opens of a free fuzzy domain). *For every fuzzy preorder B , there is a canonical isomorphism of fuzzy frames*

$$P_\Lambda(B^{\text{op}}) \cong \text{Scott}_\Lambda(\text{Idl}_\Lambda(B)).$$

An upper fuzzy predicate

$$U : B \rightarrow L$$

corresponds to the fuzzy Scott open

$$\widehat{U} : \text{Idl}_\Lambda(B) \rightarrow L$$

defined by

$$\widehat{U}(I) = \bigvee_{b \in B} I(b) \otimes U(b).$$

Proof. Let $U \in P_\Lambda(B^{\text{op}})$. Define

$$\widehat{U}(I) = \bigvee_b I(b) \otimes U(b).$$

The calculation in Proposition 3.10.1 shows that $\widehat{U}$ is an upper fuzzy predicate on $\text{Idl}_\Lambda(B)$.

For

$$\mathcal{I} \in \text{Idl}_\Lambda(\text{Idl}_\Lambda(B)),$$

we compute

$$\begin{aligned} \widehat{U}(\mu_B(\mathcal{I})) &= \bigvee_b \mu_B(\mathcal{I})(b) \otimes U(b) \\ &= \bigvee_b \left(\bigvee_I \mathcal{I}(I) \otimes I(b) \right) \otimes U(b) \\ &= \bigvee_I \mathcal{I}(I) \otimes \left(\bigvee_b I(b) \otimes U(b) \right) \\ &= \bigvee_I \mathcal{I}(I) \otimes \widehat{U}(I). \end{aligned}$$

Thus $\widehat{U}$ is Scott open.

Conversely, let

$$W \in \text{Scott}_\Lambda(\text{Idl}_\Lambda(B)).$$

Restrict along the unit and define

$$U_W(b) = W(\eta_B b).$$

Then U_W is an upper fuzzy predicate on B . For U , Yoneda gives

$$\widehat{U}(\eta_B b) = \bigvee_c B(c, b) \otimes U(c) = U(b),$$

so $U_{\widehat{U}} = U$.

It remains to show that $\widehat{U_W} = W$. For $I \in \text{Idl}_\Lambda(B)$, the upper-predicate law for W gives

$$\text{Idl}_\Lambda(B)(\eta_B b, I) \otimes W(\eta_B b) \leq W(I).$$

By Yoneda,

$$\text{Idl}_\Lambda(B)(\eta_B b, I) = I(b).$$

Hence

$$\widehat{U_W}(I) = \bigvee_b I(b) \otimes W(\eta_B b) \leq W(I).$$

For the converse, use

$$\mu_B T_\Lambda(\eta_B) = 1_{\text{Idl}_\Lambda(B)}.$$

Since W is Scott open,

$$\begin{aligned} W(I) &= W(\mu_B T_\Lambda(\eta_B)(I)) \\ &= \bigvee_J T_\Lambda(\eta_B)(I)(J) \otimes W(J). \end{aligned}$$

Moreover,

$$T_\Lambda(\eta_B)(I)(J) = \bigvee_b I(b) \otimes \text{Idl}_\Lambda(B)(J, \eta_B b).$$

Using the upper-predicate law for W ,

$$\text{Idl}_\Lambda(B)(J, \eta_B b) \otimes W(J) \leq W(\eta_B b).$$

Therefore

$$\begin{aligned} W(I) &\leq \bigvee_J \bigvee_b I(b) \otimes \text{Idl}_\Lambda(B)(J, \eta_B b) \otimes W(J) \\ &\leq \bigvee_b I(b) \otimes W(\eta_B b) \\ &= \widehat{U_W}(I). \end{aligned}$$

Thus $W = \widehat{U_W}$. The two constructions are inverse and preserve the fuzzy-frame structure because all operations are computed pointwise and each evaluation functional is a character. $\square$

Example 3.10.1 (The crisp Scott condition). For $\Lambda = 2$, if X is a dcpo with algebra

$$\alpha : \text{Idl}(X) \rightarrow X, \quad I \mapsto \bigvee I,$$

then a fuzzy Scott open is an upper set $U \subseteq X$ satisfying

$$\bigvee I \in U \iff I \cap U \neq \emptyset$$

for every ideal I . This is the ordinary Scott-open condition expressed using ideals.

Example 3.10.2 (Upper sets as Scott opens of ideal completions). *For a preorder B ,*

$$\text{Scott}(\text{Idl}(B)) \cong \text{Up}(B).$$

The correspondence sends an upper set $U \subseteq B$ to

$$\widehat{U} = \{I \in \text{Idl}(B) \mid I \cap U \neq \emptyset\}.$$

Thus the upper Alexandrov frame of B is the Scott frame of the free dcpo generated by B .

Example 3.10.3 (Standard fuzzy Scott formulas). *For Gödel truth values, the fuzzy Scott condition is*

$$U(\alpha(I)) = \bigvee_x \min(I(x), U(x)).$$

For Łukasiewicz truth values, it is

$$U(\alpha(I)) = \bigvee_x \max(0, I(x) + U(x) - 1).$$

For product truth values, it is

$$U(\alpha(I)) = \bigvee_x I(x)U(x).$$

For the Lawvere quantale, it becomes

$$U(\alpha(I)) = \inf_x (I(x) + U(x)).$$

3.11 The Alexandrov character comonad

3.11.1 The enriched comonad

Definition 3.11.1 (Alexandrov character comonad). *The **Alexandrov character comonad** is the comonad*

$$G_{\mathbb{V}} = \mathcal{P}_{\mathbb{V}}^{\uparrow} \text{Char}_{\mathbb{V}}$$

on fuzzy locales induced by

$$\mathcal{P}_{\mathbb{V}}^{\uparrow} \dashv \text{Char}_{\mathbb{V}}.$$

For a fuzzy locale $\mathcal{X}$,

$$G_{\mathbb{V}}\mathcal{X} = \mathcal{P}_{\mathbb{V}}^{\uparrow}(\text{Char}_{\mathbb{V}}(\mathcal{X})),$$

so

$$\text{Opens}_{\mathbb{V}}(G_{\mathbb{V}}\mathcal{X}) = P_{\mathbb{V}}(\text{Char}_{\mathbb{V}}(\mathcal{X})^{\text{op}}).$$

Proposition 3.11.1 (Counit of the character comonad). *The counit*

$$G_{\mathbb{V}}\mathcal{X} \rightarrow \mathcal{X}$$

is frame-dual to the evaluation fuzzy-frame morphism

$$\text{Opens}_{\mathbb{V}}(\mathcal{X}) \longrightarrow P_{\mathbb{V}}(\text{Char}_{\mathbb{V}}(\mathcal{X})^{\text{op}}),$$

given by

$$a \longmapsto \widehat{a}, \quad \widehat{a}(\chi) = \chi(a).$$

Proof. This is the counit of the adjunction, evaluated at $\mathcal{X}$. On frames it is the transpose of the identity functor on $\text{Char}_\vee(\mathcal{X})$, namely evaluation. $\square$

Definition 3.11.2 (Coalgebra for the Alexandrov character comonad). *A $G_\vee$ -coalgebra is a fuzzy locale $\mathcal{X}$ equipped with a fuzzy-locale map*

$$\mathcal{X} \rightarrow G_\vee \mathcal{X}$$

satisfying the usual comonad coalgebra laws. Frame-dually, this says that

$$\text{Opens}_\vee(\mathcal{X})$$

is a comonad-coherent retract of the upper fuzzy predicate frame on its character object.

3.11.2 The posetal specialization

Proposition 3.11.2 (The posetal character comonad). *For a fuzzy locale $\mathcal{X}$,*

$$\mathcal{O}_\Lambda(G_\Lambda \mathcal{X}) = P_\Lambda(\text{Char}_\Lambda(\mathcal{O}_\Lambda(\mathcal{X}))^{\text{op}}).$$

The counit is frame-dual to the evaluation map

$$a \mapsto \hat{a}, \quad \hat{a}(\chi) = \chi(a).$$

A G_Λ -coalgebra is frame-dually a fuzzy-frame homomorphism

$$P_\Lambda(\text{Char}_\Lambda(\mathcal{O}_\Lambda(\mathcal{X}))^{\text{op}}) \rightarrow \mathcal{O}_\Lambda(\mathcal{X})$$

compatible with the counit and comultiplication.

Proof. This is the posetal specialization of the enriched comonad and its counit. $\square$

Example 3.11.1 (The crisp case). *For $\Lambda = 2$, the character object of a locale is its preorder of points, ordered by specialization according to the convention*

$$p \leq q \iff \forall a, p(a) \leq q(a).$$

Thus

$$G_2 \mathcal{X}$$

is the upper Alexandrov locale of the specialization preorder of points of $\mathcal{X}$. The counit is frame-dual to

$$\mathcal{O}(\mathcal{X}) \rightarrow \text{Up}(\text{Pt}(\mathcal{X})), \quad a \mapsto \{p \mid p(a) = \top\}.$$

Example 3.11.2 (Presheaf fuzzy locales). *If*

$$\mathcal{X} = \mathcal{P}_\Lambda^\dagger X,$$

then

$$\text{Char}_\Lambda(\mathcal{O}_\Lambda(\mathcal{X})) = \text{Char}_\Lambda(P_\Lambda(X^{\text{op}})) \cong \text{Idl}_\Lambda(X).$$

Therefore

$$G_\Lambda(\mathcal{P}_\Lambda^\dagger X)$$

has opens given by upper fuzzy predicates on $\text{Idl}_\Lambda(X)$.

3.12 The co-Kleisli category

3.12.1 The enriched co-Kleisli category

Theorem 3.12.1 (Co-Kleisli morphisms). *For fuzzy locales $\mathcal{X}, \mathcal{Y}$,*

$$\text{CoKl}(G_{\mathbb{V}})(\mathcal{X}, \mathcal{Y}) = \mathbf{FLoc}_{\mathbb{V}}(G_{\mathbb{V}}\mathcal{X}, \mathcal{Y})$$

is naturally equivalent to

$$\mathbb{V}\text{-Cat}(\text{Char}_{\mathbb{V}}(\mathcal{X}), \text{Char}_{\mathbb{V}}(\mathcal{Y})).$$

Thus a co-Kleisli morphism

$$\mathcal{X} \rightarrow \mathcal{Y}$$

is equivalently a $\mathbb{V}$ -functor between the character objects.

Proof. By definition,

$$G_{\mathbb{V}}\mathcal{X} = \mathcal{P}_{\mathbb{V}}^{\uparrow} \text{Char}_{\mathbb{V}}(\mathcal{X}).$$

The upper fuzzy character adjunction gives

$$\begin{aligned} \mathbf{FLoc}_{\mathbb{V}}(G_{\mathbb{V}}\mathcal{X}, \mathcal{Y}) &= \mathbf{FLoc}_{\mathbb{V}}(\mathcal{P}_{\mathbb{V}}^{\uparrow} \text{Char}_{\mathbb{V}}(\mathcal{X}), \mathcal{Y}) \\ &\simeq \mathbb{V}\text{-Cat}(\text{Char}_{\mathbb{V}}(\mathcal{X}), \text{Char}_{\mathbb{V}}(\mathcal{Y})). \end{aligned}$$

□

3.12.2 The posetal specialization

Corollary 3.12.1 (Posetal co-Kleisli morphisms). *In the posetal specialization, a co-Kleisli morphism*

$$\mathcal{X} \rightarrow \mathcal{Y}$$

is a Λ -functor

$$\text{Char}_{\Lambda}(\mathcal{O}_{\Lambda}(\mathcal{X})) \rightarrow \text{Char}_{\Lambda}(\mathcal{O}_{\Lambda}(\mathcal{Y})).$$

Such a map need not arise from a fuzzy-locale map

$$\mathcal{X} \rightarrow \mathcal{Y}.$$

Proof. This is the posetal specialization of Theorem 3.12.1. □

Example 3.12.1 (The crisp case). *For $\Lambda = 2$, co-Kleisli morphisms are monotone maps between the specialization preorders of points:*

$$\text{Pt}(\mathcal{X}) \rightarrow \text{Pt}(\mathcal{Y}).$$

These need not be continuous maps of locales or spaces.

Example 3.12.2 (Presheaf fuzzy locales). *Between upper presheaf fuzzy locales, co-Kleisli morphisms are fuzzy functors between fuzzy ideal completions:*

$$\text{Idl}_{\Lambda}(X) \rightarrow \text{Idl}_{\Lambda}(Y).$$

Example 3.12.3 (Lawvere co-Kleisli maps). *For the Lawvere quantale, co-Kleisli morphisms are nonexpansive maps between metric spaces of finite-flat weights.*

3.13 Summary

The upper fuzzy predicate frame of X is

$$P_{\mathbb{V}}(X^{\text{op}}) = [X, \mathbb{V}],$$

and in the posetal specialization it is

$$P_{\Lambda}(X^{\text{op}}) = [X, \Lambda].$$

The character adjunction

$$\mathcal{P}_{\mathbb{V}}^{\uparrow} \dashv \text{Char}_{\mathbb{V}}$$

induces the fuzzy ideal monad

$$T_{\Lambda}(X) = \text{Idl}_{\Lambda}(X),$$

whose unit is the principal-ideal embedding and whose multiplication is fuzzy flattening.

Weak fuzzy domains carry chosen candidate fuzzy ideal-joins, while fuzzy domains are strict algebras for this monad. Free fuzzy domains

$$\text{Idl}_{\Lambda}(B)$$

are the presented fuzzy algebraic domains. Fuzzy continuous domains are those for which

$$\alpha : \text{Idl}_{\Lambda}(X) \rightarrow X$$

has an enriched left adjoint

$$\beta : X \rightarrow \text{Idl}_{\Lambda}(X),$$

with fuzzy way-below degree

$$\text{wb}_X(y, x) = \beta(x)(y).$$

Kleisli morphisms for the fuzzy ideal monad are fuzzy ideal-valued maps

$$X \rightsquigarrow Y,$$

or fuzzy approximable maps. Fuzzy Scott opens of a weak fuzzy domain

$$\alpha : \text{Idl}_{\Lambda}(X) \rightarrow X$$

are exactly the upper fuzzy predicates U satisfying

$$U(\alpha(I)) = \bigvee_x I(x) \otimes U(x).$$

For $\Lambda = 2$, these constructions recover ordinary ideals, ideal completion, dcpos, algebraic dcpos, continuous dcpos, approximable maps, and ordinary Scott opens.

Chapter 4

The Fuzzy Distributor Equipment and the Fuzzy Way-Below Relation

In the previous chapter we constructed the fuzzy ideal monad

$$T_{\mathbb{V}}X = \text{Idl}_{\mathbb{V}}(X)$$

from the upper fuzzy character adjunction. Its strict algebras were called fuzzy domains. Continuous fuzzy domains were defined by the existence of an enriched left adjoint

$$\beta_X : X \rightarrow \text{Idl}_{\mathbb{V}}(X)$$

to the ideal-algebra map

$$\alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X.$$

Thus

$$\beta_X \dashv \alpha_X.$$

The basic algebra and continuity data are summarized by the diagrams

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & \text{Idl}_{\mathbb{V}}(X) \\ & \searrow 1_X & \downarrow \alpha_X \\ & & X \end{array} \quad \begin{array}{ccc} & \xrightarrow{\beta_X} & \\ X & & \text{Idl}_{\mathbb{V}}(X) \\ & \xleftarrow{\alpha_X} & \end{array}$$

where the first triangle expresses the strict unit law

$$\alpha_X \eta_X = 1_X.$$

In the posetal specialization this gave the fuzzy way-below degree

$$\text{wb}_X(a, b) = \beta_X(b)(a).$$

The purpose of this chapter is to explain the equipment-theoretic origin of this formula. The way-below relation is not added as further structure. It is constructed from the ideal-algebra map and the principal-ideal embedding by a right lifting in the fuzzy distributor equipment.

For a fuzzy domain X , the available maps are

$$\eta_X : X \rightarrow \text{Idl}_{\mathbb{V}}(X), \quad \eta_X(x) = X(-, x),$$

and

$$\alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X.$$

The conjoint of α_X and the companion of η_X are distributors with the same source and the same target,

$$X \nrightarrow \text{Idl}_{\mathbb{V}}(X).$$

They fit into the following equipment diagram:

$$\begin{array}{ccc} & & \text{Idl}_{\mathbb{V}}(X) \\ & \nearrow \alpha_X^* & \\ X & \xrightarrow{(\eta_X)_*} & \end{array}$$

where

$$\alpha_X^*(b, I) = X(b, \alpha_X I), \quad (\eta_X)_*(a, I) = \text{Idl}_{\mathbb{V}}(X)(\eta_X a, I).$$

The fuzzy way-below distributor is defined by the right lifting

$$\ll_X := [\alpha_X^*, (\eta_X)_*] : X \nrightarrow X.$$

Equivalently, in the enriched distributor bicategory it represents the object of distributors $\Theta : X \nrightarrow X$ equipped with a comparison

$$\alpha_X^* \circ \Theta \rightarrow (\eta_X)_*.$$

In the locally posetal reflection, it is the largest distributor Θ for which

$$\alpha_X^* \circ \Theta \leq (\eta_X)_*.$$

Diagrammatically, a comparison

$$\alpha_X^* \circ \Theta \rightarrow (\eta_X)_*$$

has the form

$$\begin{array}{ccccc} X & \xrightarrow{\Theta} & X & \xrightarrow{\alpha_X^*} & \text{Idl}_{\mathbb{V}}(X) \\ & & & \searrow & \\ & & & & \xrightarrow{(\eta_X)_*} \end{array}$$

and the universal property of $\ll_X$ says that such comparisons are represented by comparisons

$$\Theta \rightarrow \ll_X.$$

Unwinding the right lifting gives

$$\ll_X(a, b) \cong \int_{I \in \text{Idl}_{\mathbb{V}}(X)} [X(b, \alpha_X I), \text{Idl}_{\mathbb{V}}(X)(\eta_X a, I)].$$

By enriched Yoneda,

$$\text{Idl}_{\mathbb{V}}(X)(\eta_X a, I) \cong I(a),$$

and hence

$$\ll_X (a, b) \cong \int_{I \in \text{Idl}_V(X)} [X(b, \alpha_X I), I(a)].$$

In the posetal reflection this becomes

$$\ll_X (a, b) = \bigwedge_{I \in \text{Idl}_\Lambda(X)} (X(b, \alpha_X I) \multimap I(a)).$$

If X is continuous, with

$$\beta_X \dashv \alpha_X,$$

then this right-lifted distributor is represented by the approximant map:

$$\ll_X (a, b) \cong \text{Idl}_V(X)(\eta_X a, \beta_X b) \cong \beta_X(b)(a).$$

Thus the way-below degree of the previous chapter is the value of a right-lifted fuzzy distributor.

Throughout this chapter

$$(\mathbb{V}, \otimes, I, [-, -])$$

is the fuzzy cosmos fixed in the previous chapters. We write

$$\underline{\mathbb{V}}$$

for its self-enriched category, and we use the same universe conventions as before: all ends and coends are formed in a universe in which their indexing fuzzy category is small and the corresponding limits or colimits in $\mathbb{V}$ exist.

All distributor categories below are understood in this universe-bounded sense. Thus, when $\text{Idl}_V(X)$ is not small in the original universe, we regard it as small in the next universe and form the corresponding distributor homs, ends, and coends there.

We write

$$\Phi : X \rightrightarrows Y$$

for a fuzzy distributor from X to Y . The arrow $\rightrightarrows$ denotes a proarrow, not a negated function arrow.

4.1 Inherited data from the fuzzy ideal monad

This section records the data from the previous chapter that will be used below. No new structure is imposed on fuzzy domains in this chapter.

4.1.1 The enriched construction

For a small fuzzy category X , the fuzzy ideal object is the full $\mathbb{V}$ -subcategory

$$\text{Idl}_V(X) \subseteq [X^{\text{op}}, \mathbb{V}]$$

spanned by finite-flat weights. The unit of the fuzzy ideal monad is the principal-weight embedding

$$\eta_X : X \rightarrow \text{Idl}_V(X), \quad \eta_X(x) = X(-, x).$$

This is the restricted Yoneda embedding:

$$\begin{array}{ccc} X & \xrightarrow{y_X} & [X^{\text{op}}, \mathbb{V}] \\ & \searrow \eta_X \quad \swarrow i_X & \\ & \text{Idl}_{\mathbb{V}}(X) & \end{array}$$

The multiplication is flattening:

$$\mu_X : \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X)) \rightarrow \text{Idl}_{\mathbb{V}}(X),$$

with value

$$\mu_X(\mathcal{I})(x) = \int^{I \in \text{Idl}_{\mathbb{V}}(X)} \mathcal{I}(I) \otimes I(x).$$

The two unit laws are

$$\mu_X \eta_{\text{Idl}_{\mathbb{V}}(X)} = 1_{\text{Idl}_{\mathbb{V}}(X)}, \quad \mu_X \text{Idl}_{\mathbb{V}}(\eta_X) = 1_{\text{Idl}_{\mathbb{V}}(X)}.$$

The first of these is displayed by

$$\begin{array}{ccc} \text{Idl}_{\mathbb{V}}(X) & \xrightarrow{\eta_{\text{Idl}_{\mathbb{V}}(X)}} & \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X)) \\ & \searrow 1_{\text{Idl}_{\mathbb{V}}(X)} \quad \downarrow \mu_X & \\ & \text{Idl}_{\mathbb{V}}(X). & \end{array}$$

The second is displayed by

$$\begin{array}{ccc} \text{Idl}_{\mathbb{V}}(X) & \xrightarrow{\text{Idl}_{\mathbb{V}}(\eta_X)} & \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X)) \\ & \searrow 1_{\text{Idl}_{\mathbb{V}}(X)} \quad \downarrow \mu_X & \\ & \text{Idl}_{\mathbb{V}}(X). & \end{array}$$

The associativity law is

$$\mu_X \text{Idl}_{\mathbb{V}}(\mu_X) = \mu_X \mu_{\text{Idl}_{\mathbb{V}}(X)},$$

displayed by

$$\begin{array}{ccc} \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X))) & \xrightarrow{\text{Idl}_{\mathbb{V}}(\mu_X)} & \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X)) \\ \mu_{\text{Idl}_{\mathbb{V}}(X)} \downarrow & & \downarrow \mu_X \\ \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X)) & \xrightarrow{\mu_X} & \text{Idl}_{\mathbb{V}}(X). \end{array}$$

A fuzzy domain is a strict algebra for this monad. Thus it is a small fuzzy category X equipped with a fuzzy functor

$$\alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X$$

such that

$$\alpha_X \eta_X = 1_X$$

and

$$\alpha_X \mu_X = \alpha_X \text{Idl}_{\mathbb{V}}(\alpha_X).$$

The algebra laws are encoded by the diagrams

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & \text{Idl}_{\mathbb{V}}(X) \\ & \searrow 1_X & \downarrow \alpha_X \\ & & X \end{array}$$

and

$$\begin{array}{ccc} \text{Idl}_{\mathbb{V}}(\text{Idl}_{\mathbb{V}}(X)) & \xrightarrow{\mu_X} & \text{Idl}_{\mathbb{V}}(X) \\ \text{Idl}_{\mathbb{V}}(\alpha_X) \downarrow & & \downarrow \alpha_X \\ \text{Idl}_{\mathbb{V}}(X) & \xrightarrow{\alpha_X} & X. \end{array}$$

The map α_X evaluates a finite-flat weight by its chosen ideal colimit.

A fuzzy domain X is continuous when

$$\alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X$$

has an enriched left adjoint

$$\beta_X : X \rightarrow \text{Idl}_{\mathbb{V}}(X), \quad \beta_X \dashv \alpha_X.$$

The object $\beta_X(b)$ is the finite-flat weight of approximants of b . The adjunction supplies unit and counit maps

$$1_X \rightarrow \alpha_X \beta_X, \quad \beta_X \alpha_X \rightarrow 1_{\text{Idl}_{\mathbb{V}}(X)}.$$

These may be represented by

$$\begin{array}{ccc} X & \xrightarrow{\beta_X} & \text{Idl}_{\mathbb{V}}(X) \\ & \searrow \eta^{\beta \dashv \alpha} & \downarrow \alpha_X \\ & & X \end{array} \quad \begin{array}{ccc} \text{Idl}_{\mathbb{V}}(X) & \xrightarrow{\alpha_X} & X \\ & \searrow \varepsilon^{\beta \dashv \alpha} & \downarrow \beta_X \\ & & \text{Idl}_{\mathbb{V}}(X) \end{array}$$

4.1.2 The posetal reflection

Let

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp_L, \top_L)$$

be a fuzzy truth-value object. For a fuzzy preorder X , the object

$$\text{Idl}_{\Lambda}(X)$$

is a full subpreorder of

$$[X^{\text{op}}, \Lambda].$$

Its elements are the finite-flat fuzzy ideals

$$I : X^{\text{op}} \rightarrow \Lambda.$$

The principal ideal at x is

$$\eta_X(x)(y) = X(y, x).$$

A fuzzy domain over Λ has an algebra map

$$\alpha_X : \text{Idl}_\Lambda(X) \rightarrow X$$

satisfying the strict algebra laws

$$\alpha_X \eta_X = 1_X$$

and

$$\alpha_X \mu_X = \alpha_X \text{Idl}_\Lambda(\alpha_X).$$

These are strict algebra laws for the transported ideal monad. In the posetal specialization they are literal equalities of Λ -functors, equivalently objectwise equalities compatible with the fuzzy preorder structure.

If X is continuous, then there is a Λ -adjunction

$$\beta_X \dashv \alpha_X,$$

and the previous chapter defined the fuzzy way-below degree by

$$\text{wb}_X(a, b) = \beta_X(b)(a).$$

The goal of the present chapter is to reconstruct this value as a right lifting.

4.1.3 Examples

Example 4.1.1 (The crisp case). *For*

$$\Lambda = 2,$$

fuzzy preorders are ordinary preorders and fuzzy ideals are ordinary ideals, namely directed lower sets. The principal ideal embedding is

$$\eta_X(x) = \downarrow x,$$

and a strict algebra

$$\alpha_X : \text{Idl}(X) \rightarrow X$$

assigns to each ideal its supremum:

$$\alpha_X(I) = \bigvee I.$$

Thus separated strict algebras recover dcpos. If X is continuous, then

$$\beta_X(b) = \{a \in X \mid a \ll b\}.$$

The crisp data fit the diagram

$$\begin{array}{ccc} X & \xrightarrow{x \mapsto \downarrow x} & \text{Idl}(X) \\ & \searrow 1_X & \downarrow I \mapsto \bigvee I \\ & & X. \end{array}$$

Example 4.1.2 (Gödel truth values). *For Gödel truth values,*

$$r \otimes s = r \wedge s.$$

A fuzzy ideal is a finite-flat lower predicate

$$I : X^{\text{op}} \rightarrow [0, 1],$$

and $\alpha_X(I)$ is the chosen Gödel-valued ideal colimit of I .

Example 4.1.3 (Łukasiewicz truth values). *For Łukasiewicz truth values,*

$$r \otimes s = \max(0, r + s - 1), \quad r \multimap s = \min(1, 1 - r + s).$$

Finite-flat fuzzy ideals are lower predicates whose flatness is computed using this bounded-additive tensor.

Example 4.1.4 (Product truth values). *For the product tensor,*

$$r \otimes s = rs,$$

and

$$r \multimap s = \begin{cases} 1, & r \leq s, \\ s/r, & r > s. \end{cases}$$

The ideal monad is the finite-flat completion for multiplicative fuzzy weights.

Example 4.1.5 (Lawvere truth values). *For the Lawvere truth-value object, the order is*

$$a \leq_L b \iff a \geq b$$

in the usual numerical order, the tensor is addition, the top element is 0, and the bottom element is ∞ . The residual is

$$a \multimap c = c \ominus a.$$

Fuzzy ideals are finite-flat cost weights. The algebra map

$$\alpha_X : \text{Idl}_\Lambda(X) \rightarrow X$$

evaluates such weights by their Lawvere-enriched weighted colimits.

4.2 Fuzzy distributors

We now construct the horizontal arrows of the fuzzy distributor equipment.

4.2.1 The enriched construction

Definition 4.2.1 (Fuzzy distributor). *Let X and Y be small fuzzy categories. A **fuzzy distributor***

$$\Phi : X \multimap Y$$

is a fuzzy functor

$$\Phi : X^{\text{op}} \otimes Y \rightarrow \underline{\mathbb{V}}.$$

We write its value at (x, y) as

$$\Phi(x, y).$$

Thus a fuzzy distributor is contravariant in its first variable and covariant in its second variable. This variance is indicated by the diagram

$$X^{\text{op}} \otimes Y \xrightarrow{\Phi} \underline{\mathbb{V}}.$$

Definition 4.2.2 (Composition of fuzzy distributors). *Let*

$$\Phi : X \multimap Y, \quad \Psi : Y \multimap Z$$

be fuzzy distributors. Their composite

$$\Psi \circ \Phi : X \multimap Z$$

is defined by

$$(\Psi \circ \Phi)(x, z) = \int^{y \in Y} \Phi(x, y) \otimes \Psi(y, z).$$

The identity distributor on X is

$$1_X : X \multimap X, \quad 1_X(x, x') = X(x, x').$$

The composite is represented by the coend-convolution diagram

$$\begin{array}{ccccc} X & \xrightarrow{\quad \Phi \quad} & Y & \xrightarrow{\quad \Psi \quad} & Z. \\ & \searrow & & \nearrow & \\ & \Psi \circ \Phi & & & \end{array}$$

The identity distributor is the horizontal shadow of the hom-object:

$$X \xrightarrow{1_X} X.$$

Proposition 4.2.1 (The fuzzy distributor bicategory). *Small fuzzy categories in a fixed universe, fuzzy distributors, and enriched distributor transformations form a bicategory*

$$\text{Dist}_{\mathbb{V}}.$$

For small fuzzy categories X and Y , the hom $\mathbb{V}$ -category of distributors from X to Y is

$$\text{Dist}_{\mathbb{V}}(X, Y) = [X^{\text{op}} \otimes Y, \underline{\mathbb{V}}].$$

Composition is the coend convolution of Definition 4.2.2, and identities are the hom-distributors.

Proof. The coend formula defines a fuzzy functor

$$X^{\text{op}} \otimes Z \rightarrow \underline{\mathbb{V}}$$

by the functoriality of enriched coends and the functoriality of $\otimes$. Associativity follows from Fubini for enriched coends together with associativity of the tensor product. In diagrammatic form, both routes below evaluate to the same iterated coend:

$$\begin{array}{ccccccc} & & \Xi \circ (\Psi \circ \Phi) & & & & \\ & \searrow & \text{---} & \nearrow & & & \\ X & \xrightarrow{\quad \Phi \quad} & Y & \xrightarrow{\quad \Psi \quad} & Z & \xrightarrow{\quad \Xi \quad} & W. \\ & \searrow & & \nearrow & \searrow & \nearrow & \\ & & \Xi \circ \Psi & & & & \\ & \searrow & \text{---} & \nearrow & & & \\ & & (\Xi \circ \Psi) \circ \Phi & & & & \end{array}$$

More explicitly,

$$\begin{aligned}\Xi \circ (\Psi \circ \Phi) &\cong \int^z \left(\int^y \Phi(-, y) \otimes \Psi(y, z) \right) \otimes \Xi(z, -) \\ &\cong \int^{y, z} \Phi(-, y) \otimes \Psi(y, z) \otimes \Xi(z, -) \\ &\cong (\Xi \circ \Psi) \circ \Phi.\end{aligned}$$

The unit laws are the enriched co-Yoneda formulas

$$\int^{y \in Y} Y(y, y') \otimes \Phi(x, y) \cong \Phi(x, y')$$

and

$$\int^{x' \in X} X(x, x') \otimes \Phi(x', y) \cong \Phi(x, y).$$

These are expressed by

$$\begin{array}{ccc} X & \xrightarrow{\Phi} & Y \\ & \searrow \Phi & \nearrow 1_Y \\ & & Y \end{array} \quad \begin{array}{ccc} X & \xrightarrow{1_X} & X \\ & \searrow \Phi & \nearrow \Phi \\ & & Y \end{array}$$

□

Together with fuzzy functors as vertical arrows and the companion-conjoint assignments below, this bicategory underlies the fuzzy distributor equipment.

A cell in the equipment

$$\begin{array}{ccc} X & \xrightarrow{\Phi} & Y \\ f \downarrow & \Downarrow & \downarrow g \\ X' & \xrightarrow{\Psi} & Y' \end{array}$$

is a distributor transformation

$$\Phi(x, y) \rightarrow \Psi(fx, gy),$$

natural in $x \in X$ and $y \in Y$. In the locally posetal reflection, this is the pointwise inequality

$$\Phi(x, y) \leq \Psi(fx, gy).$$

4.2.2 The posetal reflection

In the posetal specialization determined by

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp_L, \top_L),$$

a fuzzy distributor

$$\Phi : X \multimap Y$$

is a map

$$\Phi : X \times Y \rightarrow L$$

satisfying

$$X(x', x) \otimes \Phi(x, y) \otimes Y(y, y') \leq \Phi(x', y')$$

for all $x, x' \in X$ and $y, y' \in Y$. Thus Φ is lower in the first variable and upper in the second variable.

Composition is

$$(\Psi \circ \Phi)(x, z) = \bigvee_{y \in Y} \Phi(x, y) \otimes \Psi(y, z),$$

and the identity distributor is

$$1_X(x, x') = X(x, x').$$

4.2.3 Examples

Example 4.2.1 (The crisp case). *For $\Lambda = 2$, fuzzy categories are preorders and a distributor*

$$\Phi : X \multimap Y$$

is a relation

$$\Phi \subseteq X \times Y$$

such that

$$x' \leq_X x, \quad \Phi(x, y), \quad y \leq_Y y' \implies \Phi(x', y').$$

Composition is ordinary relational composition:

$$(\Psi \circ \Phi)(x, z) \iff \exists y \in Y, \Phi(x, y) \wedge \Psi(y, z).$$

Example 4.2.2 (Gödel-valued distributors). *For Gödel truth values,*

$$(\Psi \circ \Phi)(x, z) = \bigvee_{y \in Y} \min(\Phi(x, y), \Psi(y, z)).$$

Example 4.2.3 (Łukasiewicz-valued distributors). *For Łukasiewicz truth values,*

$$(\Psi \circ \Phi)(x, z) = \bigvee_{y \in Y} \max(0, \Phi(x, y) + \Psi(y, z) - 1).$$

Example 4.2.4 (Product-valued distributors). *For product truth values,*

$$(\Psi \circ \Phi)(x, z) = \bigvee_{y \in Y} \Phi(x, y) \Psi(y, z).$$

Example 4.2.5 (Lawvere-valued distributors). *For the Lawvere quantale, joins in the truth-value order are infima in the usual numerical order, and tensor is addition. Hence distributor composition is infimal convolution:*

$$(\Psi \circ \Phi)(x, z) = \inf_{y \in Y} (\Phi(x, y) + \Psi(y, z)).$$

4.3 Companions and conjoints

Fuzzy functors determine distinguished fuzzy distributors. These are the companion and conjoint constructions. They place vertical arrows inside the horizontal distributor equipment.

4.3.1 The enriched construction

Definition 4.3.1 (Companion and conjoint). *Let*

$$f : X \rightarrow Y$$

*be a fuzzy functor. Its **companion** is the distributor*

$$f_* : X \rightharpoonup Y$$

defined by

$$f_*(x, y) = Y(fx, y).$$

*Its **conjoint** is the distributor*

$$f^* : Y \rightharpoonup X$$

defined by

$$f^*(y, x) = Y(y, fx).$$

The companion and conjoint sit on opposite sides of f :

$$\begin{array}{ccc} X & \xrightarrow{f_*} & Y \\ f \downarrow & & \downarrow 1_Y \\ Y & \xrightarrow{1_Y} & Y \end{array} \quad \begin{array}{ccc} Y & \xrightarrow{f^*} & X \\ 1_Y \downarrow & & \downarrow f \\ Y & \xrightarrow{1_Y} & Y. \end{array}$$

Equivalently,

$$X \begin{array}{c} \xrightarrow{f_*} \\ \xleftarrow{f^*} \end{array} Y$$

is an adjoint pair in the distributor bicategory.

Proposition 4.3.1 (Companion–conjoint adjunction). *For every fuzzy functor*

$$f : X \rightarrow Y,$$

there is an adjunction in the distributor bicategory

$$f_* \dashv f^*.$$

Proof. The unit is a distributor transformation

$$1_X \rightarrow f^* \circ f_*.$$

At (x, x') , the target is

$$(f^* \circ f_*)(x, x') = \int^{y \in Y} Y(fx, y) \otimes Y(y, fx').$$

The required map is obtained by composing the functoriality map

$$X(x, x') \rightarrow Y(fx, fx')$$

with the coend coprojection at $y = fx'$:

$$Y(fx, fx') \cong Y(fx, fx') \otimes I \rightarrow Y(fx, fx') \otimes Y(fx', fx') \rightarrow \int^{y \in Y} Y(fx, y) \otimes Y(y, fx').$$

The counit is a distributor transformation

$$f_* \circ f^* \rightarrow 1_Y.$$

At (y, y') , it is induced by composition in Y :

$$\int^{x \in X} Y(y, fx) \otimes Y(fx, y') \rightarrow Y(y, y').$$

The triangle identities follow from associativity and unitality of composition in Y , equivalently from the standard companion–conjoint construction in the enriched distributor equipment. $\square$

Proposition 4.3.2 (Functoriality of companions and conjoints). *For fuzzy functors*

$$X \xrightarrow{f} Y \xrightarrow{g} Z,$$

there are canonical isomorphisms of distributors

$$(gf)_* \cong g_* \circ f_*$$

and

$$(gf)^* \cong f^* \circ g^*.$$

Moreover,

$$(1_X)_* \cong 1_X, \quad (1_X)^* \cong 1_X.$$

Proof. For companions,

$$\begin{aligned} (g_* \circ f_*)(x, z) &= \int^{y \in Y} Y(fx, y) \otimes Z(gy, z) \\ &\cong Z(gfx, z) \\ &= (gf)_*(x, z), \end{aligned}$$

by enriched Yoneda. Diagrammatically:

$$\begin{array}{ccccc} X & \xrightarrow{\quad f_* \quad} & Y & \xrightarrow{\quad g_* \quad} & Z. \\ & \searrow & & \nearrow & \\ & & (gf)_* & & \end{array}$$

For conjoints,

$$\begin{aligned} (f^* \circ g^*)(z, x) &= \int^{y \in Y} Z(z, gy) \otimes Y(y, fx) \\ &\cong Z(z, gfx) \\ &= (gf)^*(z, x). \end{aligned}$$

This is represented by the reverse diagram

$$\begin{array}{ccccc} Z & \xrightarrow{\quad g^* \quad} & Y & \xrightarrow{\quad f^* \quad} & X. \\ & \searrow & & \nearrow & \\ & & (gf)^* & & \end{array}$$

The identity statements follow directly from the definitions. $\square$

4.3.2 The posetal reflection

For a Λ -functor

$$f : X \rightarrow Y,$$

the companion and conjoint are

$$f_*(x, y) = Y(fx, y)$$

and

$$f^*(y, x) = Y(y, fx).$$

The adjunction

$$f_* \dashv f^*$$

amounts to the distributor inequalities

$$1_X \leq f^* \circ f_*$$

and

$$f_* \circ f^* \leq 1_Y.$$

The first inequality follows from functoriality of f , and the second follows from transitivity in Y .

4.3.3 Examples

Example 4.3.1 (The crisp case). *For a monotone map*

$$f : X \rightarrow Y$$

between preorders,

$$f_*(x, y) \iff f(x) \leq_Y y,$$

and

$$f^*(y, x) \iff y \leq_Y f(x).$$

Example 4.3.2 (Standard fuzzy truth values). *For Gödel, Łukasiewicz, and product truth values, the formulas remain*

$$f_*(x, y) = Y(fx, y), \quad f^*(y, x) = Y(y, fx).$$

The tensor and residual of the chosen truth-value object determine the distributor composition and the adjunction inequalities.

Example 4.3.3 (Lawvere truth values). *For the Lawvere quantale, a fuzzy functor is a nonexpansive map. The companion and conjoint of*

$$f : X \rightarrow Y$$

are

$$f_*(x, y) = d_Y(fx, y), \quad f^*(y, x) = d_Y(y, fx).$$

Thus companions and conjoints record the forward and backward distance kernels induced by f .

4.4 Right liftings of fuzzy distributors

Right liftings are the categorical operation that will produce the fuzzy way-below distributor.

4.4.1 The enriched construction

Let

$$\Phi : A \multimap C, \quad \Psi : B \multimap C$$

be fuzzy distributors with common target C .

Definition 4.4.1 (Right lifting). *The **right lifting** of Ψ through Φ is the fuzzy distributor*

$$[\Phi, \Psi] : B \multimap A$$

defined by

$$[\Phi, \Psi](b, a) = \int_{c \in C} [\Phi(a, c), \Psi(b, c)].$$

The right lifting is characterized by the following universal diagram:

$$\begin{array}{ccccc} B & \xrightarrow{\Theta} & A & \xrightarrow{\downarrow} & C \\ & \searrow \Psi & & \nearrow & \end{array}$$

where comparisons

$$\Theta \rightarrow [\Phi, \Psi]$$

correspond to comparisons

$$\Phi \circ \Theta \rightarrow \Psi.$$

In the locally posetal reflection, this says that $[\Phi, \Psi]$ is the largest Θ satisfying

$$\Phi \circ \Theta \leq \Psi.$$

Proposition 4.4.1 (Universal property of right liftings). *For every fuzzy distributor*

$$\Theta : B \multimap A,$$

there is a natural isomorphism

$$\text{Dist}_{\mathbb{V}}(B, A)(\Theta, [\Phi, \Psi]) \cong \text{Dist}_{\mathbb{V}}(B, C)(\Phi \circ \Theta, \Psi).$$

Equivalently, postcomposition by Φ has right adjoint

$$[\Phi, -].$$

Proof. Using the enriched hom-object in the distributor category, the left-hand side is

$$\begin{aligned} \text{Dist}_{\mathbb{V}}(B, A)(\Theta, [\Phi, \Psi]) &= \int_{b, a} [\Theta(b, a), [\Phi, \Psi](b, a)] \\ &= \int_{b, a} \left[\Theta(b, a), \int_c [\Phi(a, c), \Psi(b, c)] \right] \\ &\cong \int_{b, a, c} [\Theta(b, a), [\Phi(a, c), \Psi(b, c)]] \\ &\cong \int_{b, a, c} [\Theta(b, a) \otimes \Phi(a, c), \Psi(b, c)]. \end{aligned}$$

By the end-coend adjunction, this is isomorphic to

$$\int_{b,c} \left[\int^a \Theta(b, a) \otimes \Phi(a, c), \Psi(b, c) \right].$$

The coend is

$$(\Phi \circ \Theta)(b, c).$$

Hence the displayed object is

$$\text{Dist}_V(B, C)(\Phi \circ \Theta, \Psi).$$

□

4.4.2 The posetal reflection

In the posetal reflection, the right lifting is

$$[\Phi, \Psi](b, a) = \bigwedge_{c \in C} (\Phi(a, c) \multimap \Psi(b, c)).$$

Its universal property becomes

$$\Theta \leq [\Phi, \Psi] \iff \Phi \circ \Theta \leq \Psi.$$

Thus $[\Phi, \Psi](b, a)$ is the degree to which every instance of $\Phi(a, c)$ entails the corresponding instance of $\Psi(b, c)$.

The posetal lifting adjunction can be displayed as

$$\begin{array}{ccc} & \xrightarrow{\Phi \circ -} & \\ \text{Dist}_\Lambda(B, A) & & \text{Dist}_\Lambda(B, C) \\ & \xleftarrow{[\Phi, -]} & \end{array}$$

with order-enriched adjunction condition

$$\Phi \circ \Theta \leq \Psi \iff \Theta \leq [\Phi, \Psi].$$

4.4.3 Examples

Example 4.4.1 (The crisp case). For $\Lambda = 2$,

$$[\Phi, \Psi](b, a)$$

holds precisely when

$$\forall c \in C, \Phi(a, c) \Rightarrow \Psi(b, c).$$

The universal property is

$$\Theta \subseteq [\Phi, \Psi] \iff \Phi \circ \Theta \subseteq \Psi.$$

Example 4.4.2 (Gödel right liftings). *For Gödel truth values,*

$$[\Phi, \Psi](b, a) = \bigwedge_{c \in C} (\Phi(a, c) \multimap_{\mathbf{G}} \Psi(b, c)),$$

where

$$u \multimap_{\mathbf{G}} v = \begin{cases} 1, & u \leq v, \\ v, & u > v. \end{cases}$$

Example 4.4.3 (Łukasiewicz right liftings). *For Łukasiewicz truth values,*

$$[\Phi, \Psi](b, a) = \bigwedge_{c \in C} \min(1, 1 - \Phi(a, c) + \Psi(b, c)).$$

Example 4.4.4 (Product right liftings). *For product truth values,*

$$[\Phi, \Psi](b, a) = \bigwedge_{c \in C} (\Phi(a, c) \multimap_{\Pi} \Psi(b, c)),$$

where

$$u \multimap_{\Pi} v = \begin{cases} 1, & u \leq v, \\ v/u, & u > v. \end{cases}$$

Example 4.4.5 (Lawvere right liftings). *For the Lawvere quantale, the internal hom is truncated subtraction. Written in the usual numerical order,*

$$[\Phi, \Psi](b, a) = \sup_{c \in C} (\Psi(b, c) \ominus \Phi(a, c)),$$

where

$$v \ominus u = \max(0, v - u)$$

with the usual extended-value conventions. Thus a Lawvere right lifting measures the largest residual excess needed to transform $\Phi(a, -)$ into $\Psi(b, -)$.

4.5 Ideals as distributors and approximable kernels

We next relate the ideal monad from the previous chapter to the distributor equipment.

4.5.1 The enriched construction

A fuzzy ideal of X is a finite-flat weight

$$I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

Equivalently, it is a fuzzy distributor

$$I : X \multimap \mathbf{1},$$

because

$$X^{\text{op}} \otimes \mathbf{1} \cong X^{\text{op}}.$$

Thus

$$\text{Idl}_{\mathbb{V}}(X)$$

is the full $\mathbb{V}$ -subcategory of

$$\text{Dist}_{\mathbb{V}}(X, \mathbf{1}) = [X^{\text{op}}, \underline{\mathbb{V}}]$$

spanned by finite-flat distributors. This inclusion is represented by

$$\text{Idl}_{\mathbb{V}}(X) \xleftarrow{i_X} \text{Dist}_{\mathbb{V}}(X, \mathbf{1}) \xrightarrow{\cong} [X^{\text{op}}, \underline{\mathbb{V}}].$$

Definition 4.5.1 (Fuzzy approximable map). *Let X and Y be small fuzzy categories. A **fuzzy approximable map** from X to Y is a Kleisli morphism for the fuzzy ideal monad:*

$$R : X \rightarrow \text{Idl}_{\mathbb{V}}(Y).$$

Equivalently, it is a kernel

$$R : X \otimes Y^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

such that, for every $x \in X$, the fibre

$$R(x, -) : Y^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

is finite-flat.

The distributor variance of this kernel is opposite to the variance of a distributor $X \rightharpoonup Y$. Under the convention

$$\Phi : A \rightharpoonup B \quad \text{means} \quad \Phi : A^{\text{op}} \otimes B \rightarrow \underline{\mathbb{V}},$$

the kernel R corresponds to a distributor

$$R^{\dagger} : Y \rightharpoonup X, \quad R^{\dagger}(y, x) = R(x, y).$$

The variance conversion is summarized by

$$X \xrightarrow{R} \text{Idl}_{\mathbb{V}}(Y) \xleftarrow{i_Y} [Y^{\text{op}}, \underline{\mathbb{V}}] \quad Y \xleftarrow{R^{\dagger}} X.$$

The upper row is vertical/Kleisli data, while the lower row is distributor data.

Proposition 4.5.1 (Kleisli extension as distributor convolution). *Let*

$$R : X \rightsquigarrow Y$$

be a fuzzy approximable map. Its Kleisli extension

$$\overline{R} : \text{Idl}_{\mathbb{V}}(X) \rightarrow \text{Idl}_{\mathbb{V}}(Y)$$

is given by

$$\overline{R}(I)(y) = \int^{x \in X} I(x) \otimes R(x, y).$$

Equivalently, if

$$R^{\dagger} : Y \rightharpoonup X$$

is the associated oppositely oriented distributor, then

$$\overline{R}(I) = I \circ R^{\dagger}.$$

Proof. The Kleisli extension is

$$\mathrm{Idl}_{\mathbb{V}}(X) \xrightarrow{\mathrm{Idl}_{\mathbb{V}}(R)} \mathrm{Idl}_{\mathbb{V}}(\mathrm{Idl}_{\mathbb{V}}(Y)) \xrightarrow{\mu_Y} \mathrm{Idl}_{\mathbb{V}}(Y).$$

The multiplication μ_Y is flattening, so

$$\overline{R}(I)(y) = \int^{x \in X} I(x) \otimes R(x, y).$$

On the other hand,

$$(I \circ R^\dagger)(y) = \int^{x \in X} R^\dagger(y, x) \otimes I(x) = \int^{x \in X} R(x, y) \otimes I(x),$$

which is canonically isomorphic to the same coend by symmetry of $\otimes$. □

The convolution picture is

$$\begin{array}{ccccc} Y & \xrightarrow{\quad R^\dagger \quad} & X & \xrightarrow{\quad I \quad} & \mathbf{1}. \\ & \searrow & & \nearrow & \\ & & I \circ R^\dagger & & \end{array}$$

4.5.2 The posetal reflection

In the posetal specialization, a fuzzy approximable map

$$R : X \rightsquigarrow Y$$

is a map

$$R : X \times Y \rightarrow L$$

such that each fibre

$$R(x, -) : Y^{\mathrm{op}} \rightarrow \Lambda$$

is a fuzzy ideal of Y , and such that the assignment

$$x \longmapsto R(x, -)$$

is a Λ -functor

$$X \rightarrow \mathrm{Idl}_{\Lambda}(Y).$$

Its extension is

$$\overline{R}(I)(y) = \bigvee_{x \in X} I(x) \otimes R(x, y).$$

Kleisli composition is therefore

$$(S \circ R)(x, z) = \bigvee_{y \in Y} R(x, y) \otimes S(y, z).$$

4.5.3 Examples

Example 4.5.1 (The crisp case). *For $\Lambda = 2$, a fuzzy approximable map*

$$R : X \rightsquigarrow Y$$

is a monotone map

$$X \rightarrow \text{Idl}(Y).$$

Equivalently, it is a relation

$$R \subseteq X \times Y$$

such that every fibre

$$R[x] = \{y \in Y \mid R(x, y)\}$$

is an ideal of Y , and

$$x \leq x' \implies R[x] \subseteq R[x'].$$

The extension formula becomes

$$\overline{R}(I) = \bigcup_{x \in I} R[x].$$

Example 4.5.2 (Gödel approximable maps). *For Gödel truth values,*

$$\overline{R}(I)(y) = \bigvee_{x \in X} \min(I(x), R(x, y)).$$

Example 4.5.3 (Łukasiewicz approximable maps). *For Łukasiewicz truth values,*

$$\overline{R}(I)(y) = \bigvee_{x \in X} \max(0, I(x) + R(x, y) - 1).$$

Example 4.5.4 (Product approximable maps). *For product truth values,*

$$\overline{R}(I)(y) = \bigvee_{x \in X} I(x)R(x, y).$$

Example 4.5.5 (Lawvere approximable maps). *For the Lawvere quantale, the extension formula is infimal convolution:*

$$\overline{R}(I)(y) = \inf_{x \in X} (I(x) + R(x, y)),$$

written in the usual numerical order.

4.6 The ideal algebra map inside the distributor equipment

We now place the ideal-algebra map and the principal-ideal embedding inside the distributor equipment.

4.6.1 The enriched construction

Let X be a fuzzy domain. We have the algebra map

$$\alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X$$

and the principal-ideal embedding

$$\eta_X : X \rightarrow \text{Idl}_{\mathbb{V}}(X).$$

The conjoint of α_X is the distributor

$$\alpha_X^* : X \nrightarrow \text{Idl}_{\mathbb{V}}(X)$$

given by

$$\alpha_X^*(b, I) = X(b, \alpha_X I).$$

The companion of η_X is the distributor

$$(\eta_X)_* : X \nrightarrow \text{Idl}_{\mathbb{V}}(X)$$

given by

$$(\eta_X)_*(a, I) = \text{Idl}_{\mathbb{V}}(X)(\eta_X a, I).$$

The two constructions arise from opposite sides of the algebra triangle:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & \text{Idl}_{\mathbb{V}}(X) \\ & \searrow 1_X & \downarrow \alpha_X \\ & & X \end{array} \quad \begin{array}{ccc} & & \text{Idl}_{\mathbb{V}}(X) \\ \alpha_X^* \nearrow & & \uparrow \\ X & \xrightarrow{(\eta_X)_*} & \end{array}$$

Thus

$$\alpha_X^* \quad \text{and} \quad (\eta_X)_*$$

are two fuzzy distributors with the same source and the same target:

$$X \nrightarrow \text{Idl}_{\mathbb{V}}(X).$$

Consequently, the right lifting

$$[\alpha_X^*, (\eta_X)_*]$$

is a fuzzy distributor

$$X \nrightarrow X.$$

This is the distributor that will be defined as the fuzzy way-below relation.

4.6.2 The posetal reflection

For a fuzzy domain over Λ , the two distributors are

$$\alpha_X^*(b, I) = X(b, \alpha_X I),$$

and

$$(\eta_X)_*(a, I) = \text{Idl}_{\Lambda}(X)(\eta_X a, I).$$

By the enriched Yoneda lemma in the posetal setting,

$$\text{Idl}_{\Lambda}(X)(\eta_X a, I) = I(a).$$

Hence

$$(\eta_X)_*(a, I) = I(a).$$

4.6.3 Examples

Example 4.6.1 (The crisp case). For $\Lambda = 2$,

$$\alpha_X^*(b, I) \iff b \leq \bigvee I,$$

and

$$(\eta_X)_*(a, I) \iff \downarrow a \subseteq I.$$

Since I is lower,

$$\downarrow a \subseteq I \iff a \in I.$$

Thus the right lifting expresses the ordinary implication

$$b \leq \bigvee I \implies a \in I$$

uniformly over all ideals I .

Example 4.6.2 (Standard fuzzy truth values). For Gödel, Łukasiewicz, and product truth values, the formulas are

$$\alpha_X^*(b, I) = X(b, \alpha_X I), \quad (\eta_X)_*(a, I) = I(a).$$

The later right-lifting formula compares these two degrees using the residual appropriate to the chosen tensor.

Example 4.6.3 (Lawvere truth values). For the Lawvere quantale,

$$\alpha_X^*(b, I) = d_X(b, \alpha_X I)$$

is the cost from b to the ideal colimit of I , while

$$(\eta_X)_*(a, I) = I(a)$$

is the cost-degree with which a belongs to the weight I .

4.7 The fuzzy way-below distributor

The fuzzy way-below relation is now obtained by applying the right-lifting construction to the two distributors built from α_X and η_X .

4.7.1 The enriched construction

Definition 4.7.1 (Fuzzy way-below distributor). Let X be a fuzzy domain. The **fuzzy way-below distributor** of X is the right lifting

$$\ll_X := [\alpha_X^*, (\eta_X)_*] : X \multimap X.$$

We write

$$\ll_X (a, b)$$

for its value at (a, b) , where a is the approximant variable and b is the element being approximated.

With the right-lifting convention

$$[\Phi, \Psi](b, a) = \int_c [\Phi(a, c), \Psi(b, c)],$$

the first variable of $\ll_X (a, b)$ comes from the source variable of $(\eta_X)_*$, while the second variable comes from the source variable of α_X^* . Thus

$$\ll_X (a, b) = [\alpha_X^*, (\eta_X)_*](a, b) = \int_I [\alpha_X^*(b, I), (\eta_X)_*(a, I)].$$

The universal right-lifting triangle is

$$\begin{array}{ccccc} X & \xrightarrow{\ll_X} & X & \xrightarrow{\ulcorner^*} & \text{Idl}_V(X) \\ & & & \searrow & \\ & & & & \text{Idl}_V(X) \\ & & & \nwarrow & \\ & & & & \text{Idl}_V(X) \end{array}$$

(The diagram shows a triangle with vertices X , X , and $\text{Idl}_V(X)$. The top edge is $\ll_X$, the right edge is $\ulcorner^*$, and the bottom edge is $(\eta_X)_*$.)

and its universal property says that every compatible triangle factors through $\ll_X$:

$$\text{Dist}_V(X, X)(\Theta, \ll_X) \cong \text{Dist}_V(X, \text{Idl}_V(X))(\alpha_X^* \circ \Theta, (\eta_X)_*).$$

Equivalently, a comparison

$$\alpha_X^* \circ \Theta \rightarrow (\eta_X)_*$$

is the same as a comparison

$$\Theta \rightarrow \ll_X.$$

Proposition 4.7.1 (Formula for the fuzzy way-below distributor). *For every fuzzy domain X ,*

$$\ll_X (a, b) \cong \int_{I \in \text{Idl}_V(X)} [X(b, \alpha_X I), \text{Idl}_V(X)(\eta_X a, I)].$$

By enriched Yoneda, this is equivalently

$$\ll_X (a, b) \cong \int_{I \in \text{Idl}_V(X)} [X(b, \alpha_X I), I(a)].$$

Proof. This is the right-lifting formula applied to

$$\Phi = \alpha_X^*, \quad \Psi = (\eta_X)_*.$$

Thus

$$\ll_X (a, b) = [\alpha_X^*, (\eta_X)_*](a, b)$$

is

$$\int_{I \in \text{Idl}_V(X)} [\alpha_X^*(b, I), (\eta_X)_*(a, I)].$$

Using the definitions of α_X^* and $(\eta_X)_*$, this becomes

$$\int_{I \in \text{Idl}_V(X)} [X(b, \alpha_X I), \text{Idl}_V(X)(\eta_X a, I)].$$

Finally,

$$\text{Idl}_V(X)(\eta_X a, I) \cong I(a)$$

by enriched Yoneda. □

Proposition 4.7.2 (Universal property of way-below). *For every fuzzy distributor*

$$\Theta : X \multimap X,$$

there is a natural isomorphism

$$\text{Dist}_{\mathbb{V}}(X, X)(\Theta, \ll_X) \cong \text{Dist}_{\mathbb{V}}(X, \text{Idl}_{\mathbb{V}}(X))(\alpha_X^* \circ \Theta, (\eta_X)_*).$$

In the locally posetal reflection this becomes

$$\Theta \leq \ll_X \iff \alpha_X^* \circ \Theta \leq (\eta_X)_*.$$

Proof. This is the universal property of the right lifting

$$[\alpha_X^*, (\eta_X)_*].$$

□

The universal property can be visualized as the adjunction

$$\begin{array}{ccc} & \alpha_X^* \circ - & \\ \text{Dist}_{\mathbb{V}}(X, X) & \xrightarrow{\quad} & \text{Dist}_{\mathbb{V}}(X, \text{Idl}_{\mathbb{V}}(X)) \\ & \xleftarrow{\quad [\alpha_X^*, -] \quad} & \end{array}$$

evaluated at $(\eta_X)_*$.

Proposition 4.7.3 (Way-below implies below). *For every fuzzy domain X , there is a canonical comparison*

$$\ll_X (a, b) \rightarrow X(a, b).$$

In the posetal reflection this says

$$\ll_X (a, b) \leq X(a, b).$$

Proof. By Proposition 4.7.1, the end defining $\ll_X (a, b)$ maps to its component at the principal ideal $\eta_X b$:

$$\ll_X (a, b) \rightarrow [X(b, \alpha_X \eta_X b), \text{Idl}_{\mathbb{V}}(X)(\eta_X a, \eta_X b)].$$

The strict algebra unit law gives

$$\alpha_X \eta_X b = b.$$

The Yoneda embedding is fully faithful, so

$$\text{Idl}_{\mathbb{V}}(X)(\eta_X a, \eta_X b) \cong X(a, b).$$

Thus we have a map

$$\ll_X (a, b) \rightarrow [X(b, b), X(a, b)].$$

The unit map

$$I \rightarrow X(b, b)$$

induces, by contravariance of the first variable of the internal hom, a morphism

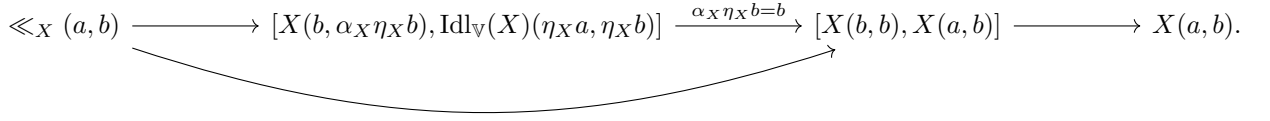
$$[X(b, b), X(a, b)] \rightarrow [I, X(a, b)] \cong X(a, b).$$

The composite gives

$$\ll_X (a, b) \rightarrow X(a, b).$$

□

The proof follows the diagram

$$\ll_X (a, b) \longrightarrow [X(b, \alpha_X \eta_X b), \text{Idl}_V(X)(\eta_X a, \eta_X b)] \xrightarrow{\alpha_X \eta_X b = b} [X(b, b), X(a, b)] \longrightarrow X(a, b).$$


4.7.2 The posetal reflection

For a fuzzy domain over Λ ,

$$\ll_X (a, b) = \bigwedge_{I \in \text{Idl}_\Lambda(X)} (X(b, \alpha_X I) \multimap I(a)).$$

The distributor law for

$$\ll_X: X \multimap X$$

gives fuzzy monotonicity:

$$X(a', a) \otimes \ll_X (a, b) \otimes X(b, b') \leq \ll_X (a', b').$$

The comparison of Proposition 4.7.3 becomes

$$\ll_X (a, b) \leq X(a, b).$$

Thus the way-below degree is always bounded above by the ordinary fuzzy comparison degree.

In the Lawvere case this inequality is in the truth-value order. Since the truth-value order is the reverse of the usual numerical order, it reads numerically as

$$\ll_X (a, b) \geq d_X(a, b).$$

4.7.3 Examples

Example 4.7.1 (The crisp case). For $\Lambda = 2$, the formula becomes

$$a \ll_X b \iff \forall I \in \text{Idl}(X), b \leq \bigvee I \Rightarrow a \in I.$$

This is the ordinary way-below relation on a dcpo.

Example 4.7.2 (Gödel way-below degrees). For Gödel truth values,

$$\ll_X (a, b) = \bigwedge_I (X(b, \alpha_X I) \multimap_G I(a)).$$

This is the degree to which every ideal whose join lies above b contains a .

Example 4.7.3 (Łukasiewicz way-below degrees). For Łukasiewicz truth values,

$$\ll_X (a, b) = \bigwedge_I \min(1, 1 - X(b, \alpha_X I) + I(a)).$$

Example 4.7.4 (Product way-below degrees). For product truth values,

$$\ll_X (a, b) = \bigwedge_I (X(b, \alpha_X I) \multimap_\Pi I(a)).$$

Example 4.7.5 (Lawvere way-below costs). *For the Lawvere quantale, the meet in the truth-value order is a supremum in the usual numerical order. Hence*

$$\ll_X (a, b) = \sup_{I \in \text{Idl}_\Lambda(X)} (I(a) \ominus X(b, \alpha_X I)),$$

where

$$v \ominus u = \max(0, v - u)$$

with the extended-value conventions. Thus $\ll_X (a, b)$ measures the worst-case residual cost needed to turn evidence that b lies below an ideal colimit into evidence that a belongs to that ideal.

4.8 Continuity as representability of way-below

We now show that the previous chapter's way-below formula is precisely the representable form of the right-lifted distributor.

4.8.1 The enriched construction

Definition 4.8.1 (Continuous fuzzy domain, recalled). *A fuzzy domain X is **continuous** if its ideal-algebra map*

$$\alpha_X : \text{Idl}_V(X) \rightarrow X$$

has an enriched left adjoint

$$\beta_X : X \rightarrow \text{Idl}_V(X), \quad \beta_X \dashv \alpha_X.$$

The continuity data form the adjoint diagram

$$\begin{array}{ccc} & \beta_X & \\ X & \xrightarrow{\quad} & \text{Idl}_V(X) \\ & \alpha_X & \end{array}$$

and hence give natural isomorphisms

$$X(b, \alpha_X I) \cong \text{Idl}_V(X)(\beta_X b, I).$$

Theorem 4.8.1 (Continuity represents the way-below distributor). *Let X be a continuous fuzzy domain. Then*

$$\ll_X (a, b) \cong \text{Idl}_V(X)(\eta_X a, \beta_X b).$$

Consequently,

$$\ll_X (a, b) \cong \beta_X(b)(a).$$

Equivalently, as distributors,

$$\ll_X \cong \beta_X^* \circ (\eta_X)_*.$$

Proof. By Proposition 4.7.1,

$$\ll_X (a, b) \cong \int_I [X(b, \alpha_X I), \text{Idl}_V(X)(\eta_X a, I)].$$

Since

$$\beta_X \dashv \alpha_X,$$

we have

$$X(b, \alpha_X I) \cong \text{Idl}_{\mathbb{V}}(X)(\beta_X b, I).$$

Therefore

$$\ll_X(a, b) \cong \int_I [\text{Idl}_{\mathbb{V}}(X)(\beta_X b, I), \text{Idl}_{\mathbb{V}}(X)(\eta_X a, I)].$$

By enriched Yoneda in $\text{Idl}_{\mathbb{V}}(X)$,

$$\int_I [\text{Idl}_{\mathbb{V}}(X)(\beta_X b, I), \text{Idl}_{\mathbb{V}}(X)(\eta_X a, I)] \cong \text{Idl}_{\mathbb{V}}(X)(\eta_X a, \beta_X b).$$

Applying Yoneda again to the ideal

$$\beta_X b : X^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

gives

$$\text{Idl}_{\mathbb{V}}(X)(\eta_X a, \beta_X b) \cong \beta_X(b)(a).$$

For the distributor-level statement, compute

$$\begin{aligned} (\beta_X^* \circ (\eta_X)_*)(a, b) &= \int^{I \in \text{Idl}_{\mathbb{V}}(X)} \text{Idl}_{\mathbb{V}}(X)(\eta_X a, I) \otimes \text{Idl}_{\mathbb{V}}(X)(I, \beta_X b) \\ &\cong \text{Idl}_{\mathbb{V}}(X)(\eta_X a, \beta_X b) \\ &\cong \ll_X(a, b), \end{aligned}$$

by enriched co-Yoneda and the calculation above. □

Thus the approximant map

$$\beta_X : X \rightarrow \text{Idl}_{\mathbb{V}}(X)$$

represents the right-lifted way-below distributor.

4.8.2 The posetal reflection

For a continuous fuzzy domain over Λ ,

$$\ll_X(a, b) = \text{Idl}_{\Lambda}(X)(\eta_X a, \beta_X b) = \beta_X(b)(a).$$

Thus the way-below degree

$$\text{wb}_X(a, b) = \beta_X(b)(a)$$

from the previous chapter is exactly the value of the right-lifted distributor constructed in this chapter.

If X is separated, then the reconstruction theorem from the previous chapter gives

$$\alpha_X \beta_X = 1_X.$$

For completeness, we recall the argument. The unit of

$$\beta_X \dashv \alpha_X$$

gives

$$1_X \leq \alpha_X \beta_X.$$

The counit gives

$$\beta_X \alpha_X \leq 1_{\text{Idl}_\Lambda(X)}.$$

Applying this to the principal ideal $\eta_X b$, and using the algebra law

$$\alpha_X \eta_X b = b,$$

gives

$$\beta_X b \leq \eta_X b.$$

Applying α_X gives

$$\alpha_X \beta_X b \leq \alpha_X \eta_X b = b.$$

Together with

$$b \leq \alpha_X \beta_X b,$$

this gives equality in the induced crisp preorder. Since X is separated,

$$\alpha_X \beta_X b = b.$$

The reconstruction is captured by

$$\begin{array}{ccc} X & \xrightarrow{\beta_X} & \text{Idl}_\Lambda(X) \\ & \searrow 1_X & \downarrow \alpha_X \\ & & X \end{array}$$

in the separated posetal reflection.

Hence every object b of a separated continuous fuzzy domain is recovered as the fuzzy ideal colimit of its approximant ideal:

$$b = \alpha_X(\beta_X b).$$

4.8.3 Examples

Example 4.8.1 (The crisp case). *For $\Lambda = 2$, continuity means that*

$$\alpha_X : \text{Idl}(X) \rightarrow X$$

has a left adjoint

$$\beta_X : X \rightarrow \text{Idl}(X).$$

Then

$$\beta_X(b) = \{a \in X \mid a \ll b\},$$

and

$$a \ll b \iff a \in \beta_X(b).$$

The reconstruction equation

$$\alpha_X \beta_X(b) = b$$

says that every element is the directed supremum of its way-below approximants.

Example 4.8.2 (Gödel continuous fuzzy domains). *For Gödel truth values,*

$$\text{wb}_X(a, b) = \beta_X(b)(a)$$

is the Gödel-valued degree to which a approximates b .

Example 4.8.3 (Łukasiewicz continuous fuzzy domains). *For Łukasiewicz truth values,*

$$\text{wb}_X(a, b) = \beta_X(b)(a)$$

is an MV-valued approximation degree. The right-lifting formula computes it using the Łukasiewicz residual.

Example 4.8.4 (Product continuous fuzzy domains). *For product truth values,*

$$\text{wb}_X(a, b) = \beta_X(b)(a),$$

and the comparison between ideal-colimit evidence and ideal-membership evidence is governed by the product residual.

Example 4.8.5 (Lawvere continuous fuzzy domains). *For the Lawvere quantale,*

$$\beta_X(b)(a)$$

is the cost with which a approximates b . Continuity says that the right-lifted approximation cost is represented by the finite-flat weight

$$\beta_X(b).$$

4.9 Scott opens through way-below approximants

The previous chapter defined fuzzy Scott opens as upper fuzzy predicates preserving fuzzy ideal joins. The way-below distributor reconstructs Scott opens from their values on approximants.

4.9.1 The enriched construction

Let X be a weak fuzzy domain with structure map

$$\alpha_X : \text{Idl}_V(X) \rightarrow X.$$

A fuzzy Scott open is an upper fuzzy predicate

$$U : X \rightarrow \underline{V}$$

satisfying

$$U(\alpha_X I) \cong \int^{x \in X} I(x) \otimes U(x)$$

for every fuzzy ideal I .

The right-hand side is the enriched pairing of the lower weight

$$I : X^{\text{op}} \rightarrow \underline{V}$$

with the upper predicate

$$U : X \rightarrow \underline{\mathbb{V}}.$$

Thus Scott openness says that U sends the chosen ideal colimit of I to the corresponding weighted colimit of its values. The pairing is represented by

$$\begin{array}{ccc} X^{\text{op}} & \xrightarrow{I} & \underline{\mathbb{V}} \xleftarrow{U} X \\ & & \uparrow \\ & & \int^{x \in X} I(x) \otimes U(x) \end{array}$$

and the Scott condition by

$$\begin{array}{ccc} \text{Idl}_{\underline{\mathbb{V}}}(X) & \xrightarrow{\alpha_X} & X \\ & \searrow I \mapsto \int^x I(x) \otimes U(x) & \downarrow U \\ & & \underline{\mathbb{V}}. \end{array}$$

If an enriched reconstruction equivalence

$$\alpha_X \beta_X \cong 1_X$$

is available, then the Scott-open condition gives

$$U(b) \cong U(\alpha_X(\beta_X b)) \cong \int^{a \in X} \beta_X(b)(a) \otimes U(a) \cong \int^{a \in X} \ll_X(a, b) \otimes U(a).$$

In the posetal separated case, the reconstruction equality was obtained in the previous section, so the formula becomes an equality of truth values.

The reconstruction path is

$$\begin{array}{ccccc} & & 1_X & & \\ & \nearrow & & \searrow & \\ X & \xrightarrow{\beta_X} & \text{Idl}_{\underline{\mathbb{V}}}(X) & \xrightarrow{\alpha_X} & X \\ & \searrow I \mapsto \int^a I(a) \otimes U(a) & \downarrow & & \downarrow U \\ & & \underline{\mathbb{V}} & \xlongequal{\quad} & \underline{\mathbb{V}}. \end{array}$$

4.9.2 The posetal reflection

In the posetal specialization, a fuzzy Scott open is an upper fuzzy predicate

$$U : X \rightarrow L$$

satisfying

$$U(\alpha_X I) = \bigvee_{x \in X} I(x) \otimes U(x)$$

for every fuzzy ideal I .

Proposition 4.9.1 (Scott opens are reconstructed from way-below approximants). *Let X be a separated continuous fuzzy domain over Λ , and let*

$$U : X \rightarrow L$$

be a fuzzy Scott open. Then for every $b \in X$,

$$U(b) = \bigvee_{a \in X} \ll_X (a, b) \otimes U(a).$$

Proof. Since X is separated and continuous,

$$b = \alpha_X(\beta_X b).$$

Since U is Scott open,

$$U(b) = U(\alpha_X(\beta_X b)) = \bigvee_{a \in X} \beta_X(b)(a) \otimes U(a).$$

By Theorem 4.8.1,

$$\beta_X(b)(a) = \ll_X (a, b).$$

Substituting gives

$$U(b) = \bigvee_{a \in X} \ll_X (a, b) \otimes U(a).$$

□

Equivalently,

$$U(b) = \bigvee_{a \in X} \beta_X(b)(a) \otimes U(a).$$

4.9.3 Examples

Example 4.9.1 (The crisp case). *For $\Lambda = 2$, if X is a continuous dcpo and $U \subseteq X$ is Scott open, then*

$$b \in U \iff \exists a \in X, a \ll b \text{ and } a \in U.$$

Thus Scott opens are determined by way-below approximants.

Example 4.9.2 (Gödel Scott opens). *For Gödel truth values, the reconstruction formula is*

$$U(b) = \bigvee_{a \in X} \min(\ll_X (a, b), U(a)).$$

Example 4.9.3 (Łukasiewicz Scott opens). *For Łukasiewicz truth values,*

$$U(b) = \bigvee_{a \in X} \max(0, \ll_X (a, b) + U(a) - 1).$$

Example 4.9.4 (Product Scott opens). *For product truth values,*

$$U(b) = \bigvee_{a \in X} \ll_X (a, b) U(a).$$

Example 4.9.5 (Lawvere Scott opens). *For the Lawvere quantale, the Scott reconstruction formula becomes*

$$U(b) = \inf_{a \in X} (\ll_X (a, b) + U(a)),$$

written in the usual numerical order. Thus the cost of satisfying the Scott predicate at b is obtained by optimizing over approximants a of b .

4.10 Summary

This chapter constructed the fuzzy distributor equipment and used it to explain the origin of the fuzzy way-below relation.

The construction starts with the data already present in the fuzzy ideal monad:

$$\eta_X : X \rightarrow \text{Idl}_{\mathbb{V}}(X), \quad \alpha_X : \text{Idl}_{\mathbb{V}}(X) \rightarrow X.$$

The first is the principal-ideal embedding, and the second is the algebra map of a fuzzy domain. Their strict algebra interaction is

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & \text{Idl}_{\mathbb{V}}(X) \\ & \searrow 1_X & \downarrow \alpha_X \\ & & X. \end{array}$$

Fuzzy distributors

$$\Phi : X \multimap Y$$

are enriched profunctors

$$X^{\text{op}} \otimes Y \rightarrow \underline{\mathbb{V}}.$$

They compose by coends, admit companions and conjoiners for fuzzy functors, and have right liftings

$$[\Phi, \Psi](b, a) = \int_c [\Phi(a, c), \Psi(b, c)].$$

The distributor equipment has the schematic form

$$\begin{array}{ccc} X & \overset{\Phi}{\dashrightarrow} & Y \\ f \downarrow & \Downarrow & \downarrow g \\ X' & \overset{\Psi}{\dashrightarrow} & Y' \end{array}$$

where vertical arrows are fuzzy functors, horizontal arrows are fuzzy distributors, and cells are distributor transformations.

For a fuzzy domain X , the conjoiner of α_X and the companion of η_X are distributors

$$\alpha_X^*, \quad (\eta_X)_* : X \multimap \text{Idl}_{\mathbb{V}}(X).$$

Their right lifting defines the fuzzy way-below distributor:

$$\ll_X = [\alpha_X^*, (\eta_X)_*] : X \multimap X.$$

The defining triangle is

$$\begin{array}{ccccc} X & \overset{\ll_X}{\dashrightarrow} & X & \overset{\alpha_X^*}{\dashrightarrow} & \text{Idl}_{\mathbb{V}}(X). \\ & \searrow & & \nearrow & \\ & & & & (\eta_X)_* \end{array}$$

Its objectwise formula is

$$\ll_X(a, b) \cong \int_{I \in \text{Idl}_{\mathbb{V}}(X)} [X(b, \alpha_X I), I(a)].$$

In the posetal reflection this becomes

$$\ll_X (a, b) = \bigwedge_{I \in \text{Idl}_\Lambda(X)} (X(b, \alpha_X I) \multimap I(a)).$$

Finally, if X is continuous, with

$$\beta_X \dashv \alpha_X,$$

then the right-lifted way-below distributor is represented by β_X :

$$\ll_X (a, b) \cong \text{Idl}_\mathbb{V}(X)(\eta_X a, \beta_X b) \cong \beta_X(b)(a).$$

Equivalently, as distributors,

$$\ll_X \cong \beta_X^* \circ (\eta_X)_*.$$

Thus the way-below degree introduced in the previous chapter is not extra structure: it is the value of the right-lifted distributor represented by the approximant map of a continuous fuzzy domain.

Chapter 5

Recursive Domain Equations by Finite-Flat Cofinal Approximation

In the preceding chapters we constructed the fuzzy ideal monad

$$T_{\mathbb{V}}X = \text{Idl}_{\mathbb{V}}(X),$$

where $\text{Idl}_{\mathbb{V}}(X)$ is the full $\mathbb{V}$ -subcategory of

$$[X^{\text{op}}, \underline{\mathbb{V}}]$$

spanned by the finite-flat weights

$$I : X^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

Here $\mathbb{V}$ denotes the fuzzy cosmos and $\underline{\mathbb{V}}$ denotes its self-enriched $\mathbb{V}$ -category of truth values.

Strict algebras for this monad are the fuzzy domains. Equivalently, a fuzzy domain is a small $\mathbb{V}$ -category equipped with a strict coherent choice of finite-flat weighted colimits, and a fuzzy-domain morphism is a finite-flat-colimit-preserving $\mathbb{V}$ -functor.

We also use the symmetric monoidal closed structure on fuzzy domains:

$$A \boxtimes_{\text{ff}} B \dashv [B, -]_{\text{ff}}.$$

Thus

$$\mathbf{FDom}_{\mathbb{V}}(A \boxtimes_{\text{ff}} B, C) \cong \mathbf{FDom}_{\mathbb{V}}(A, [B, C]_{\text{ff}}),$$

where $[B, C]_{\text{ff}}$ is the fuzzy domain of finite-flat-colimit-preserving $\mathbb{V}$ -functors $B \rightarrow C$.

The purpose of this chapter is to formulate recursive domain equation constructions inside the finite-flat doctrine. The classical construction by ω -chain bilimits in dcpos is recovered as the crisp separated specialization. The general replacement for an ordinary directed indexing chain is a finite-flat approximation shape

$$(A, W), \quad W : A^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

Cofinality is expressed by lower direct image of weights. Tails and common successors are not chosen arbitrarily; they are given by canonical comma-style constructions, and the relevant exactness conditions say that these canonical constructions are cofinal in the finite-flat sense.

The semantic target in the general resource-sensitive setting is not an ordinary cartesian reflexive object

$$D \simeq [D, D]_{\text{ff}}.$$

In an affine enriched setting there is no canonical diagonal functor

$$A \rightarrow A \otimes A,$$

since such a functor would require canonical contraction morphisms

$$A(a, a') \rightarrow A(a, a') \otimes A(a, a').$$

Affinity gives weakening morphisms

$$A(a, a') \rightarrow I,$$

not contraction into tensor squares.

Consequently the general internal-hom approximation theorem is external-product-indexed:

$$(A \otimes A, W \boxtimes W).$$

The same-index collapse is recovered only in the crisp separated/cartesian specialization, where the ordinary diagonal

$$\omega \rightarrow \omega \times \omega$$

exists and is cofinal for the ordinary directed ideal.

5.1 Finite-flat approximation shapes

Throughout this chapter, finite-flatness is understood relative to the finite-limit doctrine used in the definition of categorical fuzzy frames. Thus in a predicate category $[X, \underline{\mathbb{V}}]$, the specified finite limits are computed pointwise. In the posetal scalar-frame specialization this means preservation of the terminal predicate and binary meets.

Definition 5.1.1 (Finite-flat approximation shape). *A finite-flat approximation shape is a pair*

$$(A, W),$$

where A is a small $\mathbb{V}$ -category and

$$W : A^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

is a finite-flat weight. Equivalently,

$$W \in \text{Idl}_{\mathbb{V}}(A).$$

If X is a fuzzy domain and

$$D : A \rightarrow X$$

is a $\mathbb{V}$ -functor, the corresponding approximation object is the finite-flat weighted colimit

$$W * D.$$

Thus the passage from ordinary directed joins to fuzzy-domain approximation is the passage

$$\bigvee_{a \in A} D_a \rightsquigarrow W * D.$$

Definition 5.1.2 (Lower direct image of a weight). *Let*

$$h : A \rightarrow B$$

be a $\mathbb{V}$ -functor, and let

$$W : A^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be a weight. The lower direct image of W along h is the weight

$$h_! W : B^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

defined by

$$(h_! W)(b) = \int^{a \in A} W(a) \otimes B(b, ha).$$

Equivalently, $h_! W$ is the weighted left Kan extension of W along

$$h^{\text{op}} : A^{\text{op}} \rightarrow B^{\text{op}}.$$

By the finite-flat direct-image theorem for the ideal doctrine, if W is finite-flat, then $h_! W$ is finite-flat.

Definition 5.1.3 (Cofinal map of approximation shapes). *Let*

$$(A, W), \quad (B, U)$$

be finite-flat approximation shapes. A $\mathbb{V}$ -functor

$$h : A \rightarrow B$$

is cofinal from (A, W) to (B, U) if the canonical comparison exhibits an isomorphism of weights

$$h_! W \cong U.$$

When $A = B$ and $U = W$, we say that h is W -cofinal.

Thus cofinality is equality of approximation weights after lower direct image, not merely a point-wise finality condition on objects.

Proposition 5.1.1 (Cofinality for finite-flat weighted colimits). *Let*

$$h : (A, W) \rightarrow (B, U)$$

be a cofinal map of finite-flat approximation shapes. Then for every fuzzy domain X and every diagram

$$D : B \rightarrow X,$$

there is a canonical isomorphism

$$U * D \cong W * (Dh).$$

Proof. Since h is cofinal,

$$U \cong h_! W.$$

For every $x \in X$, the universal property of weighted colimits and the weighted left-Kan-extension adjunction give

$$\begin{aligned} X((h_!W) * D, x) &\cong [B^{\text{op}}, \mathbb{V}](h_!W, X(D-, x)) \\ &\cong [A^{\text{op}}, \mathbb{V}](W, X(Dh-, x)) \\ &\cong X(W * (Dh), x). \end{aligned}$$

By enriched Yoneda,

$$(h_!W) * D \cong W * (Dh).$$

Using $U \cong h_!W$, this gives

$$U * D \cong W * (Dh).$$

□

Definition 5.1.4 (External product of approximation shapes). *Let*

$$(A, W), \quad (B, Q)$$

be finite-flat approximation shapes. Their external product is

$$(A, W) \boxtimes (B, Q) = (A \otimes B, W \boxtimes Q),$$

where

$$(W \boxtimes Q)(a, b) = W(a) \otimes Q(b).$$

By closure of finite-flat weights under external products,

$$W \boxtimes Q$$

is finite-flat.

Lemma 5.1.1 (Constant diagrams). *Let*

$$(A, W)$$

be a finite-flat approximation shape. For every fuzzy domain X and every object $x \in X$, the W -weighted colimit of the constant diagram at x is canonically x :

$$W * \Delta x \cong x.$$

Proof. Let

$$! : A \rightarrow \mathbb{K}$$

be the terminal functor, and let

$$\bar{x} : \mathbb{K} \rightarrow X$$

classify the object x . The constant diagram at x is

$$\Delta x = \bar{x}!.$$

Since W is finite-flat, its character

$$\chi_W : [A, \mathbb{V}] \rightarrow \mathbb{V}$$

preserves the terminal predicate. Let

$$\top_A : A \rightarrow \mathbb{V}$$

be the constant upper predicate at I . Then

$$\begin{aligned} (!W)(*) &= \int^{a \in A} W(a) \otimes \mathbb{K}(*, !a) \\ &\cong \int^{a \in A} W(a) \otimes I \\ &= \chi_W(\top_A) \\ &\cong I. \end{aligned}$$

Thus

$$!W \cong \mathbb{K}(-, *).$$

By Proposition 5.1.1,

$$W * (\bar{x}!) \cong (!W) * \bar{x} \cong \mathbb{K}(-, *) * \bar{x} \cong x.$$

□

Lemma 5.1.2 (Functoriality of finite-flat colimits in diagrams). *Let C be a fuzzy domain, let*

$$R : P^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be finite-flat, and let

$$H, H' : P \rightarrow C$$

be $\mathbb{V}$ -functors. A $\mathbb{V}$ -natural transformation

$$\gamma : H \Rightarrow H'$$

induces a canonical morphism

$$R * H \longrightarrow R * H'$$

in C . This construction is functorial in γ .

Proof. For each object $c \in C$, the universal property of the weighted colimit gives natural isomorphisms

$$C(R * H, c) \cong [P^{\text{op}}, \underline{\mathbb{V}}](R, C(H-, c))$$

and

$$C(R * H', c) \cong [P^{\text{op}}, \underline{\mathbb{V}}](R, C(H'-, c)).$$

The natural transformation

$$\gamma : H \Rightarrow H'$$

induces by precomposition a $\mathbb{V}$ -natural transformation

$$C(H'-, c) \Rightarrow C(H-, c).$$

Transporting across the representing isomorphisms gives a morphism

$$C(R * H', c) \longrightarrow C(R * H, c)$$

natural in c . By enriched Yoneda this family is represented by a unique morphism

$$R * H \longrightarrow R * H'.$$

Functoriality follows from functoriality of precomposition and uniqueness in the representing property. □

5.2 Support categories and projection-pair systems

Projection-pair systems indexed by an enriched category use global support arrows for the actual transition projection pairs. The remaining enriched hom-structure enters through transition-strength morphisms.

Definition 5.2.1 (Support category of an enriched index). *Let A be a small $\mathbb{V}$ -category. Its support category*

$$A_0$$

is the ordinary category with the same objects as A , and hom-set

$$A_0(a, a') = \mathbb{V}(I, A(a, a')).$$

If

$$u : I \rightarrow A(a, a') \quad \text{and} \quad v : I \rightarrow A(a', a'')$$

are arrows in A_0 , their composite is the global element

$$I \cong I \otimes I \xrightarrow{v \otimes u} A(a', a'') \otimes A(a, a') \longrightarrow A(a, a'')$$

induced by composition in A . Identities are the enriched identity maps

$$I \rightarrow A(a, a).$$

In the posetal specialization, $A_0(a, a')$ has a unique arrow precisely when

$$A(a, a') = \top.$$

Lemma 5.2.1 (Support projections from an enriched tensor product). *Let A and B be small $\mathbb{V}$ -categories. There is a canonical ordinary functor*

$$(A \otimes B)_0 \rightarrow A_0 \times B_0.$$

On objects it sends

$$(a, b) \mapsto (a, b).$$

On arrows, a global element

$$\xi : I \rightarrow A(a, a') \otimes B(b, b')$$

is sent to the pair

$$\xi_A : I \rightarrow A(a, a'), \quad \xi_B : I \rightarrow B(b, b'),$$

where ξ_A is obtained by composing ξ with

$$A(a, a') \otimes B(b, b') \xrightarrow{1 \otimes !} A(a, a') \otimes I \cong A(a, a'),$$

and ξ_B is obtained similarly using

$$A(a, a') \otimes B(b, b') \xrightarrow{! \otimes 1} I \otimes B(b, b') \cong B(b, b').$$

Proof. The terminality of I gives the weakening maps

$$B(b, b') \rightarrow I, \quad A(a, a') \rightarrow I.$$

Thus the two displayed projections are canonical.

Compatibility with identities is immediate from the unit constraints. For composition, the composition in $A \otimes B$ is induced by the tensor of the compositions in A and B . Applying the two weakening projections sends this composite to the ordinary composites in A_0 and B_0 . Hence the assignment defines an ordinary functor. $\square$

Definition 5.2.2 (The locally ordinary 2-category of fuzzy domains). *Let*

$$\mathbf{FDom}_{\mathbb{V}}^{(2)}$$

be the locally ordinary 2-category defined as follows.

Its objects are fuzzy domains. Its 1-cells are fuzzy-domain morphisms. For fuzzy domains X, Y , the hom-category is the support category of the internal-hom fuzzy domain:

$$\mathbf{FDom}_{\mathbb{V}}^{(2)}(X, Y) = ([X, Y]_{\text{ff}})_0.$$

Thus a 2-cell

$$\alpha : f \Rightarrow g$$

is a global element

$$I \rightarrow [X, Y]_{\text{ff}}(f, g).$$

Horizontal composition is induced by the finite-flat-continuous composition bimorphism

$$[Y, Z]_{\text{ff}} \boxtimes_{\text{ff}} [X, Y]_{\text{ff}} \longrightarrow [X, Z]_{\text{ff}}.$$

Vertical composition is inherited from the support categories of the internal homs.

Definition 5.2.3 (Fuzzy projection pair). *Let X and Y be fuzzy domains. A fuzzy projection pair*

$$(e, p) : X \triangleleft Y$$

consists of fuzzy-domain morphisms

$$e : X \rightarrow Y, \quad p : Y \rightarrow X,$$

together with an adjunction

$$e \dashv p$$

in $\mathbf{FDom}_{\mathbb{V}}^{(2)}$, whose unit

$$\eta : \text{id}_X \Rightarrow pe$$

is invertible.

Equivalently, after choosing the inverse of the unit, one has

$$pe \cong \text{id}_X$$

and a counit comparison

$$ep \Rightarrow \text{id}_Y$$

satisfying the triangle identities. The morphism e is called the embedding and p the projection.

Lemma 5.2.2 (Composition of fuzzy projection pairs). *Fuzzy projection pairs compose. If*

$$(e, p) : X \triangleleft Y$$

and

$$(e', p') : Y \triangleleft Z$$

are fuzzy projection pairs, then

$$(e'e, pp') : X \triangleleft Z$$

is a fuzzy projection pair.

Proof. The composite of adjunctions

$$e \dashv p, \quad e' \dashv p'$$

is an adjunction

$$e'e \dashv pp'.$$

Its unit is the composite

$$\text{id}_X \Rightarrow pe \Rightarrow pp'e'e.$$

Both displayed unit maps are invertible, hence the composite unit is invertible. The counit and the triangle identities are those of the composite adjunction. $\square$

Definition 5.2.4 (Projection-pair 2-cells). *Let*

$$(e, p), (e', p') : X \triangleleft Y$$

be fuzzy projection pairs. A 2-cell

$$(\alpha, \beta) : (e, p) \Rightarrow (e', p')$$

consists of enriched natural transformations

$$\alpha : e \Rightarrow e', \quad \beta : p' \Rightarrow p$$

which are mates under the adjunctions

$$e \dashv p, \quad e' \dashv p'.$$

Definition 5.2.5 (Projection-pair 2-category). *The locally ordinary 2-category*

$$\mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

has fuzzy domains as objects, fuzzy projection pairs as 1-cells, and mate-compatible pairs of enriched natural transformations as 2-cells. The identity 1-cell on X is

$$(\text{id}_X, \text{id}_X) : X \triangleleft X,$$

and composition is

$$(e', p') \circ (e, p) = (e'e, pp').$$

Vertical and horizontal composition of 2-cells are inherited from $\mathbf{FDom}_{\mathbb{V}}^{(2)}$; mate-compatibility is preserved by the calculus of mates.

Definition 5.2.6 (Split A -indexed projection-pair system). *Let A be a small $\mathbb{V}$ -category. A split A -indexed projection-pair system*

$$D : A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

consists of:

1. *for every object $a \in A$, a fuzzy domain D_a ;*
2. *for every arrow*

$$u : a \rightarrow a'$$

in the support category A_0 , a fuzzy projection pair

$$(e_u, p_u) : D_a \triangleleft D_{a'};$$

3. *normality isomorphisms*

$$(e_{1_a}, p_{1_a}) \cong (\text{id}_{D_a}, \text{id}_{D_a});$$

4. *composition isomorphisms*

$$e_v e_u \cong e_{vu}, \quad p_u p_v \cong p_{vu}$$

for composable support arrows

$$a \xrightarrow{u} a' \xrightarrow{v} a'';$$

5. *the coherence equations for identities, composition, and associativity.*

If

$$h : A \rightarrow B$$

is a $\mathbb{V}$ -functor, it induces an ordinary functor

$$h_0 : A_0 \rightarrow B_0.$$

Thus a split B -indexed system

$$D : B \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

has a reindexing

$$Dh : A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

defined on objects by

$$(Dh)_a = D_{h_a}$$

and on support arrows by applying D to their images under h_0 .

Definition 5.2.7 (Compatible projection-pair cone). *Let*

$$D : A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be a split A -indexed projection-pair system, and let Z be a fuzzy domain.

A compatible projection-pair cone from D to Z consists of projection pairs

$$(\varepsilon_a, \pi_a) : D_a \triangleleft Z \quad (a \in A),$$

together with:

1. for every support arrow

$$u : a \rightarrow a'$$

in A_0 , coherent isomorphisms

$$\varepsilon_{a'} e_u \cong \varepsilon_a, \quad p_u \pi_{a'} \cong \pi_a;$$

2. writing

$$r_a = \varepsilon_a \pi_a : Z \rightarrow Z,$$

enriched transition-strength morphisms

$$\theta_{a,a'}^Z : A(a, a') \longrightarrow [Z, Z]_{\text{ff}}(r_a, r_{a'})$$

satisfying the identity and composition axioms for a $\mathbb{V}$ -functor

$$r : A \rightarrow [Z, Z]_{\text{ff}};$$

3. for every support arrow

$$u : I \rightarrow A(a, a'),$$

the composite

$$I \xrightarrow{u} A(a, a') \xrightarrow{\theta_{a,a'}^Z} [Z, Z]_{\text{ff}}(r_a, r_{a'})$$

is the canonical comparison

$$\varepsilon_a \pi_a \cong \varepsilon_{a'} e_u p_u \pi_{a'} \Rightarrow \varepsilon_{a'} \pi_{a'}$$

induced by the counit

$$e_u p_u \Rightarrow \text{id}_{D_{a'}}.$$

A morphism of compatible cones is a family of mate-compatible enriched natural transformations between the corresponding projection pairs, compatible with the support-arrow cone isomorphisms and with the transition-strength morphisms. The resulting category of compatible cones is denoted

$$\text{Cone}_{\triangleleft}(D; Z).$$

Lemma 5.2.3 (The approximant functor of a compatible cone). *Let*

$$D : A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be a split A -indexed projection-pair system, and let

$$(\varepsilon_a, \pi_a) : D_a \triangleleft Z$$

be a compatible projection-pair cone. Define

$$r_a = \varepsilon_a \pi_a : Z \rightarrow Z.$$

Then the objects r_a assemble canonically to a $\mathbb{V}$ -functor

$$r : A \rightarrow [Z, Z]_{\text{ff}}.$$

Proof. This is precisely the content of the transition-strength data in Definition 5.2.7. □

5.3 Canonical tails and common successors

The ordinary ω -chain construction uses two elementary cofinality facts: every tail of ω is cofinal, and any two indices have a common later index. In the finite-flat enriched setting these facts are represented by canonical support-comma constructions.

5.3.1 Canonical tails

Definition 5.3.1 (Canonical support tail). *Let A be a small $\mathbb{V}$ -category and let $a_0 \in A$. The canonical support tail of A at a_0 is the enriched category*

$$\text{Tail}_A(a_0)$$

defined as follows.

Its objects are pairs

$$(a, u), \quad u : a_0 \rightarrow a$$

where u is an arrow in the support category A_0 .

The hom-object from

$$(a, u)$$

to

$$(a', u')$$

is the equalizer in $\mathbb{V}$

$$\text{Tail}_A(a_0)((a, u), (a', u')) \longrightarrow A(a, a') \rightrightarrows A(a_0, a').$$

The first parallel map is composition with the global element

$$u : I \rightarrow A(a_0, a).$$

The second is the constant map determined by the global element

$$u' : I \rightarrow A(a_0, a'),$$

using the terminality of I . Composition and identities are inherited from A , by the universal property of the equalizers.

The evident projection

$$\tau_{a_0} : \text{Tail}_A(a_0) \rightarrow A, \quad (a, u) \longmapsto a,$$

is a $\mathbb{V}$ -functor.

In the crisp preorder case, $\text{Tail}_A(a_0)$ is the full tail suborder on the elements a satisfying $a_0 \leq a$.

Definition 5.3.2 (Canonical tail weight). *Let*

$$(A, W)$$

be a finite-flat approximation shape and let $a_0 \in A$. The canonical tail weight at a_0 is

$$W^{a_0} = \tau_{a_0}^* W : \text{Tail}_A(a_0)^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

Thus

$$W^{a_0}(a, u) = W(a).$$

Definition 5.3.3 (Tail-cofinal approximation shape). *A finite-flat approximation shape*

$$(A, W)$$

is tail-cofinal if, for every object $a_0 \in A$, the canonical tail weight

$$W^{a_0} = \tau_{a_0}^* W$$

is finite-flat and the canonical direct-image comparison is an isomorphism:

$$(\tau_{a_0})_! W^{a_0} \cong W.$$

Lemma 5.3.1 (Constant diagrams over canonical tails). *Let*

$$(A, W)$$

be tail-cofinal, and let $a_0 \in A$. For every fuzzy domain X and every object $x \in X$,

$$W^{a_0} * \Delta x \cong x.$$

Proof. By tail-cofinality, W^{a_0} is finite-flat. Hence the result is Lemma 5.1.1 applied to

$$(\text{Tail}_A(a_0), W^{a_0}).$$

□

5.3.2 Canonical common successors

Definition 5.3.4 (Canonical common-successor category). *Let P be a small $\mathbb{V}$ -category. The canonical common-successor category*

$$\text{Com}(P)$$

is defined as follows.

An object of $\text{Com}(P)$ is a tuple

$$(i, j, k, u, v),$$

where $i, j, k \in P$, and

$$u : i \rightarrow k, \quad v : j \rightarrow k$$

are arrows in the support category P_0 .

For two objects

$$c = (i, j, k, u, v), \quad c' = (i', j', k', u', v'),$$

the hom-object

$$\text{Com}(P)(c, c')$$

is the equalizer in $\mathbb{V}$ of the compatibility maps expressing

$$u' r = t u, \quad v' s = t v$$

for triples

$$r : i \rightarrow i', \quad s : j \rightarrow j', \quad t : k \rightarrow k'.$$

Explicitly, it is the equalizer of the two pairs of maps from

$$P(i, i') \otimes P(j, j') \otimes P(k, k')$$

to

$$P(i, k') \times P(j, k')$$

obtained respectively from the composites

$$u'r, \quad tu, \quad v's, \quad tv.$$

Composition and identities are inherited from those of P .

There are canonical $\mathbb{V}$ -functors

$$c_P : \text{Com}(P) \rightarrow P \otimes P, \quad m_P : \text{Com}(P) \rightarrow P$$

defined on objects by

$$c_P(i, j, k, u, v) = (i, j), \quad m_P(i, j, k, u, v) = k.$$

In the crisp preorder case, an object of $\text{Com}(P)$ is exactly a common upper bound

$$i \leq k, \quad j \leq k.$$

Definition 5.3.5 (Canonical common-successor weight). *Let*

$$(P, R)$$

be a finite-flat approximation shape. The canonical common-successor weight is

$$R^{\text{com}} = m_P^* R : \text{Com}(P)^{\text{op}} \rightarrow \underline{\mathbb{V}}.$$

Thus, on objects,

$$R^{\text{com}}(i, j, k, u, v) = R(k).$$

Definition 5.3.6 (Common-successor-cofinal approximation shape). *A finite-flat approximation shape*

$$(P, R)$$

is common-successor-cofinal if the canonical common-successor weight

$$R^{\text{com}} = m_P^* R$$

is finite-flat and the canonical maps

$$c_P : \text{Com}(P) \rightarrow P \otimes P, \quad m_P : \text{Com}(P) \rightarrow P$$

satisfy

$$(c_P)_! R^{\text{com}} \cong R \boxtimes R, \quad (m_P)_! R^{\text{com}} \cong R.$$

Thus common-successor-cofinality is the finite-flat analogue of directedness: double approximation indexed by $P \otimes P$ can be compared with approximation over canonical common successors, and then with single approximation over P .

Definition 5.3.7 (Linear-cofinal finite-flat approximation shape). *A finite-flat approximation shape*

$$(A, W)$$

is linear-cofinal if:

1. it is tail-cofinal;
2. its external product

$$(A \otimes A, W \boxtimes W)$$

is common-successor-cofinal.

This deliberately does not include a diagonal comparison

$$A \rightarrow A \otimes A.$$

The diagonal is a cartesian feature and appears only in the crisp separated collapse.

5.4 Weight-cofinal iteration shapes

Definition 5.4.1 (Weight-cofinal iteration shape). *A weight-cofinal iteration shape is a triple*

$$(A, W, s),$$

where (A, W) is a finite-flat approximation shape and

$$s : A \rightarrow A$$

is a W -cofinal endofunctor:

$$s_! W \cong W.$$

Lemma 5.4.1 (Shift invariance for weighted approximation). *Let*

$$(A, W, s)$$

be a weight-cofinal iteration shape. For every fuzzy domain X and every diagram

$$D : A \rightarrow X,$$

there is a canonical isomorphism

$$W * (Ds) \cong W * D.$$

Proof. Apply Proposition 5.1.1 to the W -cofinal map

$$s : (A, W) \rightarrow (A, W).$$

□

5.5 Finite-flat cofinal bilimits

Definition 5.5.1 (*W-bilimit cone*). *Let*

$$(A, W)$$

be a finite-flat approximation shape, and let

$$D : A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be a split A -indexed projection-pair system.

A W -bilimit cone over D consists of a fuzzy domain

$$D_{\infty}$$

and a compatible projection-pair cone

$$(\varepsilon_a, \pi_a) : D_a \triangleleft D_{\infty} \quad (a \in A),$$

such that:

1. *the cone represents compatible projection-pair cones: for every fuzzy domain Z , composition with the cone induces an equivalence*

$$\mathbf{FDom}_{\mathbb{V}}^{\triangleleft}(D_{\infty}, Z) \simeq \text{Cone}_{\triangleleft}(D; Z);$$

2. *if*

$$r : A \rightarrow [D_{\infty}, D_{\infty}]_{\text{ff}}$$

is the approximant functor

$$r_a = \varepsilon_a \pi_a,$$

then the finite-flat approximation identity holds:

$$W * r \cong \text{id}_{D_{\infty}}$$

in

$$[D_{\infty}, D_{\infty}]_{\text{ff}}.$$

Lemma 5.5.1 (Uniqueness of W -bilimits). *Any two W -bilimit cones over the same split projection-pair system are equivalent, and the equivalence is unique up to invertible compatible 2-cell.*

Proof. This is the uniqueness part of the representing equivalence in Definition 5.5.1. The approximation identity is transported along the resulting compatible equivalence because it is expressed in the internal hom of endomorphisms. $\square$

Definition 5.5.2 (*Bilimit-cofinal reindexing*). *Let*

$$h : (A, W) \rightarrow (B, U)$$

be a cofinal map of finite-flat approximation shapes, and let

$$D : B \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be a split B -indexed projection-pair system.

We say that h is bilimit-cofinal for D if restriction along h induces, for every fuzzy domain Z , an equivalence of compatible cone categories

$$\mathbf{Cone}_{\triangleleft}(D; Z) \simeq \mathbf{Cone}_{\triangleleft}(Dh; Z).$$

Under this equivalence, the derived approximant functor for a reindexed cone is the precomposition of the original derived approximant functor with h .

Lemma 5.5.2 (Cofinal reindexing of W -bilimits). *Let*

$$h : (A, W) \rightarrow (B, U)$$

be cofinal and bilimit-cofinal for

$$D : B \rightsquigarrow \mathbf{FDom}_{\nabla}^{\triangleleft}.$$

Then D has a U -bilimit with vertex D_{∞} if and only if the reindexed system

$$Dh : A \rightsquigarrow \mathbf{FDom}_{\nabla}^{\triangleleft}$$

has a W -bilimit with the corresponding transported cone on the same object D_{∞} .

Proof. The projection-pair universal property is transported along the equivalence of cone categories.

It remains to compare approximation identities. Let

$$r : B \rightarrow [D_{\infty}, D_{\infty}]_{\text{ff}}$$

be the approximant functor for the B -indexed cone. The approximant functor for the reindexed cone is

$$rh : A \rightarrow [D_{\infty}, D_{\infty}]_{\text{ff}}.$$

Since

$$h_! W \cong U,$$

Proposition 5.1.1 gives

$$W * (rh) \cong U * r.$$

Thus

$$U * r \cong \text{id}_{D_{\infty}}$$

if and only if

$$W * (rh) \cong \text{id}_{D_{\infty}}.$$

□

Theorem 5.5.1 (Shifted bilimits). *Let*

$$(A, W, s)$$

be a weight-cofinal iteration shape. Let

$$D : A \rightsquigarrow \mathbf{FDom}_{\nabla}^{\triangleleft}$$

be a split A -indexed projection-pair system with W -bilimit D_{∞} . If

$$s : (A, W) \rightarrow (A, W)$$

is bilimit-cofinal for D , then D_{∞} is also a W -bilimit of the shifted system

$$Ds : A \rightsquigarrow \mathbf{FDom}_{\nabla}^{\triangleleft}.$$

Proof. Apply Lemma 5.5.2 to the map

$$s : (A, W) \rightarrow (A, W).$$

Equivalently, if

$$r : A \rightarrow [D_\infty, D_\infty]_{\text{ff}}$$

is the approximant functor for the original cone, then the approximant functor for the shifted cone is

$$rs : A \rightarrow [D_\infty, D_\infty]_{\text{ff}},$$

and

$$W * (rs) \cong W * r \cong \text{id}_{D_\infty}$$

by Lemma 5.4.1. □

5.6 Recursive domain equations by cofinal approximation

Definition 5.6.1 (*W*-bilimit-preserving endopseudofunctor). *Let*

$$(A, W)$$

be a finite-flat approximation shape. A projection-pair endopseudofunctor

$$F : \mathbf{FDom}_V^\triangleleft \rightarrow \mathbf{FDom}_V^\triangleleft$$

is W-bilimit-preserving if it sends W-bilimit cones of split A-indexed projection-pair systems to W-bilimit cones.

Theorem 5.6.1 (Finite-flat cofinal recursive-domain-equation theorem). *Let*

$$(A, W, s)$$

be a weight-cofinal iteration shape. Let

$$F : \mathbf{FDom}_V^\triangleleft \rightarrow \mathbf{FDom}_V^\triangleleft$$

be a W-bilimit-preserving projection-pair endopseudofunctor.

Let

$$D : A \rightsquigarrow \mathbf{FDom}_V^\triangleleft$$

be a split A-indexed projection-pair system equipped with an isomorphism of split projection-pair systems

$$Ds \cong FD.$$

If s is bilimit-cofinal for D and D_∞ is a W-bilimit of D, then there is an equivalence of fuzzy domains

$$D_\infty \simeq F(D_\infty).$$

Proof. By Theorem 5.5.1, D_∞ is a W-bilimit of the shifted system Ds . Since

$$Ds \cong FD$$

as split projection-pair systems, D_∞ is also a W-bilimit of FD .

Since F preserves W-bilimits,

$$F(D_\infty)$$

is also a W-bilimit of FD . By Lemma 5.5.1,

$$D_\infty \simeq F(D_\infty).$$

□

5.7 Composition and external-product internal-hom approximation

Lemma 5.7.1 (Separate continuity of composition). *For fuzzy domains A, B, C , composition determines a finite-flat bimorphism*

$$[B, C]_{\text{ff}} \otimes [A, B]_{\text{ff}} \longrightarrow [A, C]_{\text{ff}}, \quad (g, f) \longmapsto gf.$$

Equivalently, it induces a fuzzy-domain morphism

$$[B, C]_{\text{ff}} \boxtimes_{\text{ff}} [A, B]_{\text{ff}} \longrightarrow [A, C]_{\text{ff}}.$$

Proof. Finite-flat colimits in internal homs are computed pointwise.

Fix

$$f : A \rightarrow B.$$

For a finite-flat weight

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

and a diagram

$$H : K \rightarrow [B, C]_{\text{ff}},$$

one has pointwise

$$\begin{aligned} (W * H) \circ f = a &\longmapsto (W * H)(fa) \\ &\cong a \longmapsto W * (k \mapsto Hk(fa)) \\ &\cong W * (k \mapsto Hk \circ f). \end{aligned}$$

Hence precomposition by f preserves finite-flat colimits.

Fix

$$g : B \rightarrow C.$$

Since g is a fuzzy-domain morphism, it preserves finite-flat colimits in B . Therefore, for every finite-flat

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

and every diagram

$$H : K \rightarrow [A, B]_{\text{ff}},$$

one has pointwise

$$g(W * (k \mapsto Hk(a))) \cong W * (k \mapsto g(Hk(a))).$$

Thus postcomposition by g preserves finite-flat colimits. Therefore composition is finite-flat-continuous separately in both variables. $\square$

Let

$$X : A \rightsquigarrow \mathbf{FDom}_{\underline{\mathbb{V}}}^{\triangleleft}, \quad Y : A \rightsquigarrow \mathbf{FDom}_{\underline{\mathbb{V}}}^{\triangleleft}$$

be split A -indexed projection-pair systems. Their associated *external-product internal-hom system*

$$[X, Y]_{\text{ext}} : A \otimes A \rightsquigarrow \mathbf{FDom}_{\underline{\mathbb{V}}}^{\triangleleft}$$

is defined on objects by

$$[X, Y]_{\text{ext}}(a, a') = [X_{a'}, Y_a]_{\text{ff}}.$$

For a support arrow

$$\xi : (a, a') \rightarrow (b, b')$$

in $(A \otimes A)_0$, let

$$\xi_1 : a \rightarrow b, \quad \xi_2 : a' \rightarrow b'$$

be its two support projections. The transition projection pair

$$[X_{a'}, Y_a]_{\text{ff}} \triangleleft [X_{b'}, Y_b]_{\text{ff}}$$

has embedding

$$E_\xi(f) = e_{\xi_1}^Y f p_{\xi_2}^X,$$

and projection

$$P_\xi(g) = p_{\xi_1}^Y g e_{\xi_2}^X.$$

Lemma 5.7.2 (Internal-hom transitions are projection pairs). *For every support arrow ξ in $(A \otimes A)_0$, the pair*

$$(E_\xi, P_\xi)$$

defined above is a fuzzy projection pair.

Proof. The two maps preserve finite-flat colimits by Lemma 5.7.1.

For

$$f : X_{a'} \rightarrow Y_a,$$

one has

$$\begin{aligned} P_\xi E_\xi(f) &= p_{\xi_1}^Y e_{\xi_1}^Y f p_{\xi_2}^X e_{\xi_2}^X \\ &\cong f, \end{aligned}$$

using the unit isomorphisms of the projection pairs for X and Y .

For

$$g : X_{b'} \rightarrow Y_b,$$

one has

$$E_\xi P_\xi(g) = e_{\xi_1}^Y p_{\xi_1}^Y g e_{\xi_2}^X p_{\xi_2}^X.$$

The counit comparisons

$$e_{\xi_1}^Y p_{\xi_1}^Y \Rightarrow \text{id}_{Y_b}, \quad e_{\xi_2}^X p_{\xi_2}^X \Rightarrow \text{id}_{X_{b'}}$$

induce

$$E_\xi P_\xi(g) \Rightarrow g.$$

The triangle identities follow from the triangle identities of the projection pairs in X and Y . $\square$

Now suppose

$$(\varepsilon_a^X, \pi_a^X) : X_a \triangleleft X_\infty, \quad (\varepsilon_a^Y, \pi_a^Y) : Y_a \triangleleft Y_\infty$$

are compatible projection-pair cones. Define

$$E_{a,a'} : [X_{a'}, Y_a]_{\text{ff}} \rightarrow [X_\infty, Y_\infty]_{\text{ff}}$$

and

$$\Pi_{a,a'} : [X_\infty, Y_\infty]_{\text{ff}} \rightarrow [X_{a'}, Y_a]_{\text{ff}}$$

by

$$E_{a,a'}(f) = \varepsilon_a^Y f \pi_{a'}^X,$$

and

$$\Pi_{a,a'}(g) = \pi_a^Y g \varepsilon_{a'}^X.$$

Lemma 5.7.3 (Internal-hom cone components are projection pairs). *For every a, a' , the pair*

$$(E_{a,a'}, \Pi_{a,a'}) : [X_{a'}, Y_a]_{\text{ff}} \triangleleft [X_\infty, Y_\infty]_{\text{ff}}$$

is a fuzzy projection pair.

Proof. The maps $E_{a,a'}$ and $\Pi_{a,a'}$ preserve finite-flat colimits by Lemma 5.7.1.

For $f : X_{a'} \rightarrow Y_a$,

$$\begin{aligned} \Pi_{a,a'} E_{a,a'}(f) &= \pi_a^Y \varepsilon_a^Y f \pi_{a'}^X \varepsilon_{a'}^X \\ &\cong f, \end{aligned}$$

using the unit isomorphisms of the projection pairs in the two bilimit cones.

For $g : X_\infty \rightarrow Y_\infty$,

$$E_{a,a'} \Pi_{a,a'}(g) = \varepsilon_a^Y \pi_a^Y g \varepsilon_{a'}^X \pi_{a'}^X.$$

The counit comparisons

$$\varepsilon_a^Y \pi_a^Y \Rightarrow \text{id}_{Y_\infty}, \quad \varepsilon_{a'}^X \pi_{a'}^X \Rightarrow \text{id}_{X_\infty}$$

induce

$$E_{a,a'} \Pi_{a,a'}(g) \Rightarrow g.$$

The triangle identities follow from the triangle identities of the two projection-pair cones. $\square$

Theorem 5.7.1 (External-product internal-hom approximation theorem). *Let*

$$(A, W)$$

be a finite-flat approximation shape. Suppose

$$X_\infty$$

and

$$Y_\infty$$

are W -bilimits of split A -indexed projection-pair systems

$$X \quad \text{and} \quad Y.$$

Then the external-product internal-hom cone

$$(E_{a,a'}, \Pi_{a,a'}) : [X_{a'}, Y_a]_{\text{ff}} \triangleleft [X_\infty, Y_\infty]_{\text{ff}}$$

has approximants satisfying

$$(W \boxtimes W) * ((a, a') \mapsto E_{a,a'} \Pi_{a,a'}) \cong \text{id}_{[X_\infty, Y_\infty]_{\text{ff}}}.$$

Proof. Let

$$x_{a'} = \varepsilon_{a'}^X \pi_{a'}^X \in [X_\infty, X_\infty]_{\text{ff}}, \quad y_a = \varepsilon_a^Y \pi_a^Y \in [Y_\infty, Y_\infty]_{\text{ff}}.$$

Then

$$E_{a,a'} \Pi_{a,a'}(g) = y_a g x_{a'}.$$

By separate finite-flat-continuity of composition and finite-flat Fubini, the $(W \boxtimes W)$ -weighted colimit of these endofunctors is

$$g \mapsto (W * (a \mapsto y_a)) g (W * (a' \mapsto x_{a'})).$$

Since X_∞ and Y_∞ are W -bilimits,

$$W * (a' \mapsto x_{a'}) \cong \text{id}_{X_\infty}, \quad W * (a \mapsto y_a) \cong \text{id}_{Y_\infty}.$$

Thus the displayed endofunctor is naturally isomorphic to the identity on

$$[X_\infty, Y_\infty]_{\text{ff}}.$$

□

5.8 Canonical common-successor comparisons

The general enriched setting permits canonical comparison maps from double approximations to single approximations. These comparisons are not generally invertible; this is the precise point at which the absence of contraction matters.

Let

$$(P, R)$$

be common-successor-cofinal. Let

$$H : P \rightsquigarrow \mathbf{FDom}_V^{\triangleleft}$$

be a split P -indexed projection-pair system. Let

$$(E_i, \Pi_i) : H_i \triangleleft H_\infty \quad (i \in P)$$

be a compatible cone, and let

$$(\delta_i, \rho_i) : H_i \triangleleft Z \quad (i \in P)$$

be another compatible cone.

Write

$$L_i = \delta_i \Pi_i : H_\infty \rightarrow Z, \quad M_i = E_i \rho_i : Z \rightarrow H_\infty.$$

Lemma 5.8.1 (Canonical tail restrictions). *Assume (P, R) is tail-cofinal. For every $i \in P$,*

$$R * (j \mapsto L_j E_i) \cong \delta_i,$$

and

$$R * (j \mapsto \Pi_i M_j) \cong \rho_i.$$

Proof. Let

$$\tau_i : \text{Tail}_P(i) \rightarrow P$$

be the canonical support tail. Since (P, R) is tail-cofinal,

$$(\tau_i)_! R^i \cong R.$$

Therefore, by Proposition 5.1.1,

$$R * (j \mapsto L_j E_i) \cong R^i * ((j, u) \mapsto L_j E_i).$$

For an object $(j, u : i \rightarrow j)$ of $\text{Tail}_P(i)$, cone compatibility gives

$$E_j e_u \cong E_i, \quad p_u \Pi_j \cong \Pi_i, \quad \delta_j e_u \cong \delta_i.$$

Hence

$$\begin{aligned} L_j E_i &= \delta_j \Pi_j E_i \\ &\cong \delta_j \Pi_j E_j e_u \\ &\cong \delta_j e_u \\ &\cong \delta_i. \end{aligned}$$

Thus the restricted diagram is constant at δ_i , and

$$R^i * ((j, u) \mapsto L_j E_i) \cong R^i * \Delta(\delta_i) \cong \delta_i$$

by Lemma 5.3.1.

The proof of

$$R * (j \mapsto \Pi_i M_j) \cong \rho_i$$

is analogous. For $(j, u : i \rightarrow j)$, cone compatibility gives

$$p_u \rho_j \cong \rho_i$$

and

$$\begin{aligned} \Pi_i M_j &= \Pi_i E_j \rho_j \\ &\cong p_u \Pi_j E_j \rho_j \\ &\cong p_u \rho_j \\ &\cong \rho_i. \end{aligned}$$

□

Lemma 5.8.2 (Canonical double-to-single comparison). *Assume (P, R) is common-successor-cofinal. Then there is a canonical counit comparison*

$$(R \boxtimes R) * ((i, j) \mapsto M_i L_j) \Rightarrow R * (k \mapsto E_k \Pi_k).$$

Moreover, there is a canonical counit comparison

$$(R \boxtimes R) * ((i, j) \mapsto L_i M_j) \Rightarrow \text{id}_Z.$$

Proof. Let

$$R^{\text{com}} = m_P^* R$$

be the canonical common-successor weight. By common-successor-cofinality,

$$(c_P)_! R^{\text{com}} \cong R \boxtimes R, \quad (m_P)_! R^{\text{com}} \cong R.$$

Hence

$$(R \boxtimes R) * ((i, j) \mapsto M_i L_j) \cong R^{\text{com}} * ((i, j, k, u, v) \mapsto M_i L_j).$$

For an object

$$(i, j, k, u : i \rightarrow k, v : j \rightarrow k)$$

of $\text{Com}(P)$, cone compatibility gives

$$E_k e_u \cong E_i, \quad p_u \rho_k \cong \rho_i,$$

and

$$\delta_k e_v \cong \delta_j, \quad p_v \Pi_k \cong \Pi_j.$$

Therefore

$$\begin{aligned}
 M_i L_j &= E_i \rho_i \delta_j \Pi_j \\
 &\cong E_k e_u p_u \rho_k \delta_k e_v p_v \Pi_k \\
 &\cong E_k e_u p_u e_v p_v \Pi_k \\
 &\Rightarrow E_k \Pi_k.
 \end{aligned}$$

The final comparison is induced by the counits

$$e_u p_u \Rightarrow \text{id}_{H_k}, \quad e_v p_v \Rightarrow \text{id}_{H_k}.$$

These comparisons are coherent in $\text{Com}(P)$. Taking the R^{com} -weighted colimit and using

$$(m_P)_! R^{\text{com}} \cong R$$

gives the first comparison.

For the second comparison, again use cofinality of c_P . For an object

$$(i, j, k, u : i \rightarrow k, v : j \rightarrow k)$$

there is a canonical comparison

$$L_i M_j \Rightarrow \text{id}_Z.$$

Indeed,

$$\begin{aligned}
 L_i M_j &= \delta_i \Pi_i E_j \rho_j \\
 &\cong \delta_k e_u p_u \Pi_k E_k e_v p_v \rho_k \\
 &\Rightarrow \delta_k \rho_k \\
 &\Rightarrow \text{id}_Z.
 \end{aligned}$$

The final comparison is the counit of

$$(\delta_k, \rho_k) : H_k \triangleleft Z.$$

Taking the R^{com} -weighted colimit and using Lemma 5.1.1 for the constant diagram at id_Z gives the second comparison. $\square$

Theorem 5.8.1 (Lax reconstruction of external-product cones). *Let*

$$(A, W)$$

be linear-cofinal. Put

$$P = A \otimes A, \quad R = W \boxtimes W.$$

Let

$$H_\infty = [X_\infty, Y_\infty]_{\text{ff}}$$

with external-product approximants

$$(E_i, \Pi_i) : H_i \triangleleft H_\infty \quad (i \in P).$$

Let

$$(\delta_i, \rho_i) : H_i \triangleleft Z$$

be a compatible cone. Define

$$\delta = R * (i \mapsto \delta_i \Pi_i) : H_\infty \rightarrow Z$$

and

$$\rho = R * (i \mapsto E_i \rho_i) : Z \rightarrow H_\infty.$$

Then

$$\delta E_i \cong \delta_i, \quad \Pi_i \rho \cong \rho_i,$$

and there are canonical comparisons

$$\rho \delta \Rightarrow \text{id}_{H_\infty}, \quad \delta \rho \Rightarrow \text{id}_Z.$$

Proof. The restriction equations follow from Lemma 5.8.1.

For the first comparison, separate finite-flat-continuity of composition and finite-flat Fubini give

$$\rho \delta \cong (R \boxtimes R) * ((i, j) \mapsto M_i L_j).$$

By Lemma 5.8.2,

$$(R \boxtimes R) * ((i, j) \mapsto M_i L_j) \Rightarrow R * (k \mapsto E_k \Pi_k).$$

By Theorem 5.7.1,

$$R * (k \mapsto E_k \Pi_k) \cong \text{id}_{H_\infty}.$$

Thus

$$\rho \delta \Rightarrow \text{id}_{H_\infty}.$$

The comparison

$$\delta \rho \Rightarrow \text{id}_Z$$

is the second comparison of Lemma 5.8.2. □

5.9 Linear application in the general finite-flat setting

The preceding theorem reconstructs canonical lax application data from external-product finite-stage cones. It does not produce a projection-pair retract in full generality; the missing ingredient is precisely a diagonal or contraction comparison.

Definition 5.9.1 (Linear reflexive fuzzy domain). *A linear reflexive fuzzy domain is a fuzzy domain D equipped with a fuzzy projection pair*

$$[D, D]_{\text{ff}} \triangleleft D.$$

Equivalently, it consists of fuzzy-domain morphisms

$$\text{encode} : [D, D]_{\text{ff}} \rightarrow D, \quad \text{decode} : D \rightarrow [D, D]_{\text{ff}},$$

together with an adjunction

$$\text{encode} \dashv \text{decode}$$

whose unit is invertible. Thus

$$\text{decode encode} \cong \text{id}_{[D, D]_{\text{ff}}},$$

and there is a counit comparison

$$\text{encode decode} \Rightarrow \text{id}_D$$

satisfying the triangle identities.

It is extensional if the counit is also invertible.

The associated application morphism is

$$D \boxtimes_{\text{ff}} D \xrightarrow{\text{decode} \boxtimes 1_D} [D, D]_{\text{ff}} \boxtimes_{\text{ff}} D \xrightarrow{\text{ev}} D.$$

Definition 5.9.2 ((G, T)-linear reflexive fuzzy domain). *Let*

$$G, T : \mathbf{FDom}_{\mathbb{V}}^{\triangleleft} \rightarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be projection-pair endopseudofunctors. A (G, T)-linear reflexive fuzzy domain is a fuzzy domain D equipped with a fuzzy projection pair

$$[GD, TD]_{\text{ff}} \triangleleft D.$$

The associated application morphism has type

$$D \boxtimes_{\text{ff}} GD \rightarrow TD$$

and is the composite

$$D \boxtimes_{\text{ff}} GD \xrightarrow{\text{decode} \boxtimes 1_{GD}} [GD, TD]_{\text{ff}} \boxtimes_{\text{ff}} GD \xrightarrow{\text{ev}} TD.$$

Definition 5.9.3 (External finite-stage endomorphism system). *Let*

$$D : A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be a split A -indexed projection-pair system. Its external finite-stage endomorphism system is the split $(A \otimes A)$ -indexed projection-pair system

$$\text{End}_{\text{ext}}(D) : A \otimes A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

defined by

$$\text{End}_{\text{ext}}(D)(a, a') = [D_{a'}, D_a]_{\text{ff}}.$$

Definition 5.9.4 (Linear self-application cone). *Let*

$$(A, W)$$

be linear-cofinal, and let

$$D : A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be a split A -indexed projection-pair system with W -bilimit

$$D_{\infty}.$$

A linear self-application cone on D is a compatible projection-pair cone

$$(\kappa_{a, a'}, \lambda_{a, a'}) : [D_{a'}, D_a]_{\text{ff}} \triangleleft D_{\infty}$$

over the external finite-stage endomorphism system

$$\text{End}_{\text{ext}}(D).$$

Theorem 5.9.1 (Pure lax linear application theorem). *Let*

$$(A, W)$$

be linear-cofinal, and suppose

$$D_\infty$$

is a W -bilimit of a split A -indexed projection-pair system D . Let

$$H_\infty = [D_\infty, D_\infty]_{\text{ff}}.$$

If D carries a linear self-application cone, then there are canonical fuzzy-domain morphisms

$$\text{encode} : H_\infty \rightarrow D_\infty, \quad \text{decode} : D_\infty \rightarrow H_\infty$$

defined by

$$\text{encode} = (W \boxtimes W) * ((a, a') \mapsto \kappa_{a,a'} \Pi_{a,a'}),$$

and

$$\text{decode} = (W \boxtimes W) * ((a, a') \mapsto E_{a,a'} \lambda_{a,a'}).$$

They satisfy canonical comparisons

$$\text{decode} \text{ encode} \Rightarrow \text{id}_{H_\infty}, \quad \text{encode} \text{ decode} \Rightarrow \text{id}_{D_\infty}.$$

Consequently there is a canonical resource-sensitive application morphism

$$D_\infty \boxtimes_{\text{ff}} D_\infty \longrightarrow D_\infty.$$

Proof. Apply Theorem 5.8.1 to the external endomorphism system

$$(a, a') \longmapsto [D_{a'}, D_a]_{\text{ff}}$$

and to the compatible cone

$$(\kappa_{a,a'}, \lambda_{a,a'}) : [D_{a'}, D_a]_{\text{ff}} \triangleleft D_\infty.$$

The resulting maps are precisely the displayed encode and decode. The application morphism is

$$D_\infty \boxtimes_{\text{ff}} D_\infty \xrightarrow{\text{decode} \boxtimes 1} [D_\infty, D_\infty]_{\text{ff}} \boxtimes_{\text{ff}} D_\infty \xrightarrow{\text{ev}} D_\infty.$$

□

Definition 5.9.5 (Mixed external finite-stage function-space system). *Let*

$$G, T : \mathbf{FDom}_{\mathbb{V}}^{\triangleleft} \rightarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be projection-pair endopseudofunctors, and let

$$D : A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be a split A -indexed projection-pair system. The mixed external finite-stage function-space system is the split $(A \otimes A)$ -indexed system

$$(a, a') \longmapsto [GD_{a'}, TD_a]_{\text{ff}}.$$

Definition 5.9.6 (Mixed linear self-application cone). *Let*

$$(A, W)$$

be linear-cofinal. Let

$$G, T : \mathbf{FDom}_V^{\triangleleft} \rightarrow \mathbf{FDom}_V^{\triangleleft}$$

be projection-pair endopseudofunctors preserving W -bilimits. Let

$$D : A \rightsquigarrow \mathbf{FDom}_V^{\triangleleft}$$

be a split A -indexed projection-pair system with W -bilimit

$$D_{\infty}.$$

A mixed linear self-application cone is a compatible projection-pair cone

$$[GD_{a'}, TD_a]_{\text{ff}} \triangleleft D_{\infty}$$

over the mixed external finite-stage function-space system.

Theorem 5.9.2 (Mixed lax linear application theorem). *Let*

$$(A, W)$$

be linear-cofinal. Let

$$G, T : \mathbf{FDom}_V^{\triangleleft} \rightarrow \mathbf{FDom}_V^{\triangleleft}$$

be projection-pair endopseudofunctors preserving W -bilimits. Suppose

$$D_{\infty}$$

is a W -bilimit of a split A -indexed projection-pair system D . If D carries a mixed linear self-application cone, then there are canonical morphisms

$$\text{encode} : [GD_{\infty}, TD_{\infty}]_{\text{ff}} \rightarrow D_{\infty},$$

and

$$\text{decode} : D_{\infty} \rightarrow [GD_{\infty}, TD_{\infty}]_{\text{ff}},$$

with canonical comparisons

$$\text{decode encode} \Rightarrow \text{id}_{[GD_{\infty}, TD_{\infty}]_{\text{ff}}}, \quad \text{encode decode} \Rightarrow \text{id}_{D_{\infty}}.$$

The associated application morphism has type

$$D_{\infty} \boxtimes_{\text{ff}} GD_{\infty} \rightarrow TD_{\infty}.$$

Proof. Since G and T preserve W -bilimits,

$$GD_{\infty}$$

is a W -bilimit of GD , and

$$TD_{\infty}$$

is a W -bilimit of TD . Apply Theorem 5.8.1 to the mixed external finite-stage function-space system. $\square$

5.9.1 Finite-stage generation of self-application cones

The preceding lax application theorem requires a compatible cone from the finite-stage function-space system into the bilimit. The following construction supplies such data over the canonical common-successor category.

Definition 5.9.7 (Finite-stage linear self-application data). *Let*

$$(A, W)$$

be linear-cofinal, and let

$$D : A \rightsquigarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be a split A -indexed projection-pair system. Finite-stage linear self-application data consist of:

1. *a support endofunctor*

$$s : A \rightarrow A;$$

2. *for every $k \in A$, a fuzzy projection pair*

$$(\sigma_k, \tau_k) : [D_k, D_k]_{\text{ff}} \triangleleft D_{sk};$$

3. *coherence isomorphisms saying that, for every support arrow*

$$w : k \rightarrow k',$$

the square of projection pairs comparing

$$[D_k, D_k]_{\text{ff}} \triangleleft D_{sk}$$

with

$$[D_{k'}, D_{k'}]_{\text{ff}} \triangleleft D_{sk'}$$

is compatible with the internal-hom transition maps and with the transition projection pair

$$D_{sk} \triangleleft D_{sk'}.$$

Construction 5.9.1 (Common-successor finite-stage encoding). *Let*

$$(i, j, k, u : i \rightarrow k, v : j \rightarrow k)$$

be an object of $\text{Com}(A)$. The transition projection pairs in D induce a projection pair

$$[D_j, D_i]_{\text{ff}} \triangleleft [D_k, D_k]_{\text{ff}}.$$

Its embedding sends

$$f : D_j \rightarrow D_i$$

to

$$e_u f p_v : D_k \rightarrow D_k,$$

and its projection sends

$$g : D_k \rightarrow D_k$$

to

$$p_u g e_v : D_j \rightarrow D_i.$$

Composing with

$$[D_k, D_k]_{\text{ff}} \triangleleft D_{sk}$$

and then with the bilimit cone component

$$D_{sk} \triangleleft D_\infty$$

gives a projection pair

$$[D_j, D_i]_{\text{ff}} \triangleleft D_\infty.$$

The coherence axioms for finite-stage self-application data make these projection pairs compatible over $\text{Com}(A)$.

Theorem 5.9.3 (Common-successor finite-stage encodings over $\text{Com}(A)$). *Let*

$$(A, W)$$

be linear-cofinal, and let

$$D : A \rightsquigarrow \mathbf{FDom}_{\nabla}^{\triangleleft}$$

be a split A -indexed system with W -bilimit D_∞ . Suppose D is equipped with finite-stage linear self-application data. Then the construction above gives a compatible projection-pair cone over the pullback of the external endomorphism system along

$$c_A : \text{Com}(A) \rightarrow A \otimes A.$$

Proof. For each object

$$(i, j, k, u, v)$$

of $\text{Com}(A)$, Construction 5.9.1 gives a projection pair

$$[D_j, D_i]_{\text{ff}} \triangleleft D_\infty.$$

Compatibility with arrows of $\text{Com}(A)$ follows from the coherence of the split system D , the coherence of the finite-stage self-application data, and the calculus of mates. $\square$

5.10 Constant-map seeds

The following construction supplies standard finite-stage seeds.

Lemma 5.10.1 (Constant finite-flat-continuous maps). *Let X and Y be fuzzy domains, and let $y \in Y$. Since ∇ is affine, y determines a constant ∇ -functor*

$$\text{const}_y : X \rightarrow Y.$$

This constant functor preserves finite-flat weighted colimits.

Proof. The constant assignment is a $\mathbb{V}$ -functor because, for every $x, x' \in X$, affinity gives a canonical morphism

$$X(x, x') \rightarrow I,$$

and the identity of y gives

$$I \rightarrow Y(y, y).$$

Their composite supplies the hom-map

$$X(x, x') \rightarrow Y(y, y).$$

Let

$$W : K^{\text{op}} \rightarrow \underline{\mathbb{V}}$$

be finite-flat and let

$$D : K \rightarrow X$$

be a diagram. Since W is finite-flat, Lemma 5.1.1 gives

$$W * (\text{const}_y D) \cong y.$$

Therefore

$$\text{const}_y(W * D) = y \cong W * (\text{const}_y D).$$

Thus const_y preserves finite-flat weighted colimits. $\square$

Proposition 5.10.1 (Pure constant-map seed). *Suppose a fuzzy domain D_0 has an enriched initial object*

$$\perp \in D_0.$$

Define

$$e : D_0 \rightarrow [D_0, D_0]_{\text{ff}}$$

by

$$e(a) = \text{const}_a,$$

and define

$$p : [D_0, D_0]_{\text{ff}} \rightarrow D_0$$

by

$$p(f) = f(\perp).$$

Then

$$(e, p) : D_0 \triangleleft [D_0, D_0]_{\text{ff}}$$

is a fuzzy projection pair.

Proof. By Lemma 5.10.1, each const_a lies in $[D_0, D_0]_{\text{ff}}$. The assignment

$$a \longmapsto \text{const}_a$$

is a $\mathbb{V}$ -functor because the end defining the internal hom admits the evident pointwise wedge

$$D_0(a, a') \rightarrow D_0(a, a').$$

It preserves finite-flat colimits by Lemma 5.10.1 and Lemma 5.1.1.

The functor p is evaluation at $\perp$. Evaluation is a $\mathbb{V}$ -functor and preserves pointwise finite-flat colimits. Thus p is a fuzzy-domain morphism.

Moreover,

$$p(e(a)) = \text{const}_a(\perp) = a.$$

Thus

$$pe = \text{id}_{D_0}.$$

For $f : D_0 \rightarrow D_0$, the composite $ep(f)$ is

$$\text{const}_{f(\perp)}.$$

The enriched initiality of $\perp$ gives a canonical enriched natural transformation

$$\Delta \perp \Rightarrow 1_{D_0}.$$

Applying f yields

$$\Delta f(\perp) \Rightarrow f.$$

This is precisely

$$ep(f) \Rightarrow f.$$

The triangle identities follow from $pe = \text{id}_{D_0}$ and from functoriality of the canonical initial-object comparison. Therefore $e \dashv p$ with invertible unit. $\square$

Proposition 5.10.2 (Mixed constant-map seed). *Let*

$$G, T : \mathbf{FDom}_{\mathbb{V}}^{\triangleleft} \rightarrow \mathbf{FDom}_{\mathbb{V}}^{\triangleleft}$$

be projection-pair endopseudofunctors. Suppose GD_0 has an enriched initial object

$$\perp_{GD_0} \in GD_0,$$

and suppose there is a fuzzy projection pair

$$(u, v) : D_0 \triangleleft TD_0.$$

Define

$$e : D_0 \rightarrow [GD_0, TD_0]_{\text{ff}}$$

by

$$e(a) = \text{const}_{u(a)},$$

and define

$$p : [GD_0, TD_0]_{\text{ff}} \rightarrow D_0$$

by

$$p(f) = v(f(\perp_{GD_0})).$$

Then

$$(e, p) : D_0 \triangleleft [GD_0, TD_0]_{\text{ff}}$$

is a fuzzy projection pair.

Proof. Constant maps preserve finite-flat colimits by Lemma 5.10.1. Since u preserves finite-flat colimits, the functor

$$e(a) = \text{const}_{u(a)}$$

preserves finite-flat colimits.

The functor p is the composite of evaluation at $\perp_{GD_0}$ with

$$v : TD_0 \rightarrow D_0.$$

Evaluation preserves pointwise finite-flat colimits, and v is a fuzzy-domain morphism. Hence p is a fuzzy-domain morphism.

For $a \in D_0$,

$$p(e(a)) = v(u(a)) \cong a$$

by the unit isomorphism of the projection pair (u, v) .

For $f : GD_0 \rightarrow TD_0$, the composite $ep(f)$ is the constant map at

$$u(v(f(\perp_{GD_0}))).$$

The counit comparison

$$uv \Rightarrow \text{id}_{TD_0}$$

gives

$$u(v(f(\perp_{GD_0}))) \rightarrow f(\perp_{GD_0}).$$

The enriched initiality of $\perp_{GD_0}$ gives

$$\Delta \perp_{GD_0} \Rightarrow 1_{GD_0}.$$

Applying f gives

$$\Delta f(\perp_{GD_0}) \Rightarrow f.$$

Composing these transformations gives

$$ep(f) \Rightarrow f.$$

The triangle identities follow from the triangle identities of (u, v) and from functoriality of the canonical initial-object comparison. $\square$

5.11 The posetal reflection

Let

$$\Lambda = (L, \leq, \wedge, \vee, \otimes, \multimap, \perp_L, \top_L)$$

be a fuzzy truth-value object. In the posetal reflection, finite-flat approximation shapes are fuzzy ideal weights

$$W : A^{\text{op}} \rightarrow \Lambda.$$

For a Λ -functor

$$h : A \rightarrow B,$$

the lower direct image is computed by

$$(h_! W)(b) = \bigvee_{a \in A} W(a) \otimes B(b, ha).$$

Cofinality means

$$h_!W \cong U.$$

Since the truth-value fuzzy preorder Λ is separated, the presheaf fuzzy preorder

$$[B^{\text{op}}, \Lambda]$$

is separated. Hence an isomorphism of weights may be written as equality:

$$h_!W = U.$$

For every finite-flat weight W , the terminal direct image satisfies

$$!_!W \cong \top_L.$$

Equivalently,

$$\bigvee_{a \in A} W(a) = \top_L.$$

This is the posetal reflection of preservation of the terminal finite-limit predicate by the character χ_W .

A projection pair

$$(e, p) : X \triangleleft Y$$

is a pair of fuzzy-domain morphisms satisfying

$$pe \cong \text{id}_X, \quad ep \leq \text{id}_Y$$

in the induced crisp preorder on fuzzy-domain morphisms, where

$$f \leq g \quad :\Longleftrightarrow \quad [X, Y]_{\text{ff}}(f, g) = \top_L.$$

A W -bilimit cone has approximation identity

$$W * (a \mapsto \varepsilon_a \pi_a) \cong \text{id}_{D_\infty}$$

inside

$$[D_\infty, D_\infty]_{\text{ff}}.$$

The finite-flat cofinal recursive-domain-equation theorem therefore gives: if

$$Ds \cong FD,$$

if s is bilimit-cofinal for D , and if F preserves the corresponding W -bilimit, then

$$D_\infty \simeq F(D_\infty).$$

The general enriched internal-hom theorem gives external-product approximation

$$(W \boxtimes W) * ((a, a') \mapsto E_{a, a'} \Pi_{a, a'}) \cong \text{id}_{[X_\infty, Y_\infty]_{\text{ff}}}.$$

It does not generally collapse to a same-index equation, because there is no canonical diagonal

$$A \rightarrow A \otimes A.$$

5.12 Pointed finite-flat minimal invariants

The preceding recursive-domain-equation theorem produces solutions from finite-flat bilimits. We now isolate the pointed form of the construction, which is the finite-flat analogue of the classical pointed-dcpo minimal solution theorem.

The point of pointedness is not that every solution of

$$X \simeq F(X)$$

is unique. That statement is false already in ordinary pointed dcpos: for $F = \text{id}$, every pointed dcpo is a solution. Rather, pointedness supplies a canonical initial approximation process. The resulting bilimit is the canonical generated invariant of F .

Definition 5.12.1 (Pointed fuzzy domain). *A pointed fuzzy domain is a fuzzy domain X equipped with an enriched initial object*

$$\perp_X \in X.$$

Thus

$$X(\perp_X, x) \cong I$$

naturally in $x \in X$.

A morphism of pointed fuzzy domains is a fuzzy-domain morphism

$$f : X \rightarrow Y.$$

It is called strict if

$$f(\perp_X) \cong \perp_Y.$$

Lemma 5.12.1 (Projection pairs between pointed domains are strict). *Let*

$$(e, p) : X \triangleleft Y$$

be a fuzzy projection pair between pointed fuzzy domains. Then both

$$e : X \rightarrow Y \quad p : Y \rightarrow X$$

preserve the chosen initial objects:

$$e(\perp_X) \cong \perp_Y, \quad p(\perp_Y) \cong \perp_X.$$

Proof. Since $e \dashv p$, for every $y \in Y$ there are natural isomorphisms

$$Y(e\perp_X, y) \cong X(\perp_X, py) \cong I.$$

Thus $e\perp_X$ is an enriched initial object of Y , so

$$e(\perp_X) \cong \perp_Y.$$

Using the invertible unit of the projection pair,

$$pe \cong \text{id}_X,$$

we obtain

$$p(\perp_Y) \cong p(e\perp_X) \cong \perp_X.$$

□

Definition 5.12.2 (Pointed projection-pair category). *Let*

$$\mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

denote the locally ordinary 2-category whose objects are pointed fuzzy domains, whose 1-cells are fuzzy projection pairs between pointed fuzzy domains, and whose 2-cells are mate-compatible pairs of enriched natural transformations.

By Lemma 5.12.1, every 1-cell in

$$\mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

is automatically strict on the chosen initial objects.

Definition 5.12.3 (The one-point pointed fuzzy domain). *Let*

$$\mathbf{1}_{\perp}$$

denote the one-object fuzzy domain. Its unique object is written

$$\perp.$$

It is pointed by its unique object.

Lemma 5.12.2 (The one-point domain embeds into every pointed fuzzy domain). *For every pointed fuzzy domain X , there is a canonical fuzzy projection pair*

$$\mathbf{1}_{\perp} \triangleleft X.$$

Its embedding sends the unique object of $\mathbf{1}_{\perp}$ to

$$\perp_X,$$

and its projection is the unique functor

$$X \rightarrow \mathbf{1}_{\perp}.$$

Proof. Let

$$e : \mathbf{1}_{\perp} \rightarrow X$$

send the unique object to $\perp_X$, and let

$$p : X \rightarrow \mathbf{1}_{\perp}$$

be the unique functor. Since $\perp_X$ is enriched initial, for every $x \in X$ one has

$$X(e*, x) = X(\perp_X, x) \cong I \cong \mathbf{1}_{\perp}(*, px).$$

Thus

$$e \dashv p.$$

The unit is the identity, and the counit is the enriched natural transformation

$$\Delta \perp_X \Rightarrow \text{id}_X$$

induced by initiality of $\perp_X$. □

Definition 5.12.4 (Sequential finite-flat approximation shape). *A sequential finite-flat approximation shape is a linear-cofinal finite-flat approximation shape*

$$(\omega_{\mathbb{V}}, W_{\omega})$$

whose support category is the ordinary preorder ω , together with a $\mathbb{V}$ -endofunctor

$$s : \omega_{\mathbb{V}} \rightarrow \omega_{\mathbb{V}}$$

whose action on support objects is

$$n \mapsto n + 1,$$

such that

$$s!W_{\omega} \cong W_{\omega}.$$

Definition 5.12.5 (Sequential finite-flat cofinality). *The fuzzy cosmos $\mathbb{V}$ is sequentially finite-flat cofinal if there is a sequential finite-flat approximation shape*

$$(\omega_{\mathbb{V}}, W_{\omega}, s)$$

such that:

1. s is W_{ω} -cofinal:

$$s!W_{\omega} \cong W_{\omega};$$

2. s is bilimit-cofinal for the pointed sequential projection-pair systems considered below;

3. the canonical tails and canonical external-product common-successor comparisons are cofinal in the senses of Definitions 5.3.3, 5.3.6, and 5.3.7.

Definition 5.12.6 (Pointed projection-pair endopseudofunctor). *A pointed projection-pair endopseudofunctor is a pseudofunctor*

$$F : \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft} \longrightarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}.$$

Definition 5.12.7 (Pointed initial iteration). *Let*

$$F : \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft} \rightarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

be a pointed projection-pair endopseudofunctor.

The pointed initial F -iteration is the sequential split projection-pair system

$$D : \omega_{\mathbb{V}} \rightsquigarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

defined on objects by

$$D_0 = \mathbf{1}_{\perp}, \quad D_{n+1} = F(D_n).$$

The adjacent transition projection pair

$$D_n \triangleleft D_{n+1}$$

is obtained by applying F^n to the canonical projection pair

$$\mathbf{1}_{\perp} \triangleleft F(\mathbf{1}_{\perp}).$$

For $n \leq m$, the transition projection pair

$$D_n \triangleleft D_m$$

is the composite of the adjacent transition projection pairs.

Definition 5.12.8 (Pointed generated invariant). *Let*

$$F : \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft} \rightarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

be a pointed projection-pair endopseudofunctor, and let

$$D : \omega_{\mathbb{V}} \rightsquigarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

be its pointed initial iteration.

A pointed generated invariant of F consists of a pointed fuzzy domain X , an equivalence

$$X \simeq F(X),$$

and a compatible projection-pair cone

$$(\varepsilon_n, \pi_n) : D_n \triangleleft X \quad (n \in \omega_{\mathbb{V}})$$

which exhibits X as the W_{ω} -bilimit of the pointed initial iteration D .

Theorem 5.12.1 (Pointed finite-flat minimal-invariant theorem). *Assume that $\mathbb{V}$ is sequentially finite-flat cofinal. Let*

$$F : \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft} \rightarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

be a pointed projection-pair endopseudofunctor preserving W_{ω} -bilimits of pointed sequential projection-pair systems.

Let

$$D : \omega_{\mathbb{V}} \rightsquigarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

be the pointed initial F -iteration. Suppose D has a W_{ω} -bilimit

$$D_{\infty}.$$

Then there is a canonical equivalence of pointed fuzzy domains

$$D_{\infty} \simeq F(D_{\infty}).$$

Moreover, D_{∞} is unique as a pointed generated invariant: if X is any other pointed generated invariant of F , then there is an equivalence

$$X \simeq D_{\infty}$$

unique up to invertible compatible 2-cell.

Proof. Let

$$s : \omega_{\mathbb{V}} \rightarrow \omega_{\mathbb{V}}$$

be the successor endofunctor. By construction of the pointed initial iteration,

$$Ds \cong FD$$

as split projection-pair systems.

Since s is W_{ω} -cofinal and bilimit-cofinal for D , Theorem 5.5.1 shows that

$$D_{\infty}$$

is a W_ω -bilimit of

$$Ds.$$

Since F preserves W_ω -bilimits,

$$F(D_\infty)$$

is a W_ω -bilimit of

$$FD.$$

Using

$$Ds \cong FD,$$

both D_∞ and $F(D_\infty)$ are W_ω -bilimits of the same system. By uniqueness,

$$D_\infty \simeq F(D_\infty).$$

If X is another pointed generated invariant, then X is also a W_ω -bilimit of the pointed initial iteration D . Hence uniqueness of W_ω -bilimits gives

$$X \simeq D_\infty$$

uniquely up to invertible compatible 2-cell. □

5.13 Pointed finite-flat generated algebraic compactness

The minimal-invariant theorem gives an equivalence

$$D_\infty \simeq F(D_\infty).$$

To express the algebraic-compactness content, we record the associated initial-algebra and final-coalgebra behavior inside the generated projection-pair fragment.

Throughout this section F is assumed to act locally on fuzzy-domain morphisms and to preserve finite-flat colimits on hom fuzzy domains.

Definition 5.13.1 (Locally finite-flat-continuous pointed endopseudofunctor). *A pointed projection-pair endopseudofunctor*

$$F : \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft} \rightarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

is locally finite-flat-continuous if it is the restriction to projection pairs of a pseudofunctor

$$\tilde{F} : \mathbf{FDom}_{\mathbb{V}, \perp}^{(2)} \rightarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{(2)}$$

on the locally ordinary 2-category of pointed fuzzy domains and fuzzy-domain morphisms, and for every pair of pointed fuzzy domains X, Y , the induced hom-action

$$\tilde{F}_{X, Y} : [X, Y]_{\mathbb{F}} \rightarrow [FX, FY]_{\mathbb{F}}$$

preserves finite-flat weighted colimits.

Definition 5.13.2 (Generated projection-pair F -algebra). *Let*

$$F : \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft} \rightarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

be locally finite-flat-continuous, and let

$$D : \omega_{\mathbb{V}} \rightsquigarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

be its pointed initial iteration.

A generated projection-pair F -algebra is a pointed fuzzy domain X equipped with a projection pair

$$a = (a_e, a_p) : FX \triangleleft X.$$

The embedding

$$a_e : FX \rightarrow X$$

is the algebra structure, and the projection

$$a_p : X \rightarrow FX$$

is the associated coalgebra structure.

The projection pair a generates a compatible cone

$$\lambda_n^a : D_n \triangleleft X$$

by recursion:

$$\lambda_0^a : \mathbf{1}_{\perp} \triangleleft X$$

is the canonical projection pair, and

$$\lambda_{n+1}^a = a \circ F(\lambda_n^a).$$

Definition 5.13.3 (Generated algebra and coalgebra categories). The locally ordinary 2-category of generated projection-pair F -algebras is denoted

$$\mathbf{Alg}_{\triangleleft}^{\text{gen}}(F).$$

The same data may be read coalgebraically: the projection half

$$a_p : X \rightarrow FX$$

of

$$a = (a_e, a_p) : FX \triangleleft X$$

is an F -coalgebra structure. With the same objects, but with the coalgebra-half compatibility emphasized, we write

$$\mathbf{Coalg}_{\triangleleft}^{\text{gen}}(F).$$

Construction 5.13.1 (The canonical generated algebra-coalgebra). Let

$$D_{\infty}$$

be the W_{ω} -bilimit of the pointed initial iteration. By Theorem 5.12.1, there is an equivalence

$$D_{\infty} \simeq F(D_{\infty}).$$

Choosing the corresponding projection-pair representative gives a canonical generated projection-pair algebra

$$a_{\infty} = (a_{\infty, e}, a_{\infty, p}) : F(D_{\infty}) \triangleleft D_{\infty}.$$

Lemma 5.13.1 (Generated folds from the bilimit). *Let*

$$a = (a_e, a_p) : FX \triangleleft X$$

be a generated projection-pair F -algebra. Then the generated cone

$$\lambda_n^a : D_n \triangleleft X$$

induces a projection pair

$$\text{fold}_a : D_\infty \triangleleft X.$$

Moreover, its embedding half is an algebra morphism

$$(D_\infty, a_\infty) \rightarrow (X, a),$$

and its projection half is a coalgebra morphism

$$(X, a_p) \rightarrow (D_\infty, a_{\infty, p}).$$

Proof. Since D_∞ represents compatible cones over the pointed initial iteration, the generated compatible cone

$$\lambda_n^a : D_n \triangleleft X$$

is represented by a projection pair

$$\text{fold}_a : D_\infty \triangleleft X.$$

The algebra and coalgebra compatibility equations are checked on finite approximants. At stage $n + 1$, both sides are obtained from stage n by applying F and then composing with a . Detection by the bilimit representation gives the required compatible 2-cells. $\square$

Lemma 5.13.2 (Uniqueness of generated folds). *Let*

$$a = (a_e, a_p) : FX \triangleleft X$$

be a generated projection-pair F -algebra. Any two generated algebra morphisms

$$(D_\infty, a_\infty) \rightarrow (X, a)$$

are uniquely isomorphic by an invertible compatible 2-cell. Dually, any two generated coalgebra morphisms

$$(X, a_p) \rightarrow (D_\infty, a_{\infty, p})$$

are uniquely isomorphic by an invertible compatible 2-cell.

Proof. The restrictions of a generated algebra morphism to finite approximants are forced recursively. At stage 0 they are the canonical map from $\mathbf{1}_\perp$. If the restrictions agree at stage n , algebra compatibility forces agreement at stage $n + 1$. Thus any two such morphisms represent the same compatible cone, and uniqueness follows from the representing property of the bilimit. The coalgebra statement is the mate dual. $\square$

Theorem 5.13.1 (Pointed finite-flat generated algebraic compactness). *Assume that $\mathbb{V}$ is sequentially finite-flat cofinal. Let*

$$F : \mathbf{FDom}_{\mathbb{V}, \perp}^\triangleleft \rightarrow \mathbf{FDom}_{\mathbb{V}, \perp}^\triangleleft$$

be a pointed projection-pair endopseudofunctor which preserves W_ω -bilimits of pointed sequential projection-pair systems and is locally finite-flat-continuous.

Let

$$D : \omega_{\mathbb{V}} \rightsquigarrow \mathbf{FDom}_{\mathbb{V}, \perp}^{\triangleleft}$$

be the pointed initial F -iteration, and suppose it has a W_ω -bilimit

$$D_\infty.$$

Then the canonical projection pair

$$a_\infty = (a_{\infty, e}, a_{\infty, p}) : F(D_\infty) \triangleleft D_\infty$$

exhibits D_∞ as the generated algebraically compact solution of F :

1. the embedding half

$$a_{\infty, e} : F(D_\infty) \rightarrow D_\infty$$

is initial in

$$\mathbf{Alg}_{\triangleleft}^{\text{gen}}(F);$$

2. the projection half

$$a_{\infty, p} : D_\infty \rightarrow F(D_\infty)$$

is final in

$$\mathbf{Coalg}_{\triangleleft}^{\text{gen}}(F);$$

3. the algebra and coalgebra structures are inverse halves of the same projection-pair equivalence

$$F(D_\infty) \triangleleft D_\infty.$$

Proof. Theorem 5.12.1 gives

$$D_\infty \simeq F(D_\infty).$$

Choosing the corresponding projection-pair representative gives

$$a_\infty : F(D_\infty) \triangleleft D_\infty.$$

For any generated projection-pair F -algebra

$$a : FX \triangleleft X,$$

Lemma 5.13.1 gives a generated fold

$$\text{fold}_a : D_\infty \triangleleft X.$$

Its embedding half is an algebra morphism, and its projection half is a coalgebra morphism. Lemma 5.13.2 gives uniqueness up to invertible compatible 2-cell. Hence the embedding half of a_∞ is initial in the generated algebra category, and the projection half is final in the generated coalgebra category.

The two structures are inverse halves of the same projection-pair equivalence by construction. $\square$

5.14 The crisp separated case

For

$$\Lambda = 2,$$

finite-flat weights are ordinary ideals. On separated fuzzy domains, strict algebras for the ideal monad are precisely dcpos, and fuzzy-domain morphisms are precisely Scott-continuous maps.

Let

$$\omega = 0 \leq 1 \leq 2 \leq \dots$$

and let

$$W_\omega : \omega^{\text{op}} \rightarrow 2$$

be the constant-true weight:

$$W_\omega(n) = 1.$$

This is the ordinary directed ideal $\omega \subseteq \omega$.

For a dcpo system

$$D_0 \triangleleft D_1 \triangleleft D_2 \triangleleft \dots$$

of projection pairs, a W_ω -bilimit is exactly the usual bilimit: there are projection pairs

$$(\varepsilon_n, \pi_n) : D_n \triangleleft D_\infty$$

such that the approximants satisfy

$$\bigvee_{n < \omega} \varepsilon_n \pi_n = \text{id}_{D_\infty}$$

in the dcpo of Scott-continuous endomorphisms of D_∞ .

In the crisp case, the ordinary diagonal

$$\Delta_\omega : \omega \rightarrow \omega \times \omega, \quad n \mapsto (n, n),$$

is a monotone map. It is cofinal for the ideal weight because for every pair

$$(m, n) \in \omega \times \omega$$

there is

$$k \geq m, n.$$

Hence

$$(\Delta_\omega)!W_\omega = W_\omega \boxtimes W_\omega.$$

This diagonal is the cartesian feature that permits same-index function-space equations.

Theorem 5.14.1 (Crisp separated same-index function-space collapse). *Let*

$$\Lambda = 2.$$

Let

$$D_0 \triangleleft D_1 \triangleleft D_2 \triangleleft \dots$$

be a sequential dcpo projection-pair system with bilimit D_∞ , and suppose

$$D_{n+1} \cong [D_n, D_n]$$

as projection-pair systems. Then

$$D_\infty \cong [D_\infty, D_\infty].$$

Proof. For $\Lambda = 2$, the finite-flat weight

$$W_\omega : \omega^{\text{op}} \rightarrow 2$$

is the ordinary directed ideal. The product weight

$$W_\omega \boxtimes W_\omega$$

is the ordinary directed ideal on $\omega \times \omega$.

The diagonal

$$\Delta_\omega : \omega \rightarrow \omega \times \omega$$

is cofinal, so restriction along Δ_ω preserves the relevant bilimit. The external function-space system is

$$(m, n) \mapsto [D_n, D_m].$$

Restricting along the diagonal gives

$$n \mapsto [D_n, D_n] \cong D_{n+1}.$$

The successor shift is cofinal, so the shifted chain has the same bilimit D_∞ . On the other hand, the standard bilimit theorem for dcpo projection-pair chains gives that

$$[D_\infty, D_\infty]$$

is the bilimit of the same external function-space system. By uniqueness of bilimits,

$$D_\infty \cong [D_\infty, D_\infty].$$

□

Similarly, for a locally Scott-continuous bilimit-preserving endofunctor T on dcpos, the effectful constructor

$$\Phi_T(D) = [D, TD]$$

gives, after diagonal collapse, the equation

$$D_\infty \cong [D_\infty, TD_\infty].$$

For a bilimit-preserving coeffect functor G , the coeffectful form is

$$D_\infty \cong [GD_\infty, D_\infty],$$

and for mixed effect/coeffect structure,

$$D_\infty \cong [GD_\infty, TD_\infty].$$

In the crisp separated case the Kelly tensor product of fuzzy domains is the cartesian product of dcpos, so the application maps have their usual cartesian types:

$$D_\infty \times D_\infty \rightarrow D_\infty,$$

$$D_\infty \times D_\infty \rightarrow TD_\infty,$$

and

$$D_\infty \times GD_\infty \rightarrow TD_\infty$$

in the pure, effectful, and mixed cases respectively.

Proposition 5.14.1 (Crisp separated pointed minimal invariants). *For*

$$\Lambda = 2,$$

the pointed finite-flat minimal-invariant and generated algebraic-compactness theorems specialize to the classical pointed-dcpo bilimit construction.

More explicitly:

1. *pointed fuzzy domains are precisely pointed dcpos;*
2. *fuzzy projection pairs are precisely embedding-projection pairs*

$$e : D \rightarrow E, \quad p : E \rightarrow D, \quad pe = \text{id}_D, \quad ep \leq \text{id}_E;$$

3. *the finite-flat weight*

$$W_\omega : \omega^{\text{op}} \rightarrow 2$$

is the ordinary directed ideal on ω ;

4. *the W_ω -bilimit approximation identity becomes*

$$\bigvee_{n < \omega} \varepsilon_n \pi_n = \text{id}_{D_\infty}$$

in the dcpo of Scott-continuous endomorphisms of D_∞ ;

5. *a locally finite-flat-continuous endopseudofunctor becomes a locally Scott-continuous endofunctor on the projection-pair category of pointed dcpos;*

6. *the canonical projection pair*

$$F(D_\infty) \triangleleft D_\infty$$

gives the generated coincidence of the initial algebra and final coalgebra.

Proof. For $\Lambda = 2$, finite-flat weights are ordinary ideals and strict algebras for the ideal monad on separated preorders are precisely dcpos. An enriched initial object is exactly a least element, so pointed fuzzy domains are pointed dcpos.

A fuzzy projection pair

$$(e, p) : D \triangleleft E$$

is an adjunction

$$e \dashv p$$

with invertible unit. In order-enriched notation this is precisely

$$pe = \text{id}_D, \quad ep \leq \text{id}_E.$$

Thus projection pairs are the usual embedding-projection pairs.

The weight

$$W_\omega : \omega^{\text{op}} \rightarrow 2$$

is constant true, hence is the ordinary directed ideal on ω . Therefore a W_ω -weighted colimit in an internal hom dcpo is an ordinary directed join. The approximation identity

$$W_\omega * (n \mapsto \varepsilon_n \pi_n) \cong \text{id}_{D_\infty}$$

therefore becomes

$$\bigvee_{n < \omega} \varepsilon_n \pi_n = \text{id}_{D_\infty}.$$

The generated projection-pair algebra

$$F(D_\infty) \triangleleft D_\infty$$

has embedding half

$$F(D_\infty) \rightarrow D_\infty$$

and projection half

$$D_\infty \rightarrow F(D_\infty).$$

These are the classical inverse algebra and coalgebra maps obtained from the bilimit of the initial embedding-projection chain. $\square$

5.15 Summary

Recursive domain equations in fuzzy domains are governed by the finite-flat doctrine. The primitive approximation objects are finite-flat approximation shapes

$$(A, W), \quad W \in \text{Idl}_V(A).$$

Cofinality of weights is expressed by lower direct image:

$$h : (A, W) \rightarrow (B, U) \iff h_! W \cong U.$$

An iteration shape is a finite-flat approximation shape equipped with a weight-cofinal self-map

$$s : A \rightarrow A, \quad s_! W \cong W.$$

Tails are canonical support-comma categories

$$\text{Tail}_A(a_0),$$

equipped with their canonical projection

$$\tau_{a_0} : \text{Tail}_A(a_0) \rightarrow A.$$

Tail-cofinality says that

$$\tau_{a_0}^* W$$

is finite-flat and

$$(\tau_{a_0})_! \tau_{a_0}^* W \cong W.$$

Common successors are encoded by

$$\text{Com}(P),$$

whose objects are tuples

$$(i, j, k, u : i \rightarrow k, v : j \rightarrow k).$$

The canonical maps

$$c_P : \text{Com}(P) \rightarrow P \otimes P, \quad m_P : \text{Com}(P) \rightarrow P$$

are cofinal when

$$(c_P)_! m_P^* R \cong R \boxtimes R, \quad (m_P)_! m_P^* R \cong R.$$

For bilimits, weight-cofinality is paired with bilimit-cofinality: restriction along the indexing map must preserve compatible projection-pair cone categories. With this distinction in place, a split projection-pair system

$$D : A \rightsquigarrow \mathbf{FDom}_V^{\triangleleft}$$

satisfying

$$Ds \cong FD$$

has a generated solution whenever D has a W -bilimit, s is bilimit-cofinal for D , and F preserves that bilimit:

$$D_\infty \simeq F(D_\infty).$$

The general internal-hom theorem is external-product-indexed. If

$$X_\infty \quad \text{and} \quad Y_\infty$$

are W -bilimits of split systems X and Y , then

$$[X_\infty, Y_\infty]_{\text{ff}}$$

has external-product approximants

$$(a, a') \longmapsto [X_{a'}, Y_a]_{\text{ff}},$$

and these satisfy

$$(W \boxtimes W) * ((a, a') \mapsto E_{a, a'} \Pi_{a, a'}) \cong \text{id}_{[X_\infty, Y_\infty]_{\text{ff}}}.$$

Compatible cones over the external-product system reconstruct canonically to lax application data. They do not generally reconstruct to projection pairs, because the common-successor comparison supplies

$$\rho\delta \Rightarrow \text{id}$$

rather than an invertible unit

$$\text{id} \Rightarrow \rho\delta.$$

This is exactly the resource-sensitive obstruction caused by the absence of a canonical diagonal

$$A \rightarrow A \otimes A.$$

In the pointed case, a sequential finite-flat approximation shape

$$(\omega_{\mathbb{V}}, W_\omega, s)$$

replaces the ordinary chain

$$0 \leq 1 \leq 2 \leq \cdots.$$

The pointed initial iteration

$$D_0 = \mathbf{1}_\perp, \quad D_{n+1} = F(D_n)$$

has a W_ω -bilimit D_∞ . If F preserves this bilimit, then

$$D_\infty \simeq F(D_\infty).$$

If F is locally finite-flat-continuous, the resulting projection pair

$$F(D_\infty) \triangleleft D_\infty$$

is the generated algebraically compact solution: its embedding half is the initial generated algebra, its projection half is the final generated coalgebra, and the two structures are inverse halves of the same projection-pair equivalence.

In the crisp separated case, taking the ordinary ideal

$$W_\omega = \omega \subseteq \omega$$

recovers the standard bilimit method for solving recursive domain equations of dcpos . The ordinary reflexive-domain equation

$$D_\infty \cong [D_\infty, D_\infty]$$

is recovered precisely because the cartesian diagonal

$$\omega \rightarrow \omega \times \omega$$

collapses the external-product function-space approximation to a same-index chain.

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